

# **Health Data Science: AI and Knowledge Translation**

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# The Project

Welcome to the course notes for **Health Data Science: AI and Knowledge Translation**.

This platform bridges the gap between traditional epidemiology and modern data science. Unlike standard textbooks that focus on a single language, this resource adopts a **“Polyglot”** approach—teaching R and Python side-by-side—and integrates **Artificial Intelligence** directly into the workflow.

Whether you are a health researcher looking to modernize your skills or a data scientist seeking to understand epidemiological rigor, you are in the right place.

Here, we offer:

- **“Zero to AI Co-Pilot” Training:** Learn to use Large Language Models (LLMs) to generate, debug, and translate code while critically auditing them for bias and hallucinations.
- **Real-World Data:** Guides on navigating “messy” open data sources such as NHANES and WHO, rather than sanitized “toy” datasets.
- **Knowledge Translation:** Step-by-step walkthroughs on building interactive dashboards (Shiny/Dash) and dynamic reports to communicate findings to policymakers.

## What We Aim to Achieve

We are on a mission to:

- **Democratize Compute Access:** Eliminate hardware barriers by using cloud-native environments (Google Colab, Kaggle) that run on any device.

- **Bridge the “Polyglot” Gap:** Provide “Rosetta Stone” tutorials where concepts are implemented in both R and Python, allowing learners to be fluent in both.
- **Teach “AI-Resilience”:** Shift the focus from manual syntax generation to **rigorous code auditing**, security validation, and reproducibility.
- **Enable Professional Portfolios:** Help students transform static analyses into public-facing data science portfolios.

## Dive into Our Modules

Embark on a journey through our structured modules, designed to take you from data literacy to advanced AI workflows.

### Note

The tutorials are designed with a consistent structure to provide a cohesive learning experience. Here is what you can expect in each chapter:

1. **Overview:** A concise summary of learning objectives and topics.
2. **Concepts:** Core theoretical materials (slides, video lessons).
3. **Polyglot Tutorials:** In-depth walkthroughs with tabbed code blocks allowing you to toggle between **R** and **Python** solutions.
4. **AI Auditing:** Specific sections dedicated to verifying and “stress-testing” AI-generated code for the chapter’s topic.
5. **Knowledge Check:** Ungraded self-assessments to reinforce key concepts.
6. **Practice Exercises:** Hands-on tasks using real-world health data to build your portfolio.

## How Our Content is Presented

All our resources are hosted on an easy-to-access GitHub page. The format? Engaging text, reproducible code, clear outputs, and crisp videos that distill complex topics.

This document is created using [Quarto](#), allowing us to weave analysis, code, and interpretation into a single reproducible document.

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## How to Cite

The BibTeX format can be downloaded from [here](#).



# **Course Operations**

# Course Operations

This chapter is the single maintainer-facing home for course planning, assistant guidance, and release-readiness checks. Use it before editing the book, preparing a week for release, or asking an instructional reviewer to review course materials.

## Operating Principles

- Treat the syllabus as the source of truth for weeks, deliverables, and deadlines.
- Keep student-facing content beginner-accessible and cloud-first.
- Treat the fork-based repository workflow as the official student path unless Canvas explicitly announces an instructor-managed alternative.
- Prefer reproducible Quarto, R, Python, and GitHub Codespaces workflows over local-machine assumptions.
- Keep generated output separate from source content.
- Add content where the normalized structure says it belongs: weeks, assignments, milestones, references, examples, or assets.

## Editing Rules

- Update `_quarto.yml` whenever a page should appear in the book.
- Keep paths relative and test them after moving files.
- Use `assets/` for shared images and data that support teaching pages.
- Use `examples/` for runnable code and sample datasets.

- Do not bury assignment requirements only inside the syllabus; add a student-facing assignment or milestone brief.

## Reproducibility Expectations

- Every activity should state expected inputs, outputs, and submission location.
- Every code activity should run in a fresh GitHub Codespace.
- Every dataset activity should record source, license, access date, and limitations.
- Every AI-assisted activity should include an audit note: what AI helped with, what changed, and how the result was verified.

## Review Cadence

- Before term: verify links, screenshots, Codespace setup, dependencies, and all assignment briefs.
- Weekly: update troubleshooting notes based on student issues.
- Before each milestone: test the milestone instructions from a fresh repository.
- After term: archive changes, refresh screenshots, and update the completion matrix.

## Done Checklists

### Course Page

- The page appears in `_quarto.yml` if it is student-facing.
- The title matches the week or reference purpose.
- Learning goals, activity steps, and expected outputs are clear.
- Links and image paths render from the Quarto project root.
- Open work is clearly scoped and assigned.

## Assignment

- The assignment has a student-facing brief.
- Required files and submission location are explicit.
- The work can be completed in GitHub Codespaces.
- Packages, data, and outputs are documented.
- The grading checklist matches the syllabus assessment logic.
- A1-A3 use Complete/Incomplete language; A4-A10 use the best-5-of-7 scored assignment structure.
- Every assignment brief states the Canvas repository-link submission route and AI-use expectation.
- Assignment 1 and Week 2 should teach the default sequence: open template repository, fork, launch Codespaces from the fork, commit and sync, then submit the fork link on Canvas.

## Milestone

- The milestone states the exact artifact students submit.
- The milestone maps to the final portfolio.
- The rubric includes reproducibility, auditability, data stewardship, and KT value.
- Peer review and instructional review instructions are actionable.

## Reproducibility

- A fresh Codespace can render or run the project.
- Paths are relative.
- Dependencies are declared with `renv`, `requirements.txt`, or a clearly documented equivalent.
- No required data exists only on one person's local computer.
- Randomness uses a seed when results depend on it.
- The README explains how to reproduce the output.

## AI Audit

- The AI-use note says what tool was used and for what task.
- Generated code was checked against real column names and expected outputs.
- Generated summaries are grounded in actual results.
- Security, privacy, and bias risks are considered.
- The final text does not cite sources invented by AI.

## NHANES Case Study

The NHANES Health Equity Dashboard case study is the course spine. Students revisit the same app, data, and workflow from different angles as their skills mature. It should not become a second term project.

## Implemented Structure

- `examples/nhanes-equity/app/app.R`
- `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- `examples/nhanes-equity/scripts/build_nhanes_equity.R`
- `examples/nhanes-equity/snippets/case-study-data-only.qmd`
- `examples/nhanes-equity/snippets/case-study-data-analysis.qmd`
- `examples/nhanes-equity/snippets/case-study-dashboard.qmd`
- `examples/nhanes-equity/README.md`

The cached data live in `examples/nhanes-equity/data/` so the app, script, README, and weekly snippets all use one consistent location.

## Editor Decisions

- The NHANES sequence is data-first through Week 7 and app-forward starting in Week 8.

- The official setup path is fork-first. GitHub Classroom is not the default in this version of the course book; if used in a future offering, it should be labeled as an instructor-managed alternative in Canvas and in any local handout.
- Week 8's required pathway is local and Codespaces-friendly: run the Shiny exemplar, make one audience-facing text or documentation improvement, and record a test result.
- Static dashboard and deployment paths are optional demonstrations unless a later assignment explicitly asks for them.
- Weeks 6-8 now have complete exemplar packs; instructional reviewers use the provided keys and briefs for facilitation rather than drafting new NHANES materials.

## Teaching Progression

- Weeks 1-2: provenance, ethics, folder structure, relative paths, and rendering; no app run instructions.
- Weeks 3-5: data loading, missingness checks, AI audit comments, documentation-only Git changes, and R/Python parity checks using the CSV snapshot.
- Weeks 6-7: visualization critique, flawed-plot audit, EDA, Table 1, and planted-error audit.
- Week 8: first app-forward week, with the Shiny dashboard exemplar used for a small KT adaptation.
- Weeks 9-10: slides, reports, source notes, limitations, and audience-specific communication.
- Week 11: fresh-session reproducibility testing and optional rebuild from CDC if internet access is available.

## RA Watchpoints

- Students may confuse the cached case-study dataset with raw NHANES microdata.
- The app's Download CSV button exports the currently summarized table output, not raw NHANES microdata.
- The CDC rebuild path requires internet access and the `nhanesA` package.

- Week pages should include the shared snippet rather than duplicating instructions.
- Weeks 1-2 should not instruct students to run the app.
- Week 8 is the first week where app running is central.
- Student changes should stay small unless the instructor explicitly assigns a dashboard feature.
- Relative paths are part of the lesson; discourage absolute machine-specific paths.
- Weeks 6-8 should be treated as complete NHANES examples; updates should maintain the existing worked examples, activities, answer keys, and assignment briefs.

## **Case Study Done**

- The book renders from the repo root.
- Weeks 1-11 each include a short case-study touchpoint.
- The case-study folder contains runnable app code, cached data, rebuild script, shared snippet, and README.
- Weeks 6-7 include worked examples, student-facing activities, and instructor-only keys.
- Students can complete the default classroom path without internet.
- CDC rebuilds are documented as optional, network-dependent reproducibility work.

# Syllabus

## ## Acknowledgement

UBC's Point Grey Campus is located on the traditional, ancestral, and unceded territory of the x mə k əyəm (Musqueam) people. The land it is situated on has always been a place of learning for the Musqueam people, who for millennia have passed on their culture, history, and traditions from one generation to the next on this site.

## Course Information

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|                      |                                                                    |
|----------------------|--------------------------------------------------------------------|
| <b>Course Title:</b> | Health Data Science: AI and Knowledge Translation                  |
| <b>Credit Value:</b> | 3                                                                  |
| <b>Format:</b>       | Integrated Seminar & Hands-on Activities (1 session/week, 3 hours) |
| <b>Time:</b>         | Thu 11:00 AM - 2:00 PM (Sept-Dec 2026)                             |
| <b>Room:</b>         | Room posted in Canvas (in-person)                                  |

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## Course at a Glance

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|                                 |                                                                                                         |
|---------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Core workspace</b>           | GitHub Codespaces (GitHub, Quarto, & VS Code in the browser)                                            |
| <b>Core language</b>            | R, Quarto, and reproducible project workflows                                                           |
| <b>Secondary exposure</b>       | Short Python examples for polyglot awareness; optional extension pathway                                |
| <b>Main project product</b>     | Reproducible Quarto report + required dashboard-style KT product + auditable repository                 |
| <b>Primary assessment logic</b> | Foundational gates, best-5 weekly assignments, staged project milestones, presentation, final portfolio |

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## Prerequisites

**None required.** This course adopts a “Zero to AI Co-Pilot” pedagogy. We welcome students from all disciplines. You do not need prior experience with R, Python, statistics, or programming. We teach foundational concepts and tools from the ground up.

### What “no prerequisites” means in practice

This is a beginner-accessible course, not a no-work course. You will learn a substantial but carefully scaffolded toolkit at a steady pace: R, GitHub Codespaces, Quarto, Git, reproducible workflows, AI-assisted coding, basic Python awareness, and dashboard-style knowledge translation. Because coding is a cumulative skill, the weekly assignments are carefully scaffolded to serve as your core practice. If you are new to programming, completing these assignments is how you will steadily build your foundational skills during the first month. Comfort with web browsers, basic file management, and willingness to troubleshoot are assumed; everything else is taught and supported through templates, in-class studio time, peer debugging, and TA support.

## Primary Language: R, with Python Awareness

This course uses **R** as the primary programming language for all core graded work. R is widely used in epidemiology, biostatistics, public health, and reproducible research, and the tidyverse ecosystem (`dplyr`, `ggplot2`, `plotly`, Quarto, and dashboard-style reporting tools) provides a coherent pipeline from data import to publication.

The course remains **polyglot-aware** without becoming a full second programming course. Students will read and run short **Python/pandas** examples in GitHub Codespaces, compare Python and R solutions with AI assistance, and learn when Python may be the more appropriate tool. A Python-flavoured option may be offered for one activity or project extension, but students can complete all required outcomes in R.

Students will create a small dashboard-style knowledge translation product as part of the portfolio. The required core will focus on approaches that work reliably in GitHub Codespaces and Quarto-based publishing. Shiny, Shinylive, and AI-generated static HTML/JavaScript dashboards may be demonstrated as optional or advanced deployment pathways rather than required technologies for every student.

## Computing Environment: Fork-Based Workflow

All supported coding takes place in **GitHub Codespaces**, a cloud computer accessible from any modern web browser. No local installation is required. Students **fork** a course template repository into their own GitHub account, then launch Codespaces from their fork.

Tools required for graded work will be selected and tested for use in Codespaces. Optional demonstrations may show additional deployment pathways, but students will not be penalized if an advanced demo technology is unsuitable for their device or project.

## Contacts

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|                |                                                                 |
|----------------|-----------------------------------------------------------------|
| <b>Role:</b>   | Course Instructor                                               |
| <b>Name:</b>   | Dr. M. Ehsan Karim                                              |
| <b>Email:</b>  | ehsan.karim@ubc.ca                                              |
| <b>Office:</b> | Appointments by request                                         |
| <b>Hours:</b>  | Weekly instructor and TA support times will be posted on Canvas |

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*Typical response time: within 48 hours on weekdays.*

**Other Instructional Staff:** TBA. To communicate with the computing TA, attend weekly classes and specified drop-in hours (as announced in Canvas).

## **Instructor Biographical Statement**

Dr. M. Ehsan Karim is an Associate Professor in Health Data Science at the UBC School of Population and Public Health (SPPH), a Scientist at the Centre for Advancing Health Outcomes, an associate member of the Department of Statistics (UBC), and a previous Michael Smith Foundation for Health Research (MSFHR) Scholar. He obtained his PhD in Statistics from the University of British Columbia and completed his postgraduate training in the Department of Epidemiology, Biostatistics, and Occupational Health at McGill University. His current research focuses on causal inference, real-world observational data analyses, and applications of machine learning approaches in epidemiologic studies.

## **Course Description**

This interdisciplinary course bridges traditional epidemiology, modern data science, responsible AI use, and knowledge translation. Using a “Zero to AI Co-Pilot” pedagogy, students learn how to work with health-related data in a cloud-based environment, use generative AI as a coding assistant, and critically audit AI-generated code and outputs. The course is beginner-accessible and assumes no prior coding or statistics background, while still requiring steady hands-on practice.

The curriculum emphasizes reproducible research and the “last mile” of knowledge translation: transforming real-world health data into clear, auditable, and audience-appropriate products. By the end of the course, students will have built a portfolio that includes documented code, a reproducible Quarto report, clear visualizations, a dashboard-style knowledge translation product, and a reflection on AI use and auditability.

## **Course Structure**

This course integrates theoretical foundations with the practical realities of modern health data science. Through short

seminars, guided setup time, hands-on studio work, and structured peer support, you will move from basic data literacy to reproducible, portfolio-ready communication products: reports, dashboards, visual stories, and project repositories that can be audited by others.

## Course Pillars

1. **The Full Data Lifecycle:** From ethical data access and cleaning to secure cloud computing and final publication.
2. **Reproducible Science:** Git-based workflows, environment management (`renv`), and literate programming in Quarto - ensuring analyses are transparent, shareable, and robust.
3. **AI-Augmented Workflows:** Ethically leveraging LLMs as coding co-pilots to debug, refactor, and translate code, while auditing AI-generated outputs for coding errors, hallucinated claims, security risks, health bias, and reproducibility failures.
4. **Knowledge Translation (KT):** The “last mile” of data science - transforming results into dashboard-style products, visual stories, plain-language summaries, and policy-ready reports. Shiny and static web approaches are introduced as optional or advanced pathways rather than assumed for every project.
5. **Real-World Health Data:** Hands-on activities using messy, real-world datasets (e.g., NHANES, Canadian PUMFs) rather than toy examples.

## Session Structure (3-Hour Block)

Each weekly session generally follows the following segments:

| Segment                          | Duration | Description                                           |
|----------------------------------|----------|-------------------------------------------------------|
| <b>Seminar and demonstration</b> | 45 min   | Concepts, examples, and live coding by the instructor |

| Segment                        | Duration | Description                                                                                    |
|--------------------------------|----------|------------------------------------------------------------------------------------------------|
| <b>Guided setup</b>            | 20 min   | Everyone completes the first step of the activity together; TAs confirm students are unblocked |
| <b>Break / technical reset</b> | 10 min   | Short break and time to resolve login, rendering, or package issues                            |
| <b>Hands-on studio</b>         | 90 min   | Independent or group work on the weekly activity; TA circulation and peer debugging            |
| <b>Wrap-up and next steps</b>  | 15 min   | Expected output demo, common errors, submission expectations, and preview of next week         |

## Peer Support: Debugging Partners

To scale support with high enrollment, each student is assigned a **debugging partner**. Before raising a hand for TA help, try your partner first. This builds collaboration skills and reduces wait times for everyone.

## Asynchronous Onboarding (Week 0)

**Due before the first class session (Week 1).**

A short Canvas module to complete before the term begins:

1. Create a **GitHub account** at [github.com](https://github.com) using your UBC email.
2. Choose a **professional username** (this will appear on your public portfolio).

3. Watch a **5-minute orientation video** (posted on Canvas) introducing the course workflow.
4. Complete the **pre-course survey** (anonymous; helps the teaching team calibrate to the cohort's baseline experience).

*This takes approximately 15-20 minutes and ensures Week 1 can focus entirely on health data concepts rather than account setup.*

## Schedule of Topics

| Week                                       | Seminar Topics                                                                                                                                                                                                                               | Activities                                                                                                                                                               | Deadlines                   |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Part 1: Foundations &amp; Workflows</b> |                                                                                                                                                                                                                                              |                                                                                                                                                                          |                             |
| <b>0 (Async)</b>                           | -                                                                                                                                                                                                                                            | Create GitHub account; watch orientation video; pre-course survey.                                                                                                       | Complete before first class |
| <b>1 (Sept 10)</b>                         | Health Data, KT & Ethics: research lifecycle overview; KT 'last mile'; privacy/security; Indigenous Data Sovereignty (OCAP, CARE); AI hallucinations & bias (AI fluency does not equal correctness).                                         | Explore open data portals; complete Data Intake Card; discuss data formats, provenance, stewardship, and responsible AI use; course workflow overview (Canvas + GitHub). | (No graded submission)      |
| <b>2 (Sept 17)</b>                         | Modern Workflows: IDE vs notebook vs script; literate programming; reproducible project structure; Codespaces as the shared lab bench; Quarto basics.                                                                                        | Fork course template repo into personal GitHub account; launch Codespace; tour VS Code; edit + render Quarto notebook; stage, commit, sync; stop Codespace.              | Assignment 1 (Mon 4 PM)     |
| <b>3 (Sept 24)</b>                         | R with AI (Code runs does not mean code is correct): data import/export; core constructs (objects, functions, control flow); the tidyverse pipeline; debugging with LLMs; writing custom functions; intro to renv for dependency management. | Run R in Codespace; load dataset; inspect & transform with dplyr; write a custom function; initialize renv; document in Quarto; commit incrementally.                    | Assignment 2 (Mon 4 PM)     |

(continued)

| Week                                                    | Seminar Topics                                                                                                                                                                                                                                       | Activities                                                                                                                                                                                         | Deadlines                                                  |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 4 (Oct 1)                                               | Git & Collaboration: commits as analysis provenance; branching/merging; Pull Requests & peer review mindset; repo hygiene (.gitignore, relative paths).                                                                                              | Git basics via GUI; meaningful commits; .gitignore & relative paths; create branch, open Pull Request; one member forks group project template, adds teammates as collaborators.                   | Assignment 3 (Mon 4 PM); Milestone 0: form project group   |
| 5 (Oct 8)                                               | Polyglot Awareness & R Deepening: R as the core course language; when Python is useful; reading short Python/pandas examples; translation traps; AI-assisted parity checks; R mastery exercises on dplyr and functions.                              | Read and run a short Python/pandas notebook in Codespaces; compare with an R/tidyverse solution; use AI to translate a small task; verify parity with checks; complete R deepening exercises.      | Assignment 4 (Mon 4 PM)                                    |
| <b>Part 2: Analysis, Visualization &amp; Dashboards</b> |                                                                                                                                                                                                                                                      |                                                                                                                                                                                                    |                                                            |
| 6 (Oct 15)                                              | Data Visualization: visual reasoning in AI-assisted workflows; steering AI-generated plots (ggplot2, plotly); evaluating clarity, bias, aggregation, and misrepresentation.                                                                          | Generate & refine visualizations with AI (ggplot2/plotly); evaluate AI-generated plots for clarity, bias, aggregation choices, and misleading design; embed visuals in Quarto with captions.       | Assignment 5 (Mon 4 PM); Milestone 1: project proposal     |
| 7 (Oct 22)                                              | EDA, Table 1 & AI Auditing: descriptive analysis workflow; what summary statistics claim; auditing AI-generated summaries/plots for planted coding, statistical, and stewardship errors; revisiting data provenance and Indigenous Data Sovereignty. | Perform EDA on project data; create Table 1; complete an AI audit exercise with planted errors; add or revise a data provenance and stewardship statement; checkpoint commits.                     | Assignment 6 (Mon 4 PM); Milestone 2: Preliminary Analysis |
| 8 (Oct 29)                                              | Dashboard Prototypes for KT: Quarto dashboard-style products and Shiny concepts; building a minimal dashboard from EDA outputs; deployment options introduced as demonstrations only (e.g., Shinylive/GitHub Pages).                                 | Build a minimal dashboard-style KT product from EDA outputs; test in Codespaces; view optional Shiny/Shinylive deployment demo; identify which dashboard pathway is feasible for the term project. | Assignment 7 (Mon 4 PM)                                    |

(continued)

| Week                                                         | Seminar Topics                                                                                                                                                                                                                                                          | Activities                                                                                                                                                                                      | Deadlines                                                         |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>9 (Nov 5)</b>                                             | Communication of Scientific Findings: audience-specific KT; plain language summaries; Quarto revealjs slides; citations & references; peer review practices.                                                                                                            | Create Quarto revealjs slides (guided intro to slide format); integrate figures; structured peer feedback on clarity & KT.                                                                      | Assignment 8 (Mon 4 PM)                                           |
| <b>Part 3: Communication, Portfolios &amp; Presentations</b> |                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                 |                                                                   |
| <b>10 (Nov 12)</b>                                           | Writing & Publishing Reports: scientific formatting in Quarto; dynamic reporting; embedding interactive elements; enabling GitHub Pages; publishing your portfolio; citation/grounding audit. Note: UBC Mid-Term Break is Nov 9-11; class meets as usual on Thu Nov 12. | Draft report in Quarto; add citations (Zotero/BibTeX); embed visuals; enable GitHub Pages and publish portfolio site (or private equivalent).                                                   | Assignment 9 and Milestone 3: Project Update due Mon Nov 16, 4 PM |
| <b>11 (Nov 19)</b>                                           | Portfolio Surgery & Optional Advanced Deployment: reproducibility stress-test in a fresh Codespace; fix broken paths, missing dependencies, unclear READMEs; optional demo of AI-generated static HTML/JS dashboards and optional Python-flavoured project pathway.     | Swap repositories with a partner for reproducibility stress-test; run in a fresh Codespace; fix failures; optional advanced demo comparing Shiny, Quarto dashboard, and static HTML approaches. | Assignment 10 and Milestone 4: Peer Review due Mon Nov 23, 4 PM   |
| <b>12 (Nov 26)</b>                                           | Term Project Presentations (Part 1): live Q&A and peer review.                                                                                                                                                                                                          | Milestone 5 presentations; live Q&A; peer review feedback.                                                                                                                                      | Presentation slides due before class                              |
| <b>13 (Dec 3)</b>                                            | Term Project Presentations (Part 2): live Q&A and peer review.                                                                                                                                                                                                          | Milestone 5 presentations; live Q&A; peer review feedback.                                                                                                                                      | Presentation slides due before class                              |
| -                                                            | -                                                                                                                                                                                                                                                                       | -                                                                                                                                                                                               | Final Portfolio due: Dec 11 (Fri 4 PM)                            |

## Learning Outcomes

By the end of this course, students will be able to:

1. Explain key ethical, privacy, security, Indigenous Data Sovereignty, and knowledge translation issues in health



data science, including responsible use of generative AI tools.

2. Use GitHub Codespaces, Quarto, Git, and GitHub to organize, document, version, and reproduce a small health data analysis.
3. Import, clean, summarize, and visualize health-related data using R and the tidyverse.
4. Read and compare short Python/pandas examples with AI assistance, identify common translation traps between R and Python, and explain when Python may be an appropriate tool.
5. Use generative AI as a coding assistant while critically auditing AI-generated code and outputs for coding errors, hallucinated claims, bias, security risks, and reproducibility problems.
6. Conduct a bounded exploratory data analysis and communicate descriptive findings using clear tables, figures, narrative interpretation, and appropriate limitations.
7. Create a dashboard-style knowledge translation product and a reproducible Quarto report that translate health data findings for a defined audience.
8. Manage reproducible software environments (`renv`) and project documentation so that analyses can be rerun in a fresh Codespace.
9. Provide constructive peer feedback on the clarity, reproducibility, auditability, and knowledge translation value of another project.

## Learning Activities

### Weekly Assignments (Weeks 2-11)

Each week, students complete short applied assignments that reinforce the tools and concepts from class. These assignments are scaffolded and practice-oriented, but they also provide an important measure of steady engagement and skill development. Unless otherwise noted, assignments are due the Monday after class at 4 PM, giving students time to revisit the activity,

troubleshoot technical issues, and use office hours or discussion boards.

### Grading policy:

- **Assignments 1-3 (Weeks 2-4) are required but worth 0% numerically.** These foundational assignments (Codespace/Quarto, R basics, Git) are graded on a Complete/Incomplete basis. Each must be completed to pass the course. Incomplete submissions may be resubmitted once within one week. A student who completes A1-A3 and otherwise earns at least 50% on the graded components will pass.
- **Assignments 4-10 (Weeks 5-11): Best 5 of 7 count.** Only your best 5 scores count toward your final grade. This provides flexibility for difficult weeks while ensuring consistent engagement after the foundation is established. Assignment 6 includes a dedicated AI-audit task involving planted errors in AI-generated code, summaries, and/or visualizations.

### Reproducibility requirement

All assignments must render/run in a GitHub Codespace without manual intervention. Assignments that violate this rule will be returned for a "Reproducibility Fix" and may incur a grade deduction.

## Term Project / Data Science Portfolio

A capstone project synthesized across the full term, broken into milestones:

| Milestone                  | Due             | Description                                                                                       |
|----------------------------|-----------------|---------------------------------------------------------------------------------------------------|
| <b>M0: Group Formation</b> | Week 4<br>(Mon) | Form project group of 3-4 students; one member forks template and adds teammates as collaborators |

| Milestone                       | Due                     | Description                                                                                                                |
|---------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>M1: Proposal</b>             | Week 6<br>(Mon)         | Research question, dataset selection, feasibility check, and one-paragraph data provenance/stewardship statement           |
| <b>M2: Preliminary Analysis</b> | Week 7<br>(Mon)         | Initial data cleaning, exploratory visualizations, Table 1 or equivalent descriptive summary, runnable repository          |
| <b>M3: Project Update</b>       | Week 10<br>(Mon Nov 16) | Draft dashboard-style KT product + report draft submitted for peer review; polish is expected later in the Final Portfolio |
| <b>M4: Peer Review</b>          | Week 11<br>(Mon Nov 23) | Constructive, specific, actionable feedback on a peer's M3, including reproducibility and auditability checks              |
| <b>M5: Presentation</b>         | Weeks 12-13             | In-class scientific seminar; knowledge translation focus                                                                   |
| <b>Final Portfolio</b>          | Dec 11 (Fri)            | Complete report, required dashboard-style KT product, and fully auditable code repository                                  |

### Analytic Scope of the Term Project

The project is graded on the data lifecycle, reproducibility, visualization clarity, AI auditability, and knowledge translation - not on the sophistication of statistical inference. Descriptive summaries, well-chosen visualizations, clear documentation of data decisions, and clearly stated limitations are expected. Formal hypothesis tests, regression models, predictive models, or causal claims are out of scope unless agreed with the instructor in the proposal.

The required knowledge translation product is a small dashboard-style product suitable for the chosen audience. This may be implemented through Quarto dashboard-style reporting, R-generated interactive elements, or another instructor-approved pathway that runs reliably in GitHub Codespaces. Advanced deployment options such as Shiny hosting, Shinylive, or AI-generated HTML/JavaScript may be demonstrated, but are not required for every student.

## Reproducibility Contract

### The Reproducibility Checklist

A significant portion of the final grade depends on reproducibility. The following checklist serves as both a learning guide and the grading rubric for the Final Portfolio. Refer to it every week:

1. **Fresh Codespace test:** The project launches and runs start-to-finish in a newly created GitHub Codespace without manual intervention.
2. **Relative paths only:** No absolute paths anywhere in the codebase.
3. **Dependencies declared:** All R packages recorded via `renv` (introduced in Week 3, reinforced throughout).
4. **No manual steps:** The pipeline from raw data to final output is fully scripted.

5. **README quality:** README.md explains the research question, data provenance, how to reproduce the analysis, and known limitations.
6. **.gitignore configured:** Rendered outputs, OS junk, and sensitive files are excluded from version control.
7. **Commit history:** Incremental, meaningful commits throughout the term.
8. **Documentation:** Code is commented; Quarto documents interleave code with interpretive narrative.
9. **Dashboard-style KT product works:** The project includes a small dashboard-style knowledge translation product that renders or runs in the supported course workflow. If the portfolio is public, it should be viewable through GitHub Pages or another instructor-approved route; if the project is private, the teaching team must be able to render or preview it in a fresh Codespace.

*This contract is introduced in Week 1 and reinforced in every subsequent week.*

## Assessments of Learning

### Scope & Expectations

- **Focus:** This project prioritizes descriptive health data analysis, visualization, dashboard-style knowledge translation, reproducible reporting, and AI auditability.
- **Audience:** SPPH 381H is an undergraduate course. No separate graduate-level requirements are assumed unless a graduate section is formally approved. Advanced students may pursue optional extensions with instructor approval, but full credit is achievable through the core R-based pathway.
- **Analytic boundaries:** Students are expected to clearly describe their data, workflow, results, and limitations. Formal modelling, prediction, causal inference, or hypothesis testing is optional only with instructor approval and must be described cautiously.

| Component                                    | Weight                             | Notes                                                                                                      |
|----------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------|
| Foundational Assignments (Weeks 2-4)         | Required (0%; Complete/Incomplete) | Must complete all 3 to pass; does not contribute to numeric grade                                          |
| Weekly Assignments (Best 5 of 7, Weeks 5-11) | 35%                                | Best 5 counted; each counted assignment worth 7%                                                           |
| Milestone 1: Project Proposal                | 5%                                 | Dataset, question, feasibility, provenance/stewardship statement                                           |
| Milestone 2: Preliminary Analysis            | 5%                                 | Cleaning, EDA, descriptive summary, runnable repo                                                          |
| Milestone 3: Project Update                  | 5%                                 | Draft dashboard-style KT product + report draft for peer review                                            |
| Milestone 4: Peer-Review Contribution        | 5%                                 | Specific, actionable feedback on a peer's work; includes reproducibility check                             |
| Milestone 5: Presentation                    | 10%                                | In-class scientific seminar (Weeks 12-13)                                                                  |
| Final Portfolio                              | 35%                                | Report, required dashboard-style KT product, auditable repository; graded against Reproducibility Contract |
| Total                                        | 100%                               |                                                                                                            |

- **Python option:** A small Python/pandas component may be used for an optional extension if it runs in Codespaces and is documented clearly; the required project can be completed entirely in R.
- **Auditability:** Graded against the Reproducibility Contract (above). The project must run in a fresh Codespace.

## Late Submissions

- **Foundational Assignments (1-3):** May be resubmitted once within one week of the original deadline.

- **Weekly Assignments (4-10):** Due Mondays at 4 PM unless otherwise noted. Late submissions are not accepted. Missed assignments count as one of your dropped grades (Best 5 of 7 policy).
- **Final Portfolio:** Late submissions penalized 10% per day. Submissions more than 5 days late not accepted without prior arrangement.
- **Academic Concession:** Requests handled in accordance with UBC Policy V-135. Contact the instructor as soon as the need arises.

## Course Resources

### Online Textbook (OER)

The course textbook is a free, open-access **\*\*Quarto Book\*\*** published on GitHub Pages. Each chapter corresponds to one week of the course. The textbook serves as both pre-reading and reference material. It is version-controlled, searchable, and updated each term.

\*Link will be posted on Canvas before the first class.\*

## Survival Guide & FAQ

A living document maintained on Canvas (and in the course GitHub repository) addressing the most common technical issues: Codespace launch failures, Git merge conflicts, rendering errors, AI tool access, package installation, rendering problems, and Codespaces usage management. Updated weekly by the teaching team based on issues encountered in class.

## Required Hardware (BYOD Policy)

This course adopts a **Cloud-First** approach. Because we use GitHub Codespaces, you do not need a high-performance computer. Any device capable of running a modern web browser (including Chromebooks, older laptops, or tablets with a keyboard) is sufficient. GitHub Codespaces is the required and

supported computing environment; other platforms (e.g., Colab, Kaggle) are not supported for graded work unless explicitly approved.

**Bring your device to every class.** Come fully charged, as classroom outlets may be limited.

**Need a device?** Students can borrow from the [UBC Library](#).

## Codespaces: Hours and Best Practices

GitHub Codespaces usage is measured in **core-hours**, not clock hours. At the time this outline was prepared, GitHub Free personal accounts included a monthly quota of Codespaces use; on a typical 2-core Codespace this is approximately **60 hours of active runtime per month**. GitHub may change quotas, so the Canvas Survival Guide will link to the current GitHub documentation and show students how to check their remaining usage.

This is sufficient for the course if managed well:

- **Always stop your Codespace** when you finish working ([github.com/codespaces](https://github.com/codespaces) -> three dots -> Stop). Leaving a tab open wastes credits.
- **Reopen existing Codespaces** rather than creating new ones each session.
- **Use the default course machine type** unless instructed otherwise; larger machines consume core-hours faster.
- **If you run out of hours**, contact the teaching team for guidance before adding billing information. Plan your work accordingly.
- **If Codespaces is slow or unavailable** (rare), the Survival Guide includes contingency steps. We will communicate via Canvas if a platform-wide outage affects class.



## Open Science & Portfolio Policy

A key goal of this course is to ensure you leave with more than just a grade. By the end of the term, you will have a public-facing **Data Science Portfolio** hosted on GitHub Pages featuring real-world health data analysis - a tangible asset for your CV or LinkedIn.

- **Default:** We encourage all students to publish final projects as public repositories.
- **Your privacy:** You may submit as a private repository (shared only with the teaching team) without any penalty. Communicate with the instructor by Week 4 if you opt for this.

## University Policies

UBC provides resources to support student learning and to maintain healthy lifestyles but recognizes that sometimes crises arise and so there are additional resources to access including those for survivors of sexual violence. UBC values respect for the person and ideas of all members of the academic community. Harassment and discrimination are not tolerated nor is suppression of academic freedom. UBC provides appropriate accommodation for students with disabilities and for religious observances. UBC values academic honesty and students are expected to acknowledge the ideas generated by others and to uphold the highest academic standards in all of their actions.

Details of the policies and how to access support are available on the [UBC Senate website](#).

## Other Course Policies

### Plagiarism

Students are expected to review the Student Discipline section of the UBC Calendar and know what constitutes plagiarism and academic misconduct. Such activities are subject to penalty.

## Generative AI Policy (Permissive but Regulated)

Generative AI tools (e.g., GitHub Copilot) are **permitted and encouraged** in this course. However, you must adhere to the AI-Augmented Workflow guidelines:

- **Attribution:** Acknowledge the use of AI tools in your submissions (e.g., “Code block X was generated by Copilot and debugged by me”).
- **Accountability:** You are fully responsible for any errors, biases, or security flaws in the code you submit. “The AI made a mistake” is not a valid excuse.
- **Verification:** Always audit AI output against official documentation, the course notes, and your own data.
- **Audit trail:** For major assignments and the final portfolio, include a brief AI-use note describing what the tool helped with, what you changed, and how you verified the result.

## Learning Analytics

This course uses Canvas, which captures data about student activity. The instructor plans to use analytics data to: view overall class progress, track students’ progress for personalized feedback, review content access statistics, and assess participation.

## Copyright

All materials of this course (handouts, slides, assessments, readings, etc.) are the intellectual property of the Course Instructor or licensed for use in this course by the copyright owner. Redistribution without permission constitutes a breach of copyright and may lead to academic discipline.

Recording of class sessions by students is not permitted. Selected recordings by the instructor/TAs will be released on Canvas for viewing outside class. Contact the instructor immediately with any objections.

# Syllabus Inventory Completion Matrix

Last updated after milestone/presentation/final-portfolio completion pass: 2026-05-14.

## Summary

- Week/module pages ready: 15 of 15 tracked entries (Week 0, Weeks 1-13, Final Portfolio).
- Weekly assignments ready: A1-A10.
- Project milestones ready: M0-M5 plus Final Portfolio.
- Navigation state: student-facing assignments, milestones, and final portfolio brief are surfaced in `_quarto.yml`; instructor-only keys remain hidden.
- Workflow decision: fork-based repository setup is the official student workflow. GitHub Classroom is not the default route for this book.

## Remaining Human-Only Release Items

These items require live course infrastructure rather than more drafting:

- Canvas orientation video link.
- Canvas pre-course survey link.
- Canvas course template repository link.
- Final presentation order, room, and live schedule.

## Week And Module Matrix

| Entry  | Topic                    | Primary source                                                                                    | Status | Evidence and notes                                                                                                                 |
|--------|--------------------------|---------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------|
| Week 0 | Asynchronous onboarding  | <a href="#">weeks/week0/ready-to-board/index.qmd</a>                                              | Ready  | Canvas-posted orientation video/survey instructions, GitHub account guidance, fork workflow, support path, and Week 1 preparation. |
| Week 1 | Health Data, KT & Ethics | <a href="#">weeks/week0/ready-to-board/health-data-activity-data-intake-data-ethics/index.qmd</a> | Ready  | Card and NHANES data spine frame provenance, ethics, stewardship, privacy, and KT.                                                 |

| Entry  | Topic                    | Primary source                                                                                                                                           | Status | Evidence and notes                                                                                                                                                                                                                                 |
|--------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 2 | Modern Workflows         | <code>weeks/week02-modern-workflows/workflow</code><br>subpages;<br><code>assignments/assignment02-modern-workflows/</code>                              | Ready  | <code>Stufllows/index.qmd</code> ;<br>path is<br>template<br><code>OpenWorkflows/README.md</code><br>Codespaces<br>from fork,<br>com-<br>mit/sync,<br>submit<br>fork link<br>on<br>Canvas.                                                         |
| Week 3 | R with AI                | <code>weeks/week03-r-with-ai/</code><br><code>assignments/assignment03-r-with-ai/</code>                                                                 | Ready  | <code>index.qmd</code> ;<br><code>02-r-with-ai/README.md</code><br><code>renv::status()</code> ,<br><code>renv::snapshot()</code> ,<br>and<br><code>renv::restore()</code><br>awareness<br>aligned<br>with the<br>depend-<br>ency<br>note.         |
| Week 4 | Git & Col-<br>laboration | <code>weeks/week04-git-collaboration/</code><br><code>assignments/assignment04-git-collaboration/</code><br><code>milestones/m0-group-formatting/</code> | Ready  | <code>04-git-collaboration/index.qmd</code> ;<br><code>04-git-collaboration/README.md</code> ;<br><code>04-git-collaboration/README.md</code><br>a forked<br>group<br>template<br>and col-<br>laborator<br>access;<br>M0 is<br>fully<br>specified. |

| Entry  | Topic                            | Primary source                                                                                 | Status | Evidence and notes                                                                                                                                                                                                                                                                         |
|--------|----------------------------------|------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 5 | Polyglot Awareness & R Deepening | <code>weeks/week04/polyglot-assignments/assignment04/polyglot/README.md</code>                 | Ready  | <code>R-first-deepening/index.qmd</code> ; <code>04-polyglot/README.md</code> workflow, Python awareness, dependency notes, and AI audit expectations are present.                                                                                                                         |
| Week 6 | Data Visualization               | <code>weeks/week06/data-visualization/viz01-viz07/incremental pages; A5; M1 visual plan</code> | Ready  | <code>06-data-visualization/index.qmd</code> ; page sequence (overview, grammar+first plot, worked example, misleading-design audit, AI prompts, studio, A5 walk-through+reference). Instructor key files unlinked from student-facing pages; see <code>instructor/MAINTENANCE.md</code> . |

| Entry  | Topic                      | Primary source                                                     | Status             | Evidence and notes                                                                                                                                                                                                                                                                                      |
|--------|----------------------------|--------------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 7 | EDA, Table 1 & AI Auditing | <code>weeks/week07/eda01-eda07</code><br>incremental pages; A6; M2 | Ready for AI-audit | <code>Site/index.qmd</code> ; page sequence (overview, cohort+missingness, worked example, descriptive-statistic claims, four audit categories, planted-error studio, A6 walk-through+reference). Planted-error key<br>unlinked from student-facing pages; see <code>instructor/MAINTENANCE.md</code> . |

| Entry  | Topic                        | Primary source                                   | Status | Evidence and notes                                                                                                                                                                                                                                                                                 |
|--------|------------------------------|--------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 8 | Dashboard Proto-types for KT | weeks/week08-dash01-dash07 incremental pages; A7 | Ready  | Source: index.qmd; page sequence with the Quarto starter dashboard as the required core artifact (dash03); Shiny exemplar covered as demonstration only (dash05); 1-1-1-1-1 prototype rule, allowed-vs-not-allowed adaptation guidance, testing/privacy routines, and A7 walk-through are present. |



| Entry   | Topic                                | Primary source                                             | Status                 | Evidence and notes                                                                                                                                                                                              |
|---------|--------------------------------------|------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 9  | Communication of Scientific Findings | <code>weeks/week09-revealjs-starter-slides.qmd</code> ; A8 | Ready for presentation | <code>communication/index.qmd</code> ; <code>revealjs-peer-feedback.qmd</code> ; starter, three-slide structure, plain-language activity, peer feedback form, and accessibility/citation checklist are present. |
| Week 10 | Writing & Publishing Reports         | <code>weeks/week10-report-template.qmd</code> ; A9; M3     | Ready for publishing   | <code>publishing/index.qmd</code> ; template has YAML, citation workflow, generated figure/table, publishing/private preview checklist, and grounding audit.                                                    |

| Entry   | Topic                | Primary source                                                                                     | Status | Evidence and notes                                                                                                                                      |
|---------|----------------------|----------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 11 | Portfolio Surgery    | <code>weeks/week11/PortfolioSurgery/reproducibility-test.qmd</code><br>A10; M4;<br>Final Portfolio | Ready  | <code>PortfolioSurgery/index.qmd</code> ;<br>runbook, sample failure log, final portfolio checklist link, and optional Python pathway note are present. |
| Week 12 | Presentations Part 1 | <code>weeks/week12/Presentations/M5</code>                                                         | Ready  | <code>Presentations/index.qmd</code> ;<br>rules, slide submission, rubric, peer feedback form, slide checklist, and portfolio reminder are present.     |

| Entry   | Topic                | Primary source                                                                          | Status                                                                       | Evidence and notes                                                                                        |
|---------|----------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Week 13 | Presentations Part 2 | weeks/week13/representations/primary-source/primary-source-qmd; M5; Final Portfolio     | Representations/primary-source-qmd;                                          | logistics, closing reflection, final portfolio reminders, and final revision priorities are present.      |
| Final   | Final Portfolio      | milestones/ready-portfolio/ready-portfolio-checklist.md, repository-sharing-guidance.md | ready-portfolio/ready-portfolio-checklist.md, repository-sharing-guidance.md | lic/private repository guidance, deliverables, Canvas route, AI-use expectations, and rubric are present. |

## Assignment Matrix

| Assignment | Week | Source                   | Status | Notes                                                                                                                                                           |
|------------|------|--------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1         | 2    | assignments/assignment02 | Yes    | fork-workflows/README.md<br>based<br>workflow,<br>render,<br>com-<br>mit/sync,<br>and fork<br>link sub-<br>mission.                                             |
| A2         | 3    | assignments/assignment03 | Yes    | run-with-ai/README.md<br>port/inspection,<br>dplyr,<br>function,<br>AI audit,<br>depen-<br>dency<br>note, and<br><code>renv::status()</code><br>aware-<br>ness. |
| A3         | 4    | assignments/assignment04 | Yes    | git/README.md<br>branches,<br>pull<br>request<br>evidence,<br>documen-<br>tation<br>edit, and<br>M0<br>readiness.                                               |
| A4         | 5    | assignments/assignment05 | Yes    | Python/README.md<br>parity,<br>helper<br>files, de-<br>pendency<br>note, and<br>AI-use<br>note.                                                                 |

| Assignment | Week | Source                   | Status | Notes                                                                                                         |
|------------|------|--------------------------|--------|---------------------------------------------------------------------------------------------------------------|
| A5         | 6    | assignments/assignment6/ | Yes    | Plot visualization/README.md plot audit, corrected plot, caption, and M1 link.                                |
| A6         | 7    | assignments/assignment7/ | Yes    | Planted-error audit, stewardship note, and M2 link.                                                           |
| A7         | 8    | assignments/assignment8/ | Yes    | Dashboard style KT product, testing, audience note, limitation, and privacy check.                            |
| A8         | 9    | assignments/assignment9/ | Yes    | AI-communication/README.md deck/outline, plain-language summary, peer feedback, source note, and AI-use note. |

| Assignment | Week | Source                                               | Status | Notes                                                                           |
|------------|------|------------------------------------------------------|--------|---------------------------------------------------------------------------------|
| A9         | 10   | assignments/assignment09-reporting/README.md         | Yes    | report, citation workflow, publishing checklist, grounding audit, and M3 link.  |
| A10        | 11   | assignments/assignment10-portfolio-surgery/README.md | Yes    | environment test, failure log, peer review, README repair, and final risk list. |

## Milestone Matrix

| Milestone | Source                                    | Status | Notes                                                                                          |
|-----------|-------------------------------------------|--------|------------------------------------------------------------------------------------------------|
| M0        | milestones/m0-group-information/README.md | Yes    | formation, shared fork, access evidence, communication plan, Codespace check, and AI-use note. |

| Milestone | Source                              | Status                         | Notes                                                                                              |
|-----------|-------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------|
| M1        | milestones/m1-proposal/             | Proposal/README.md             | Proposal, Data Intake Card, initial visual plan, reproducibility plan, and AI-use note.            |
| M2        | milestones/m2-preliminary-analysis/ | Preliminary-analysis/README.md | Preliminary analysis, rendered output, Table 1, missingness, methods note, and AI audit.           |
| M3        | milestones/m3-project-update/       | Project-update/README.md       | Project update, dashboard preview, reproducibility note, feedback request, and AI-use note.        |
| M4        | milestones/m4-ready-review/         | Ready-review/README.md         | Ready for review, reproducibility check, auditability check, KT feedback, and reviewer reflection. |

| Milestone          | Source                      | Status | Notes                                                                                                                                   |
|--------------------|-----------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------|
| M5                 | milestones/m5-presentation/ | Ready  | README.md<br>timing,<br>rubric,<br>source/limitations<br>note,<br>feedback<br>record, and<br>AI-use note.                               |
| Final<br>Portfolio | milestones/final-portfolio/ | Ready  | README.md,<br>dashboard-<br>style KT<br>product,<br>checklist,<br>repository<br>access,<br>sharing<br>guidance,<br>and AI-use<br>audit. |

## Navigation Notes

- `_quarto.yml` lists `assignments/index.qmd` and A1-A10 README pages directly.
- `_quarto.yml` lists `milestones/index.qmd`, M0-M5, and Final Portfolio.
- Instructor-only answer keys under week `instructor/` folders are intentionally not listed.
- Starter templates may be linked from assignment/week pages without being separate navigation chapters.



# Syllabus-To-Content Coverage Audit

Last updated after milestone/presentation/final-portfolio completion pass: 2026-05-14.

## Executive Summary

- Complete topic groups: 15 of 15 tracked entries.
- Weekly assignment briefs complete: A1-A10.
- Project milestones complete: M0-M5 plus Final Portfolio.
- Student-facing navigation: assignments, milestones, and final portfolio are surfaced.
- Remaining human-only release items: Canvas orientation video link, Canvas pre-course survey link, Canvas course template repository link, and final presentation order/room.

## Highest-Priority Release Checks

1. Post the orientation video in Canvas and link it from the Week 0 Canvas module.
2. Post the pre-course survey in Canvas and verify access from a student account.
3. Post the course template repository link in Canvas for the fork workflow.
4. Post final presentation order, room, and live schedule in Canvas.
5. Re-render the full book after any live Canvas wording changes.

## Coverage Matrix

### Week 0

- **Topic name:** Asynchronous Onboarding
- **Status:** Complete
- **Evidence:** weeks/week00-onboarding/index.qmd / Required Tasks: “Watch the orientation video posted in Canvas”; Course Repository Entry Point: “Open the course template repository linked from Canvas”; Support Path: Canvas support channel with error details.
- **What is missing:** Live Canvas URLs are external to the repository.
- **Recommended next additions:** Add the actual Canvas video, survey, and template-repository links inside Canvas.

### Week 1

- **Topic name:** Health Data, KT & Ethics
- **Status:** Complete
- **Evidence:** weeks/week01-health-data-ethics/index.qmd / Learning Objectives; weeks/week01-health-data-ethics/activity-data-intake-card.qmd / Data Intake Card activity and NHANES example.
- **What is missing:** No content drafting gap.
- **Recommended next additions:** Refresh portal links before term.

### Week 2

- **Topic name:** Modern Workflows
- **Status:** Complete
- **Evidence:** weeks/week02-modern-workflows/index.qmd / Getting Started: GitHub Account and Fork: “Click Fork”; Assignment 1 brief: “Submit your forked GitHub repository link on Canvas.”
- **What is missing:** No content drafting gap.

- **Recommended next additions:** Keep screenshots aligned with GitHub's current interface.

### Week 3

- **Topic name:** R with AI
- **Status:** Complete
- **Evidence:** `weeks/week03-r-with-ai/index.qmd`  
`/ Minimal renv Checkpoint: renv::status(),`  
`renv::snapshot(), renv::restore(); assignments/assignment02-r-with-ai/README.md`  
`/ Reproducibility Requirements.`
- **What is missing:** No content drafting gap.
- **Recommended next additions:** Recheck package environment before release.

### Week 4

- **Topic name:** Git & Collaboration
- **Status:** Complete
- **Evidence:** `weeks/week04-git-collaboration/index.qmd`  
`/ Setting Up Your Group Repository: group fork and`  
`collaborator workflow; milestones/m0-group-formation/README.md:`  
`access evidence and communication plan.`
- **What is missing:** No content drafting gap.
- **Recommended next additions:** Verify group template repository permissions before M0 opens.

### Week 5

- **Topic name:** Polyglot Awareness & R Deepening
- **Status:** Complete
- **Evidence:** `weeks/week05-polyglot-r-deepening/index.qmd`  
`/ R-first parity workflow; assignments/assignment04-polyglot/README.md`  
`/ parity table, notebook hygiene, dependency note, and`  
`AI-use note.`
- **What is missing:** No content drafting gap.
- **Recommended next additions:** Test Python examples in Codespaces before release.

## Week 6

- **Topic name:** Data Visualization
- **Status:** Complete
- **Evidence:** `weeks/week06-visualization/viz01-overview-and-goals.qmd` through `viz07-assignment-a5-and-reference.qmd`: incremental seven-page sequence covering visual reasoning, grammar of a ggplot, end-to-end worked example, misleading-design failure modes, AI prompt patterns, studio audit activity, and A5 walkthrough plus quick reference. `instructor/answer-key-visual-audit.md` and `instructor/corrected-plot-code.R` remain unlinked from student-facing pages; see `instructor/MAINTENANCE.md`.
- **What is missing:** No content drafting gap.
- **Recommended next additions:** Re-render the worked example if the NHANES CSV changes.

## Week 7

- **Topic name:** EDA, Table 1 & AI Auditing
- **Status:** Complete
- **Evidence:** `weeks/week07-eda-ai-audit/eda01-overview-and-goals.qmd` through `eda07-assignment-a6-and-reference.qmd`: incremental seven-page sequence covering evidence stewardship framing, cohort and missingness, end-to-end worked example, descriptive-statistic claims, four AI-audit categories, planted-error studio activity, and A6 walkthrough plus quick reference. `instructor/planted-error-key.md` remains unlinked from student-facing pages; see `instructor/MAINTENANCE.md`.
- **What is missing:** No content drafting gap.
- **Recommended next additions:** Keep the unweighted/descriptive limitation visible.

## Week 8

- **Topic name:** Dashboard Prototypes for KT
- **Status:** Complete

- **Evidence:** `weeks/week08-dashboard-kt/dash01-overview-and-pathways.qmd` through `dash07-assignment-a7-and-reference.qmd`: incremental seven-page sequence covering pathway choice (Quarto core, Shiny demo, static HTML/JS demo), the 1-1-1-1-1 prototype rule, the Quarto starter dashboard (required core artifact), allowed-vs-not-allowed adaptation changes, testing/privacy/Shiny demo, in-class studio, and A7 walkthrough plus reference. Shiny exemplar treated as demonstration only.
- **What is missing:** No content drafting gap.
- **Recommended next additions:** Test the Quarto starter dashboard render flow in Codespaces before Week 8; demo (do not require) the Shiny exemplar in the seminar segment.

## Week 9

- **Topic name:** Communication of Scientific Findings
- **Status:** Complete
- **Evidence:** `weeks/week09-science-communication/index.qmd` / Plain-Language Translation; `starter-slides.qmd`: revealjs YAML and three-slide KT story; `revealjs-peer-feedback.qmd`: peer feedback and accessibility checklist; A8 brief.
- **What is missing:** No content drafting gap.
- **Recommended next additions:** Verify slide rendering after any revealjs theme changes.

## Week 10

- **Topic name:** Writing & Publishing Reports
- **Status:** Complete
- **Evidence:** `weeks/week10-reporting-publishing/report-template.qmd`: YAML, citation to `@cdc-nhanes`, one table, one figure, limitations, publishing checklist, grounding audit; A9 brief; M3 brief.
- **What is missing:** No content drafting gap.
- **Recommended next additions:** Recheck `references.bib` if source citations change.

## Week 11

- **Topic name:** Portfolio Surgery & Optional Advanced Deployment
- **Status:** Complete
- **Evidence:** `weeks/week11-portfolio-surgery/reproducibility-test.qmd`: step-by-step runbook, sample failure log, blank failure log, optional Python pathway, final portfolio checklist link; A10 and M4 briefs.
- **What is missing:** No content drafting gap.
- **Recommended next additions:** Run one fresh Codespace smoke test before portfolio week.

## Week 12

- **Topic name:** Term Project Presentations, Part 1
- **Status:** Complete
- **Evidence:** `weeks/week12-presentations/index.qmd`: timing rules, slide submission paths, rubric, peer feedback form, slide checklist, portfolio reminder; M5 brief.
- **What is missing:** Live presentation order and room are external to the repository.
- **Recommended next additions:** Post live presentation order in Canvas.

## Week 13

- **Topic name:** Term Project Presentations, Part 2
- **Status:** Complete
- **Evidence:** `weeks/week13-presentations/index.qmd`: timing rules, slide submission, closing reflection, final portfolio reminder, revision priorities; M5 and Final Portfolio links.
- **What is missing:** Live presentation order and room are external to the repository.
- **Recommended next additions:** Post live presentation order in Canvas.

## Final Portfolio

- **Topic name:** Final Portfolio
- **Status:** Complete
- **Evidence:** `milestones/final-portfolio/README.md`: required artifacts, Canvas route, public/private repository guidance, reproducibility checklist, AI-use expectations, grading checklist; `final-portfolio-checklist.md`; `repository-sharing-guidance.md`.
- **What is missing:** Live Canvas submission link is external to the repository.
- **Recommended next additions:** Verify course staff access for private repositories before grading.

## Deliverables Alignment

| Deliverable | Status   | Evidence                                                                                                                                             |
|-------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1          | Complete | <code>assignments/assignment01-workflows/README.md</code> uses fork repository workflow and Canvas fork-link submission.                             |
| A2          | Complete | <code>assignments/assignment02-r-with-ai/README.md</code> includes R workflow, AI audit, dependency note, and <code>renv::status()</code> awareness. |
| A3          | Complete | <code>assignments/assignment03-git/README.md</code> covers commits, diffs, branches, pull request evidence, and M0 readiness.                        |

| Deliverable | Status   | Evidence                                                                                                                                                           |
|-------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A4          | Complete | <code>assignments/assignment04-polyglot/README.md</code><br>covers R/Python<br>parity and<br>dependency<br>hygiene.                                                |
| A5          | Complete | <code>assignments/assignment05-visualization/README.md</code><br>covers ggplot,<br>flawed plot audit,<br>caption, and<br>AI-use note.                              |
| A6          | Complete | <code>assignments/assignment06-eda-ai-audit/README.md</code><br>covers EDA, Table<br>1, planted-error<br>audit, and<br>stewardship note.                           |
| A7          | Complete | <code>assignments/assignment07-dashboard/README.md</code><br>covers<br>dashboard-style<br>KT product and<br>testing.                                               |
| A8          | Complete | <code>assignments/assignment08-communication/README.md</code><br>covers slide<br>deck/outline,<br>plain-language<br>summary, peer<br>feedback, and<br>source note. |
| A9          | Complete | <code>assignments/assignment09-reporting/README.md</code><br>covers Quarto<br>report, citations,<br>publishing/private<br>preview, and<br>grounding audit.         |
| A10         | Complete | <code>assignments/assignment10-portfolio-surgery/README.md</code><br>covers fresh<br>environment<br>testing, failure log,<br>peer review, and<br>final risk list.  |



| Deliverable | Status   | Evidence                                                                                                                                                                      |
|-------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M0          | Complete | <code>milestones/m0-group-formation/README.md</code><br>includes group<br>formation, access,<br>communication<br>plan, Codespace<br>check, and AI-use<br>note.                |
| M1          | Complete | <code>milestones/m1-proposal/README.md</code><br>includes proposal,<br>Data Intake Card,<br>visual plan,<br>reproducibility<br>plan, and AI-use<br>note.                      |
| M2          | Complete | <code>milestones/m2-preliminary-analysis/README.md</code><br>includes<br>preliminary<br>analysis, rendered<br>output, Table 1,<br>missingness,<br>methods, and<br>audit note. |
| M3          | Complete | <code>milestones/m3-project-update/README.md</code><br>includes report<br>draft, dashboard<br>preview, feedback<br>request,<br>reproducibility<br>note, and AI-use<br>note.   |
| M4          | Complete | <code>milestones/m4-peer-review/README.md</code><br>includes peer<br>review,<br>reproducibility<br>check, auditability<br>check, KT<br>feedback, and<br>reflection.           |

| Deliverable     | Status   | Evidence                                                                                                                                 |
|-----------------|----------|------------------------------------------------------------------------------------------------------------------------------------------|
| M5              | Complete | milestones/m5-presentation/README.md<br>includes slides,<br>timing, rubric,<br>source/limitations<br>note, feedback,<br>and AI-use note. |
| Final Portfolio | Complete | milestones/final-portfolio/README.md,<br>final-portfolio-checklist.md,<br>and<br>repository-sharing-guidance.md.                         |

## Consistency Checks

- Fork workflow mismatch is resolved: Week 0, Week 2, A1, syllabus-facing text, and Course Operations all use fork-first language.
- Student-facing assignments A1-A10 are listed in `_quarto.yml`.
- Student-facing milestones M0-M5 and Final Portfolio are listed in `_quarto.yml`.
- Instructor-only files in Week 6 and Week 7 remain outside `_quarto.yml`.
- Starter files are linked from week or assignment pages rather than promoted as main chapters unless they are core student-facing pages.

## Appendix A: Files Scanned

- `syllabus/SPPH-381H-Course-Outline-v4.qmd`
- `syllabus/syllabus-inventory-completion-matrix.md`
- `weeks/week00-onboarding/index.qmd`
- `weeks/week01-health-data-ethics/**`
- `weeks/week02-modern-workflows/**`
- `weeks/week03-r-with-ai/**`
- `weeks/week04-git-collaboration/**`
- `weeks/week05-polyglot-r-deepening/**`

- weeks/week06-visualization/\*\*
- weeks/week07-eda-ai-audit/\*\*
- weeks/week08-dashboard-kt/\*\*
- weeks/week09-science-communication/\*\*
- weeks/week10-reporting-publishing/\*\*
- weeks/week11-portfolio-surgery/\*\*
- weeks/week12-presentations/index.qmd
- weeks/week13-presentations/index.qmd
- assignments/assignment01-workflows/README.md  
through assignments/assignment10-portfolio-surgery/README.md
- milestones/\*\*
- \_quarto.yml
- course-operations/index.qmd

## Appendix B: Keyword Patterns Used

- Workflow: fork, Codespaces, Quarto, render, commit, sync, relative paths
- R and reproducibility: R, dplyr, read\_csv, renv, status, snapshot, restore
- Ethics and data: provenance, stewardship, privacy, security, OCAP, CARE, responsible AI
- Visualization and EDA: ggplot2, plotly, visual audit, misleading, Table 1, missingness, planted-error
- KT products: dashboard, Shiny, plain language, revealjs, report, citations, grounding audit
- Portfolio: fresh Codespace, failure log, peer review, presentation, final portfolio, AI-use note

## **Week 0: Asynchronous Onboarding**

# Week 0: Asynchronous Onboarding

## Purpose

Week 0 gets the account setup and orientation work out of the way before the first class. You do not need to install local software for this course. The default workflow uses a personal fork of the course template repository, GitHub Codespaces, Quarto, and Canvas.

## Required Tasks

Complete these steps before Week 1:

1. Log in to Canvas and open the Week 0 module.
2. Watch the orientation video posted in Canvas.
3. Complete the pre-course survey posted in Canvas.
4. Create or confirm access to a GitHub account.
5. Submit the Week 0 Canvas onboarding check with your GitHub username.

## GitHub Account

Use a professional GitHub username that you are comfortable sharing in a course setting. If you already have a GitHub account, you may use it. You do not need to create a second account unless your current username or profile is not appropriate for coursework.

Before submitting the onboarding check:

- confirm you can log in to GitHub
- turn on two-factor authentication if GitHub prompts you
- make sure you know which email address is attached to the account
- record your GitHub username exactly as it appears in the profile URL

## Existing-Account Help

If you already have a GitHub account but cannot access it:

- try GitHub account recovery before creating a new account
- use the email address most likely attached to the account
- check whether your browser is logged into a different personal account
- ask for help through the Canvas course support channel if you are blocked

If you create a new account, use that account consistently for course repositories.

## Course Repository Entry Point

The current course workflow uses a fork-based setup:

1. Open the course template repository linked from Canvas.
2. Click **Fork**.
3. Create a personal fork under the GitHub account you submitted in the onboarding check.
4. Launch Codespaces from your fork.
5. Commit and sync changes to your fork.
6. Submit your fork repository link on Canvas.

If a future offering uses an instructor-managed repository system, Canvas will say so explicitly. For this course book, the fork-based workflow is the default student route.

## Support Path

Use the Canvas course support channel for setup problems. Include:

- your GitHub username
- the step where you got stuck
- the exact error message or screenshot
- whether you are using GitHub, Codespaces, or Canvas

Do not post passwords, tokens, or private account recovery details.

## What To Bring To Week 1

Bring:

- a working Canvas login
- a working GitHub login
- your GitHub username
- a laptop or access plan for in-class work
- one question about health data, public data, AI, or reproducibility

## Definition Of Done

Week 0 is done when you have watched the Canvas orientation video, completed the Canvas pre-course survey, submitted the onboarding check, and can log in to GitHub with the account you will use for course work.

# **Week 1: Health Data, KT & Ethics**



# Health Data & Ethics

## Overview

Week 1 sets the ethical and communication foundation for the whole course. Health data science does not end when code runs or a p-value appears; it ends when evidence is translated responsibly for a real audience. This week introduces open health data, data provenance, stewardship, privacy, responsible AI use, Indigenous Data Sovereignty awareness, and Knowledge Translation (KT) as the “last mile” of analysis.

This is an awareness-heavy week. There is no coding, no dashboard running, and no app setup. The NHANES case study appears only as a data/provenance artifact so you can practice evaluating a dataset before using it.

## Learning Objectives

By the end of this week, you should be able to:

- identify credible open health data portals;
- distinguish microdata from aggregated indicators;
- complete a Data Intake Card for a public health dataset;
- record basic provenance, licensing, access, and citation information;
- name privacy, security, and stewardship risks that can still apply to public data;
- explain why AI summaries must be checked against source documentation;
- recognize when Indigenous Data Sovereignty, OCAP, or CARE principles may be relevant;
- frame a dataset as the start of a KT product, not just as a file to analyze.

## Connection

This first week asks, “Should we use this data, and how should we describe it?” Next week asks, “How do we organize the files so another person can reproduce our work?” Later weeks return to the same NHANES case study through analysis, visualization, EDA, and dashboard design.

## Case Study Data Spine

The **NHANES Health Equity data spine** is the recurring dataset thread for this course. In the early weeks, use it only as a documented public-health data artifact: provenance, ethics, file organization, and reproducible paths.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The cached files are provided so early work can happen without internet access.

```
1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_spine <- read_csv(nhanes_path, show_col_types = FALSE)
15
16 dim(nhanes_spine)
17 #> [1] 105626      18
```

```

18 glimpse(nhanes_spine)
19 #> Rows: 105,626
20 #> Columns: 18
21 #> $ Cycle      <chr> "1999-2000", "1999-2000", "1999-2000", "1999-2000", "19~
22 #> $ BMI        <dbl> 14.90, 24.90, 17.63, NA, 29.10, 22.56, 29.39, 15.51, 18~
23 #> $ Weight     <dbl> 12.5, 75.4, 32.9, 13.3, 92.5, 59.2, 78.0, 40.7, 45.5, 1~
24 #> $ Height     <dbl> 91.6, 174.0, 136.6, NA, 178.3, 162.0, 162.9, 162.0, 156~
25 #> $ Waist      <dbl> 45.7, 98.0, 64.7, NA, 99.9, 81.6, 90.7, 64.1, 64.6, 108~
26 #> $ WeightMEC  <dbl> 10982.899, 28325.385, 46192.257, 10251.260, 99445.066, ~
27 #> $ Strata     <dbl> 5, 1, 7, 2, 8, 2, 4, 6, 9, 7, 1, 6, 13, 12, 11, 11, 5, ~
28 #> $ PSU        <dbl> 1, 3, 2, 1, 2, 2, 2, 1, 2, 1, 2, 2, 2, 1, 2, 1, 1, 1, 2~
29 #> $ Gender     <chr> "Female", "Male", "Female", "Male", "Male", "Female", "~
30 #> $ Race       <chr> "Non-Hispanic Black", "Non-Hispanic White", "Non-Hispan~
31 #> $ Education  <chr> NA, "College Grad", NA, NA, "College Grad", NA, "HS or ~
32 #> $ Marital    <chr> NA, NA, NA, NA, "Married/Partner", "Never Married", "Ma~
33 #> $ Age        <dbl> 2, 77, 10, 1, 49, 19, 59, 13, 11, 43, 15, 37, 70, 81, 3~
34 #> $ PIR        <dbl> 0.86, 5.00, 1.47, 0.57, 5.00, 1.21, NA, 0.53, NA, NA, 1~
35 #> $ EducationClean <chr> NA, "College Grad", NA, NA, "College Grad", NA, "HS or ~
36 #> $ MaritalClean <chr> NA, NA, NA, NA, "Married/Partner", "Never Married", "Ma~
37 #> $ IncomeGroup <chr> "Low Income (<1.3)", "High Income (>3.5)", "Middle Inco~
38 #> $ WHtR       <dbl> 0.4989083, 0.5632184, 0.4736457, NA, 0.5602916, 0.50370~

```

For Week 1, treat the NHANES case study as a provenance and stewardship example only. Do not run the app. Do not rebuild the data. Do not upload row-level data to an AI tool.

## In-Class Activity

Use [Activity: Data Intake Card](#).

1. Choose one approved public health dataset or use the shared NHANES case-study example.
2. Find the dataset publisher, documentation, access page, and citation guidance.
3. Complete the Data Intake Card.
4. Write one KT framing sentence: who could use this dataset, and what decision or conversation could it support?

5. Name one ethics, privacy, security, or stewardship issue that should be checked before publishing results.

Student output: a completed Data Intake Card and one KT framing sentence.

## Open Data Portals

Open data can accelerate public health learning when the source, meaning, and limits of the data are clear. Start with reputable portals that provide documentation, terms of use, and stable links.

Recommended starter portals:

- CDC/NCHS public-use survey data, such as NHANES or NHIS;
- WHO Global Health Observatory for aggregated international indicators;
- BC Data Catalogue for provincial open data;
- Government of Canada Open Government Portal for Canadian datasets.

When you land on a dataset page, look for:

- publisher or steward;
- population, geography, and time coverage;
- data level, such as microdata or aggregated indicators;
- documentation, codebook, or methodology notes;
- license or terms of use;
- citation recommendation;
- last updated date or access date.

## Provenance

Provenance means the traceable story of where data came from and how it reached your project. A dataset is not just a filename. It has a steward, collection context, transformations, limitations, and documentation.

For every dataset, record:

- original source page;
- local file path if a copy is stored in the repository;
- date accessed;
- file format;
- variables you plan to use;
- known limitations or cautions;
- citation or attribution language.

The Data Intake Card is the course habit that makes provenance visible.

## Stewardship

Stewardship means using data in a way that respects the people, communities, and institutions represented by the data. Public data still requires care.

Ask:

- Could a group be stigmatized by a careless summary?
- Are small cells or rare combinations being exposed?
- Does the dataset include populations for whom special governance principles may apply?
- Are limitations visible enough for a non-technical audience?
- Does the planned KT product support understanding rather than overclaiming?

## Privacy And Security

In this course, Week 1 uses public, de-identified, or aggregated data only. Still, avoid treating “public” as the same as “risk-free.”

Course rules:

- Do not upload row-level health data to AI tools.
- Do not attempt re-identification.

- Do not publish small-cell claims.
- Do not copy data into public repositories unless the license and course instructions allow it.
- Do not use absolute local paths that reveal a personal machine or user name.

## Responsible AI

AI tools may help you draft plain-language summaries or identify questions to ask about documentation. They are not source documentation.

Acceptable Week 1 AI use:

- asking for a plain-language explanation of a term from a codebook;
- drafting a first pass of a dataset description;
- generating a checklist of questions to verify.

Required checks:

- cite the official source, not the AI output;
- verify variable meanings in documentation;
- include an AI-use note if AI helped draft your card;
- revise AI language that overstates certainty or hides limitations.

## Indigenous Data Sovereignty

Indigenous Data Sovereignty refers to the rights and interests of Indigenous Peoples in data about their peoples, lands, resources, and knowledges. Two frameworks students should recognize are:

- OCAP: ownership, control, access, and possession, specific to First Nations contexts in Canada;
- CARE: collective benefit, authority to control, responsibility, and ethics.

Week 1 expectation: if a dataset includes Indigenous identifiers, communities, lands, or governance-relevant content, flag that on the Data Intake Card and review the source guidance before using or publishing analysis.

## Formats And Metadata

Common formats:

- CSV: rectangular rows and columns;
- JSON: nested data, common for APIs;
- XLSX: spreadsheet files that may contain formatting and hidden assumptions;
- RDS: R-specific saved objects;
- GeoJSON or shapefiles: spatial data for maps.

Metadata explains what the file means. It may include variable definitions, units, missing-value codes, collection methods, survey design variables, and weighting instructions. A dataset without metadata is risky for analysis because column names alone rarely tell the full story.

## KT Framing

KT asks who needs the information, what decision or conversation it supports, and how the evidence should be communicated. A Week 1 KT framing sentence can be simple:

This dataset could help [audience](#) understand [health issue] so they can [decision, planning task, or conversation], with the limitation that [key caution].

Example:

The NHANES classroom dataset could help students understand how BMI summaries differ across income groups while practicing clear limitations about descriptive, unweighted survey summaries.

## What Students Leave With

By the end of Week 1, you should have:

- one completed Data Intake Card;
- one KT framing sentence;
- one ethics or stewardship caution;
- a clear understanding that the NHANES case study starts as a data artifact, not an app to run;
- readiness to place files into a reproducible project structure in Week 2.



# Activity: Data Intake Card

## Goal

Evaluate whether a health dataset is usable, ethical, documented, and appropriate for reproducible analysis. This activity is about judgment and documentation. No coding is required.

## Approved Portals

Choose one dataset or indicator page from one of these portals:

- CDC/NCHS public-use survey data, such as NHANES or NHIS;
- WHO Global Health Observatory;
- BC Data Catalogue;
- Government of Canada Open Government Portal.

The NHANES Health Equity data spine may be used as the shared classroom example.

## Student Template

Copy this template into your notes.

```
1 dataset_name: ""
2 publisher_org: ""
3 permalink_url: ""
4 date_accessed: "YYYY-MM-DD"
5 license_terms: ""
```

```

6 geography: ""
7 time_coverage: ""
8 data_level: "" # microdata / aggregated indicator / unknown
9 file_format: "" # CSV / RDS / XLSX / JSON / other
10 metadata_or_codebook: ""
11 variables_of_interest:
12   - name: ""
13     definition: ""
14     unit: ""
15     caveat: ""
16 known_limitations:
17   - ""
18 privacy_security_notes: ""
19 stewardship_notes: ""
20 indigenous_data_relevance: "no / yes / unknown"
21 kt_framing_sentence: ""
22 how_to_cite: ""
23 ai_use_note: ""

```

## Completed Example

```

1 dataset_name: "NHANES Health Equity classroom dataset"
2 publisher_org: "CDC / National Center for Health Statistics"
3 permalink_url: "https://wwwn.cdc.gov/nchs/nhanes/"
4 date_accessed: "YYYY-MM-DD"
5 license_terms: "Public-use NHANES files; cite CDC/NCHS documentation"
6 geography: "United States"
7 time_coverage: "Multiple NHANES cycles represented in the prepared class file"
8 data_level: "Microdata-derived classroom dataset"
9 file_format: "CSV and RDS cached in examples/nhanes-equity/data/"
10 metadata_or_codebook: "CDC/NCHS NHANES documentation and case-study README"
11 variables_of_interest:
12   - name: "BMI"
13     definition: "Body mass index from NHANES body measures data"
14     unit: "kg/m^2"
15     caveat: "Use descriptively unless survey design and weights are handled correctly"
16 known_limitations:
17   - "Prepared class dataset is for classroom analysis and should not be treated as a causal analysis"

```

```

18 - "Population inference requires attention to NHANES survey design and weights"
19 privacy_security_notes: "Use public-use data responsibly, avoid small-cell claims, and do not v
20 stewardship_notes: "State limitations plainly and avoid stigmatizing group comparisons"
21 indigenous_data_relevance: "unknown in the prepared class example; check source documentation :
22 kt_framing_sentence: "This dataset can help students practice describing health equity patterns
23 how_to_cite: "Cite CDC/NCHS NHANES documentation and the course case-study README"
24 ai_use_note: "AI may help draft plain-language wording, but source documentation must be check

```

## Completion Checklist

To be marked complete, your card must include:

- stable source link or access date;
- publisher or steward;
- license or terms of use;
- geography and time coverage;
- data level and file format;
- documentation or codebook location;
- at least one documented variable definition;
- one limitation or stewardship risk;
- one privacy or security note;
- Indigenous Data Sovereignty relevance marked as no, yes, or unknown;
- KT framing sentence;
- AI-use note.

## Scoring Guide

| Rating   | Evidence                                                                                                                                                      |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Complete | Source, publisher, terms, metadata, variable definition, limitation, stewardship note, KT framing, and AI-use note are present and grounded in documentation. |

| Rating     | Evidence                                                                                                                                                                          |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Needs work | Source documentation is missing, license terms are unclear, variable definitions are guessed, risks are absent, or AI-written text is not checked against official documentation. |

## Submission

Submit the completed card in Canvas under the Week 1 module. Starting in Week 2, keep a copy in your course repository as:

```
1 data-intake-card.md
```

## Reproducibility Check

Before submitting, check:

- the link opens or the access date is recorded;
- the publisher and license are named;
- variable definitions come from documentation, not guesses;
- file format and data level are named;
- the AI-use note says what AI helped with and how the text was checked.

## **Week 2: Modern Workflows**

# Modern Workflows

## Overview

**Summary:** This week you set up the computational environment you will use for the rest of the course. By the end, the technology should fade into the background so you can focus on health data science, not installation errors. We adopt a **“Zero to AI Co-Pilot”** approach: you do not need to install R, Python, or Git on your personal laptop. Everything runs in a cloud-native environment called GitHub Codespaces.

### Learning Objectives:

- Understand the “public computer vs cloud drive” analogy for cloud computing.
- Create a GitHub account, fork the course template repository, and work from your personal fork.
- Launch a Codespace and identify the three key areas of the VS Code interface.
- Distinguish between an IDE, a notebook, and a script, and explain why this course uses Quarto notebooks.
- Explain the concept of literate programming and why it matters for reproducible health research.
- Edit and render a Quarto (`.qmd`) document: YAML header, Markdown text, and code chunks.
- Maintain a reproducible project structure using the **data/** and **output/** folder convention.
- Stage, commit, and sync changes using the VS Code Source Control panel.
- Stop your Codespace to conserve your free monthly hours.

## Connection

Week 1 focused on whether a dataset is trustworthy enough to use. This week focuses on where that dataset lives in a project and how relative paths make the same work portable across machines. Next week you will start inspecting the data with R.

## Case Study Data Spine

The **NHANES Health Equity data spine** is the recurring dataset thread for this course. In the early weeks, use it only as a documented public-health data artifact: provenance, ethics, file organization, and reproducible paths.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The cached files are provided so early work can happen without internet access.

```
1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../..examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_spine <- read_csv(nhanes_path, show_col_types = FALSE)
15
16 dim(nhanes_spine)
17 #> [1] 105626      18
```

```

18 glimpse(nhanes_spine)
19 #> Rows: 105,626
20 #> Columns: 18
21 #> $ Cycle      <chr> "1999-2000", "1999-2000", "1999-2000", "1999-2000", "19~
22 #> $ BMI        <dbl> 14.90, 24.90, 17.63, NA, 29.10, 22.56, 29.39, 15.51, 18~
23 #> $ Weight     <dbl> 12.5, 75.4, 32.9, 13.3, 92.5, 59.2, 78.0, 40.7, 45.5, 1~
24 #> $ Height     <dbl> 91.6, 174.0, 136.6, NA, 178.3, 162.0, 162.9, 162.0, 156~
25 #> $ Waist      <dbl> 45.7, 98.0, 64.7, NA, 99.9, 81.6, 90.7, 64.1, 64.6, 108~
26 #> $ WeightMEC  <dbl> 10982.899, 28325.385, 46192.257, 10251.260, 99445.066, ~
27 #> $ Strata     <dbl> 5, 1, 7, 2, 8, 2, 4, 6, 9, 7, 1, 6, 13, 12, 11, 11, 5, ~
28 #> $ PSU        <dbl> 1, 3, 2, 1, 2, 2, 2, 1, 2, 1, 2, 2, 2, 1, 2, 1, 1, 1, 2~
29 #> $ Gender     <chr> "Female", "Male", "Female", "Male", "Male", "Female", "~
30 #> $ Race       <chr> "Non-Hispanic Black", "Non-Hispanic White", "Non-Hispan~
31 #> $ Education  <chr> NA, "College Grad", NA, NA, "College Grad", NA, "HS or ~
32 #> $ Marital    <chr> NA, NA, NA, NA, "Married/Partner", "Never Married", "Ma~
33 #> $ Age        <dbl> 2, 77, 10, 1, 49, 19, 59, 13, 11, 43, 15, 37, 70, 81, 3~
34 #> $ PIR        <dbl> 0.86, 5.00, 1.47, 0.57, 5.00, 1.21, NA, 0.53, NA, NA, 1~
35 #> $ EducationClean <chr> NA, "College Grad", NA, NA, "College Grad", NA, "HS or ~
36 #> $ MaritalClean <chr> NA, NA, NA, NA, "Married/Partner", "Never Married", "Ma~
37 #> $ IncomeGroup <chr> "Low Income (<1.3)", "High Income (>3.5)", "Middle Inco~
38 #> $ WHtR      <dbl> 0.4989083, 0.5632184, 0.4736457, NA, 0.5602916, 0.50370~

```

## In-Class Activity

- Locate the case-study data, README, and snippet folders in the Explorer.
- Explain why the data example uses relative paths instead of machine-specific paths.
- Check that the weekly page includes the shared snippet rather than duplicating instructions.
- Render the book from the repository root and check that this page updates in the rendered output.

Student output: a rendered Quarto page and one commit showing a small, reproducible edit. Assignment link: [Assignment 1](#).



## workflows1.qmd The Analogy: Public Computer and Cloud Drive

To understand how we work in this course, forget about “servers” and “clients.” Imagine you are working in a university computer lab.

### The Repository (Your Cloud Drive)

Think of **GitHub** as your personal **Google Drive** or **Dropbox**.

- It is where your files live permanently.
- If your laptop crashes, the files are still here.
- **Key concept:** This is your “Master Copy.”

### The Codespace (The Public Computer)

Think of a **Codespace** as a **public computer in the library**.

- It is a temporary machine you borrow to do your work.
- It comes pre-installed with all the software you need (R, Python, Quarto), so you don’t have to install anything on your own laptop.
- **Crucial difference:** Just like a public library computer, it can be wiped clean. If you close the browser without sending your work back to your Cloud Drive, your changes may be lost.

### The Workflow

1. **Log in:** Launch your Codespace (open the public computer).
2. **Work:** Edit files using the computer’s software.
3. **Save to the cloud:** Commit and push your work back to your Cloud Drive (GitHub).
4. **Log off:** Stop the Codespace to save your free hours.

This cycle — launch, work, commit, stop — is the rhythm of every week in this course.

---

## workflows2.qmd Getting Started: GitHub Account and Fork

### Step 1: Create a GitHub Account

If you do not already have one:

1. Go to [github.com](https://github.com) and click **Sign up**.
2. Use your UBC email address (this helps with student benefits).
3. Choose a professional username — this will appear on your public portfolio later in the course.

#### Tip

**Portfolio thinking:** Your GitHub profile will become part of your professional presence. Pick a username you would be comfortable putting on a CV (e.g., `jchen-epi` rather than `xXcoolcoder99Xx`).

### Step 2: Fork the Course Template

1. Open the course template repository linked from Canvas.
2. Click **Fork** in the upper-right corner of the GitHub page.
3. Create the fork under your own GitHub account.
4. Keep the default repository name unless Canvas gives a specific naming rule.
5. After the fork is created, make sure the page URL includes your GitHub username.

### Step 3: Verify Your Repository

Navigate to your forked repository. You should see:

- A file list including `practice_report.qmd`, `README.md`, and a `data/` folder.
- A formatted README below the file list explaining the project.

If you see these in your fork, you are ready to launch your Codespace.

---

### workflows3.qmd Launching Your Codespace

1. On your repository page, click the green `<>` **Code** button.
2. Select the **Codespaces** tab.
3. Click **Create codespace on main**.

The first launch takes 2–3 minutes while the environment builds. Subsequent launches are faster.

#### Note

**Reopening an existing Codespace:** After you stop or close your Codespace, do **not** create a new one each time. Go to [github.com/codespaces](https://github.com/codespaces), find your existing Codespace, and click its name to restart it. Creating multiple Codespaces wastes your hours and causes confusion.

## workflows4.qmd Tour of VS Code: The “Big Three” Zones

The VS Code interface can look intimidating — it is designed for professional software engineers. You only need to know three areas. Safely ignore 90% of the buttons.

### 1. The Explorer (Left Sidebar)

Your **file cabinet**. It shows folders and files in your project. You will use this to open `.qmd` files, navigate the `data/` folder, and create new files.

### 2. The Editor (Center)

Your **work area**. This is where you read and write code and text. You can have multiple files open in tabs.

### 3. The Terminal (Bottom Panel)

The **engine room** where R and Python actually run. You will not type commands here often — most of your work happens in the Editor — but this is where output and error messages appear.

#### Tip

**Lost a panel?** Use the top menu: **View → Explorer** or **View → Terminal** to bring them back. You can also press `Ctrl+`` (backtick) to toggle the Terminal.

## workflows5.qmd **Concepts: IDE, Notebook, Script**

The syllabus mentions three ways to write code. Understanding the difference helps you choose the right tool throughout the course.

### **IDE (Integrated Development Environment)**

VS Code is an **IDE** — a text editor with superpowers: file browsing, syntax highlighting, built-in terminal, extensions, and version control. It is the container that holds everything else.

### **Script**

A plain text file containing only code (e.g., `.R` or `.py`). Runs top-to-bottom. Best for reusable utilities and automation. You will write your first script in Week 5.

### **Notebook**

A document that **interleaves code and narrative**. You run code in chunks and see results inline. Examples include Jupyter notebooks (`.ipynb`, Week 5) and Quarto documents (`.qmd`, which you use starting today).

**This course primarily uses Quarto notebooks** because they combine the strengths of notebooks (interactivity, narrative) with the ability to produce publication-quality reports, slides, and websites — all from a single file.

---

## workflows6.qmd Concepts: Literate Programming

### The Big Idea

Traditional workflows separate code from writing: you run an analysis in one program, copy results into a Word document, and hope nothing changes. **Literate programming** weaves code, results, and interpretation into a single document. When the data changes, you re-render and the entire report updates automatically.

This is not just convenient — it is a **scientific obligation**. In health research, reviewers and regulators need to see exactly how you went from raw data to conclusions. A literate document is a transparent, auditable record of your reasoning.

### How Quarto Implements This

A `.qmd` file has three ingredients:

1. **YAML header** (metadata at the top, between `---` lines)
2. **Markdown text** (human-readable prose)
3. **Code chunks** (executable R or Python code in fenced blocks)

When you **Render**, Quarto runs every code chunk, captures the output (tables, figures, numbers), and weaves everything into a finished HTML, PDF, or Word document.

You will use literate programming in every subsequent week of this course: your assignments, your EDA (Week 7), your reports (Week 10), and your final portfolio are all Quarto documents.

## workflows7.qmd Project Structure: Files and Folders

A reproducible project needs a consistent folder structure. This course uses the following convention:

```
my-project/  
  data/           ← Raw data. Never edit files here directly.  
  output/         ← Cleaned data, figures, and tables generated by your code.  
  practice_report.qmd ← Your Quarto notebook.  
  README.md       ← Explains what this project is.  
  .gitignore      ← Tells Git which files to ignore (more in Week 4).
```

### Rules

- **data/ is read-only.** Import from it; never save into it.
- **output/ is generated.** Your code writes cleaned data and figures here. In Week 3, you will export `output/clean_data.csv`.
- **Use relative paths.** Always refer to files relative to the project root (e.g., `"data/patients.csv"`), never absolute paths (e.g., `"/home/user/Documents/patients.csv"`). This ensures your code runs on any machine.

**Activity:** In the Explorer pane, click the **New Folder** icon and create the `output/` folder if it does not already exist.

---

## workflows8.qmd Tutorial: Your First Quarto Document

### Step 1: Open the Starter File

In the Explorer, click `practice_report.qmd`. It opens in the Editor.

## Step 2: Understand the YAML Header

At the very top, between `---` lines, you will see something like:

```
1 ---
2 title: "My First Report"
3 author: "Your Name"
4 format: html
5 ---
```

Change `author:` to your actual name. The YAML header controls metadata and output format.

## Step 3: Read the Markdown

Below the YAML, you will see plain text with formatting:

- `#` Heading creates a section title.
- `**bold**` makes text **bold**.
- `- item` creates a bullet point.

This is **Markdown** — a lightweight way to format text without a word processor.

## Step 4: Read the Code Chunks

Look for grey blocks that start with ````{r}` and end with `````. These are **code chunks**. Everything inside them is R code that will execute when you render.

```
1 ```{r}
2 # This is a code chunk
3 2 + 2
4 ```
```



## Step 5: Edit a Code Chunk

Find a code chunk in the starter file. Make a small change — for example, change a number or add a comment. This is your first edit.

## Step 6: Render

Click the **Render** button (blue arrow icon at the top of the editor). A preview window appears showing your formatted report with code output embedded.



**What just happened:** Quarto ran every code chunk, captured the results, combined them with your Markdown text, and produced an HTML document. This is literate programming in action.

---

## workflows9.qmd Tutorial: Stage, Commit, and Sync

Your work currently exists only on the public computer (Codespace). To save it permanently to your Cloud Drive (GitHub), you need to **commit**.

## Step 1: Open Source Control

Click the **Source Control** icon on the left sidebar (it looks like a branching graph). You will see a list of files that have changed since your last commit.

## Step 2: Stage Your Changes

Hover over a changed file name. Click the + icon that appears. This **stages** the file — it tells Git “I want to include this change in my next snapshot.”

### Note

**Staging vs committing:** Staging selects *which* changes to include. Committing takes the snapshot. Think of staging as putting items on the conveyor belt; committing is pressing the “scan” button at checkout.

## Step 3: Write a Commit Message

In the text box at the top of the Source Control panel, type a short, descriptive message:

Updated author name and edited first code chunk  
in practice report

A good commit message answers: *what did I change and why?*

## Step 4: Commit

Click the **Commit** button (checkmark). Your changes are now saved as a snapshot in your local history.

## Step 5: Sync with GitHub

Click **Sync Changes** (in the Source Control panel or the status bar at the bottom). This pushes your commit to GitHub — your Cloud Drive.

**Verify:** Go to your repository on [github.com](https://github.com) and check that your changes appear. You should see your commit message next to the updated file.

 Warning

If you skip the **Sync step**, your commit exists only on the Codespace. If the Codespace is deleted, the commit goes with it. Always Sync.

---

## workflows10.qmd Stopping Your Codespace (Saving Credits)

GitHub Codespaces gives you a monthly allowance of free hours (approximately 60 core-hours). **If you leave the browser tab open, the meter keeps running**, even if you are not actively working.

### The Right Way to Stop

1. Go to [github.com/codespaces](https://github.com/codespaces).
2. Find your active Codespace.
3. Click the **three dots (...)** → **Stop codespace**.

### What Happens If You Just Close the Tab?

The Codespace runs for another 30 minutes before it times out automatically. That is 30 minutes of wasted credits every time.

 Warning

**Policy:** It is your responsibility to manage your compute hours. If you run out before the month resets, you will need to wait or pay out of pocket. Get in the habit of stopping your Codespace at the end of every session.

## workflows11.qmd Troubleshooting

Sometimes things break. Here is a graduated approach — try the simplest fix first.

### Level 1: Reload the Window

Press F1 (or Ctrl+Shift+P / Cmd+Shift+P) to open the **Command Palette**. Type `Reload Window` and select it. This restarts VS Code without rebuilding the entire environment.

### Level 2: Restart the R or Python Session

If R or Python is frozen, open the Command Palette → type `R: Restart R Session` (or the equivalent for Python). This clears the session without touching your files.

### Level 3: Rebuild the Container (Nuclear Option)

If nothing else works:

1. Open the **Command Palette**.
2. Type `Rebuild Container`.
3. Select **Codespaces: Rebuild Container**.

This shuts down and rebuilds the entire environment from scratch. It solves nearly all technical glitches but takes a few minutes.

#### Warning

**Before rebuilding:** Make sure you have committed and synced your work. Uncommitted changes may survive a rebuild, but there is no guarantee.

## workflows12.qmd Knowledge Check

1. In the analogy, what is the “Cloud Drive” and what is the “Public Computer”?
  2. Why don’t you need to install R or Python on your laptop for this course?
  3. Name the “Big Three” zones of the VS Code interface and what each is for.
  4. What is the difference between an IDE, a notebook, and a script?
  5. What is literate programming, and why does it matter for health research?
  6. What are the three ingredients of a .qmd file?
  7. What is the difference between **staging** and **committing**?
  8. Why must you click **Sync Changes** after committing?
  9. How do you properly stop a Codespace to save your free hours?
- 

## workflows13.qmd Assignment 1: Your First Quarto Report

Due Friday 4 pm. Submit via Canvas.

### Task

Edit and render the starter `practice_report.qmd`, demonstrating that you can navigate your Codespace, edit Quarto, and commit your work.

### Step-by-Step Workflow

1. **Launch** your Codespace from your forked repository.
2. **Edit the YAML:** Change `author:` to your name.

3. **Add a Markdown section:** Below the existing content, add a new heading (**## My Notes**) and write 2–3 sentences about what you learned this week. Use at least one Markdown feature (bold, italics, or a bullet list).
4. **Edit a code chunk:** Modify an existing code chunk or add a new one. For example:

```

1 ```{r}
2 # My first R calculation
3 2 + 2
4 ```

```

5. **Render** the document. Check that the output looks correct.
6. **Commit and Sync:**
  - Open Source Control.
  - Stage your changed files (+).
  - Write a commit message: e.g., "Complete Week 2 assignment: edit YAML, add notes, render report".
  - Click **Commit**, then **Sync Changes**.
7. **Stop** your Codespace.
8. **Verify** on github.com that your commit appears in the repository.
9. **Submit** your forked repository link on Canvas.

---

## workflows14.qmd Quick Reference

| Action           | How                                                                                         |
|------------------|---------------------------------------------------------------------------------------------|
| Launch Codespace | Repository → green <><br><b>Code</b> → Codespaces →<br><b>Create codespace on main</b>      |
| Reopen Codespace | <a href="https://github.com/codespaces">github.com/codespaces</a> →<br>click Codespace name |

| Action            | How                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------|
| Stop Codespace    | <a href="https://github.com/codespaces">github.com/codespaces</a> → ...<br>→ <b>Stop codespace</b> |
| Open Explorer     | <b>View</b> → <b>Explorer</b> or click<br>file icon (left sidebar)                                 |
| Open Terminal     | <b>View</b> → <b>Terminal</b> or<br><b>Ctrl+`</b>                                                  |
| Render Quarto     | Click <b>Render</b> button (blue<br>arrow) at top of editor                                        |
| Stage a file      | Source Control → hover file<br>→ click +                                                           |
| Commit            | Type message → click                                                                               |
| Sync to GitHub    | Click <b>Sync Changes</b>                                                                          |
| Reload window     | Command Palette (F1) →<br><b>Reload Window</b>                                                     |
| Rebuild container | Command Palette (F1) →<br><b>Codespaces: Rebuild<br/>Container</b>                                 |

# The Analogy: Public Computer vs Cloud Drive

To understand the workflow, imagine working in a university computer lab.

## Repository (Cloud Drive)

**Important Distinction:** While GitHub is your “Cloud Drive,” it does **NOT** work like Google Drive.

- **Google Drive:** Auto-saves every keystroke.
- **GitHub:** Works like **Email**. You can write a draft (save in Codespace), but the cloud never sees it until you hit “Send” (Commit & Push).
- Stores your files permanently

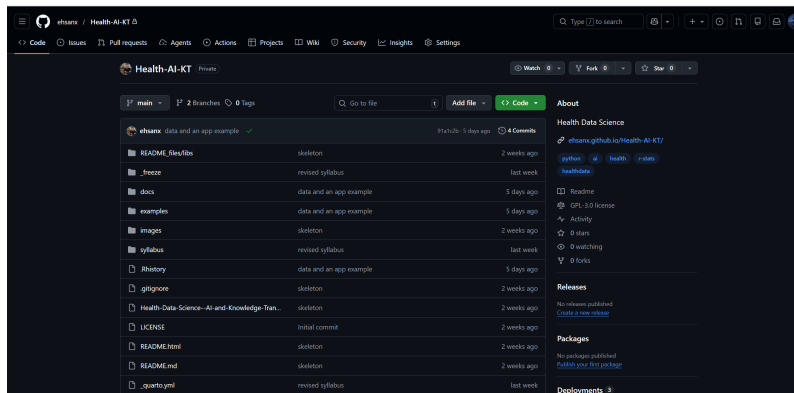


Figure 1: The code will be stored and accessible on GitHub as long as the company exists. Given that it’s owned by a large corporation like Microsoft, it will likely last for a long time.



- Safe if your computer crashes
- This is your **master copy**

## Codespace (Public Computer)

Codespaces is like a library computer:

- **Temporary machine** It is a disposable environment. If it breaks, we just delete it and get a new one.

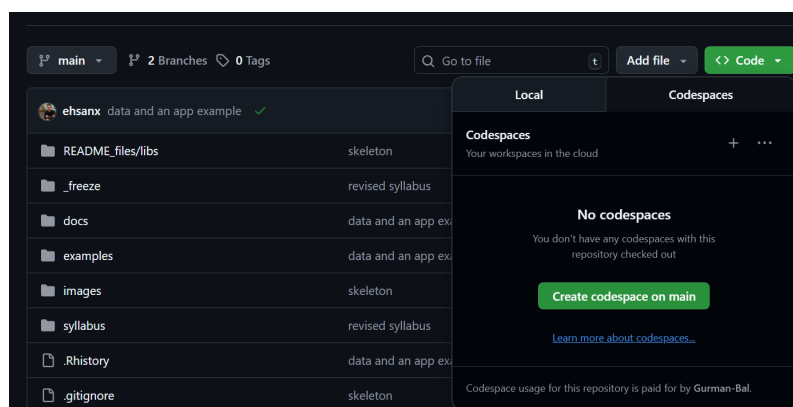


Figure 2: Can be accessed by clicking the <> Code button, navigating to the Codespaces tab, and then starting up a Codespace which contains all the file in the Repo.

- **Pre-installed software** It comes with R, Python, and Quarto installed, so you don't have to install anything on your own laptop.
- **The “Two-Save” Rule (Crucial)** Because this is a temporary computer, there are two steps to saving:
  1. **Ctrl+S (Save):** This only saves to the temporary machine. If the machine times out (closes), this is lost.
  2. **Commit & Push:** This sends your work back to the Repository (Cloud Drive). This is the only way to be safe.

**Warning:** If you close the tab without “Committing” (Sending), your work is erased.

### **Check for understanding**

Answer:

- Where are your permanent files stored?
- What happens if you press **Ctrl+S** but never commit?

# Tour of the Repository

## Step-by-step

1. Open your forked course repository on GitHub

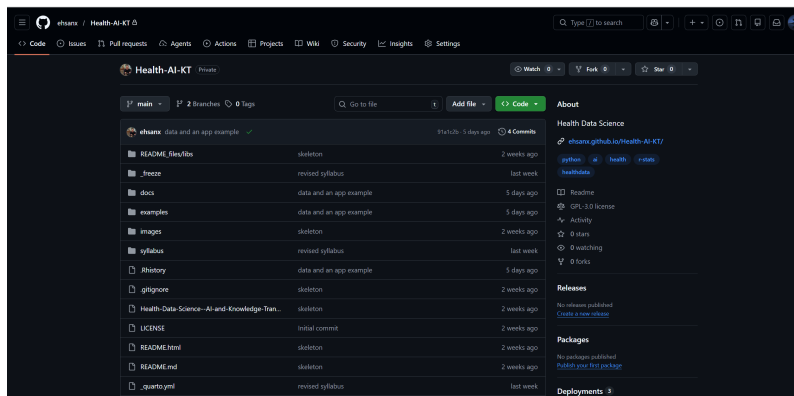
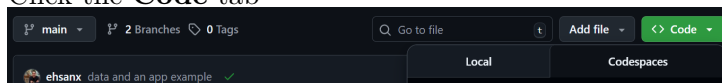


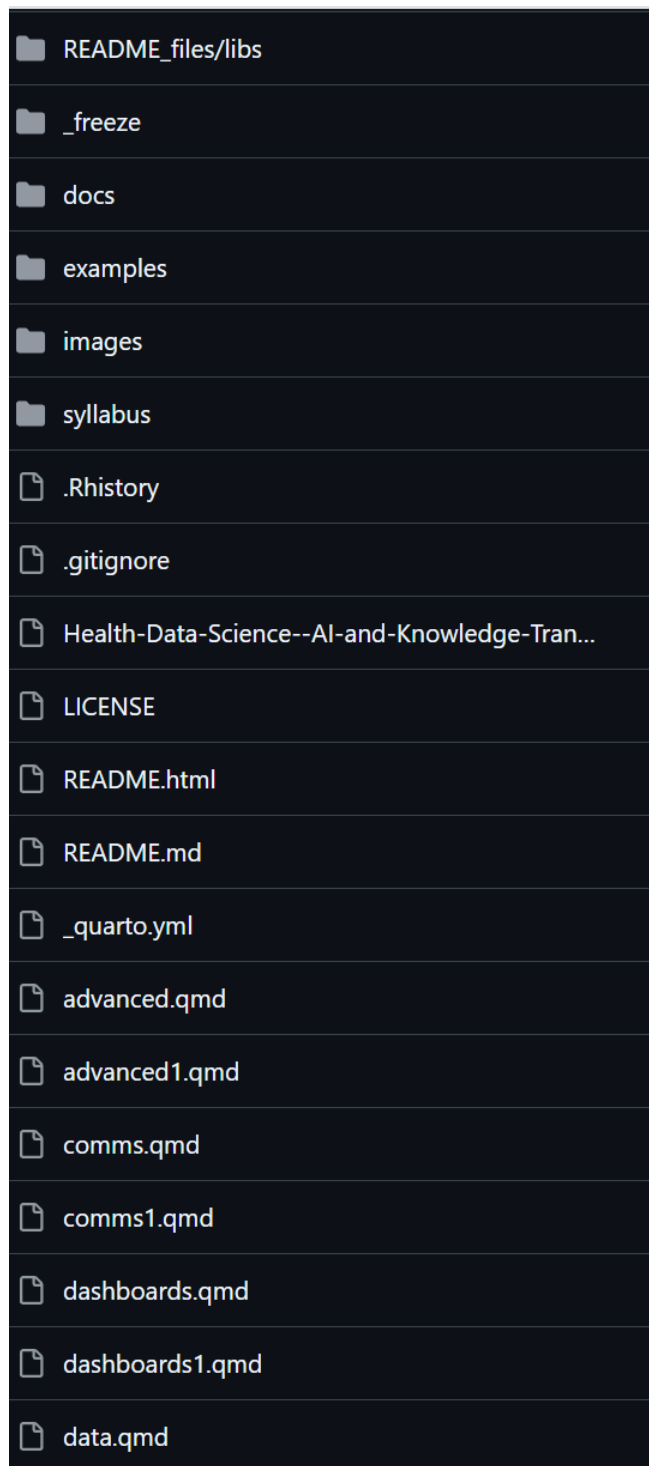
Figure 1: Main github page

2. Click the **Code** tab



- **Why?** GitHub has many tabs (Issues, Actions, Settings). The **Code** tab is your “Reset” button. If you ever get lost in a menu, click this to go back to your files.

3. Look at the file list



- This shows the **current state** of the project in your fork.
- **Crucial:** If a file is listed here, it is safe. If it is *not* here, it only exists on your temporary “Desk” (Codespace) and is at risk of being lost.

#### 4. Scroll to read the README

- The `README.md` file is displayed automatically at the bottom. Think of it as the “Cover Sheet” or “Instruction Manual” for your project.

## Task

Find this file:

`practice_report.qmd`

**Verification:** If you can see `practice_report.qmd` in this list, it means the file is safely stored in the cloud (The Vault).

If you can see it in your fork, it is safely stored in the cloud.

# Opening Your First Codespace

This process is like walking into the library and turning on a computer. Start from your forked repository, then create a fresh, temporary environment in the cloud.

## Steps

1. Click the green **Code** button
  - This button is usually for downloading files, but we are using it to launch a computer.

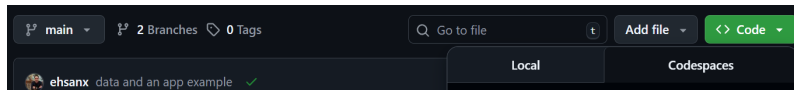


Figure 1: Code button

2. Click the **Codespaces** tab
  - **Crucial Choice:** You will see two tabs: “Local” and “Codespaces.”
  - **Local:** Means downloading to your own messy laptop (Hard mode).
  - **Codespaces:** Means using the pre-installed cloud computer (Easy mode). Always choose this.

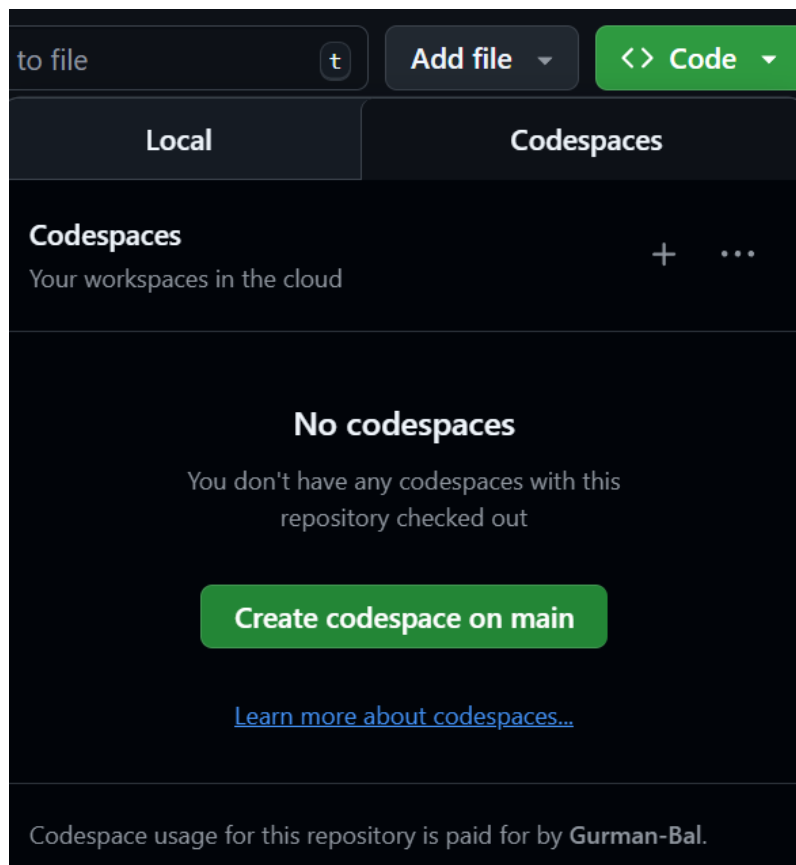


Figure 2: Codespaces tab

3. Click **Create Codespace on main**

- This tells GitHub to build a new computer for you from scratch.

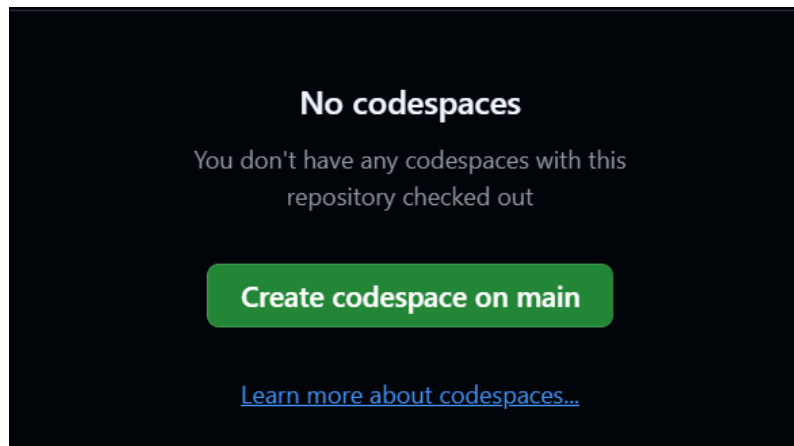


Figure 3: Create Codespace

## The “Boot Up” Phase

Wait 2–3 minutes.

- **What is happening?** GitHub is silently installing R, Python, Quarto, and all the course libraries for you.
- **The Analogy:** Think of this as the “ventilation system” starting up in a lab. It takes a moment to get the environment ready.

**Success:** You should see **VS Code** (a dark-themed code editor) appear in your browser window.



# Reopening a Codespace

**The Rule:** Do not create a new Codespace each time you log in.

- **Why?** If you click the green “Create” button again, GitHub builds a *second* computer.
- **The Risk:** If you left uncommitted work (Drafts) on “Computer A” and you start “Computer B,” your work will appear lost. It isn’t lost: it’s just on the other computer.

To resume exactly where you left off:

## Steps

1. Go to the **Codespace Dashboard**: <https://github.com/codespaces>
  - Think of this as the “Front Desk” where all your active computers are listed.
2. Find your Codespace
  - Look for the box labeled with your course repository name.
  - **Note:** GitHub gives each Codespace a funny random name (e.g., *literate-computing-machine* or *glorious-couscous*). This is the name of your rented desk.

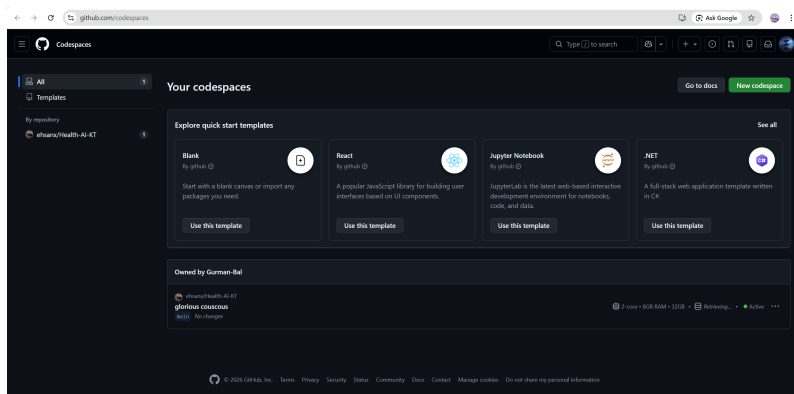


Figure 1: GitHub Codespaces

### 3. Click the **Name** of the Codespace

- Do **not** click the “New codespace” button in the corner. Click the text link of the funny name.

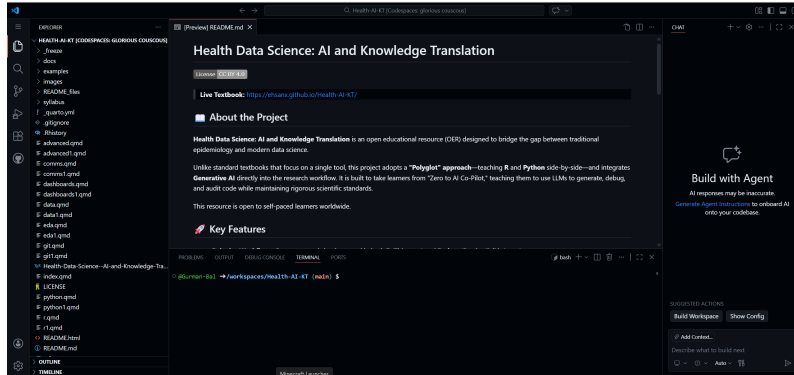


Figure 2: Your Codespace

### 4. Wait for it to resume

- **Benefit:** Resuming an existing Codespace is much faster than creating a fresh one because the software is already installed.

# VS Code Orientation



When you open a Codespace, you are using a version of **VS Code** in your browser. It has the same interface as the desktop version used by professional software engineers at Google and Microsoft.

**Don't Panic:** The interface looks complex because it has hundreds of features. **You only need to know three areas.** You can safely ignore 90% of the buttons you see.

## Explorer (left)

The “File Cabinet”

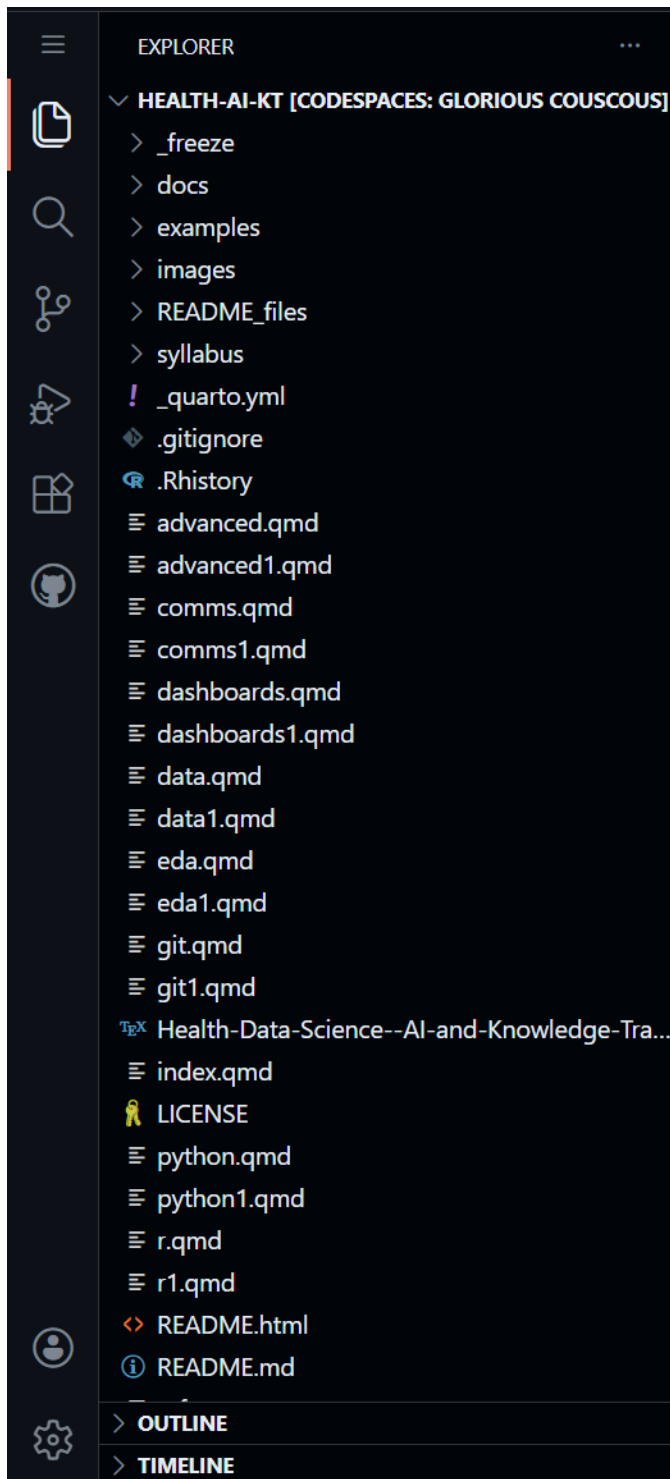


Figure 1: Shows files and folders.

- This shows your folders (directories) and files.
- Think of this as your navigation pane. Double-click a file here to open it in the Editor.

## Editor (center)

### The “Notebook”

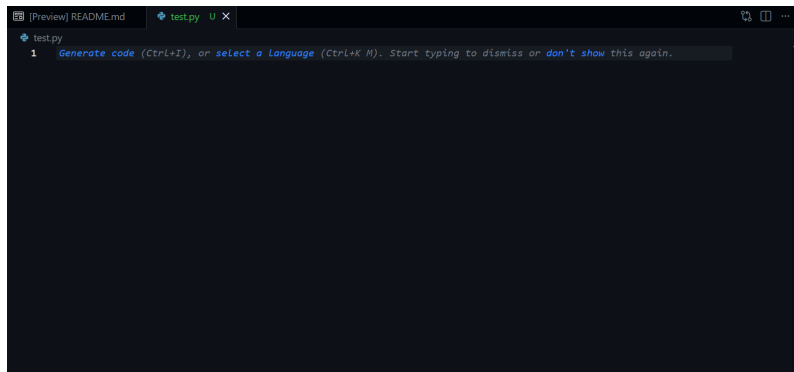


Figure 2: Where you type.

- This is where you do your actual work: typing code and writing text.
- In this course, you will mostly be editing `.qmd` (Quarto) files here.

## Terminal (bottom)

### The “Engine Room”

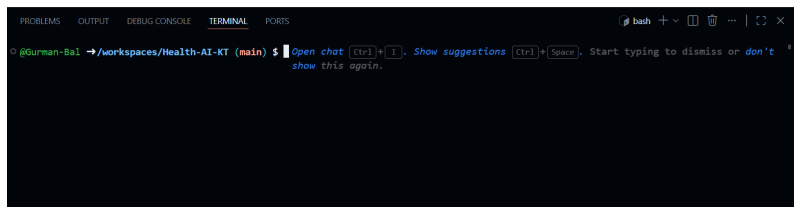


Figure 3: Terminal window

- This is where R and Python actually run.
- You will see text scrolling here when you “Render” a document. It looks like “The Matrix,” but it is usually just the computer telling you what it is working on.

If the terminal disappears (or you accidentally close it):

**View → Terminal**

### **Mini Task**

Open and close a few files in the Explorer to get comfortable with the layout.

# Files vs Folders

We use a consistent structure to ensure reproducibility. In Data Science, being “organized” isn’t just about being neat—it is a requirement for your code to work on someone else’s machine.

## The Golden Rule of Health Data

We treat data like a **Crime Scene**—never touch the evidence.

- **Start:** `data/` (The Crime Scene). **READ ONLY.**
  - *Rule:* Never open a dataset in Excel and “fix” a typo. That is tampering with evidence.
  - *Correct Way:* Load the “dirty” data into R/Python and fix the typo using code. This leaves a trail of exactly what you changed.
- **End:** `output/` (The Evidence). All graphs and cleaned tables go here.
- **The Detective:** Your code (`.qmd`) moves information from Start to End.

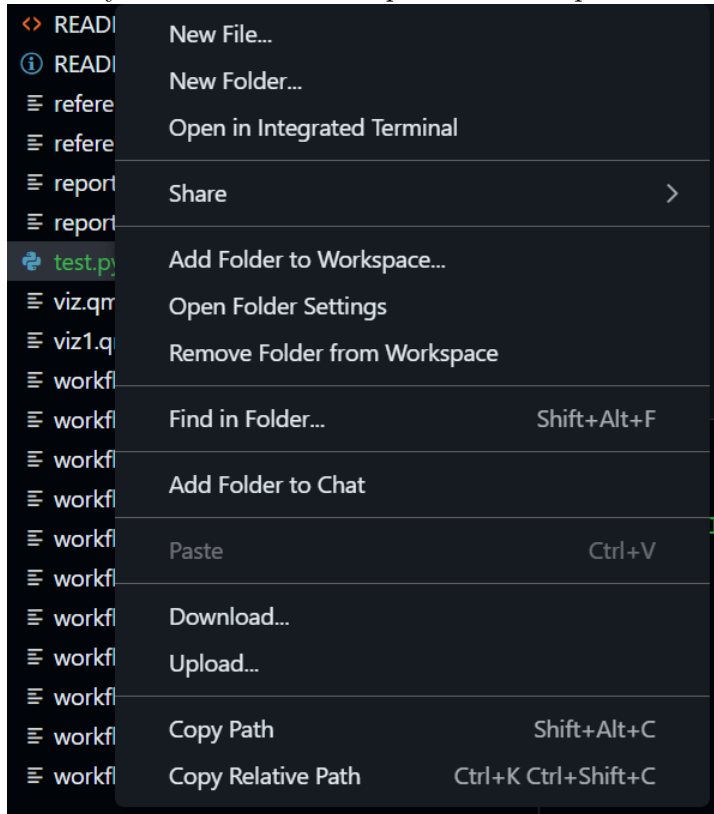
## Files vs. Folders (The Mental Model)

- **Files (e.g., `report.qmd`):** These are **sheets of paper**. You write on them.
- **Folders (e.g., `data/`):** These are **labeled boxes**. You store things in them.

## Create a folder

In Explorer:

1. Right-click anywhere in the blank space of the Explorer



pane.

2. Select **New Folder**
3. Name it:

images

**Pro Tip:** Always use **lowercase** and **no spaces** for folder names (e.g., `my_data`, not `My Data`). Computers struggle with spaces in file paths.

## Typical structure

Your project should always look like this:



|         |                   |
|---------|-------------------|
| data/   | raw data          |
| output/ | generated results |
| images/ | screenshots       |
| .qmd    | reports           |

**Remember:** The `data` folder is for *input*. The `output` folder is for *results*. They should never mix.

# The Magic Moment: Render Quarto

## Why do we “Render”?

In the old days, you would run code, copy a graph, save it as a JPG, and paste it into Word. If the data changed, you had to redo every step manually.

**The Better Way:** Think of your code (`.qmd` file) as a **Recipe** and the final report (HTML/PDF) as the **Meal**. \* **Editing:** You don’t change the meal; you change the recipe. \* **Rendering:** This is the “cooking” process. When you click **Render**, the computer follows your instructions from scratch to create a fresh meal.

This is the secret to reproducibility: if you give your recipe to a friend, they can cook the exact same meal.

## Step 1 — Open the file

1. Go to the **Explorer** pane (Left side).
2. Locate and click on the file:

```
practice_report.qmd
```

This will open the “Source Code” in the center Editor window. You will see a mix of normal text and code chunks.

## Step 2 — Edit name (The YAML Header)

Look at the very top of the file. You will see a block of text sandwiched between three dashes (`---`). This is called the **YAML Header**. It controls the document’s metadata.

Find the line that says `author:`

```
author: "Your Name"
```

**Task:** Delete “Your Name” and type your actual name inside the quotes. Do not remove the quotes!

## Step 3 — Render

1. Look for the **Render** button at the top of the Editor pane (it often looks like a Blue Arrow pointing right, or simply says “Render”).
2. **Click it.**
3. **Observe:**
  - **The Terminal (Bottom):** You will see text scrolling quickly. This is R/Python executing your code chunks.
  - **The Preview (Right):** A new window should appear showing a beautiful, formatted report with your name on it.

**Congratulations!** You just performed “Knowledge Translation”—turning raw code into a human-readable document.

## If it fails (Troubleshooting)

Rendering involves many moving parts, so it sometimes gets stuck.

- **The “Wait” Rule:** If nothing happens for 20 seconds, the background process might be initializing. Give it a moment.
- **The “Retry” Rule:** Sometimes the connection hiccups. Click **Render** one more time.
- **The “Red Text” Rule:** Look at the **Terminal** tab at the bottom. If you see red text, it is an error message. Read it—it usually tells you exactly what went wrong (e.g., “File not found” or “Syntax error”).

# Commit Your Work

Your work is currently only on the “Rented Desk” (Codespace). It is a **Draft**. To save it permanently to the “Vault” (GitHub), you must **Commit** and **Push** (Send).

**The Analogy:** Think of this like shipping a package. You have packed the box (saved the file), but now you need to put a label on it and hand it to the courier.

## Steps

1. Click the **Source Control** icon
  - It looks like a **blue bubble** or a **connect-the-dots graph** on the far left sidebar.
  - This opens the “Shipping Department.” You will see a list of every file you changed.

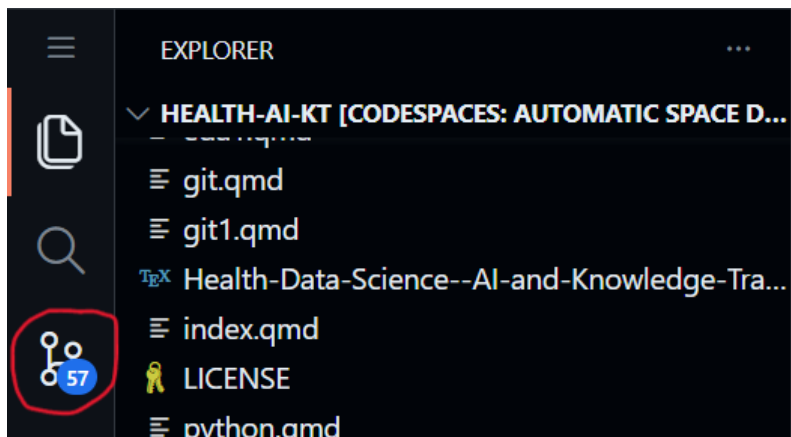


Figure 1: Source control icon circled in red

## 2. Type a **Commit Message**

- **The Label:** You cannot ship a package without a label.
- **What to write:** Be descriptive but brief. Imagine you are leaving a note for your future self about *what* you did.
- *Example:* “Updated author name in practice report”.
- *Bad Example:* “stuff” or “fix”.

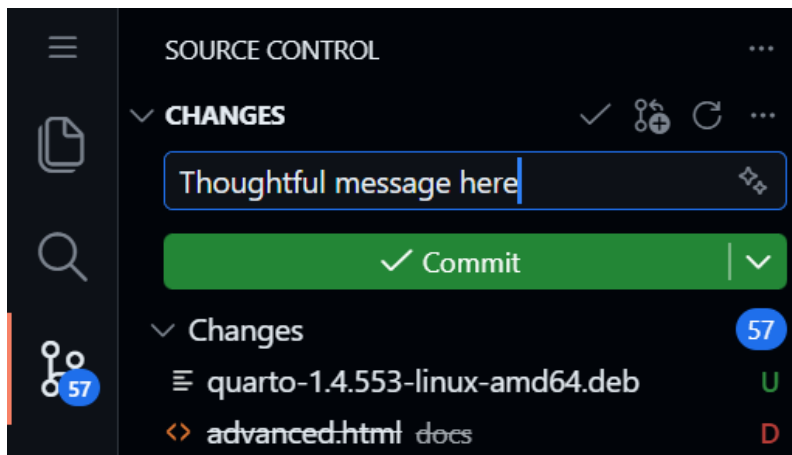


Figure 2: This is required

## 3. Click **Commit and Sync** (or Commit and Push)

- This is the “**Send**” button.
- **Action:** It wraps up your changes, stamps them with the time and your message, and teleports them to the Cloud Vault.
- *Note:* You might be asked to click “Always” or “Yes” to approve the sync.

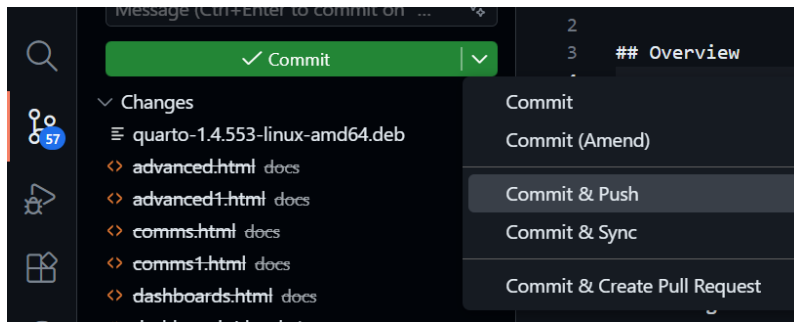


Figure 3: This will push your changes to the remote branch

## Verify (Trust but Verify)

Did it actually work?

1. Go back to your **GitHub Repository** tab (The Vault).
2. **Refresh** the page.
3. Look at the file list. You should see your “Commit Mes-  
sage” displayed next to the file you just edited.

# Troubleshooting: Rebuild Container

Sometimes the cloud computer gets “stuck.” Maybe the terminal froze, or R crashed.

**The Fix:** We don’t fix broken computers; we get new ones. “Rebuilding the Container” is the cloud equivalent of **“Have you tried turning it off and on again?”**—but better, because it gives you a fresh machine while keeping your files safe.

## When to use this?

If things break:

- **Terminal frozen:** You type, but nothing appears.
- **Render fails:** You get weird errors that make no sense.
- **Environment issues:** R or Python can’t find a package that was there yesterday.

## Steps

### 1. Open the Command Palette

- Press `F1` (or `Ctrl+Shift+P` on Windows / `Cmd+Shift+P` on Mac).
- A search bar will appear at the very top of the screen. This is the “God Mode” search bar for VS Code.

### 2. Search

- Type: Rebuild Container



### 3. Execute

- Click on the option that says: **Codespaces: Re-build Container**

### 4. Wait (~2-5 minutes)

- The screen will reload.
- **Don't Panic:** You might see a black screen with scrolling text logs. This is normal: it is reinstalling the software.
- When your files reappear in the Explorer, you are ready to go.

### Did I lose my work?

- **Saved files:** No. Anything you pressed **Ctrl+S** on is safe.
- **Unsaved typing:** Yes.
- **R Memory:** Yes. You will need to re-run your code chunks from the top.

# Stop Your Codespace

Codespaces is not free forever. You have a monthly allowance (approx. 60 hours/month for free accounts).

**The Analogy:** Think of a Codespace like a **Taxi**.

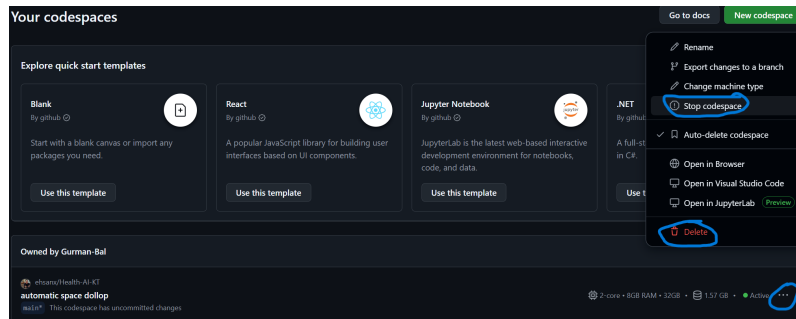
- **Working:** When you are coding, the meter is running.
- **The Trap:** If you just close your browser tab, it is like getting out of the taxi but asking the driver to “wait here.” **The meter keeps running.**
- **The Fix:** You must explicitly “end the ride” (Stop the Codespace).

## Correct shutdown

1. Go to the **Codespace Dashboard**: <https://github.com/codespaces>
  - This is your command center.
2. Find your active Codespace
  - Look for the green text that says “Active”.
3. Click ... (The “Meatball” Menu)
  - It is located on the far right side of the Codespace box.
4. Click **Stop Codespace**
  - **Stop:** This “pauses” the computer. It saves your installed packages and RAM state so you can resume quickly next time. **(Recommended)**
  - **Delete:** This destroys the computer. You will have to wait 2-3 minutes for the “Boot Up” phase next time you start.

**Critical Warning:** Closing the browser tab is **NOT** enough.

- GitHub keeps the machine running for **30 minutes** after you close the tab (in case your internet flickered).
- **The Cost:** If you just close the tab every day, you will waste ~15 hours of your monthly quota on empty time.



## Final Checklist

- ☐ I understand repository vs Codespace
- ☐ I created or located my forked course repository
- ☐ I rendered a `.qmd` file
- ☐ I committed changes
- ☐ I know how to rebuild container
- ☐ I know how to stop Codespace

## Reflection

Write 2–3 sentences:

- What was easiest?
- What was confusing?
- What questions do you still have?

## **Week 3: R with AI**

# Week 3: R with AI

## Overview

**Summary:** In Week 2 you learned how to use your Codespace, edit a Quarto document, and make your first commit. Now you learn how to speak R—not by memorizing syntax, but by directing an AI co-pilot and *auditing* what it writes. Your job is to be the pilot: you decide what the analysis should do, the AI drafts the code, and you verify every line before it flies.

### Learning Objectives:

- Understand that R is a calculator that needs specific, case-sensitive instructions.
- Load packages with `library()` and explain why this is needed each session.
- Run a minimal `renv` checkpoint so package changes are visible.
- Use core constructs: variables (assignment), functions (with arguments), and the pipe (`%>%` / `|>`).
- Perform basic data transformations using `dplyr` verbs: `filter()`, `select()`, `mutate()`, `summarise()`, `group_by()`.
- Inspect a dataset on first contact: structure, dimensions, missingness.
- Identify control flow (`if/else`, loops) in AI-generated code and audit whether it is correct.
- Write and adapt a simple custom function.
- Apply an AI safety checklist to catch hallucinations, wrong column names, and unsafe operations.
- Document your analysis in Quarto by weaving code chunks with Markdown narrative.
- Commit incrementally as you work (reinforcing Week 2 Git basics).

## Connection

Week 2 made the project structure reproducible. This week you use that structure to load, inspect, summarize, and audit R code with AI support. Next week you will put those edits under version control.

## Case Study Data Analysis

The **NHANES Health Equity data spine** now supports code-reading, EDA, visualization, and reproducibility checks. Use the cached CSV/RDS for routine class work; a CDC retrieval script exists for advanced users.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The default classroom path is to use the cached data so the analysis work is reproducible without internet access.

```
1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_analysis <- read_csv(nhanes_path, show_col_types = FALSE)
15
16 nhanes_analysis |>
17   summarise(
18     rows = n(),
```

```

19     missing_bmi = sum(is.na(BMI)),
20     missing_income = sum(is.na(IncomeGroup))
21   )
22   #> # A tibble: 1 x 3
23   #>       rows missing_bmi missing_income
24   #>   <int>         <int>         <int>
25   #> 1 105626         9356         9415
26
27   nhanes_analysis |>
28     filter(!is.na(BMI), !is.na(IncomeGroup), Age >= 20, Age <= 80) |>
29     group_by(IncomeGroup) |>
30     summarise(
31       n = n(),
32       mean_bmi = round(mean(BMI), 1),
33       .groups = "drop"
34     )
35   #> # A tibble: 3 x 3
36   #>   IncomeGroup           n mean_bmi
37   #>   <chr>             <int>   <dbl>
38   #> 1 High Income (>3.5) 16330    28.6
39   #> 2 Low Income (<1.3) 15293    29.4
40   #> 3 Middle Income     19228    29.3

```

## In-Class Activity

- Load the CSV snapshot and inspect column names, row count, and missingness.
- Summarize one numeric variable and one grouping variable with clear NA handling.
- Ask an AI assistant to comment on your code, then audit whether the comments match the actual operations.
- Note one place where an AI assistant could hallucinate a variable name or overstate what the descriptive summary means.

Student output: a short Quarto note with one data inspection, one summary, and one AI-audit comment. Assignment link: [Assignment 2](#).



---

## r1.qmd Packages: Opening the Toolbox

### Install vs Load

- `install.packages("tidyverse")` — **buying** the toolbox. Done once per environment (already done in your course Codespace).
- `library(tidyverse)` — **opening** the toolbox. Required every time you restart R or reopen your Codespace.

```
1 library(tidyverse)
```

#### Note

If you see an error like `there is no package called 'tidyverse'`, the package is not installed in your Codespace. Use the Canvas support channel or office hours and include the full error message.

### Minimal renv Checkpoint

`renv` helps record which R packages a project uses. You do not need to master `renv` this week, but you should recognize the three checkpoint commands that support reproducibility.

```
1 # Check whether the project environment has changed.
2 renv::status()
3
4 # Record package changes when you intentionally add or update packages.
5 renv::snapshot()
6
7 # Restore the recorded package environment in a fresh setup.
8 renv::restore()
```

For Assignment 2, run `renv::status()` and write a short dependency note. Only run `renv::snapshot()` if you intentionally added a package.

---

## r2.qmd Core Constructs: Boxes, Toasters, and the Assembly Line

### Variables: The Labeled Box

R has no memory unless you store results in a named box using `<-`.

```
1 patient_age <- 55
2 clinic_name <- "Vancouver General"
```

#### Warning

**Case sensitivity:** Age, age, and AGE are three different boxes. This is one of the most common bugs in AI-generated code. Always check that column names match your data exactly.

### Functions: The Toaster

A function takes **input** and returns **output**. Arguments (inside the parentheses) control its behaviour.

```
1 mean(c(72, 85, 90), na.rm = TRUE)
```

- `c(72, 85, 90)` is the input (a vector of numbers).
- `na.rm = TRUE` tells R to ignore missing values. **Always look for this** when auditing AI code that works with health data — missingness is the norm, not the exception.

## The Pipe: The Assembly Line

The pipe passes the result of one step into the next. Think of it as “and then...”

```
1 data %>%  
2   filter(age > 65) %>%  
3   summarise(mean_bmi = mean(bmi, na.rm = TRUE))
```

Read this as: “Take the data, and then keep only patients over 65, and then calculate the average BMI.”

### Tip

You will see two pipe symbols in the wild: %>% (from the `magrittr` / `tidyverse` packages) and `|>` (built into R 4.1+). They do the same thing for our purposes. If AI generates the other one, don’t panic — both work in your Codespace.

---

## r3.qmd Data Import and First Contact

### Loading Data

Use relative paths so your code runs on any machine (including your grader’s Codespace).

```
1 my_data <- read_csv("data/patients.csv")
```

**How to get the path right:** Right-click the file in the VS Code Explorer → **Copy Relative Path** → paste it into your code.

### AI Prompt:

Read the CSV file at “data/patients.csv” and save it as `my_data`. Use the `readr` package.

## First Contact: Inspect Before You Analyze

Never start analysis without looking at your data. Run these four checks:

```
1 # How big is it?
2 dim(my_data)          # rows x columns
3
4 # What are the column names and types?
5 glimpse(my_data)
6
7 # Quick summary statistics
8 summary(my_data)
9
10 # How much is missing?
11 colSums(is.na(my_data))
```

Open the **Environment** pane (top-right in RStudio or via the R extension) to see your variables and click a data frame to view it in a grid.

### Warning

**Health data reality:** Real datasets like NHANES have substantial missingness. Knowing *where* and *how much* is missing is not optional — it shapes every downstream decision.

---

## r4.qmd Data Transformation: The Five dplyr Verbs

These five verbs are the core of data wrangling in R. You will use them in nearly every assignment and in your term project.

### **filter() — Keep rows that match a condition**

```
1 # Patients aged 65 and older
2 seniors <- my_data %>% filter(age >= 65)
```

### **select() — Keep (or drop) columns**

```
1 # Keep only ID, age, and BMI
2 slim <- my_data %>% select(id, age, bmi)
```

### **mutate() — Create or modify a column**

```
1 # Add a BMI category column
2 my_data <- my_data %>%
3   mutate(bmi_category = case_when(
4     bmi < 18.5 ~ "Underweight",
5     bmi < 25   ~ "Normal",
6     bmi < 30   ~ "Overweight",
7     TRUE      ~ "Obese"
8   ))
```

### **summarise() — Collapse rows into a summary**

```
1 my_data %>% summarise(mean_age = mean(age, na.rm = TRUE))
```

### **group\_by() + summarise() — Summaries by group**

```
1 my_data %>%
2   group_by(sex) %>%
3   summarise(
4     n = n(),
5     mean_bmi = mean(bmi, na.rm = TRUE)
6   )
```

### AI Prompt:

Using `dplyr`, filter my dataset to patients over 65, group by sex, and calculate the mean BMI for each group. Handle missing values.



#### Tip

**Audit the AI output:** After running the AI-generated code, check that (a) the column names match your actual data, (b) `na.rm = TRUE` is present where needed, and (c) the row counts make sense (e.g., the filtered data should be smaller than the original).

**Git checkpoint:** You have now loaded, inspected, and transformed data. Stage your `.qmd`, write a commit message describing what you did, and commit. Don't wait until the end.

---

## r5.qmd Control Flow: Auditing AI Decisions

You will not write many `if/else` blocks or loops by hand. But AI frequently generates them, so you need to **read** them.

### If/Else: The Fork in the Road

```
1 if (mean_age > 50) {  
2   print("Older cohort")  
3 } else {  
4   print("Younger cohort")  
5 }
```

**Audit question:** Is the threshold (50) appropriate for my research question, or did the AI pick an arbitrary number?

## Loops: The Treadmill

```
1 for (col in c("age", "bmi", "sbp")) {  
2   print(summary(my_data[[col]]))  
3 }
```

**Audit questions:** Does the loop have a clear end condition? Could this be replaced with a simpler vectorized operation (e.g., `summarise(across(...))`)?

### Note

In R, loops are rarely the best tool. If AI gives you a loop, ask: “*Can this be done without a loop using dplyr or purrr?*” The answer is usually yes.

---

## r6.qmd Writing a Simple Function

The syllabus asks you to write at least one custom function. A function packages a repeated task so you don’t copy-paste code.

### Example: A Reusable Summary

```
1 summarise_column <- function(data, column_name) {  
2   data %>%  
3     summarise(  
4       n_total   = n(),  
5       n_missing = sum(is.na(.data[[column_name]])),  
6       mean_val  = mean(.data[[column_name]], na.rm = TRUE),  
7       sd_val    = sd(.data[[column_name]], na.rm = TRUE)  
8     )  
9 }  
10
```

```
11 # Use it
12 summarise_column(my_data, "bmi")
13 summarise_column(my_data, "age")
```

## Your Task

1. Ask your AI co-pilot to write a function that takes a data frame and a column name and returns summary statistics.
2. **Audit the code:** Does it handle NA? Does it work on a numeric column? What happens if you pass a categorical column?
3. **Adapt it:** Change one thing (e.g., add a median, or add a `group_by` argument). This is the difference between accepting code blindly and understanding it.

## AI Prompt:

Write an R function called `summarise_column` that takes a data frame and a column name (as a string) and returns the count, number of missing values, mean, and standard deviation. Then show me how to call it on the “bmi” column of `my_data`.

---

## r7.qmd Debugging: Reading the Red Text

Errors are normal. AI makes mistakes, you make typos, and data surprises you.

## The “Fix It” Loop

1. **Read** the error message. Look for keywords: `not found`, `unexpected`, `non-numeric`.
2. **Copy** the full error text.
3. **Paste** it into your AI chat and ask:



Here is an R error from my Codespace: [paste]. Explain what went wrong in one sentence and give corrected code.

## Common Errors and What They Mean

| Error message                           | Likely cause                                                      |
|-----------------------------------------|-------------------------------------------------------------------|
| object 'X' not found                    | Typo in variable name, or you forgot to run an earlier chunk      |
| could not find function "X"             | Forgot <code>library(tidyverse)</code> or misspelled the function |
| non-numeric argument to binary operator | Trying to do math on a text/factor column                         |
| unexpected symbol                       | Missing comma, parenthesis, or pipe                               |

### Warning

**AI debugging guardrail:** When AI suggests a fix, re-read the corrected code before running it. Does the fix address the *actual* error, or did AI rewrite the whole chunk and change your logic?

## r8.qmd AI Safety: The Audit Checklist

Every time you accept AI-generated code, run through this checklist:

1. **Column names:** Are they spelled exactly as they appear in `glimpse(my_data)`?
2. **Data types:** Is the code doing math only on numeric columns?
3. **Missingness:** Is `na.rm = TRUE` present where needed?

4. **Privacy:** Does the code avoid printing individual patient rows that could be identifiable?
5. **Logic:** Does the code do what *you* intended, or what the AI *assumed* you meant?

## The Paper Trail Rule

Every time you use AI to write a code chunk, add a `#` comment above it explaining what you asked for and what you verified:

```
1 # AI prompt: "Calculate mean BMI by sex, handling NAs"
2 # Verified: column names match, na.rm = TRUE present, output has 2 rows (M/F)
3 my_data %>%
4   group_by(sex) %>%
5   summarise(mean_bmi = mean(bmi, na.rm = TRUE))
```

This comment trail is your audit log. It demonstrates accountability (required by the course AI policy) and helps you debug later.

---

## r9.qmd Exporting Clean Data

After cleaning and transforming, save your work so future weeks can pick up where you left off.

```
1 # Save a clean version for Week 4 and beyond
2 write_csv(my_data, "output/clean_data.csv")
```

### Tip

**Reproducibility habit:** Your Quarto document should contain *every step* from raw data to clean export. Anyone (including your grader) should be able to render it from scratch and get the same `clean_data.csv`.

**Git checkpoint:** Your analysis is complete. Stage, commit with a clear message (e.g., "Complete Week 3: load, inspect, transform, and export patient data"), and Sync.

---

## r10.qmd Literate Analysis in Quarto

Your .qmd file is not just code — it is a **document**. Weave narrative around your code chunks to explain what you are doing and why.

### Pattern

```
1  ## Data Inspection
2
3  We begin by examining the structure of the dataset to understand
4  variable types and missingness before any transformations.
5
6  ```{r, eval=FALSE}
7  glimpse(my_data)
8  colSums(is.na(my_data))
9  ```
10
11
12  The dataset contains `nrow(my_data)` observations across
13  `ncol(my_data)` variables. Notably, BMI is missing for
14  approximately 12% of records, which we handle with `na.rm = TRUE`
15  in downstream summaries.
```

This interleaving of code and prose is **literate programming** — the methodology that underpins reproducible research in this course. Practice it now; you will need it for your term project report (Week 10) and final portfolio.

---

## r11.qmd Knowledge Check

1. What is the difference between `install.packages()` and `library()`?
  2. Why is `na.rm = TRUE` important when working with health data?
  3. What does `my_data %>% filter(age > 65) %>% summarise(n = n())` do? Read it aloud using “and then.”
  4. Name the five core `dplyr` verbs and what each does in one sentence.
  5. You see the error `object 'BMI' not found` but your column is called `bmi`. What went wrong?
  6. What is the difference between `%>%` and `|>`? Does it matter for this course?
  7. Why should you write `#` comments above AI-generated code?
  8. What four checks should you run immediately after loading a new dataset?
- 

## r12.qmd Assignment: The “Mad Libs” Report

**Task:** Build a short Quarto report that loads, inspects, transforms, and summarizes a health dataset — using your AI copilot for code generation and your own judgment for verification.

### Step-by-Step Workflow

1. Open `week3_assignment.qmd` in your Codespace.
2. **Load packages and data:** Use AI to write the `library()` and `read_csv()` calls. Verify the data loaded correctly with `dim()` and `glimpse()`.
3. **Inspect missingness:** Run `colSums(is.na(...))`. Add a Markdown paragraph interpreting what you see.

4. **Transform:** Use `dplyr` verbs to:
  - Filter to a meaningful subgroup (e.g., `age > 18`).
  - Create at least one new column with `mutate()`.
  - Produce a grouped summary with `group_by()` + `summarise()`.
5. **Write a function:** Create (or adapt from AI) a reusable summary function. Call it on at least two different columns.
6. **Export:** Save your cleaned data to `output/clean_data.csv`.
7. **Document:** Every code chunk should have:
  - A `#` comment stating what you asked the AI and what you verified.
  - A Markdown paragraph below interpreting the result.
8. **Render** the document to verify it runs cleanly from top to bottom.
9. **Git:** Stage, write a descriptive commit message, Commit, and Sync.

## Deliverable

A rendered `.qmd` that a reader can follow as a narrative: what data you started with, what you did to it, what you found, and why your decisions were justified.

## r13.qmd Quick Reference

| Concept       | R code                                  | Notes                                       |
|---------------|-----------------------------------------|---------------------------------------------|
| Load packages | <code>library(tidyverse)</code>         | Every session                               |
| Read CSV      | <code>read_csv("data/files.csv")</code> | Use relative paths                          |
| Assignment    | <code>x &lt;- 5</code>                  | The <code>&lt;-</code> arrow stores a value |

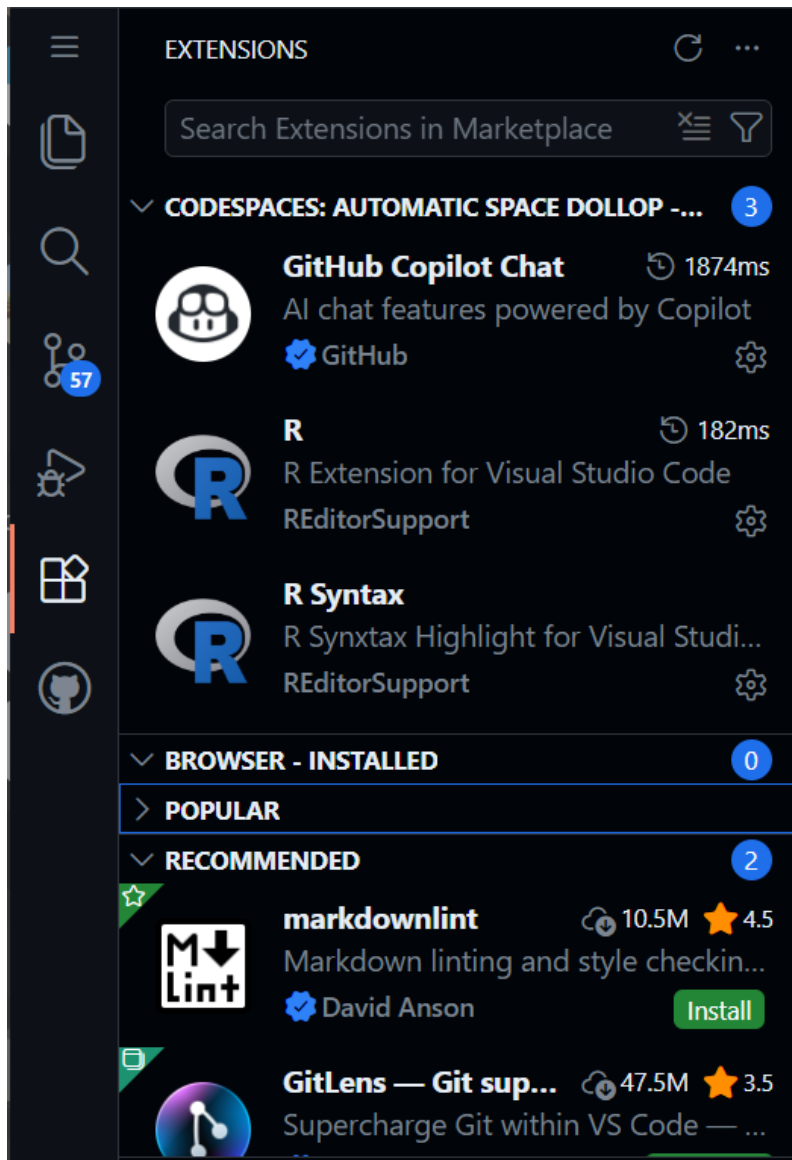
| Concept           | R code                                                | Notes                        |
|-------------------|-------------------------------------------------------|------------------------------|
| Inspect structure | <code>glimpse(df)</code>                              | Column names, types, preview |
| Dimensions        | <code>dim(df)</code>                                  | Rows $\times$ columns        |
| Summary stats     | <code>summary(df)</code>                              | Min, median, mean, max, NAs  |
| Check missingness | <code>colSums(is.na(df))</code>                       | Count NAs per column         |
| Filter rows       | <code>filter(df, age &gt; 65)</code>                  | Keep matching rows           |
| Select columns    | <code>select(df, id, age, bmi)</code>                 | Keep named columns           |
| New column        | <code>mutate(df, bmi_cat = ...)</code>                | Create or modify             |
| Summarize         | <code>summarise(df, m = mean(x, na.rm = TRUE))</code> | Collapse to one row          |
| Group + summarize | <code>group_by(df, sex) %&gt;% summarise(...)</code>  | Per-group summaries          |
| Export            | <code>write_csv(df, "output/clean.csv")</code>        | Save for later               |
| Pipe              | <code>%&gt;%</code> or <code> &gt;</code>             | “And then...”                |

# The Setup (The Cockpit)

Before we start flying, we need to check our instruments in **VS Code**. If your environment isn't set up correctly, your code will not run.

## Verify Your AI Extensions

1. Click the **Extensions** icon on the far-left sidebar (it looks like four squares).
2. In the search bar, ensure both **GitHub Copilot Chat** and **R** are installed and active.



## Set the AI Model to “0x”

1. Open the Copilot Chat sidebar (Shortcut: **Ctrl+Alt+I** or **Cmd+Option+I** on Mac).
2. Look for the model selector dropdown at the bottom of the chat window.



3. **Crucial Check:** Choose a model with a “0x” multiplier (like GPT-4o 0x). If you do not do this, you will quickly run out of your free student quota.
  - *Troubleshooting:* If the Chat asks you to “Sign In” or “Subscribe,” your Student Pack is not active. Use your institutional login at [copilot.microsoft.com](https://copilot.microsoft.com) or authenticate via your university email at <https://m365.cloud.microsoft/chat/>.

## Interface Review: Where does R actually live?

To learn R, we are going to start in the “sandbox.” Here is how your screen is laid out:

- **The Console (Bottom):** Your **Engine Room**. *Start here!* This is where you can type R commands directly next to the > prompt and press Enter to see immediate results. If you make a mistake, it will show up here in red text.
- **The Environment (Top Right):** Your **Work Bench**. Whenever you save a piece of data (like a patient’s age), it will appear in a list here so you can verify R actually “remembers” it.
- **The Explorer (Left Sidebar):** Your **File Cabinet**. This is where you find your **data/** folder containing your CSV files.
- **The Editor (Top Center):** Your **Laboratory Notebook**. Typing in the Console is great for testing, but it forgets everything when you close the window. Later in the lesson, we will use the Editor to write and save our permanent code.

## Interface Review: Working with .qmd and .Rmd files

When you open a .qmd (Quarto) or .Rmd (RMarkdown) file, you must know where to look. Here is how your coding workflow maps to your screen:

- **The Explorer (Left Sidebar): Your File Cabinet.** This is where you find your reports and your `data/` folder. You cannot run code here; you only click files to open them.
- **The Editor (Top Center): Your Laboratory Notebook.** This is where you write your text narrative and your R code.
  - *Note:* R code must be placed inside grey **Code Chunks** (starting with ````\r{` and ending with `````). To run a chunk, click the small green “Play” triangle in the top right corner of that specific grey box.
- **The Console (Bottom): Your Engine Room.** When you click the green “Play” button in the Editor, the code is sent down here to be executed. If there is an error, it will show up here in red.
  - *Pro Tip:* You can type quick math directly into the console (like `100 / 4`), but it will not be saved in your final report.
- **The Environment (Top Right): Your Work Bench.** Whenever you load data or save a calculation, it will appear in a list here so you can verify R actually “remembers” it.

# Packages (The Toolbox)

In R, tools come in “packages.” Think of `install.packages` as buying a physical toolbox, and `library()` as opening the lid.

- **To open your tools:** Type this in your code and run it:

```
library(tidyverse)
```

**Note:** You must run `library(tidyverse)` every single time you restart your Codespace.

# Basics: Boxes & Toasters

R is a system of storage (Boxes) and transformation (Toasters).

## Variables (The Box)

Use `<-` to save data into a name.

Code snippet

```
Save the number 55 into a variable called patient_age patient_age
```

## Functions (The Toaster)

Functions take input, perform an action, and return output.  
Look at the snippet below

Code snippet

```
calculate_rectangle_area <- function(length, width) {  
  area <- length * width  
  return(area)  
}
```

```
# Example usage:  
calculate_rectangle_area(10, 5) # Returns 50
```

# Control Flow

## The Pipe %>% (The Assembly Line)

The symbol %>% means “**and then...**”.

Both of these lines are identical, which would you prefer to read? (Note |> is another way to write %>%)

```
print(round(sqrt(10), 2))
```

```
10 |> sqrt() |> round(2) |> print()
```

For a real world example of how one may use the pipe

```
library(dplyr)
```

```
mtcars %>%  
  filter(wt > 3) %>%          # 1. Filter for cars heavier than 3,000 lbs  
  group_by(cyl) %>%          # 2. Group by number of cylinders  
  summarize(avg_mpg = mean(mpg)) # 3. Calculate the average MPG
```

## If / Else (The Fork in the Road)

An **if/else** statement lets the computer make a decision.

The program checks a condition:

- If it is **TRUE**, it runs one block of code.
- If it is **FALSE**, it runs another.

```
temperature <- 5
if (temperature > 10) {
  print("You can wear a t-shirt")
} else {
  print("Wear a jacket")
}
```

## What happens

1. The computer checks: `temperature > 10`
2. `5 > 10` is **FALSE**
3. So the **else** part runs.
4. Output:

```
"Wear a jacket"
```

Think of it like a **fork in the road**:

- One path if the condition is true
- Another path if it's false.

## Loops (The Treadmill)

A loop repeats code multiple times.

Instead of writing the same instruction again and again, you let the computer do it.

## Example

```
for (i in 1:5) {
  print(i)
}
```

## What happens

The computer runs the code **5 times**.

Steps:

1. `i = 1` → print 1
2. `i = 2` → print 2
3. `i = 3` → print 3
4. `i = 4` → print 4
5. `i = 5` → print 5

Output:

1 2 3 4 5

## Audit Check (Important)

When using loops, ask:

### 1. Does the loop end?

Bad example (infinite loop):

```
while (TRUE) {  print("This never stops") }
```

### 2. Could this be simpler?

Sometimes R has built-in functions that replace loops.

Instead of:

```
total <- 0 for (i in 1:5) {  total <- total + i }
```

You can write:

```
sum(1:5)
```

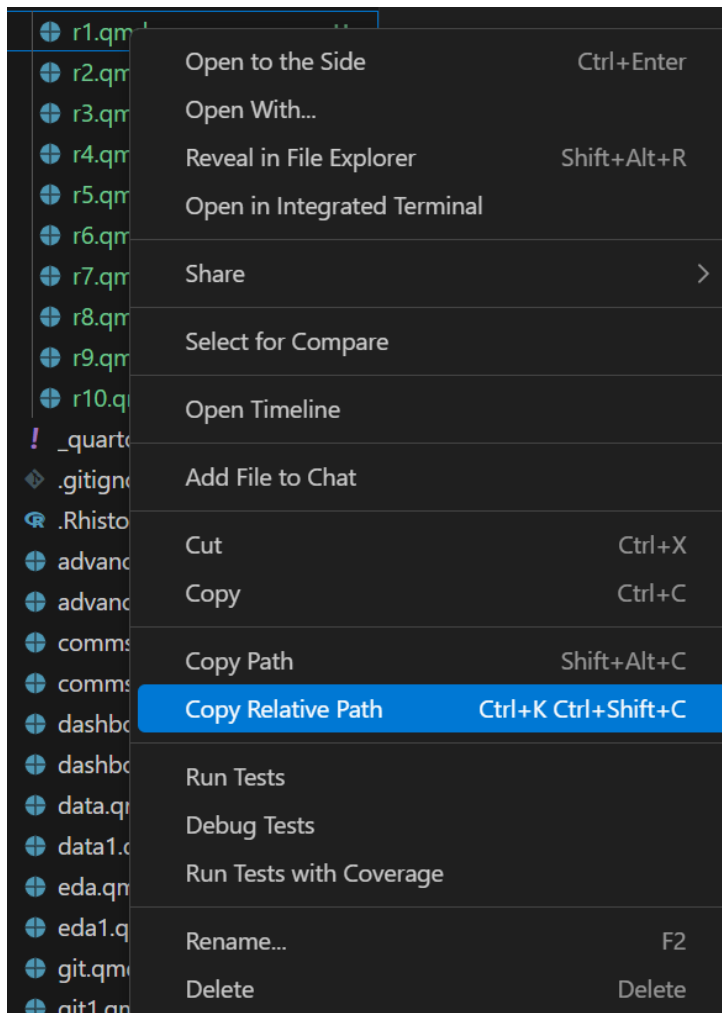
This is **shorter, safer, and faster**.

## The Loading Dock (Data Import)

Don't type file paths from memory; you will likely make a typo.

1. Find a data file in the **Explorer** (left sidebar).
2. **Right-click** the file → Select **Copy Relative Path**.





3. Ask AI: “Read the CSV at [Paste Path Here] and save it as my\_data.”

# The Detective (Debugging)

AI makes mistakes. When you see **Red Text** (errors) in the Console:

1. **Copy** the Red Text.
2. **Paste** it into Copilot Chat.
3. **Ask:** *“Explain this error and fix the code below: [Paste Code]”*

Try running this line and ask AI why it doesn't work:

```
"apple" + 5
```

# AI Safety: The Audit Checklist

Use this checklist before accepting any AI code:

- **Spelling:** Are column names spelled exactly as they appear in the data?
- **Case Sensitivity:** Is it `Age` or `age`? (R sees these as different things).
- **Missingness:** Did the AI remember `na.rm = TRUE` for `math`?
- **Privacy:** Did the AI try to print the whole dataset (potentially exposing patient IDs)?

# The Paper Trail (Comments)

**The Rule:** Every time you use AI to write code, write a # comment in plain English above it explaining what you told the AI to do.

Code snippet

```
# I asked AI to filter for patients with BMI over 30 and count them # [AI CODE BELOW]
```

Below is an example of non readable code

```
# What is x? What is 1.05? Why are we subsetting?
f <- function(x, y) {
  z <- x[x$a > y, ]
  r <- sum(z$b) * 1.05
  return(r)
}
```

```
res <- f(df, 100)
```

And here is an example of readable code

```
library(dplyr)

# Calculate the total revenue for high-value orders including a 5% tax
calculate_total_revenue <- function(order_data, minimum_price_threshold) {

  sales_tax_rate <- 1.05

  total_revenue <- order_data |>
    filter(price > minimum_price_threshold) |>
    summarize(total = sum(price)) |>
    pull(total)
```

```
    return(total_revenue * sales_tax_rate)
}
```

# Usage:

```
final_report_value <- calculate_total_revenue(current_orders, minimum_price_threshold = 100)
```

Just by reading through each line a reader should understand what is going on. One does not need to comment every single line, but rather comment whenever something complex is happening on the code line below it. And as always source work which is not yours.

# Assignment: The “Mad Libs” Report

1. Open `week3_assignment.qmd`.
2. Use Copilot to import the Week 1 dataset.
3. Use the **Pipe** to filter and summarize one variable.
4. **Audit:** Add a `#` comment explaining your prompt.
5. **Render** the file to HTML.
6. **Commit and Push** to GitHub.

## Final Summary Checklist

- ☐ Extensions are active (Copilot + R).
- ☐ Using **0x model** in Chat.
- ☐ Loaded `library(tidyverse)`.
- ☐ Data imported via **Relative Path**.
- ☐ All code chunks have an **Audit Comment (#)**.
- ☐ Handled **NA** values correctly.
- ☐ Document **Rendered** and **Pushed**.

## **Week 4: Git & Collaboration**



# Git & Collaboration

## Overview

**Summary:** You already know how to save files and make a basic commit in your Codespace. Now, learn to use Git as a *time machine* and *collaboration engine* for your analysis—tracking every change, safely undoing mistakes, working with teammates through branches and Pull Requests, and keeping a professional audit trail of your research workflow.

### Learning Objectives:

- Distinguish between saving a file and taking a version-control “snapshot” (commit).
- Read code differences (diffs) to see exactly what changed.
- Use the Codespaces Source Control UI to revert to a previous working version.
- Create and switch branches to isolate experimental work.
- Open a Pull Request on GitHub and understand its role in peer review.
- Configure `.gitignore` and use relative paths for repo hygiene.
- Use AI to translate diffs into clear, professional audit logs.
- Set up a group repository and agree on collaboration norms for the term project.

## Connection

Week 3 focused on writing and auditing small R summaries. This week treats each edit as a reviewable change with a clear history. Next week you will compare equivalent summaries across R and Python.

## Case Study Data Analysis

The **NHANES Health Equity data spine** now supports code-reading, EDA, visualization, and reproducibility checks. Use the cached CSV/RDS for routine class work; a CDC retrieval script exists for advanced users.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The default classroom path is to use the cached data so the analysis work is reproducible without internet access.

```
1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../..examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_analysis <- read_csv(nhanes_path, show_col_types = FALSE)
15
16 nhanes_analysis |>
17   summarise(
18     rows = n(),
19     missing_bmi = sum(is.na(BMI)),
20     missing_income = sum(is.na(IncomeGroup))
21   )
22 #> # A tibble: 1 x 3
23 #>   rows missing_bmi missing_income
24 #>   <int>      <int>      <int>
25 #> 1 105626      9356      9415
26
27 nhanes_analysis |>
```

```

28 filter(!is.na(BMI), !is.na(IncomeGroup), Age >= 20, Age <= 80) |>
29 group_by(IncomeGroup) |>
30 summarise(
31   n = n(),
32   mean_bmi = round(mean(BMI), 1),
33   .groups = "drop"
34 )
35 #> # A tibble: 3 x 3
36 #>   IncomeGroup      n mean_bmi
37 #>   <chr>      <int>   <dbl>
38 #> 1 High Income (>3.5) 16330    28.6
39 #> 2 Low Income (<1.3) 15293    29.4
40 #> 3 Middle Income    19228    29.3

```

## In-Class Activity

- Create a branch for a tiny dashboard documentation change.
- Edit only a label, caption, or README sentence; do not change analytic logic.
- Open a pull request and use the diff to verify the change is limited.
- In the pull request description, state how you verified the change is documentation-only.

Student output: a documentation-only pull request with a short audit note. Assignment link: [Assignment 3](#).

---

## git1.qmd Concepts: The “Save Game” of Reproducible Science

### Why Git Matters in Health Research

- **The audit trail:** Why `analysis_final_v3.qmd` is unsafe in health research; Git gives you a permanent, searchable history.

- **The timeline:** Your repository is a sequence of safe checkpoints (commits). Each commit records *who* changed *what* and *why*.
- **Reading the diff:** Red = removed. Green = added. This is the visual language of reproducibility.

## Quick Recap (from Week 2)

You already know how to stage, commit, and sync in your Codespace. This week, we go deeper: branching, collaboration, and building habits that will carry through your term project.

---

## git2.qmd Tutorial: Time Travel in Codespaces

*Scenario:* We will intentionally break last week's R script and use Git to rescue it.

### Step 1: Verify Your Baseline

Open your `.qmd` from last week. Make a small, meaningful edit (e.g., improve a comment). Stage (+), write a descriptive commit message, and commit (.). This is your safety net.

### Step 2: The Disaster

Delete a crucial line of R code in a chunk (e.g., the line that loads the data). **Save** the file (Ctrl/Cmd + S).

### Step 3: Assess the Damage

Open **Source Control** → click the modified file → view the side-by-side **diff** to see exactly what changed. Practice reading the red (removed) and green (added) lines.

## Step 4: The Undo Button

Hover over the file name in the Source Control panel and click **Discard Changes** (the curved arrow) to restore the file to the **last committed** version.



Tip

**Key takeaway:** Saving writes to disk. Committing writes to *history*. Only committed work can be recovered by Git.

---

## git3.qmd Repo Hygiene: .gitignore and Relative Paths

### Why This Matters

As your projects grow, you will generate files that should *not* be tracked: rendered HTML, temporary data caches, `.DS_Store`, API keys. Tracking these clutters your history and can leak sensitive information.

### .gitignore Walkthrough

1. Open (or create) the `.gitignore` file in the root of your repository.
2. Add common patterns:

```
# Rendered outputs
*.html
*_files/

# OS junk
.DS_Store
Thumbs.db
```

```
# Sensitive
.env
```

3. Stage, commit with a message like "Add .gitignore to exclude rendered files".

## Relative Paths

Always use **relative paths** in your code (e.g., `read.csv("data/nhanes.csv")`), never absolute paths (e.g., `read.csv("/home/user/Documents/nhanes.csv")`). This ensures your analysis runs on *any* machine—including your grader's Codespace.

### Warning

**Reproducibility rule:** Your term project must run from start to finish in a *fresh* Codespace. Absolute paths guarantee failure.

---

## git4.qmd Branching: Safe Experimentation

### Concept

A **branch** is a parallel copy of your project. You can experiment freely on a branch without affecting the stable **main** version. When your experiment works, you **merge** it back.

Think of it like a lab notebook: **main** is your clean, published record. A branch is your scratch pad.

### Tutorial: Create and Merge a Branch in VS Code

#### Step 1: Create a Branch

- Click the branch name in the bottom-left corner of VS Code (it likely says **main**).

- Select **Create new branch...** and name it something descriptive: `add-age-filter`.

### Step 2: Make Changes on the Branch

- Edit your `.qmd` (e.g., add a line of code that filters data by age).
- Stage and commit on this branch.

### Step 3: Switch Back to `main`

- Click the branch name again → select `main`. Notice your recent edits *disappear* from the file. They are safely stored on the other branch.

---

## git5.qmd Pull Requests: The Peer Review Gateway

### What Is a Pull Request?

A **Pull Request (PR)** is a GitHub feature that says: *“I have changes on a branch. Please review them before they go into `main`.”* PRs are how professional teams and researchers collaborate. You will use them for peer review later in this course.

### Tutorial: Open Your First PR

1. After committing changes to a branch (see `git4.qmd`), click **Sync Changes** to push the branch to GitHub.
2. Open your repository on **github.com**. You should see a banner: *“add-age-filter had recent pushes.”* Click **Compare & pull request**.
3. Write a title and short description explaining what you changed and why.
4. Click **Create pull request**.

5. For now, review the diff on GitHub, then click **Merge pull request** and complete the merge.

 Tip

**Collaboration norm:** In your term project, agree with your group that changes go through PRs rather than directly to main. This creates a built-in review step.

---

## git6.qmd AI Auditing for Version Control

Treat the LLM as a **Git Translator** that explains diffs and drafts audit-quality messages.

### git6a.qmd Scenario A: Diff Translation (Pre-Commit Audit)

**Prompt:**

“I am about to commit these changes. Explain what I changed in plain English. Did I delete anything important?”

Use this to spot accidental deletions or unintended edits before you commit.

### git6b.qmd Scenario B: Auto-Scribe Commit Messages

**Prompt:**

“Based on this diff, write a one-sentence, professional commit message for my Quarto document.”

Check that the message captures the **intent** (e.g., “Filter patient cohort to age 65 per protocol”), not just the action (“changed line 42”).



## git6c.qmd Scenario C: Panic Translator (Error Handling)

### Prompt:

“VS Code in my GitHub Codespace showed this Git error: [paste]. What does it mean in simple terms, and which **buttons in the VS Code UI** should I click to fix it?”

#### Warning

**Strict Guardrail:** Never run random terminal commands suggested by an AI (especially `--force` or `reset --hard`). Prefer the VS Code visual interface, which is safer. If the AI suggests a terminal command, ask it for the GUI equivalent first.

---

## git7.qmd Setting Up Your Group Repository

This week you form your term project groups (**Milestone 0**). Use this session to establish your collaboration workflow.

### Steps

1. **One member** opens the group project template repository linked from Canvas and creates a fork for the group.
2. That member adds teammates as collaborators on the group fork.
3. All other members accept the collaborator invitation and launch a Codespace from the group fork.
4. As a group, agree on norms:
  - All changes go through **branches** → **PRs** (no direct pushes to `main`).
  - Commit messages should be descriptive (use the AI auto-scribe technique).

- Keep the repo clean: use `.gitignore`, relative paths, and a clear folder structure.
5. Each member makes a small “hello” commit (e.g., add your name to the `README.md`) and pushes it to verify access.

**i** Note

**Milestone 0 due Friday:** Record your group formation on Canvas and ensure everyone can commit to the shared repository.

---

## git8.qmd Knowledge Check

1. What is the difference between **saving** a file and **committing** it?
2. What does a red line indicate in a diff?
3. Which button restores the last **committed** version of a file?
4. Why should you use a `.gitignore` file?
5. What is the purpose of a **branch**?
6. How does a **Pull Request** support peer review?
7. Why are relative paths essential for reproducibility?

---

## git9.qmd Practice Exercise: The Full Workflow Drill

1. Open your Codespace from last week.
2. Create a `.gitignore` file with sensible defaults for your project. **Commit.**
3. Create a new branch called `practice-edit`.

4. On the branch, make a small improvement to your R script (e.g., add a clarifying comment or improve a variable name).
5. Use an AI model to draft a clear commit message from your diff. **Commit** on the branch.
6. Push the branch and open a **Pull Request** on GitHub.
7. Review the PR diff on GitHub, then merge it.
8. Back on **main**, intentionally introduce a breaking typo and **Save**.
9. Use **Discard Changes** to recover the last working version.

---

## git10.qmd Quick Reference: Common Git Actions in VS Code

| Action                           | How                                                                |
|----------------------------------|--------------------------------------------------------------------|
| <b>Stage changes</b>             | Source Control → hover file → click +                              |
| <b>Commit</b>                    | Type message → click                                               |
| <b>View diff</b>                 | Click file name in Source Control                                  |
| <b>Discard uncommitted edits</b> | Hover file → click <b>curved arrow</b> (Discard Changes)           |
| <b>Create a branch</b>           | Click branch name (bottom-left) → <b>Create new branch...</b>      |
| <b>Switch branches</b>           | Click branch name → select target branch                           |
| <b>Merge a branch</b>            | Command Palette → <b>Git: Merge Branch...</b>                      |
| <b>Push / Sync</b>               | Click <b>Sync Changes</b> in status bar                            |
| <b>Open a Pull Request</b>       | Push branch → go to github.com → <b>Compare &amp; pull request</b> |

| Action                   | How                                              |
|--------------------------|--------------------------------------------------|
| <b>Add to .gitignore</b> | Edit .gitignore in repo root<br>→ stage + commit |

# Git Basics

- **The Audit Trail:** Git records *who* changed *what*.



Selecting any line will show who changed that line and when they did so.

- **The Timeline:** A sequence of safe checkpoints (commits). This shows the reason behind each change and what was change. These checkpoints can always be reverted back to if the need be.

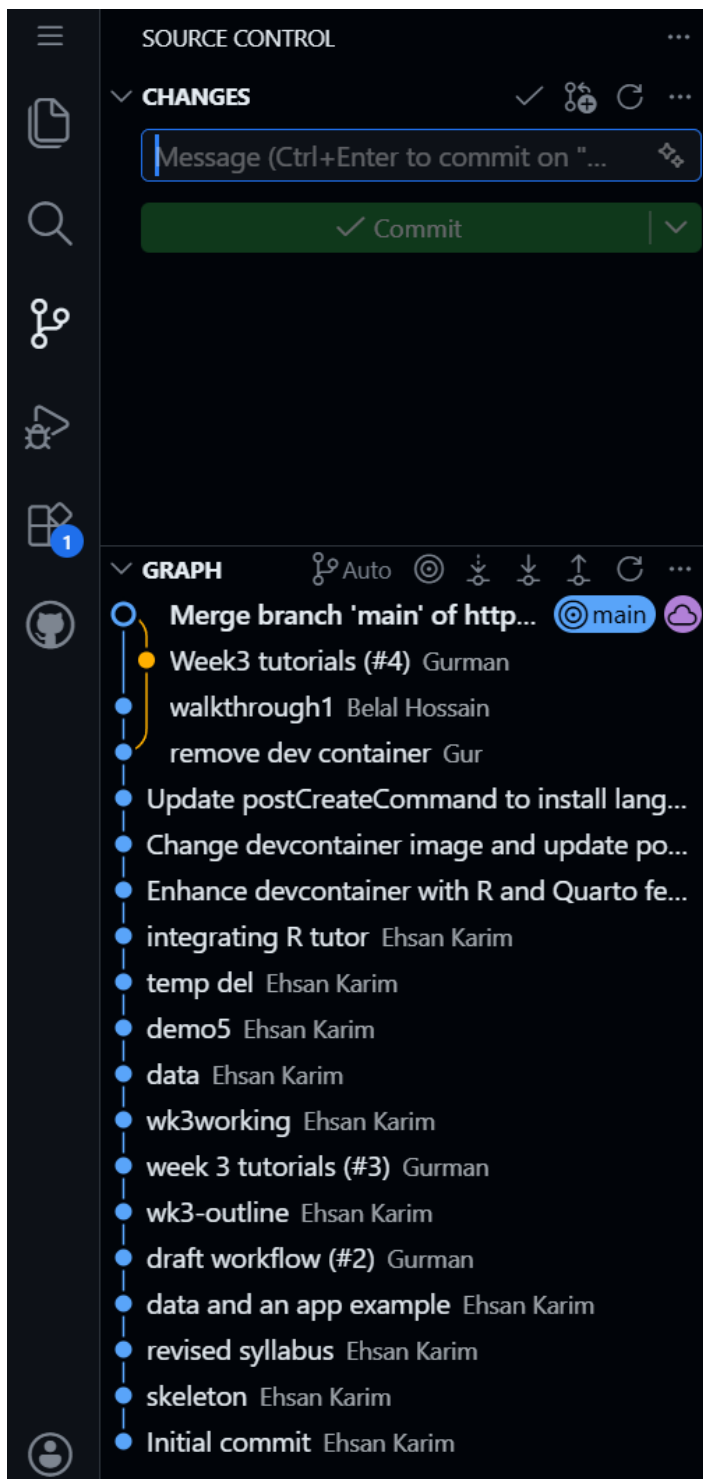
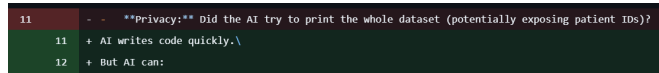


Figure 1: This tree shows every commit (checkpoints)

- **The Language of Diffs:**

- Red (-): Removed lines.
- Green (+): Added lines.



```
11 - - **Privacy:** Did the AI try to print the whole dataset (potentially exposing patient IDs)?
12 + AI writes code quickly.
13 + But AI can:
```

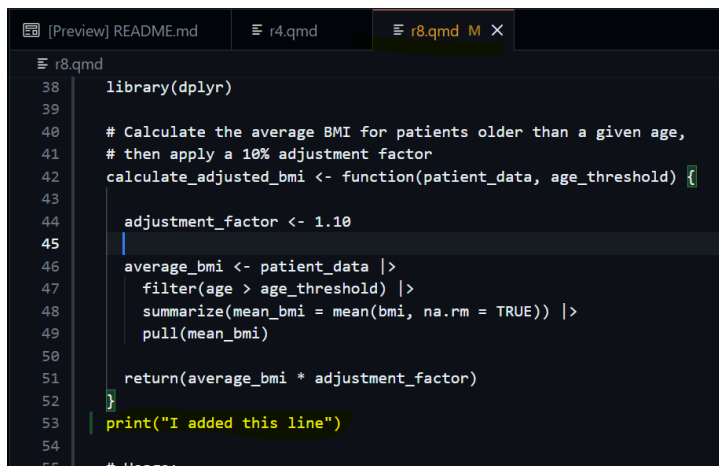
Observing this image, we see that line 11 was replaced, the old data on line 11 is colored red, while the new changes are shown in green.

**Key Rule:** Saving writes to disk on your commuter. Committing and pushing to the repo writes to history.

# Time Travel

Follow these steps

1. **Edit:** Make a change to your code, any file in the repo will do, and save the file.



```
38 library(dplyr)
39
40 # Calculate the average BMI for patients older than a given age,
41 # then apply a 10% adjustment factor
42 calculate_adjusted_bmi <- function(patient_data, age_threshold) {
43
44   adjustment_factor <- 1.10
45
46   average_bmi <- patient_data |>
47     filter(age > age_threshold) |>
48     summarize(mean_bmi = mean(bmi, na.rm = TRUE)) |>
49     pull(mean_bmi)
50
51   return(average_bmi * adjustment_factor)
52 }
53 print("I added this line")
54
55 # Usage:
```

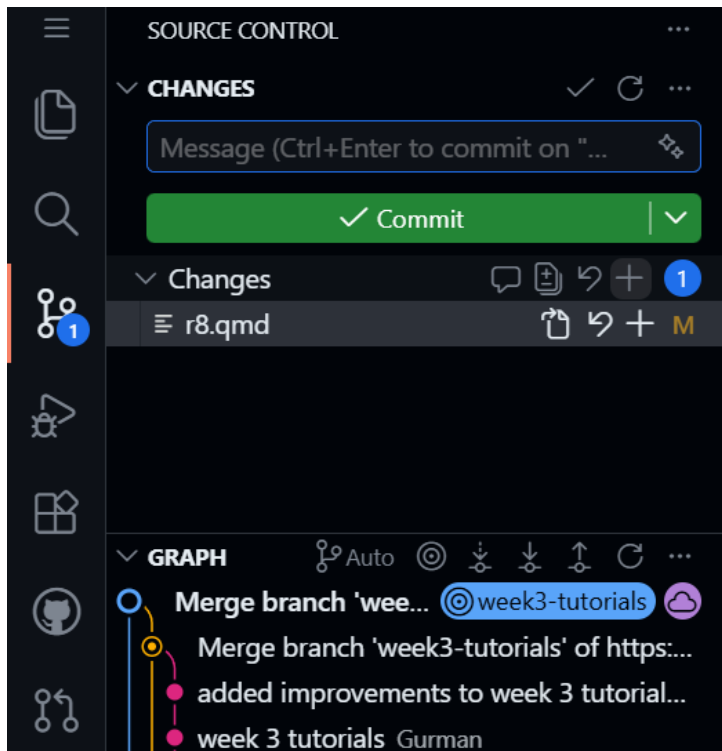
Key things to note are that the open file tab has changed colors, this shows that github has acknowledged that the file has been changed.

2. **Commit:** Stage and Commit in the Source Control tab.

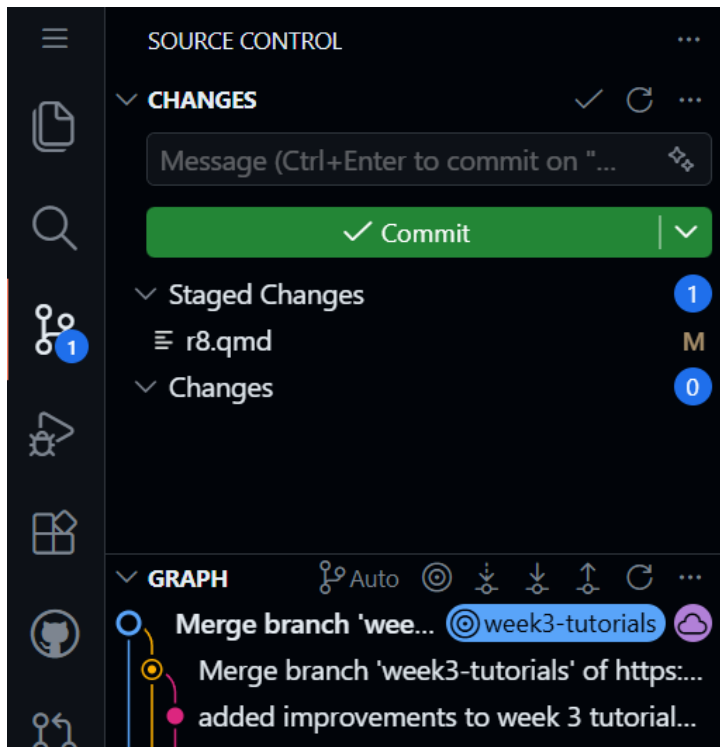
As we have seen from the previous tutorial, open up the git tab as indicated by the orange-peach highlight. When a file is changed you will be presented with the screen below.

To stage a change, press the + icon.

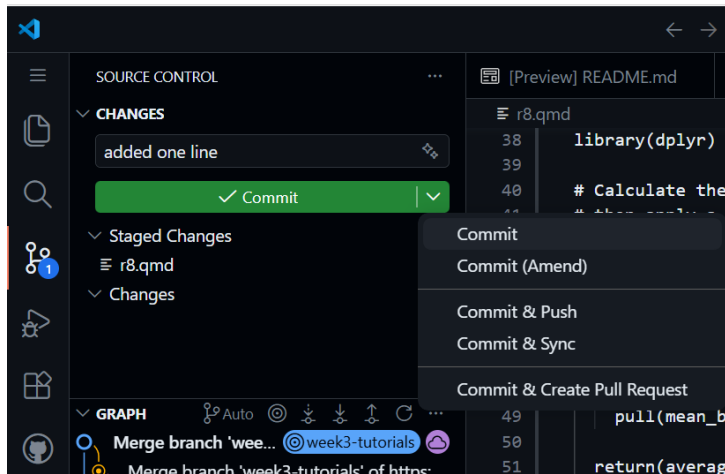




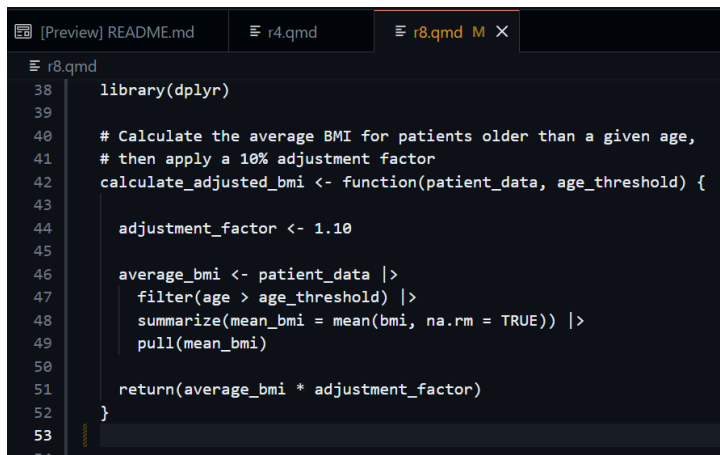
Your menu should look like this now;



Then when you are ready to commit, you can click the big green button! You can also click the down arrow on the big green button to see more options, as well as the commit option. We will discuss those other options later.

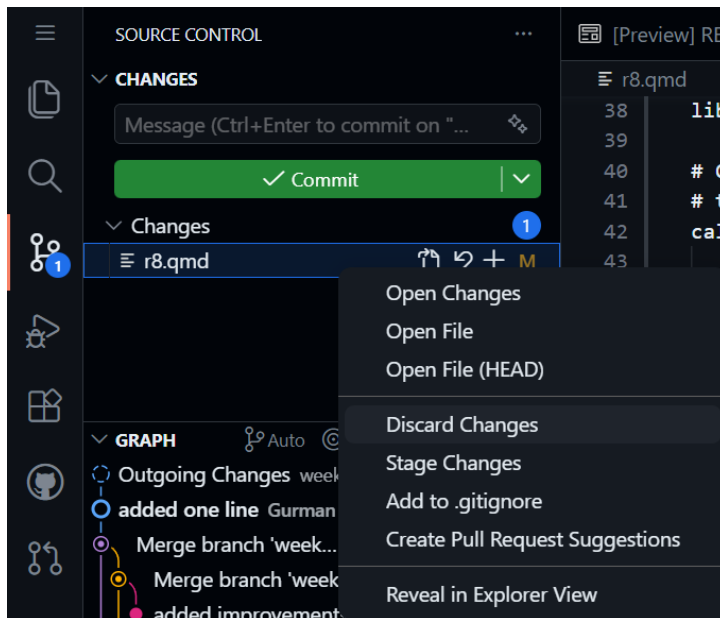


3. **Break:** Delete the code you just added and save.



```
38 library(dplyr)
39
40 # Calculate the average BMI for patients older than a given age,
41 # then apply a 10% adjustment factor
42 calculate_adjusted_bmi <- function(patient_data, age_threshold) {
43
44   adjustment_factor <- 1.10
45
46   average_bmi <- patient_data |>
47     filter(age > age_threshold) |>
48     summarize(mean_bmi = mean(bmi, na.rm = TRUE)) |>
49     pull(mean_bmi)
50
51   return(average_bmi * adjustment_factor)
52 }
53
```

4. **Rescue:** Hover over the file in Source Control again, click the **Discard Changes** option.



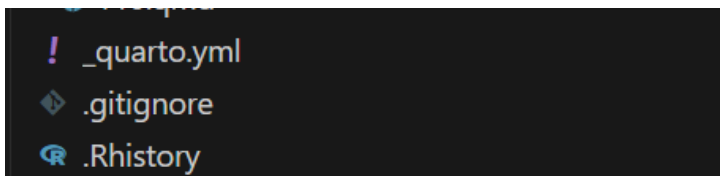
This will revert the file back to what it was before. This can recover data that your computer potentially cannot, its a very powerful in git and showcases one aspect of how git operates. It is a record of all your work and saved chunks.

# Repo Hygiene

## **.gitignore**

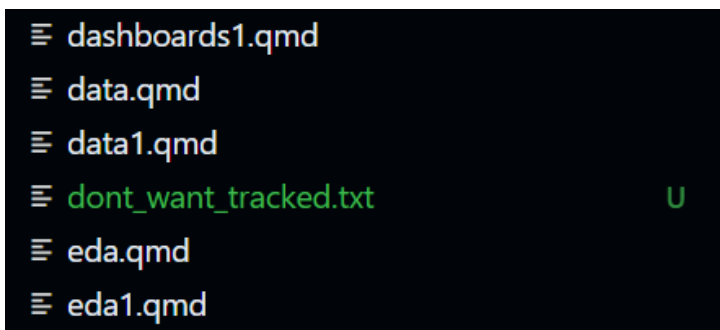
Sometimes there are files you do not want to be tracked by github. This could be anything from files specific to your computer to items like patient data which should not be stored on the cloud unless allowed so by policy.

1. Create a file named **.gitignore** in your root directory:  
It has to be this exact spelling for git to recognize it.

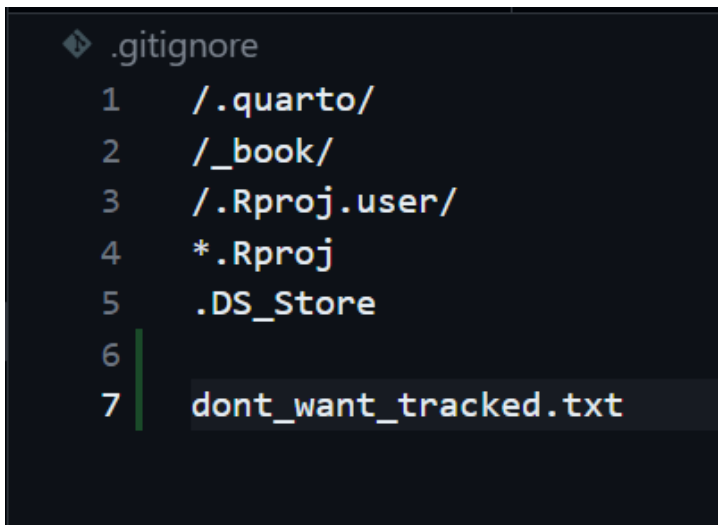


You will know its correct when it has its own icon as seen above

2. Then create a new file, any type will do but for this lesson I will create a text file called dont\_want\_tracked

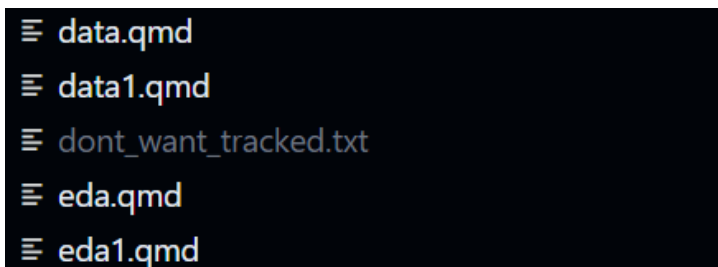


3. Then open the gitignore file and add the name of the file you dont want tracked into it.



```
.gitignore
1  /.quarto/
2  /_book/
3  /.Rproj.user/
4  *.Rproj
5  .DS_Store
6
7  dont_want_tracked.txt
```

4. Now you will see the new file, which was green is now grayed out. This shows that github is no longer tracking it and coloring it green (which it does for new files).



```
≡ data.qmd
≡ data1.qmd
≡ dont_want_tracked.txt
≡ eda.qmd
≡ eda1.qmd
```

This is simple enough right? Whenever in doubt, ask AI how to stop a file from being tracked by git and it will offer some advice. Now stage and commit this like we have done in the previous tutorial.

## Relative Paths

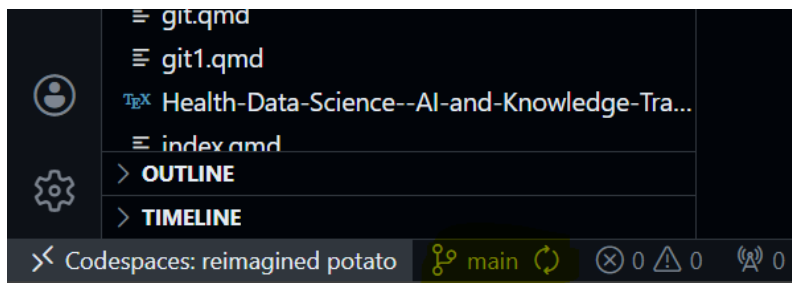
Always use relative paths in your code (e.g., `read.csv("data/nhanes.csv")`), never absolute paths (e.g., `read.csv("/home/user/Documents/nhanes.csv")`). This ensures your analysis runs on any machine—including your grader’s Codespace.

Reproducibility rule: Your term project must run from start to finish in a fresh Codespace. Absolute paths guarantee failure.

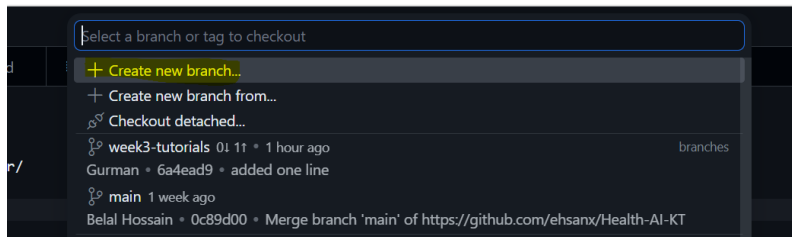
# Branching

## 1. Create:

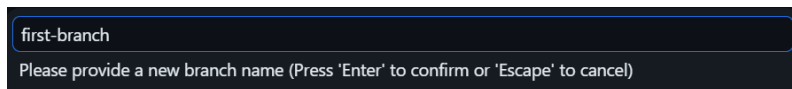
Click branch name (bottom-left).



This will pop up a menu where you have the option to **Create new branch**.

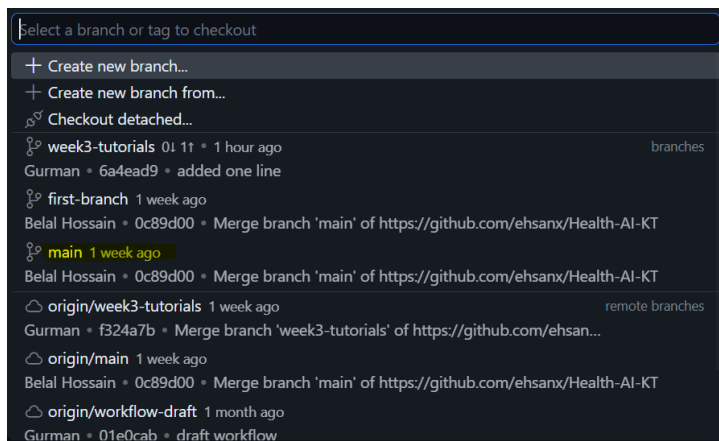


Select this and call it first-branch. Note branch names may not have any spaces and limited special characters, the system will let you know if a name is invalid.



## 2. Work: Make changes, like those in the previous tutorial and commit your changes on your branch.

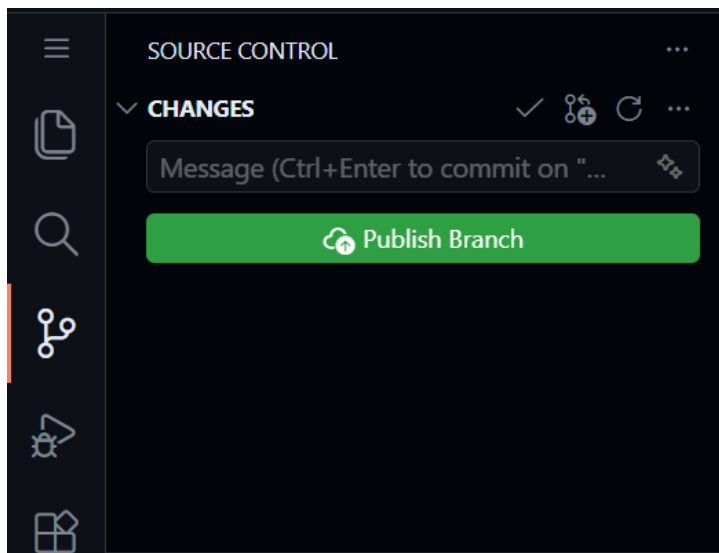
3. **Switch:** Click the branch button on the bottom left again and once this menu pops up, select **main**. Note that your changes vanish (they are safely in the other branch). If you decide to switch back to your branch, your changes will reappear. This is amazing for multitasking on different features and keeping code clean.



# Pull Requests

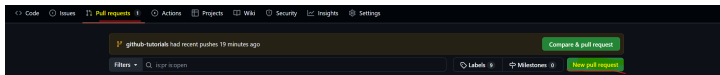
A **Pull Request (PR)** is a GitHub feature that says: *“I have changes on a branch. Please review them before they go into **main**.”* PRs are how professional teams and researchers collaborate. You will use them for peer review later in this course. This is also the best way to merge your own changes into main, as it forces you to see what has been change and allows one to check what they are submitting to the main branch, the backbone or “trunk” of the tree

1. **Sync:** Click “Sync Changes” to push your branch to GitHub. This can be blue or green or say publish branch or not. Whichever the case, clicking this button will make everything on your computer update to the online github repo.

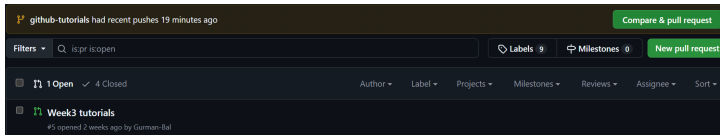


2. **PR:** Go to your repo on github online and click “Compare & pull request.”

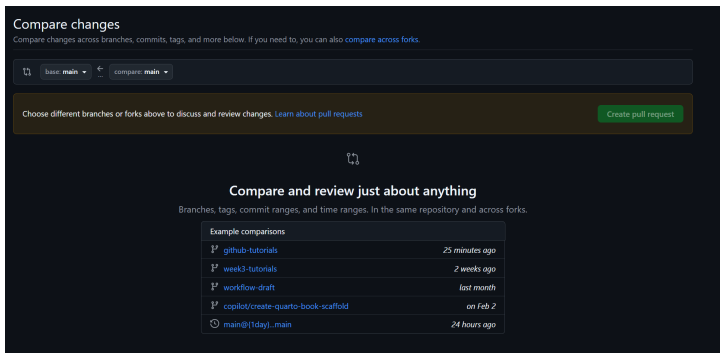




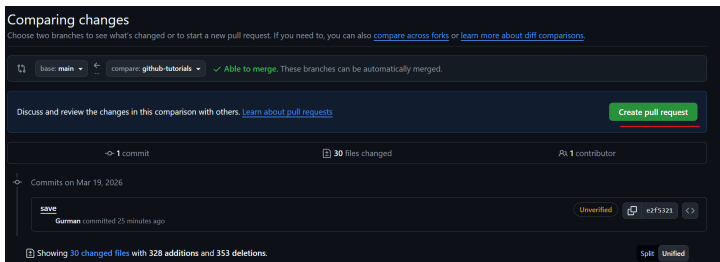
We are then presented with the screen below. Press the big green button “New Pull Request”



Then you are presented this menu. This shows all the different branches that can be merged into main. Select on the one you want to merge into main.



Once you select the one, you can make a pull request! Select this.



3. **Merge:** Review the diff, then click “Merge pull request.” If there are merge conflicts, ask AI the best way to resolve them.

Gurman-Bal and others added 6 commits [last month](#)

week 3 tutorials

Verified8794d84

added improvements to week 3 tutorials

Verified76c47bd

Merge branch 'week3-tutorials' of <https://github.com/ehsanx/Health-AI-KT>

788bc0f

address changes

479773a

Merge branch 'week3-tutorials' of <https://github.com/ehsanx/Health-AI-KT>

f324a7b

Merge branch 'main' into week3-tutorials

Verified17f0242

This branch has not been deployed

No deployments

No conflicts with base branch

Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions.](#)

Still in progress?

[Convert to draft](#)

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# AI Auditing

Use these prompts with your AI assistant:

- **Audit:** “I am about to commit these changes. Explain what I changed in plain English. Did I delete anything important?”

Use this to spot accidental deletions or unintended edits before you commit.

- **Scribe:** “Based on this diff, write a one-sentence, professional commit message for my Quarto document.”

Check that the message captures the **intent** (e.g., “Filter patient cohort to age 65 per protocol”), not just the action (“changed line 42”).

- **Panic:** “VS Code shows this error: [PASTE ERROR]. What does it mean in simple terms and what buttons should I click?”

Never run random terminal commands suggested by an AI (especially `--force` or `reset --hard`). Prefer the VS Code visual interface, which is safer. If the AI suggests a terminal command, ask it for the GUI equivalent first.

# Setting Up Your Group Repository

1. **Invite:** One member creates the repo; others accept the invite.
2. **Clone:** Everyone launches their own Codespace.
3. **Norms:** Agree: “No direct pushes to `main`.”
4. **Test:** Everyone makes a `README.md` update to verify access.

# Knowledge Check

1. **Saving vs. Committing:** Saving = disk; Committing = history.
2. **Red line:** Deletion.
3. **Restore:** Discard Changes button.
4. **.gitignore:** Prevents tracking junk/secrets.
5. **Branch:** Parallel experimental workspace.
6. **PR:** A peer review workflow for merging code.
7. **Relative Paths:** Ensures the project runs on any machine.

## Practice Exercise: Full Workflow

| Step | Task           | Done                     |
|------|----------------|--------------------------|
| 1    | Edit Code      | <input type="checkbox"/> |
| 2    | Create Branch  | <input type="checkbox"/> |
| 3    | AI Draft Msg   | <input type="checkbox"/> |
| 4    | Stage & Commit | <input type="checkbox"/> |
| 5    | Push & PR      | <input type="checkbox"/> |
| 6    | Merge          | <input type="checkbox"/> |

## Quick Reference

| Action  | How-To                                                      |
|---------|-------------------------------------------------------------|
| Stage   | Source Control (+)                                          |
| Commit  | Type msg +                                                  |
| View    | Click file in Source Control                                |
| Diff    |                                                             |
| Discard | Left click file in source control and click discard changes |
| Branch  | Bottom-left status bar                                      |
| Sync    | Sync Changes button                                         |

# **Week 5: Polyglot Awareness & R Deepening**



# Polyglot Awareness & R Deepening

## Overview

Week 5 is about becoming a careful bilingual reader of data analysis code. R remains the core language for required course work. Python is introduced so you can read short **pandas** examples, translate small tasks with AI assistance, and verify that a translated result still answers the same question.

The week has two linked goals:

- deepen R fluency with **dplyr**, functions, and clear summaries;
- compare a small R result with a Python/pandas translation using explicit parity checks.

## Learning Objectives

By the end of this week, you should be able to:

- explain when R is the better default and when Python may be useful;
- write a clear R **dplyr** pipeline for a descriptive health-data summary;
- refactor repeated R code into a small function;
- create and run a Python notebook in Codespaces;
- load and inspect a CSV with **pandas**;
- identify common R-to-Python translation traps;
- use AI one step at a time while auditing each translated output;
- document Python dependencies in **requirements.txt**;

- use Git to commit a clean translation workflow.

## Connection

Week 4 made small changes reviewable through Git. This week uses that review mindset inside the analysis itself: first build a trusted R result, then translate one step at a time and compare. Next week you will turn those checked summaries into clearer visual explanations.

## Case Study Data Analysis

The **NHANES Health Equity data spine** now supports code-reading, EDA, visualization, and reproducibility checks. Use the cached CSV/RDS for routine class work; a CDC retrieval script exists for advanced users.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The default classroom path is to use the cached data so the analysis work is reproducible without internet access.

```
1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../..examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_analysis <- read_csv(nhanes_path, show_col_types = FALSE)
```

```

15
16 nhanes_analysis |>
17   summarise(
18     rows = n(),
19     missing_bmi = sum(is.na(BMI)),
20     missing_income = sum(is.na(IncomeGroup))
21   )
22 #> # A tibble: 1 x 3
23 #>   rows missing_bmi missing_income
24 #>   <int>      <int>      <int>
25 #> 1 105626      9356      9415
26
27 nhanes_analysis |>
28   filter(!is.na(BMI), !is.na(IncomeGroup), Age >= 20, Age <= 80) |>
29   group_by(IncomeGroup) |>
30   summarise(
31     n = n(),
32     mean_bmi = round(mean(BMI), 1),
33     .groups = "drop"
34   )
35 #> # A tibble: 3 x 3
36 #>   IncomeGroup      n mean_bmi
37 #>   <chr>      <int>   <dbl>
38 #> 1 High Income (>3.5) 16330    28.6
39 #> 2 Low Income (<1.3) 15293    29.4
40 #> 3 Middle Income    19228    29.3

```

## R Deepening Studio

Start in R. Use the NHANES case-study CSV:

```
1 examples/nhanes-equity/data/nhanes_equity_v6.csv
```

If you are working from the Week 5 folder, the relative path is:

```
1 ../../examples/nhanes-equity/data/nhanes_equity_v6.csv
```

## R Pipeline

```
1 library(readr)
2 library(dplyr)
3
4 nhanes <- read_csv("../../examples/nhanes-equity/data/nhanes_equity_v6.csv")
5
6 bmi_by_income <- nhanes |>
7   filter(!is.na(BMI), !is.na(IncomeGroup), !is.na(Gender)) |>
8   group_by(IncomeGroup, Gender) |>
9   summarise(
10     n = n(),
11     mean_bmi = round(mean(BMI), 2),
12     sd_bmi = round(sd(BMI), 2),
13     .groups = "drop"
14   ) |>
15   arrange(IncomeGroup, Gender)
16
17 bmi_by_income
```

## R Function

```
1 summarise_mean_sd <- function(data, value, group1, group2) {
2   data |>
3     filter(!is.na({{ value }}), !is.na({{ group1 }}), !is.na({{ group2 }})) |>
4     group_by({{ group1 }}, {{ group2 }}) |>
5     summarise(
6       n = n(),
7       mean_value = round(mean({{ value }}), 2),
8       sd_value = round(sd({{ value }}), 2),
9       .groups = "drop"
10    ) |>
11    arrange({{ group1 }}, {{ group2 }})
12 }
13
14 bmi_function_check <- summarise_mean_sd(nhanes, BMI, IncomeGroup, Gender)
15 bmi_function_check
```

Before translating, compare the pipeline output and the function output. They should have the same grouping labels and the same number of rows.

## In-Class Activity: R First, Python Second

Work in pairs. One partner reads the R output; the other reads the Python output.

1. Load the NHANES CSV in R and Python.
2. Record raw row count, column count, and missing BMI count in both languages.
3. Produce mean and SD of BMI by `IncomeGroup` and `Gender` in R.
4. Translate the same task to Python/pandas one step at a time.
5. Fill in a parity table comparing row counts, grouping labels, and rounded summary values.
6. If something does not match, write the likely reason before changing code.

Student output: a parity-check note with the R pipeline, R function, Python translation, parity table, and a short AI-use note.

## Page Map

Use the child pages in this order:

1. [Python vs. R](#)
2. [Python Environment](#)
3. [Packages and Reproducibility](#)
4. [Translation Traps](#)
5. [Load and Inspect Data](#)
6. [Python Tracebacks](#)
7. [One Step, One Cell](#)
8. [Mixing R and Python](#)
9. [Reusable Python Scripts](#)
10. [Git Progress](#)

11. [Proposal Readiness](#)
12. [Knowledge Check](#)
13. [Bilingual Translation](#)
14. [R To Python Cheatsheet](#)

For the explicit R mastery exercise, use [R Deepening and Parity Check](#).

## Assignment 4

Complete [Assignment 4: Polyglot Awareness & R Deepening](#).

Submit a repository link on Canvas after committing:

- `assignments/assignment04-polyglot/submission/polyglot-parity.qmd;`
- `assignments/assignment04-polyglot/submission/translation.ipynb;`
- `assignments/assignment04-polyglot/submission/helpers.py.`

## Knowledge Check

Before leaving Week 5, answer these:

1. What is one analysis task you would keep in R?
2. What is one task where Python may be useful?
3. What are three parity checks you ran?
4. What did AI help translate, and how did you verify it?
5. What changed in your Git history from the beginning to the end of the week?

# R Deepening and Parity Check

Week 5 is polyglot-aware, not Python-first. Your strongest analysis path remains R and the tidyverse. Python enters as a comparison language: it helps you read code from other sources, ask better AI translation questions, and verify that a translated result still means the same thing.

## Learning Goals

By the end of this activity, you should be able to:

- write a clear R `dplyr` pipeline for a descriptive health-data summary;
- refactor a repeated summary into a small R function;
- translate the same task to Python/pandas one step at a time;
- compare outputs across languages using row counts, grouping levels, and summary values;
- document any mismatch before deciding which code to change.

## Dataset

Use the NHANES case-study snapshot:

```
1 examples/nhanes-equity/data/nhanes_equity_v6.csv
```

If you are working from this Week 5 folder, the relative path is:

```
1 ../../examples/nhanes-equity/data/nhanes_equity_v6.csv
```

## Part A: R Pipeline

Start with R. The goal is a descriptive, unweighted summary of BMI by income group and gender.

```
1 library(readr)
2 library(dplyr)
3
4 nhanes <- read_csv("../examples/nhanes-equity/data/nhanes_equity_v6.csv")
5
6 r_summary <- nhanes |>
7   filter(!is.na(BMI), !is.na(IncomeGroup), !is.na(Gender)) |>
8   group_by(IncomeGroup, Gender) |>
9   summarise(
10     n = n(),
11     mean_bmi = round(mean(BMI), 2),
12     sd_bmi = round(sd(BMI), 2),
13     .groups = "drop"
14   ) |>
15   arrange(IncomeGroup, Gender)
16
17 r_summary
```

## Part B: R Function

Refactor the summary so the value and grouping variables are easy to change later.

```
1 summarise_mean_sd <- function(data, value, group1, group2) {
2   data |>
3     filter(!is.na({{ value }}), !is.na({{ group1 }}), !is.na({{ group2 }})) |>
4     group_by({{ group1 }}, {{ group2 }}) |>
5     summarise(
6       n = n(),
7       mean_value = round(mean({{ value }}), 2),
```



```

8     sd_value = round(sd({{ value }}), 2),
9     .groups = "drop"
10  ) |>
11    arrange({{ group1 }}, {{ group2 }})
12  }
13
14  r_function_summary <- summarise_mean_sd(nhanes, BMI, IncomeGroup, Gender)
15  r_function_summary

```

Before moving to Python, check that the function output has the same number of rows and the same grouping labels as the original pipeline.

## Part C: Python Translation

Translate one step at a time. Do not ask AI for a full script until you have verified each cell.

```

1  import pandas as pd
2
3  nhanes = pd.read_csv("../examples/nhanes-equity/data/nhanes_equity_v6.csv")
4
5  py_summary = (
6      nhanes
7      .dropna(subset=["BMI", "IncomeGroup", "Gender"])
8      .groupby(["IncomeGroup", "Gender"], as_index=False)
9      .agg(
10         n=("BMI", "size"),
11         mean_bmi=("BMI", "mean"),
12         sd_bmi=("BMI", "std")
13     )
14     .sort_values(["IncomeGroup", "Gender"])
15 )
16
17 py_summary["mean_bmi"] = py_summary["mean_bmi"].round(2)
18 py_summary["sd_bmi"] = py_summary["sd_bmi"].round(2)
19
20 py_summary

```

Part D: Parity Audit

Record the following checks in a short Quarto note:

| Check                                  | R command                                                                           | Python command                                                           | What to record                  |
|----------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------|
| Raw row count                          | <code>nrow(nhanes)</code>                                                           | <code>len(nhanes)</code>                                                 | Same number of rows             |
| Column names                           | <code>names(nhanes)</code>                                                          | <code>nhanes.columns.tolist()</code>                                     | Same list of expected variables |
| Analysis rows after missingness filter | <code>nrow(filter(nhanes, !is.na(BMI), !is.na(IncomeGroup), !is.na(Gender)))</code> | <code>len(nhanes.dropna(subset=["BMI", "IncomeGroup", "Gender"]))</code> | Same row count                  |
| Group count                            | <code>nrow(r_summary)</code>                                                        | <code>len(py_summary)</code>                                             | Same number of grouped rows     |
| Summary values                         | <code>inspect mean_bmi and sd_bmi</code>                                            | <code>inspect mean_bmi and sd_bmi</code>                                 | Values match after rounding     |

AI Audit Prompt

Use this prompt only after you have a working R result:

Translate this R `dplyr` pipeline into Python/pandas one step at a time. For each step, explain the equivalent Python operation, the output I should inspect, and the parity check I should run against my R result. Do not combine steps or add new analysis choices.

Student Output

Submit a short note with:

- the R pipeline;

- the R function;
- the Python translation;
- a parity table showing the checks above;
- two sentences explaining any mismatch or confirming that the outputs agree.

# Python vs. R

## The Course Position

R remains the core language for required analysis in this course. Python is introduced so you can read short Python examples, translate small workflows with AI assistance, and recognize when Python may be useful in a health data project.

You are not choosing one language forever. You are learning how to move carefully between them.

## Strengths At A Glance

| Task                     | R is often strongest for                                        | Python is often strongest for                                              |
|--------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------|
| Descriptive epidemiology | <code>tidyverse</code> , survey workflows, reports              | possible with <code>pandas</code> , but often more verbose                 |
| Visualization            | <code>ggplot2</code> , <code>plotly</code> , Quarto integration | <code>seaborn</code> , <code>matplotlib</code> , interactive web libraries |
| Machine learning         | <code>tidymodels</code> , <code>torch</code>                    | <code>scikit-learn</code> , <code>tensorflow</code> , <code>pytorch</code> |
| Automation               | project scripts and reports                                     | general-purpose scripting and APIs                                         |
| Communication            | Quarto reports, slides, dashboards                              | web apps, APIs, automation, notebooks                                      |

## Notebooks vs. Scripts

- A notebook (`.ipynb`) is useful for exploration, teaching, and step-by-step AI translation checks.
- A script (`.py`) is useful for reusable functions and automation.
- A Quarto document (`.qmd`) is best when your analysis needs narrative, code, figures, citations, and a rendered final product.

## Check Your Understanding

Write two sentences:

1. One task where you would stay in R.
2. One task where Python might be worth considering.

# Python Environment

## Create A Notebook

In your GitHub Codespace:

1. Open the Week 5 folder.
2. Create a new file named `translation.ipynb`.
3. Select the recommended Python kernel when VS Code asks.
4. Run this first cell:

```
1 print("Hello from Python. My Week 5 notebook is running.")
```

## Use The Variables Pane

The Variables pane helps you audit Python objects the way the Environment pane helps you audit R objects.

Run this cell:

```
1 age = 25
2 course = "HDSx"
```

Then open the notebook Variables pane and check that both variables appear.

## Load A Tiny DataFrame

```

1 import pandas as pd
2
3 demo = pd.DataFrame({
4     "group": ["A", "A", "B"],
5     "value": [10, 15, 20]
6 })
7
8 demo

```

Double-click `demo` in the Variables pane to open the Data Viewer. This is where you will inspect row counts, column names, and translated outputs later in the week.

## Common Fixes

| Problem                    | First thing to try                                                 |
|----------------------------|--------------------------------------------------------------------|
| No kernel appears          | Wait for the Codespace to finish starting, then reload the window  |
| Variables pane is missing  | Check that the Jupyter extension is enabled                        |
| Notebook cell will not run | Select a Python kernel from the notebook toolbar                   |
| Package import fails       | Check whether the package belongs in <code>requirements.txt</code> |

# Packages and Reproducibility

## R `library()` vs. Python `import`

Both languages require you to load packages before using them.

| R                                | Python                                 | Purpose                           |
|----------------------------------|----------------------------------------|-----------------------------------|
| <code>library(readr)</code>      | <code>import pandas<br/>as pd</code>   | read CSV files                    |
| <code>library(dplyr)</code>      | <code>import pandas<br/>as pd</code>   | filter, group, and summarise data |
| <code>library(ggplot2)</code>    | <code>import seaborn<br/>as sns</code> | create plots                      |
| <code>library(reticulate)</code> | not applicable                         | call Python from R                |

## Why `requirements.txt` Matters

Python projects record package dependencies in `requirements.txt`. This is the Python-side equivalent of saying, “Here is what another person must install before my code will run.”

For this week, a minimal file is enough:

```
1 pandas
```

If you also create a plot in Python, add:

```
1 seaborn
2 matplotlib
```



## Course Rule

If you install or rely on a Python package for submitted work, record it in `requirements.txt`. Do not leave package installation as an undocumented memory from your current Codespace.

## R Connection

R reproducibility is handled through project documentation and, later in the course, `renv`. Week 5 is a bridge: you should be able to explain where dependencies are recorded in both languages.

# Translation Traps

AI translation is useful, but it often hides small language differences. These are the traps to check first.

## Trap 1: Indexing

R starts counting at 1. Python starts counting at 0.

```
1 values <- c(10, 20, 30)
2 values[1]
```

```
1 values = [10, 20, 30]
2 values[0]
```

Python slices are end-exclusive:

```
1 values[0:2]
```

This returns the first two values, not the first three.

## Trap 2: Indentation

R uses braces to mark code blocks. Python uses indentation.

```
1 if (age >= 20) {
2   "adult"
3 } else {
4   "not in adult cohort"
5 }
```

```

1 if age >= 20:
2     "adult"
3 else:
4     "not in adult cohort"

```

If Python indentation is inconsistent, the code may fail or run with a different meaning.

### Trap 3: Assignment vs. Comparison

| Meaning            | R                       | Python                                            |
|--------------------|-------------------------|---------------------------------------------------|
| assign a value     | <code>x &lt;- 10</code> | <code>x = 10</code>                               |
| compare values     | <code>x == 10</code>    | <code>x == 10</code>                              |
| missing/null check | <code>is.na(x)</code>   | <code>pd.isna(x)</code> or <code>x is None</code> |

### Trap 4: Missing Values

R and Python both have missing values, but they are handled by different functions. Always check missingness before comparing summaries.

```

1 sum(is.na(nhanes$BMI))

```

```

1 nhanes["BMI"].isna().sum()

```

### Audit Habit

When a translated result differs, check indexing, indentation, assignment/comparison, missingness handling, and grouping variables before assuming one language is “wrong.”

# Load and Inspect Data

## Load The NHANES Snapshot

In `translation.ipynb`, load the same CSV used in the R example:

```
1 import pandas as pd
2
3 nhanes = pd.read_csv("../examples/nhanes-equity/data/nhanes_equity_v6.csv")
```

## Inspect Before Summarising

```
1 nhanes.head()
```

```
1 nhanes.shape
```

```
1 nhanes.columns.tolist()
```

```
1 nhanes[["BMI", "IncomeGroup", "Gender"]].isna().sum()
```

## Compare To R

| Question          | R                         | Python                       |
|-------------------|---------------------------|------------------------------|
| How many rows?    | <code>nrow(nhanes)</code> | <code>len(nhanes)</code>     |
| How many columns? | <code>ncol(nhanes)</code> | <code>nhanes.shape[1]</code> |

| Question                     | R                                    | Python                                  |
|------------------------------|--------------------------------------|-----------------------------------------|
| What are the column names?   | <code>names(nhanes)</code>           | <code>nhanes.columns.tolist()</code>    |
| How many missing BMI values? | <code>sum(is.na(nhanes\$BMI))</code> | <code>nhanes["BMI"].isna().sum()</code> |

## Student Check

Before asking AI for any transformation, write down the raw row count, column count, and missingness count for BMI.

# Python Tracebacks

## The Bottom Line Is Usually At The Bottom

Python errors often print several lines called a traceback. Read from the bottom upward:

```
1 FileNotFoundError: [Errno 2] No such file or directory: 'Data/patients.csv'
```

This tells you the immediate problem: Python cannot find the file at the path you gave it.

## Common Week 5 Errors

| Error                            | Likely cause                                                        | First fix                                                                        |
|----------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <code>FileNotFoundError</code>   | path is wrong, file is in another folder, or capitalization differs | check the relative path from the notebook location                               |
| <code>ModuleNotFoundError</code> | package is missing                                                  | add the package to <code>requirements.txt</code> and install it in the Codespace |
| <code>KeyError</code>            | column name is misspelled or not present                            | run <code>nhanes.columns.tolist()</code>                                         |
| <code>IndentationError</code>    | Python block spacing is inconsistent                                | re-indent the block with four spaces                                             |
| <code>NameError</code>           | object has not been created in the current kernel                   | rerun earlier cells in order                                                     |

## AI Debugging Prompt

Read this Python traceback from the bottom up. Identify the error type, the exact object or path causing the issue, and the smallest code change that would fix it. Do not rewrite the full analysis.

## Student Check

When you fix an error, add one sentence to your notebook explaining what failed and how you knew the fix worked.

# One Step, One Cell

## Why This Rule Exists

Large AI-generated code blocks are hard to audit. A single hidden mistake in loading, filtering, grouping, or missingness handling can cascade into a polished but incorrect result.

Use one cell per analytic step:

1. load data;
2. inspect rows and columns;
3. filter missing values;
4. group and summarise;
5. compare with the R result.

## Good AI Prompt

Translate only the data-loading step from R to Python/pandas. Tell me what output I should inspect before moving to the next step.

## Risky AI Prompt

Translate my full analysis, make the plots, and write the interpretation.

The risky prompt gives up too much control. It may change the analytic question without telling you.



## Parity Audit Checks

| Check          | R                                    | Python                                              |
|----------------|--------------------------------------|-----------------------------------------------------|
| Row count      | <code>nrow(df)</code>                | <code>len(df)</code> or<br><code>df.shape[0]</code> |
| Column names   | <code>names(df)</code>               | <code>df.columns.tolist()</code>                    |
| Unique levels  | <code>unique(df\$IncomeGroup)</code> | <code>df["IncomeGroup"].unique()</code>             |
| Missing values | <code>sum(is.na(df\$BMI))</code>     | <code>df["BMI"].isna().sum()</code>                 |
| Summary table  | <code>nrow(summary_df)</code>        | <code>len(summary_df)</code><br>rows                |

## Notebook Habit

After each Python cell, add a short Markdown note:

```
1 Audit check: this cell matches the R output because ...
```

If you cannot fill in that sentence, do not move to the next cell yet.

# Mixing R and Python

## Why Interoperability Matters

Sometimes a project has one useful Python tool, but the rest of the analysis is already in R. The `reticulate` package lets R call Python code and bring Python objects back into the R session. In this course, `reticulate` is awareness-level: you should know what it does and when it might be useful, but you are not expected to build a full mixed-language pipeline.

## Minimal Demo

Run this in an R console or an R Quarto document if `reticulate` is available in your Codespace.

```
1 library(reticulate)
2
3 py_run_string("
4 import pandas as pd
5 numbers = pd.DataFrame({'group': ['A', 'A', 'B'], 'value': [10, 15, 20]})
6 summary = numbers.groupby('group', as_index=False)['value'].mean()
7 ")
8
9 py$summary
```

## Health Data Example

The same idea works with the NHANES case-study CSV:

```

1 library(reticulate)
2
3 py_run_string("
4 import pandas as pd
5 nhanes = pd.read_csv('../examples/nhanes-equity/data/nhanes_equity_v6.csv')
6 bmi_rows = nhanes[['BMI', 'IncomeGroup']].dropna()
7 ")
8
9 bmi_rows <- py$bmi_rows
10 head(bmi_rows)

```

## When To Use It

Use `reticulate` when:

- a small Python library solves a specific problem better than the R ecosystem;
- the rest of your report, tables, or dashboard are already in R;
- you can document the Python dependency clearly.

Avoid it when:

- a clean R solution already exists;
- mixing languages makes the project harder to reproduce;
- the Python step cannot run in a fresh Codespace.

## Check Your Understanding

In one sentence, explain what object moves from Python back into R in the NHANES example above.

# Reusable Python Scripts

## Notebook or Script?

Use a notebook when you are exploring, explaining, and checking outputs. Use a script when you have a helper function that should be reused without copying and pasting.

This mirrors the R pattern from Week 3: once a chunk of code has a clear job, turn it into a named function.

## Create `helpers.py`

In the Week 5 folder, create a file named `helpers.py` with this function:

```
1 from pathlib import Path
2 import pandas as pd
3
4
5 def load_and_preview(path, n_rows=5):
6     """Load a CSV, print its shape, and return the first rows."""
7     csv_path = Path(path)
8     if not csv_path.exists():
9         raise FileNotFoundError(f"Could not find file: {csv_path}")
10
11     data = pd.read_csv(csv_path)
12     print(f"Loaded {data.shape[0]} rows and {data.shape[1]} columns")
13     return data.head(n_rows)
```

## Import It In A Notebook

In `translation.ipynb`, import and use the helper:

```
1 from helpers import load_and_preview
2
3 preview = load_and_preview("../examples/nhanes-equity/data/nhanes_equity_v6.csv")
4 preview
```

## Reproducibility Check

Before committing:

- verify that `helpers.py` is saved in the same folder as your notebook;
- rerun the notebook from a fresh kernel;
- check that the relative path starts from the notebook location;
- commit both files together with a message such as **Add Week 5 Python helper**.

## Student Output

For Assignment 4, your reusable script should contain at least one function that is actually called from the notebook. A function that is written but never used does not demonstrate reproducibility.

# Git Progress

## Commit The Audit Trail

Week 4 introduced Git as analysis provenance. Week 5 uses that skill to make translation auditable.

Commit after meaningful checkpoints:

1. R pipeline works.
2. R function reproduces the pipeline.
3. Python notebook loads the same data.
4. Python summary matches the R summary.
5. Final parity note is complete.

## Notebook Output Hygiene

Jupyter notebooks can store rendered tables and images inside the `.ipynb` file. That makes Git diffs noisy.

Before committing:

1. Open the Command Palette with `Ctrl+Shift+P`.
2. Run Notebook: Clear All Outputs.
3. Save the notebook.
4. Stage `translation.ipynb`, `helpers.py`, and your Quarto note together.

## Example Commit Messages

```
1 Add R BMI summary for Week 5 parity check
```

```
1 Translate BMI summary to pandas and verify row counts
```

```
1 Complete Assignment 4 parity audit note
```

## Student Check

Your Git history should show the work developing in steps, not one unexplained final upload.

# Proposal Readiness

## Why This Belongs In Week 5

Before the project proposal is due, you should know whether your chosen dataset can be loaded, inspected, and explained in the course workflow. Week 5 adds one extra test: can you verify the basic data shape in both R and Python?

## Proposal Readiness Checklist

Your proposal should include:

- a focused health-data question;
- the dataset name and source;
- a brief provenance and stewardship note;
- a feasibility check showing the file can load in a Codespace;
- one or two planned descriptive summaries or visualizations;
- any known privacy, licensing, or access limitations.

## Feasibility Check

For your project dataset, record:

| Check                | What to write                                      |
|----------------------|----------------------------------------------------|
| File path            | relative path from the repo root                   |
| File format          | CSV, RDS, Excel, API, or other                     |
| Row and column count | basic shape after loading                          |
| Key variables        | names and meanings of the fields you expect to use |



| Check               | What to write                                |
|---------------------|----------------------------------------------|
| Missingness concern | one variable where missing values may matter |
| R/Python check      | whether both languages can load the file     |

## Student Output

Add this feasibility check to your project proposal notes. If your dataset does not load cleanly yet, describe the blocker honestly instead of hiding it.

# Knowledge Check

Answer these in a Markdown cell or short Quarto note.

1. Why is Week 5 called “polyglot awareness” rather than “Python replacement”?
2. Name one task where R is the better default for this course.
3. Name one task where Python might be useful.
4. Why does `my_list[1]` return the second item in Python?
5. What is the difference between `=` and `==`?
6. What does `requirements.txt` document?
7. What is the One Step, One Cell rule?
8. What are three parity checks you should run after translating R code to Python?
9. Why should notebook outputs be cleared before committing?
10. What does `reticulate` allow R to do?

## Short Applied Check

Given the R expression:

```
1 mean(nhanes$BMI, na.rm = TRUE)
```

Write the Python/pandas equivalent and name the missingness check you would run before trusting it.

# Bilingual Translation

## Purpose

This exercise turns the week into a concrete artifact: one small R analysis, one Python translation, and one audit note showing that the outputs agree.

## Required Task

Translate the Week 5 R-deepening summary into Python/pandas:

1. Start from the R pipeline in [R Deepening and Parity Check](#).
2. Create `translation.ipynb`.
3. Load the NHANES CSV from `../../examples/nhanes-equity/data/nhanes_equity_v6.csv`.
4. Recreate the BMI-by-income-and-gender summary in Python.
5. Compare the Python output to the R output using row counts, grouping labels, and rounded summary values.
6. Extract one Python helper into `helpers.py` and call it from the notebook.
7. Add a Markdown cell explaining what AI helped with and how you verified the result.
8. Clear notebook outputs before committing.

## Parity Table Template

| Check              | R result | Python result | Match? | Notes |
|--------------------|----------|---------------|--------|-------|
| Raw row count      |          |               |        |       |
| Filtered row count |          |               |        |       |

| Check                          | R result | Python result | Match? | Notes |
|--------------------------------|----------|---------------|--------|-------|
| Number of grouped rows         |          |               |        |       |
| Mean BMI values after rounding |          |               |        |       |
| Group labels                   |          |               |        |       |

## AI Prompt

I have a working R `dplyr` summary. Translate it to Python/pandas one step at a time. After each step, tell me exactly what output to compare with R. Do not add new variables, filters, plots, or interpretation.

## Definition of Done

Your exercise is complete when a reader can open the notebook, rerun the cells in order, and see why the R and Python outputs do or do not match.

# R To Python Cheatsheet

Use this as a translation aid, not as a substitute for checking outputs.

| Task           | R                                                       | Python                                                       |
|----------------|---------------------------------------------------------|--------------------------------------------------------------|
| Load package   | <code>library(dplyr)</code>                             | <code>import pandas<br/>as pd</code>                         |
| Read CSV       | <code>read_csv("file.csv")</code>                       | <code>pd.read_csv("file.csv")</code>                         |
| First rows     | <code>head(df)</code>                                   | <code>df.head()</code>                                       |
| Row count      | <code>nrow(df)</code>                                   | <code>len(df)</code> or<br><code>df.shape[0]</code>          |
| Column names   | <code>names(df)</code>                                  | <code>df.columns.tolist()</code>                             |
| Select columns | <code>select(df, BMI,<br/>Gender)</code>                | <code>df[["BMI",<br/>"Gender"]]</code>                       |
| Filter rows    | <code>filter(df, Age<br/>&gt;= 20)</code>               | <code>df[df["Age"] &gt;=<br/>20]</code>                      |
| Missing values | <code>sum(is.na(df\$BMI))</code>                        | <code>df["BMI"].isna().sum()</code>                          |
| Group summary  | <code>group_by(...)<br/> &gt;<br/>summarise(...)</code> | <code>.groupby(...).agg(...)</code>                          |
| New column     | <code>mutate(df, whtr<br/>= Waist /<br/>Height)</code>  | <code>df["whtr"] =<br/>df["Waist"] /<br/>df["Height"]</code> |
| Sort rows      | <code>arrange(df,<br/>IncomeGroup)</code>               | <code>df.sort_values("IncomeGroup")</code>                   |
| Unique values  | <code>unique(df\$Gender)</code>                         | <code>df["Gender"].unique()</code>                           |

## Translation Rule

The translation is not finished when the code runs. It is finished when the R and Python outputs have been compared and the comparison is documented.

## **Week 6: Data Visualization**

# Week 6: Data Visualization

## Overview

This week turns a descriptive summary into a visual claim. You will use `ggplot2` as the core tool, learn how to spot the most common ways a plot misleads a reader, practice prompting an AI co-pilot for plot code you can defend, and finish with a polished figure for Assignment 5.

The emphasis is not decoration. A useful plot makes the statistic, grouping, exclusions, and limitation visible enough that a reader can understand what the figure does and does not support.

## Objectives

By the end of the week you can:

- build a rerunnable `ggplot2` visualization from the NHANES Health Equity CSV snapshot;
- identify visual choices that can exaggerate, hide, or mislabel a descriptive result;
- revise title, scale, color, grouping, and caption choices to reduce misinterpretation;
- document whether a plot is weighted or unweighted, descriptive or causal, and complete-case or missingness-aware;
- use AI assistance as a draft partner while checking the code and claims yourself.

## Connection

Week 5 checked whether simple summaries agree across R and Python. Week 6 asks whether a visual summary is faithful to the evidence. Week 7 then documents the cohort, missingness, and Table 1 behind those same descriptive claims.

## Case Study Data Analysis

The **NHANES Health Equity data spine** now supports code-reading, EDA, visualization, and reproducibility checks. Use the cached CSV/RDS for routine class work; a CDC retrieval script exists for advanced users.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The default classroom path is to use the cached data so the analysis work is reproducible without internet access.

```
1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_analysis <- read_csv(nhanes_path, show_col_types = FALSE)
15
16 nhanes_analysis |>
17   summarise(
18     rows = n(),
```



```

19     missing_bmi = sum(is.na(BMI)),
20     missing_income = sum(is.na(IncomeGroup))
21   )
22   #> # A tibble: 1 x 3
23   #>       rows missing_bmi missing_income
24   #>   <int>         <int>         <int>
25   #> 1 105626         9356         9415
26
27   nhanes_analysis |>
28     filter(!is.na(BMI), !is.na(IncomeGroup), Age >= 20, Age <= 80) |>
29     group_by(IncomeGroup) |>
30     summarise(
31       n = n(),
32       mean_bmi = round(mean(BMI), 1),
33       .groups = "drop"
34     )
35   #> # A tibble: 3 x 3
36   #>   IncomeGroup      n mean_bmi
37   #>   <chr>         <int>   <dbl>
38   #> 1 High Income (>3.5) 16330    28.6
39   #> 2 Low Income (<1.3) 15293    29.4
40   #> 3 Middle Income    19228    29.3

```

## Reading order

Work through the pages in order. Each page is short and self-contained.

1. [viz01 — Why Visualize, and What For](#)
2. [viz02 — Grammar of a ggplot, and Your First Plot](#)
3. [viz03 — Worked Example: Descriptive BMI Visualization](#)
4. [viz04 — How a Plot Can Mislead](#)
5. [viz05 — AI Prompts You Can Defend](#)
6. [viz06 — In-Class Studio: AI Visual Audit](#)
7. [viz07 — Assignment 5 Walkthrough and Reference](#)

## Class plan

1. Read the worked example ([viz03](#)) and name the statistic, grouping variable, exclusions, and limitation.
2. Inspect the flawed plot code referenced in the [studio](#).
3. Identify at least three ways the plot could mislead a reader.
4. Revise the plot so the scale, geometry, title, labels, and caption match the evidence.
5. Pair-review the corrected caption and check that it does not imply causality or population inference.

## Student output

By the end of class each student has a corrected plot, a caption, a short audit note, and an AI-use note if AI helped draft code or wording.

## Links

- Assignment: [Assignment 5 — Visualization Audit](#)
- Milestone: [M1 — Project Proposal](#) (`initial-visual-plan.md` is the connection point)

## Definition of done

- The worked example renders from source using a relative path.
- The revised figure has clear labels, an honest scale, and a limitation-aware caption.
- The audit note identifies at least three issues across correctness, design, interpretation, or reproducibility.
- The M1 proposal uses the same standard of transparency for dataset choice, feasibility, and stewardship.

## **What students leave with**

One polished descriptive visualization and a reusable checklist for auditing AI-assisted plots before they appear in a report, dashboard, or slide.

# Why Visualize, and What For

## Goal

Before you write any `ggplot()` code, get clear on what a plot is *for* in a health-data project. A figure is a **claim about evidence** — a reader looks at it, decides what it says, and walks away with a belief. Your job for the next two weeks is to make sure that belief matches what the data can actually support.

## The four-step pipeline you will follow

This week’s pages all support the same short pipeline. Every visualization you make for the course should run through it:

1. **Load and inspect** the data so you know what is actually there.
2. **Plot** a clear descriptive summary — one figure, one comparison.
3. **Audit** the figure against the data and the question.
4. **Caption** the figure so a reader can interpret it without reading your code.

If any step is skipped, the figure is fragile. If all four are present, the figure can survive an AI review, a peer review, or a project grader.

## What a “good” descriptive plot does

A descriptive plot in this course does three things:

- It names the **statistic** (a count, mean, proportion, median).
- It names the **comparison** (across what groups, time, or condition).
- It names the **limitation** (what the figure does *not* support — e.g., not a causal claim, not a national estimate).

A “bad” plot quietly drops one of these and lets the reader fill in the gap. Your audit habit is what catches that.

## Where AI fits

You will use an AI co-pilot to draft `ggplot2` code. You will not let it choose your scale, your title, or your caption without review. The Week 3 rule still holds: every AI-drafted chunk gets a `# AI prompt: ... # Verified: ...` comment line, and you check the column names, the missingness handling, and the claim the figure makes.

## What you produce this week

Two artifacts:

- One revised, captioned `ggplot2` figure based on the NHANES Health Equity CSV (the Assignment 5 deliverable).
- A short M1 visual plan paragraph naming the figure direction for your term project (one part of the Milestone 1 proposal due the same week).

### Tip

The plot you make for A5 does not need to be the same as the figure you sketch for M1. A5 is practice on a known dataset; M1 is your team’s plan for your own project. The audit standard is the same in both places.

## Where this goes next

[viz02](#) teaches the grammar of a ggplot from scratch, with one minimal guided figure on the NHANES CSV.

# Grammar of a ggplot, and Your First Plot

## Where this fits

[viz01](#) framed visualization as a claim about evidence. This page teaches the five moving parts of any `ggplot2` figure and walks you through your first plot on the NHANES Health Equity CSV. Everything builds from these five parts.

## The five moving parts

Every ggplot is built from the same five ingredients. You will see all five in the worked example on the next page, so the names matter.

| Part                         | What it answers                                 | A beginner-friendly analogy                  |
|------------------------------|-------------------------------------------------|----------------------------------------------|
| <b>Data</b>                  | What table are we plotting?                     | The contents of one box                      |
| <b>Mapping (aes)</b>         | Which variable goes where (x, y, color, group)? | Telling R which column to draw on which axis |
| <b>Geometry (geom_*)</b>     | What kind of mark (point, line, bar)?           | The shape of the ink on the page             |
| <b>Scale and coordinates</b> | How are values stretched along an axis?         | The ruler under the ink                      |
| <b>Labels and theme</b>      | What does the reader see in plain text?         | The caption around the picture               |

If a reader cannot answer “which statistic? across what groups? with what limitation?” from your figure, one of these five parts is doing too little.

## Setup

Open your Codespace and start a new code chunk in a `.qmd` file. Run:

```
1 library(readr)
2 library(dplyr)
3 library(ggplot2)
```

### Warning

`library(ggplot2)` is required every session. If you see “could not find function `ggplot`”, run the `library()` line first.

## Load the cached NHANES CSV

This is the same file you will use for the worked example and your A5 deliverable. The candidate-path pattern below works whether your `.qmd` is at the repository root or under `weeks/week06-visualization/`.

```
1 candidate_paths <- c(
2   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
3   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
4 )
5 data_path <- candidate_paths[file.exists(candidate_paths)][1]
6 nhanes <- read_csv(data_path, show_col_types = FALSE)
7
8 glimpse(nhanes)
```

`glimpse()` shows the column names and types. Read them carefully. Misspelling a column name (e.g., `bmi` when the data has `BMI`) is the #1 source of broken ggplots in this course.



### **i** Tester note

Capture: a screenshot of the Codespaces Quarto preview after `glimpse(nhanes)` runs, showing the column names. New students rely on this to spot case-sensitive column names like BMI and IncomeGroup.

## Your first plot

Here is the smallest meaningful ggplot you can build from this dataset: a scatter of one numeric variable against age, with no grouping and no transformation. Try it as is, then change Weight to BMI or Height and re-render.

```
1 ggplot(nhanes, aes(x = Age, y = Weight)) +  
2   geom_point(alpha = 0.2) +  
3   labs(  
4     title = "Weight by age, NHANES Health Equity classroom dataset",  
5     subtitle = "One point per record; unweighted, descriptive only",  
6     x = "Age (years)",  
7     y = "Weight (kg)"  
8   ) +  
9   theme_minimal()
```

Map the five parts onto this code so the grammar feels concrete:

- **Data:** `nhanes`
- **Mapping:** `aes(x = Age, y = Weight)`
- **Geometry:** `geom_point(alpha = 0.2)` (translucent points so overlap is visible)
- **Scale and coordinates:** the default linear axes (you have not changed them)
- **Labels and theme:** `labs(...) + theme_minimal()`

### **i** Tester note

Capture: a screenshot of the rendered scatter plot. Confirm the axis labels read “Age (years)” and “Weight (kg)”

so students can compare against their own first attempt.

## Three things to check on every plot

After you run any ggplot, ask:

1. **Column names match the data.** Did `glimpse()` confirm BMI is uppercase, not `bmi`?
2. **Axes tell a clear story.** Are the units readable? Is the range honest?
3. **Title and subtitle match what is plotted.** Does the title overstate (“BMI rises with age across Canada”)? If yes, soften it (“Weight by age in the prepared classroom dataset”).

## Where this goes next

[viz03](#) is the first end-to-end worked example: load the CSV, prepare a small descriptive summary, render a labeled figure, and write a caption.

# Worked Example: Descriptive BMI Visualization

## Where this fits

You learned the grammar of a ggplot and made one minimal scatter on the [previous page](#). This page is the first end-to-end visualization of the term: load the cached NHANES Health Equity CSV, prepare a small descriptive summary, render one labeled figure, and write a caption that does not overclaim. We will return to this same figure in [viz04](#) to see how easy it is to spoil with a few “improvements.”

## Goal

This worked example uses the NHANES Health Equity CSV snapshot to build one descriptive visualization. The plot summarizes mean BMI by income group across NHANES cycle. It is **unweighted** and should be interpreted as a classroom visualization exercise, not a population estimate.

```
1 library(readr)
2 library(dplyr)
3 library(ggplot2)
4
5 candidate_paths <- c(
6   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
7   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
8 )
9
10 data_path <- candidate_paths[file.exists(candidate_paths)][1]
11 if (is.na(data_path)) {
```

```

12   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
13 }
14
15 nhanes <- read_csv(data_path, show_col_types = FALSE)

```

## Prepare a Descriptive Summary

Keep the summary intentionally simple. We are describing the prepared class dataset, not estimating a national trend.

```

1  bmi_by_income_cycle <- nhanes |>
2    filter(
3      !is.na(BMI),
4      !is.na(IncomeGroup),
5      !is.na(Cycle),
6      Age >= 20,
7      Age <= 80
8    ) |>
9    group_by(Cycle, IncomeGroup) |>
10   summarise(
11     n = n(),
12     mean_bmi = mean(BMI),
13     .groups = "drop"
14   ) |>
15   filter(n >= 30)
16
17 bmi_by_income_cycle
18 #> # A tibble: 33 x 4
19 #>   Cycle      IncomeGroup      n mean_bmi
20 #>   <chr>      <chr>      <int>   <dbl>
21 #> 1 1999-2000 High Income (>3.5) 1123    27.9
22 #> 2 1999-2000 Low Income (<1.3) 1091    28.8
23 #> 3 1999-2000 Middle Income      1327    29.0
24 #> 4 2001-2002 High Income (>3.5) 1491    28.1
25 #> 5 2001-2002 Low Income (<1.3) 1089    28.7
26 #> 6 2001-2002 Middle Income      1561    28.3
27 #> 7 2003-2004 High Income (>3.5) 1284    28.4
28 #> 8 2003-2004 Low Income (<1.3) 1183    28.6

```

```

29 #> 9 2003-2004 Middle Income      1573      28.8
30 #> 10 2005-2006 High Income (>3.5) 1480      28.5
31 #> # i 23 more rows

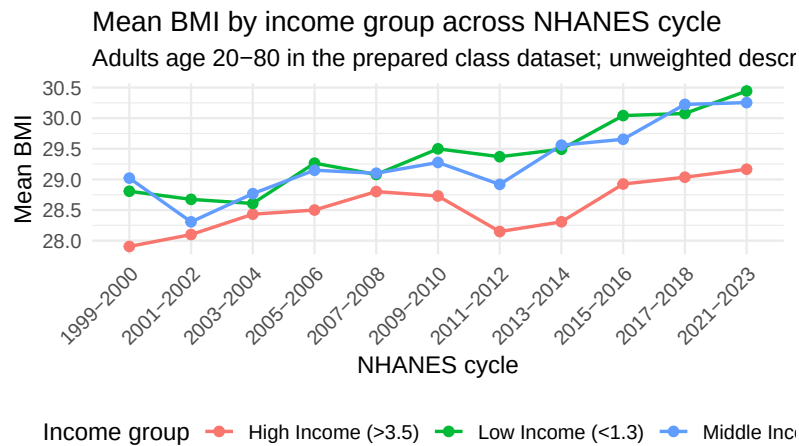
```

## Plot

```

1  bmi_plot <- ggplot(
2    bmi_by_income_cycle,
3    aes(x = Cycle, y = mean_bmi, color = IncomeGroup, group = IncomeGroup)
4  ) +
5    geom_line(linewidth = 0.8) +
6    geom_point(size = 2) +
7    labs(
8      title = "Mean BMI by income group across NHANES cycle",
9      subtitle = "Adults age 20-80 in the prepared class dataset; unweighted descriptive summary",
10     x = "NHANES cycle",
11     y = "Mean BMI",
12     color = "Income group",
13     caption = "Caption template: Mean BMI is summarized descriptively from the cached class data",
14   ) +
15   theme_minimal(base_size = 12) +
16   theme(
17     axis.text.x = element_text(angle = 45, hjust = 1),
18     legend.position = "bottom"
19   )
20
21 bmi_plot

```



## Optional Plotly Check

The core assignment uses the static `ggplot2` figure above. If `plotly` is installed, you can preview the same plot interactively while checking labels and tooltips. This extension is optional because static Quarto output is the supported submission path.

```
1 # plotly::ggplotly(bmi_plot)
```

## Caption Template

Use this structure when revising your own plot:

This figure shows [measure] by [group] across [time/category] using the prepared NHANES Health Equity classroom dataset. The summary is descriptive and unweighted. Records with missing [variables] were excluded. The plot supports comparison of patterns, not causal claims or population-level estimates.

## Mini Visual Audit Checklist

- Does the title describe what is plotted without overstating the claim?
- Is it clear whether the statistic is weighted or unweighted?
- Are axes, groups, and units labeled plainly?
- Are missing values or exclusions disclosed?
- Is the scale chosen to support honest comparison rather than exaggeration?
- Does the caption state the limitation that this is descriptive, not causal?

## Where this goes next

Hold onto this figure. The [next page](#) walks through six common ways a plot like this can mislead a reader, and [viz06](#) is the in-class studio where you apply the audit to a deliberately flawed version of it.

# How a Plot Can Mislead

## Where this fits

You have a clean, captioned worked example ([viz03](#)). This page is the audit lens: six common ways a figure that *looks fine* can quietly mislead a reader. Each one shows up in real research, and each one is something an AI co-pilot will produce if you do not push back.

## Six failure modes

For each failure mode, you will see the problem in one sentence and the fix in one sentence. Studio time in [viz06](#) revisits these on a deliberately flawed plot.

### 1. Truncated y-axis exaggeration

A small difference in means looks huge when the y-axis starts at 27 instead of 0.

- **Symptom:** `coord_cartesian(ylim = c(27.5, 31.5))` on a BMI mean plot, no narrative explanation.
- **Fix:** Start the axis at 0, or at a value justified in the caption. Use `expand_limits(y = 0)` when in doubt.



## 2. Causal title on a descriptive plot

A title that says “X causes Y” is a causal claim. Descriptive summaries cannot support it.

- **Symptom:** “Income level causes BMI to rise across NHANES cycles.”
- **Fix:** “Mean BMI by income group across NHANES cycles in the prepared classroom dataset.”

## 3. Smoothing across categorical time

`geom_smooth(method = "lm")` draws a line that *implies* a continuous linear trend. NHANES cycles are categorical labels, not a continuous axis.

- **Symptom:** A linear regression line layered across cycle bins.
- **Fix:** Use `geom_line()` plus `geom_point()` to connect descriptive cycle summaries, with no regression overlay.

## 4. Weighted-vs-unweighted ambiguity

If the subtitle says “weighted” but the code does not use a survey weight variable, the reader has been misled.

- **Symptom:** Subtitle: “Weighted BMI trend.” Code: `mean(BMI, na.rm = TRUE)` with no weight.
- **Fix:** State plainly that the summary is unweighted, or compute a weighted estimate using the documented weight variable.

## 5. Silent missingness

Rows with missing BMI or income disappear without a caption.

- **Symptom:** `filter(!is.na(BMI), !is.na(IncomeGroup))` deep inside a pipe with no note in the caption.

- **Fix:** Disclose missing-value exclusions in the caption. Better: report missingness counts before the figure (you will do this on [eda03](#) next week).

## 6. Hidden small-cell counts

A group with  $n = 7$  plots the same way as a group with  $n = 700$ , but the reader cannot tell them apart.

- **Symptom:** A small income group plotted as a line with no indication of sparse cells.
- **Fix:** Filter out groups below a minimum count, *or* label them visibly with  $n$  annotations, *or* report the counts beside the figure.

## Why six and not sixty

You will see other failure modes in the wild — duplicated colors, default rainbow palettes, dual axes, missing legends. The six above account for most of the issues you will catch when auditing an AI-drafted plot for descriptive health data work in this course. Master these first.

## A quick audit habit

When you finish a plot, ask three questions out loud:

1. Could a reader make a causal claim from this title or caption?
2. Could a small difference look bigger than it is because of the axis?
3. Is missingness, exclusion, or small-cell count visible to the reader?

If any answer is “yes” or “I don’t know,” revise the figure or the caption before moving on.

## Where this goes next

[viz05](#) shows how to write AI prompts that produce plots you can defend, and how to audit what the AI gives you.

# AI Prompts You Can Defend

## Where this fits

You can build a ggplot from scratch ([viz02](#)) and you can spot six common failure modes ([viz04](#)). This page connects the two: how to ask an AI co-pilot for plot code without inheriting its bad defaults, and how to audit what it gives you.

## Why this matters

If you paste “make me a nice plot of NHANES” into a chat window, the model will produce something — possibly with a flashy palette, a trend line, and a confident title. None of those choices were yours. None of them were audited. By the time the figure is in your report, you own them anyway.

The fix is to write prompts that **constrain the model** instead of decorate the request.

## Three prompt templates that work

### Template 1 — Start from the data, not the picture

```
I have a data frame called bmi_by_income_cycle
with columns Cycle, IncomeGroup, n, and
mean_bmi. Write a ggplot2 figure that shows
mean_bmi on the y-axis and Cycle on the x-axis,
with one line per IncomeGroup. The summary is
unweighted and descriptive. Do not add a trend
line or smoothing.
```

What this prompt does well: it names the data, the variables, the geometry, and what *not* to add.

## Template 2 — Ask for an honest scale and labels

Add `labs(...)` to the figure above so the title describes the comparison without making a causal claim, the subtitle states that the summary is unweighted and uses adults age 20-80, and the caption discloses that records with missing BMI, income, or cycle were excluded.

What this prompt does well: it tells the model what each label should *do*, not just that there should be labels.

## Template 3 — Ask for a caption that names the limitation

Write a one-sentence caption for this figure that names the statistic (mean BMI), the comparison (across NHANES cycles, by income group), the data source (prepared NHANES Health Equity classroom CSV), the exclusions (missing values, ages outside 20-80), and one limitation (descriptive only, not a population estimate).

What this prompt does well: the model now has a checklist of what the caption must contain.

## Auditing what the model gives you

After you paste in any AI-drafted plot code, walk through this checklist before you accept it:

1. **Column names match the data.** Run `glimpse(your_data)` and compare.
2. **Geometry is appropriate.** No smoothing across categorical cycles. No regression lines on a descriptive summary.

3. **Scale is honest.** Does the y-axis start at 0, or is there a stated reason it does not?
4. **Title and subtitle are descriptive.** No “X causes Y” wording.
5. **Caption discloses exclusions and limitation.** Missing values, cohort restrictions, weighted vs unweighted.
6. **na.rm = TRUE is present** wherever the summary uses a numeric column that may have NA.

If any item fails, edit the code or the labels before rendering.

## The paper-trail rule (still in force)

From Week 3: every AI-drafted chunk in your .qmd gets a two-line comment.

```
1 # AI prompt: "Plot mean_bmi by Cycle, one line per IncomeGroup, no smoothing"
2 # Verified: columns match glimpse(); no smoother; descriptive title; caption lists exclusions
3 bmi_plot <- ggplot(bmi_by_income_cycle, aes(...)) + ...
```

This is the same audit log you used in Week 3 for data wrangling. The course AI policy requires it for graded work.

## A note on what NOT to ask the AI

- Do not paste row-level NHANES data into an external chat window. Aggregate first.
- Do not ask the AI to “decide” whether income causes BMI. The figure is descriptive; causal language is your responsibility to remove.
- Do not accept a “looks fine to me” from the model as audit evidence. The audit is yours.

## Where this goes next

[viz06](#) is the in-class studio where you apply all of this to a deliberately flawed plot.

# In-Class Studio: AI Visual Audit

## Where this fits

You have the worked example ([viz03](#)), the audit categories ([viz04](#)), and the AI-prompt pattern ([viz05](#)). This page is the in-class studio where you put all three to work: read a deliberately flawed plot, identify at least three problems, revise the plot, and write a short audit note. The same workflow is what Assignment 5 expects, so think of this as the dress rehearsal.

## Before you start

Take 2 minutes to check that your Codespace is ready:

1. `library(readr); library(dplyr); library(ggplot2)` runs without error.
2. The cached data file resolves from the repository root: `examples/nhanes-equity/data/nhanes_equity_v6.csv`.
3. Your `viz03-worked-example.qmd` renders cleanly.

If any of those three fail, raise it with your debugging partner before moving on.

## Goal

Evaluate one AI-generated visualization for correctness, clarity, bias, aggregation choices, and misleading design. Then revise it so the visual matches the evidence.

## Inputs

- Worked example to compare against: `weeks/week06-visualization/viz03-worked-example.qmd`
- Dataset: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Flawed plot code (the artifact you will audit):  
`weeks/week06-visualization/instructor/flawed-plot-code.R`

### Tester note

Capture: a screenshot of the rendered flawed plot in the Codespaces Quarto preview, with the misleading title clearly visible. This will be referenced by other students during the studio.

## Activity Steps

1. Review the worked example and note how it labels the data source, statistic, and limitation.
2. Open `instructor/flawed-plot-code.R` and read the comments and output labels before you run it.
3. Run the script and inspect the rendered figure.
4. List at least three issues that could mislead a reader. Tag each one with one of the audit categories below.
5. Revise the plot so the title, scale, grouping, and caption match the evidence. Keep the cohort restriction explicit.
6. Write a short note explaining what you changed and why.

## Audit Categories

- **Correctness:** Does the plot show what the title and labels claim?
- **Design:** Do scale, color, ordering, and geometry support fair comparison?
- **Interpretation:** Does the caption avoid causal or population-level claims that the plot cannot support?
- **Bias and aggregation:** Could grouping, missingness, or small counts hide important variation?



- **Reproducibility:** Can another person rerun the code from the repository root using relative paths?

## Required Outputs (studio draft)

These are the draft files you produce in class. The Assignment 5 deliverable list ([viz07](#)) is the same set, polished.

- `visual-audit-note.md` — at least three issues with category, why it matters, and the corrected approach.
- `corrected-plot.qmd` — a rerunnable Quarto file that creates the corrected plot.
- `corrected-plot.png` — the exported corrected plot.
- `caption.md` — a caption that states the statistic, data source, exclusions, and limitations.
- `ai-use-note.md` — if you used AI, explain what it helped with and how you checked the result.

## Completion Check

You are done with the studio when a peer can rerun your corrected plot, identify the main comparison without reading your code, and see at least one explicit limitation in the caption.

## Where this goes next

Polish the studio drafts into the Assignment 5 submission. [viz07](#) walks through the seven A5 files one by one and connects the work to the M1 visual plan due the same week.

# Assignment 5 Walkthrough and Reference

## Where this fits

You have the worked example, the failure modes, the prompt templates, and the studio drafts. This page is the bridge from in-class work to the Assignment 5 submission. The deliverables listed here are the same set in the [A5 README](#) — this page explains what goes into each one and how to verify it.

## Deadline and link to M1

Assignment 5 is due the Monday after Week 6 class, 4 PM. The Milestone 1 (Project Proposal) `initial-visual-plan.md` is due the same Monday. The two pieces of work share a standard of transparency, but they are *different* artifacts: A5 polishes a plot from the NHANES classroom CSV; M1 sketches one planned figure for your team's project. Do not conflate them.

## The seven A5 deliverables, explained

Each file goes in `assignments/assignment05-visualization/submission/`.

### 1. visualization-audit.qmd

A rerunnable Quarto file that loads the cached CSV with a relative path, prepares a descriptive summary, renders the final corrected plot, and includes one or two short paragraphs explaining what you revised and why.

- **Verify:** open it in a fresh Codespace, render top to bottom, no errors.

### 2. visualization-audit.html

The rendered output of the `.qmd` above. Quarto writes this when you click Render.

- **Verify:** open the HTML in the Codespaces preview and confirm the figure, caption, and audit paragraph all appear.

### 3. corrected-plot.png

The exported corrected plot as a static image. Use `ggsave("corrected-plot.png", plot, width = 6, height = 4)` to write it from code rather than by screenshot.

- **Verify:** the PNG exists, opens, and matches what the rendered HTML shows.

### 4. visual-audit-note.md

A short note listing at least three issues you identified in the flawed plot from [viz06](#). For each issue, write one sentence on what it was, one sentence on why it matters, and one sentence on how you corrected it.

- **Verify:** three issues, each tagged with a category from [viz04](#) (correctness, design, interpretation, bias/aggregation, or reproducibility).

## 5. caption.md

The caption for your final figure. Use the template from [viz03](#):

This figure shows [measure] by [group] across [time/category] using the prepared NHANES Health Equity classroom dataset. The summary is descriptive and unweighted. Records with missing [variables] were excluded. The plot supports comparison of patterns, not causal claims or population-level estimates.

- **Verify:** the caption names the statistic, the data source, the exclusions, and one limitation.

## 6. m1-link-note.md

Two or three sentences connecting the visualization standard from this assignment to your M1 visual plan. What did this assignment change about the figure your team plans to make? Did it change your dataset choice, the exclusions you will document, or the caption your team will write?

- **Verify:** the note is specific to *your team's* project, not generic advice.

## 7. ai-use-note.md

If you used AI to draft code or wording, name what it helped with and how you verified the result. If you did not use AI, write one sentence saying so.

- **Verify:** specific (which chunk? which prompt template?), not vague (“I used Copilot”).

## Reproducibility checklist for A5

Before you click submit:

- The `.qmd` renders in a fresh Codespace with no manual edits.
- The data path is relative (`examples/nhanes-equity/data/nhanes_equity_v6.csv`).
- The PNG is generated by code (or a documented render step), not by screenshot.
- The caption uses the descriptive template and does not imply causality or national prevalence.
- All seven files are committed and pushed to your GitHub repo.

## Accessibility (good practice, not a graded deliverable)

A plain-language alt text sentence next to your figure helps screen-reader users, colour-blind reviewers, and busy policy-makers. The A5 rubric does not score alt text — it is encouraged practice. If you want to try it, add `fig-alt:` as a chunk option:

## ggplot2 quick reference

| Need                       | Code                                    | Note                                                              |
|----------------------------|-----------------------------------------|-------------------------------------------------------------------|
| Load packages              | <code>library(ggplot2)</code>           | Every session                                                     |
| Set data + mapping         | <code>ggplot(df, aes(x, y))</code>      | The starting point                                                |
| Points                     | <code>geom_point()</code>               | Use <code>alpha</code> for overlap                                |
| Lines connecting summaries | <code>geom_line() + geom_point()</code> | Better than <code>geom_smooth()</code> for descriptive cycle data |
| Bars / columns             | <code>geom_col()</code>                 | For pre-aggregated counts                                         |

| Need          | Code                                                         | Note                                       |
|---------------|--------------------------------------------------------------|--------------------------------------------|
| Labels        | <code>labs(title, subtitle, x, y, caption)</code>            | Caption goes under the figure              |
| Honest y-axis | <code>expand_limits(y = 0)</code>                            | Start at zero unless you justify otherwise |
| Theme         | <code>theme_minimal()</code>                                 | Clean default                              |
| Save figure   | <code>ggsave("plot.png", plot, width = 6, height = 4)</code> | Code-generated, not screenshot             |

## Visual audit checklist

A short version of the five questions from [viz04](#):

- Title and caption do not imply causation.
- Y-axis starts at 0 or has a stated reason.
- Geometry suits descriptive data (no smoother across categorical x).
- Weighted vs unweighted is stated.
- Missingness and exclusions are disclosed.
- Small-cell counts are visible or filtered out.

## Five quick knowledge checks

1. What are the five moving parts of a ggplot?
2. Name two failure modes from [viz04](#) and one fix for each.
3. Why is `geom_smooth(method = "lm")` rarely appropriate for NHANES cycle data?
4. What two lines of comment should appear above any AI-drafted chunk?
5. What is the difference in scope between Assignment 5 and the M1 `initial-visual-plan.md`?

## Where this goes next

Next week ([Week 7](#)) builds the EDA evidence behind the descriptive claims you have been plotting: cohort definition, missingness reporting, Table 1, and the AI audit categories.

## **Week 7: EDA, Table 1 & AI Auditing**



# Week 7: EDA, Table 1 & AI Auditing

## Overview

This week builds the audit trail behind a descriptive analysis. You will define an analysis cohort, check missingness, create a Table 1-style summary, and audit a planted-error starter that mimics common AI-assisted mistakes.

EDA is treated as **evidence stewardship**: every exclusion, statistic, and interpretation should be visible enough for another person to rerun and challenge.

## Objectives

By the end of the week you can:

- define and document an analysis cohort before summarizing;
- report missingness for key variables rather than dropping rows silently;
- create a Table 1-style descriptive summary with N, mean, and SD;
- distinguish what descriptive statistics claim from what they cannot claim;
- audit code and prose for correctness, reproducibility, interpretation, and stewardship errors.

## Connection

Week 6 focused on whether visual design makes a descriptive claim honest and clear. Week 7 documents the cohort, missingness, and summary statistics behind those claims. Week 8 then turns one audited descriptive finding into a dashboard-style KT prototype.

## Case Study Data Analysis

The **NHANES Health Equity data spine** now supports code-reading, EDA, visualization, and reproducibility checks. Use the cached CSV/RDS for routine class work; a CDC retrieval script exists for advanced users.

- Cached RDS: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Case-study README: `examples/nhanes-equity/README.md`

The default classroom path is to use the cached data so the analysis work is reproducible without internet access.

```
1 library(readr)
2 library(dplyr)
3
4 candidate_paths <- c(
5   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
6   "../..examples/nhanes-equity/data/nhanes_equity_v6.csv"
7 )
8
9 nhanes_path <- candidate_paths[file.exists(candidate_paths)][1]
10 if (is.na(nhanes_path)) {
11   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
12 }
13
14 nhanes_analysis <- read_csv(nhanes_path, show_col_types = FALSE)
15
16 nhanes_analysis |>
17   summarise(
```

```

18     rows = n(),
19     missing_bmi = sum(is.na(BMI)),
20     missing_income = sum(is.na(IncomeGroup))
21   )
22 #> # A tibble: 1 x 3
23 #>   rows missing_bmi missing_income
24 #>   <int>      <int>      <int>
25 #> 1 105626      9356      9415
26
27 nhanes_analysis |>
28   filter(!is.na(BMI), !is.na(IncomeGroup), Age >= 20, Age <= 80) |>
29   group_by(IncomeGroup) |>
30   summarise(
31     n = n(),
32     mean_bmi = round(mean(BMI), 1),
33     .groups = "drop"
34   )
35 #> # A tibble: 3 x 3
36 #>   IncomeGroup      n mean_bmi
37 #>   <chr>      <int>   <dbl>
38 #> 1 High Income (>3.5) 16330    28.6
39 #> 2 Low Income (<1.3)  15293    29.4
40 #> 3 Middle Income    19228    29.3

```

## Reading order

1. [eda01 — EDA as Evidence Stewardship](#)
2. [eda02 — Cohort and Missingness](#)
3. [eda03 — Worked Example: EDA and Table 1](#)
4. [eda04 — What Each Statistic Claims](#)
5. [eda05 — Four Audit Categories for AI-Drafted EDA](#)
6. [eda06 — In-Class Studio: Planted-Error Audit](#)
7. [eda07 — Assignment 6 Walkthrough and Reference](#)

## Class plan

1. Work through the [worked example](#) using the CSV snapshot.

2. Name the cohort definition before inspecting any group summaries.
3. Build a missingness table for the variables used in the summary.
4. Produce a Table 1-style summary with N, BMI mean (SD), and age mean (SD).
5. Audit the planted-error starter in the [studio](#) and categorize at least five issues.
6. Revise the methods note so provenance, missingness, and non-causal framing are explicit.

## Student output

By the end of class each student has a rerunnable EDA note, a Table 1-style summary, a planted-error audit, a provenance/stewardship note, and an AI-use note if AI helped draft code or prose.

## Links

- Assignment: [Assignment 6 — EDA and AI Audit](#)
- Milestone: [M2 — Preliminary Analysis](#)

## Definition of done

- Cohort restrictions are stated before results are interpreted.
- Missingness is checked and reported before complete-case summaries are used.
- Table labels match the statistic actually computed.
- The audit identifies at least five planted issues across correctness, reproducibility, interpretation, and stewardship.
- M2 preliminary analysis renders from source and explains what the descriptive results do not prove.

## **What students leave with**

A reproducible EDA note and a habit of treating AI-generated summaries as drafts that must be checked against the data, code, and stewardship context.

# EDA as Evidence Stewardship

## Goal

Last week you turned a descriptive summary into a clear figure. This week you build the evidence trail behind that summary. The phrase “exploratory data analysis” often sounds casual — *“just look at the data”*. In a health-data project it is not casual at all: every exclusion, every statistic, and every interpretation should be visible enough that another person can rerun your work and challenge it.

Think of EDA as the **paperwork** that defends a finding. If the paperwork is missing, the finding cannot survive an audit.

## The four artifacts this week produces

Every page this week supports one of these four:

1. A **cohort definition** that names exactly who is in the analysis sample.
2. A **missingness table** that says how much of each key variable is missing.
3. A **Table 1** that summarizes the cohort with N, mean (SD), and equivalents.
4. A **methods note** that states what the table can and cannot claim.

If any one of the four is absent, the work is not ready for Milestone 2 or Assignment 6.

## Why this matters for AI work

When you ask an AI co-pilot for “a Table 1,” the model will produce one. It will not check whether the cohort makes sense, whether missingness was reported first, or whether the labels match the statistics. Those are your responsibilities. The four-artifact workflow is what makes that responsibility manageable: each artifact is a small piece you can audit on its own.

The four audit categories you will use this week — **correctness, reproducibility, interpretation, stewardship** — are introduced on [eda05](#) and practiced in the [studio](#).

## What “descriptive, unweighted, non-causal” means

You will see this phrase repeatedly. Each word matters:

- **Descriptive:** the table reports what is in the data, not what the population looks like.
- **Unweighted:** we are not using NHANES survey weights, so the numbers describe the prepared classroom dataset, not US adults.
- **Non-causal:** the table cannot tell you that one variable *causes* another. Descriptive comparison is not causal inference.

Saying this in your methods note is how you defend the table from being misread.

## Where AI fits

Same rule as Week 3 and Week 6: every AI-drafted chunk in your `.qmd` gets a two-line `# AI prompt: ... # Verified: ...` comment. The verification step on Table 1 work specifically checks: cohort filter, missingness handling, label/statistic match, and interpretation.

## What you produce this week

- A rerunnable `eda-note.qmd` with cohort definition, missingness table, and Table 1 (the Assignment 6 deliverable, also the M2 preliminary analysis).
- A short audit note identifying at least five issues in a planted-error starter.
- A revised provenance/stewardship statement suitable for reuse in M2.

## Where this goes next

[eda02](#) walks through cohort definition and missingness reporting — the first two of the four artifacts.



# Cohort and Missingness

## Where this fits

[eda01](#) named the four artifacts your EDA work produces. This page covers the first two: how to define and document an analysis cohort, and how to report missingness *before* you summarize anything. Both habits are the foundation of every Table 1 you will make for the rest of the term.

## Why cohort definition comes first

A dataset is not an analysis sample. The NHANES Health Equity CSV contains records of all ages, with some variables intentionally coded as NA. If you summarize without restricting, you are answering a question the data cannot answer cleanly.

A cohort definition is a one-sentence answer to “*who is in this analysis?*” — written down before you compute any statistics. Common examples:

- “Adults age 20-80 in the prepared classroom dataset.”
- “Female respondents with non-missing BMI.”
- “Records from NHANES cycles 2013-2014 and 2015-2016.”

Each restriction has a reason. The reason goes in the methods note.

## Documenting the cohort in code

This pattern shows up in the [worked example](#). You will reuse it for A6 and M2.

```
1 library(readr)
2 library(dplyr)
3 library(knitr)
4
5 candidate_paths <- c(
6   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
7   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
8 )
9 data_path <- candidate_paths[file.exists(candidate_paths)][1]
10 nhanes <- read_csv(data_path, show_col_types = FALSE)
11
12 analysis_df <- nhanes |>
13   filter(Age >= 20, Age <= 80)
14
15 cohort_counts <- tibble(
16   step = c("Prepared CSV rows", "After age 20-80 restriction"),
17   n = c(nrow(nhanes), nrow(analysis_df))
18 )
19
20 kable(cohort_counts)
```

The `cohort_counts` table is your **flow diagram in text form**. It shows N before and after each restriction. If a peer asks “how many people are in your analysis?”, this table is the answer.

### **i** Tester note

Capture: a screenshot of the rendered `cohort_counts` table in the Quarto preview. Beginners often skip this step; seeing it labeled clearly helps them recognize what they are missing.

## Why missingness comes before the summary

A complete-case summary silently drops rows. If you compute `mean(BMI, na.rm = TRUE)` without first showing how many BMI values were missing, your reader has no idea how representative the mean is.

The audit habit is: **report missingness first, then summarize, then interpret.** Never the other order.

## Counting missingness

The cleanest beginner pattern:

```
1 key_vars <- c("BMI", "Age", "IncomeGroup", "Gender", "Race", "Cycle")
2
3 missingness <- tibble(variable = key_vars) |>
4   mutate(
5     missing_n = vapply(analysis_df[key_vars], function(x) sum(is.na(x)), numeric(1)),
6     missing_percent = round(100 * missing_n / nrow(analysis_df), 1)
7   )
8
9 kable(missingness)
```

This produces a small table: one row per key variable, with the missing count and the percent missing. Three rules apply when you write or accept code like this:

1. **Pick `key_vars` deliberately.** Only include variables you will actually use in the Table 1 or downstream summary.
2. **Use the denominator from the cohort, not the raw dataset.** Missingness as a percent of *your cohort* tells the right story.
3. **Report it, even if it is small.** A row that says “BMI missing: 2 (0.1%)” is still informative.

**i** Tester note

Capture: a screenshot of the rendered `missingness` table for the adult cohort. Beginners need to see what “percent missing per variable” actually looks like.

## What “complete-case” means and why it can bias the summary

After you see the missingness table, you have a choice. The simplest path is **complete-case analysis**: drop rows where any key variable is missing. This is what most beginner tutorials show.

That choice is reasonable for a classroom Table 1, but it is not automatically unbiased. If the rows you drop are systematically different from the rows you keep (for example, people with missing income may be different from people with reported income), your summary tilts in a direction you cannot see from the table alone.

The honest move is to:

1. Report missingness *first*.
2. Filter to complete cases *with the missingness disclosed*.
3. Add a sentence in the methods note acknowledging the complete-case assumption.

You will see this exact sequence in the [worked example](#).

## A short audit habit

After you produce a cohort plus missingness table, ask three questions:

1. Could a reader tell who was excluded and why?
2. Is the percent missing reported for every variable in the downstream Table 1?
3. Did I describe the cohort *before* I described the summary?

If any answer is “no”, revise before the table.

## Where this goes next

[eda03](#) is the first end-to-end worked example: cohort, missingness, Table 1, methods note — all four artifacts in one re-runnable Quarto document.

# Worked Example: EDA and Table 1

## Where this fits

You met the four-artifact workflow on [eda01](#) and practiced cohort definition and missingness reporting on [eda02](#). This page is the first end-to-end EDA of the term: load the cached CSV, restrict to an adult cohort, report missingness, produce a Table 1, and write a short methods note that survives audit. The same workflow underwrites Assignment 6 and Milestone 2.

## Goal

This worked example creates a small Table 1-style descriptive summary from the NHANES Health Equity CSV snapshot. The goal is transparent EDA: define the cohort, check missingness, summarize key variables, and write a methods note that does not overclaim.

```
1 library(readr)
2 library(dplyr)
3 library(knitr)
4
5 candidate_paths <- c(
6   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
7   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
8 )
9
10 data_path <- candidate_paths[file.exists(candidate_paths)][1]
11 if (is.na(data_path)) {
12   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
13 }
```

```

13 }
14
15 nhanes <- read_csv(data_path, show_col_types = FALSE)

```

## Define the Analysis Cohort

Decision: restrict to adults age 20-80. This keeps the example focused on adult BMI summaries and avoids mixing children, adolescents, and the oldest age-coded records into one classroom table.

```

1 analysis_df <- nhanes |>
2   filter(Age >= 20, Age <= 80)
3
4 cohort_counts <- tibble(
5   step = c("Prepared CSV rows", "Rows after age 20-80 restriction"),
6   n = c(nrow(nhanes), nrow(analysis_df))
7 )
8
9 kable(cohort_counts)

```

| step                             | n      |
|----------------------------------|--------|
| Prepared CSV rows                | 105626 |
| Rows after age 20-80 restriction | 57137  |

## Missingness Check

```

1 key_vars <- c("BMI", "Age", "IncomeGroup", "Gender", "Race", "Cycle")
2
3 missingness <- tibble(variable = key_vars) |>
4   mutate(
5     missing_n = vapply(analysis_df[key_vars], function(x) sum(is.na(x)), numeric(1)),
6     missing_percent = round(100 * missing_n / nrow(analysis_df), 1)
7   )

```

```
8
9 kable(missingness)
```

| variable    | missing_n | missing_percent |
|-------------|-----------|-----------------|
| BMI         | 1009      | 1.8             |
| Age         | 0         | 0.0             |
| IncomeGroup | 5395      | 9.4             |
| Gender      | 0         | 0.0             |
| Race        | 0         | 0.0             |
| Cycle       | 0         | 0.0             |

## Table 1-Style Summary

This example stratifies by `IncomeGroup` and reports N plus mean (SD) for BMI and age. It is unweighted and descriptive.

```
1 mean_sd <- function(x) {
2   sprintf("%.1f (%.1f)", mean(x, na.rm = TRUE), sd(x, na.rm = TRUE))
3 }
4
5 table1 <- analysis_df |>
6   filter(!is.na(IncomeGroup)) |>
7   group_by(IncomeGroup) |>
8   summarise(
9     N = n(),
10    `BMI, mean (SD)` = mean_sd(BMI),
11    `Age, mean (SD)` = mean_sd(Age),
12    .groups = "drop"
13  )
14
15 kable(table1)
```

| IncomeGroup        | N     | BMI, mean (SD) | Age, mean (SD) |
|--------------------|-------|----------------|----------------|
| High Income (>3.5) | 16520 | 28.6 (6.3)     | 49.7 (16.0)    |
| Low Income (<1.3)  | 15646 | 29.4 (7.5)     | 47.4 (18.1)    |
| Middle Income      | 19576 | 29.3 (7.0)     | 50.0 (18.4)    |



## Methods Note Template

We used the prepared NHANES Health Equity classroom CSV snapshot. The analysis cohort included records with age 20-80. We summarized BMI and age by income group using unweighted descriptive statistics. Missing values were checked before summarizing. These summaries describe the prepared class dataset and should not be interpreted as causal effects or population estimates.

## Stewardship Note Template

The dataset is a prepared classroom snapshot derived from public-use NHANES files. It should be used for descriptive learning activities, not for individual-level inference or small-cell claims. Reports should cite the source, describe exclusions, and avoid uploading row-level records to external AI tools.

## Survey Design Note

NHANES has a complex survey design. This classroom Table 1 is intentionally unweighted so that the workflow is easy to audit. A population-representative NHANES analysis would require the appropriate weights, strata, and primary sampling units.

## Reproducibility Notes

- The data path is relative to the repository.
- The cohort restriction is stated before the table.
- Missingness is reported before complete-case summaries.
- The table labels distinguish N, mean, and SD.
- The methods note states that the analysis is descriptive and non-causal.

## Where this goes next

[eda04](#) takes a step back to ask what each statistic in this table actually *claims*. [eda05](#) introduces the four audit categories you will use to evaluate a deliberately flawed version of this analysis in the [eda06](#) studio.

# What Each Statistic Claims (and Does Not)

## Where this fits

You produced a Table 1 on the [previous page](#). Before you write an interpretation of it, pause and ask: what does each cell in that table actually *claim*? This page is a short tour through the five most common descriptive statistics you will see — what each one says, what it leaves out, and the most common ways beginners (and AI tools) mislabel them.

## The five common statistics

| Statistic                      | What it answers                            | What it does not answer                                        |
|--------------------------------|--------------------------------------------|----------------------------------------------------------------|
| <b>N (count)</b>               | How many records are in this group?        | Whether the group is representative of any population          |
| <b>Mean</b>                    | What is the average value?                 | Whether the distribution is symmetric, skewed, or has outliers |
| <b>SD (standard deviation)</b> | How spread out are values around the mean? | Whether the spread is symmetric or driven by a tail            |
| <b>Median</b>                  | What value falls in the middle?            | How extreme the tails are                                      |

| Statistic                           | What it answers                       | What it does not answer          |
|-------------------------------------|---------------------------------------|----------------------------------|
| <b>IQR</b><br>(interquartile range) | What is the spread of the middle 50%? | The tails outside the middle 50% |

A Table 1 typically reports **N** plus either **mean (SD)** or **median (IQR)** depending on the variable.

## When to prefer mean(SD) vs median(IQR)

Use **mean (SD)** when the variable is roughly symmetric and well-behaved. Age in the adult cohort is a reasonable example.

Use **median (IQR)** when the variable is skewed or has outliers. Healthcare costs, hospital length of stay, and household income are usual suspects. The mean of a skewed variable is pulled by the long tail; the median is not.

If you are unsure, plot a histogram. If you see a long tail on one side, prefer median (IQR).

### Tip

For this course's NHANES classroom Table 1, BMI and age are commonly reported as mean (SD). That is a reasonable convention for adult anthropometric variables. State the choice and the reason in your methods note.

## What “N” alone does not tell you

A cell that says **N = 47** is informative only if the reader knows:

- which cohort the 47 came from (cohort definition, [eda02](#));
- how many of those 47 were excluded for missing values (missingness table, also [eda02](#));

- what the comparison group sizes are (the rest of the Table 1).

If any of those three is invisible, the N is not yet defensible.

## Three common beginner mistakes

These are the mistakes that show up most often in AI-drafted Table 1 code. The [studio](#) revisits them on a deliberately flawed starter.

### 1. Label says “mean” but code computes median()

```
1 # WRONG: label and code disagree
2 summarise(`Mean BMI` = median(BMI, na.rm = TRUE))
```

Either change the label to `Median BMI` or change the function to `mean()`. Decide which one you want — do not let the AI choose silently.

### 2. SD reported without `na.rm = TRUE`

```
1 # WRONG: returns NA whenever any BMI value is missing
2 summarise(sd_bmi = sd(BMI))
```

If a single NA is present, `sd(BMI)` returns NA. Use `na.rm = TRUE` and report the missingness count separately (see [eda02](#)).

### 3. Percent without a denominator

```
1 # WRONG: a "60% with low income" cell with no N
2 summarise(percent_low_income = mean(IncomeGroup == "Low Income (<1.3)") * 100)
```

A percent is only meaningful if the reader can see the denominator. Always include the N in the same row or in an adjacent cell.

## Small-cell suppression: when not to report a number

If a stratified cell has very few records (say, fewer than 10), publishing the exact statistic can be misleading or, in some health data settings, a privacy risk. The convention is to either:

- combine the small group with a neighbor and label it clearly, or
- replace the cell with a “<10” marker.

Your classroom Table 1 with  $n \geq 30$  per group (as in the [worked example](#)) is already cautious. The principle still applies: do not interpret cells where  $N$  is too small to support the claim.

## The audit habit for descriptive statistics

After you build a Table 1, look at each column heading and each row:

1. Does the column heading describe the statistic *actually* computed (mean, median, count, percent)?
2. Is the denominator visible for every percent?
3. Are missing values either filtered out (with the count reported elsewhere) or handled with `na.rm = TRUE`?
4. Are small cells either combined, marked, or excluded?

If any answer is “no”, revise the table or its labels before you write the methods note.

## Where this goes next

[eda05](#) introduces the four AI-audit categories that organize every check you have seen so far this week, ready for the [studio](#) on the planted-error starter.

# Four Audit Categories for AI-Drafted EDA

## Where this fits

You have a clean Table 1 ([eda03](#)) and a clear-eyed view of what each statistic claims ([eda04](#)). This page introduces the four categories you will use to organize your audit of AI-drafted EDA work — both in this week’s [studio](#) and in every future descriptive analysis you submit.

## The four categories

| Category        | The question it asks                                                          |
|-----------------|-------------------------------------------------------------------------------|
| Correctness     | Does the code do what the prose says it does?                                 |
| Reproducibility | Could another person rerun this in a fresh Codespace and get the same output? |
| Interpretation  | Does the narrative overclaim what the descriptive table can support?          |
| Stewardship     | Does the work respect data provenance, privacy, and classroom-use boundaries? |

Each issue you find during an audit should be tagged with one of these four. A good A6 audit note hits at least three of them.

## Beginner-level examples for each

### Correctness

- **Before:** Column labeled “Mean BMI” but code uses `median()`.
- **After:** Either rename the column or change the function. Pick one and document it.
- **Before:** Cohort prose says “adults” but the filter uses `Age >= 0, Age <= 80`.
- **After:** Decide what “adults” means (typically `Age >= 20`) and make the filter match the prose.

### Reproducibility

- **Before:** A path like `"C:/Users/me/Downloads/nhanes.csv"` in the code chunk.
- **After:** A relative path like `"examples/nhanes-equity/data/nhanes_equity_v6.csv"`, with the candidate-path pattern used in the worked example so the chunk runs from either the repo root or the week folder.
- **Before:** Missing `library(...)` calls. The chunk works in your Codespace because you ran `library(tidyverse)` earlier, but a fresh render fails.
- **After:** Every chunk that uses a package names its `library(...)` calls explicitly.

### Interpretation

- **Before:** “The table proves that income group causes differences in BMI.”
- **After:** “Mean BMI differed across income groups in this prepared classroom dataset. This descriptive comparison cannot establish cause.”



- **Before:** “Because missing values were removed, the table is unbiased and ready for publication.”
- **After:** “Complete-case filtering removed X rows. If the missingness is related to BMI or income, the complete-case summary may be biased.”

## Stewardship

- **Before:** Submitted notebook pastes a row-level NHANES record into an external AI chat to “ask for a Table 1.”
- **After:** Aggregate to group counts before any external AI is used. Never share row-level health data with external tools.
- **Before:** The methods note does not name the data source, the access method, or the classroom-use context.
- **After:** The methods note says: prepared NHANES Health Equity CSV snapshot, derived from public-use NHANES files, for classroom learning only.

## What goes in an audit note

Each issue you flag should have three things:

1. **What** the problem is (one sentence).
2. **Why it matters** in terms of correctness, reproducibility, interpretation, or stewardship.
3. **How** you corrected it (one sentence, or a small code block).

This is the structure that A6 expects, and the studio is your rehearsal.

## A worked audit note line

**Issue 3.** The interpretation says “income group causes differences in BMI.” This is an **interpretation** problem because a descriptive summary cannot support a causal claim. Corrected: “Mean BMI differed across income groups in the classroom dataset; the comparison is descriptive and does not establish cause.”

## Stewardship recap: what goes in the provenance statement

The provenance/stewardship statement you produce for A6 (and reuse in M2) should name:

- the data source (prepared NHANES Health Equity classroom CSV);
- the source of the source (derived from public-use NHANES files at the CDC);
- the access method (cached in the repo, no internet access required for class work);
- the classroom-use boundary (descriptive learning only, not population inference);
- one privacy or stewardship caution (e.g., no row-level data uploaded to external AI tools).

Five sentences, written carefully once, reused everywhere.

## The audit habit

Every time you accept AI-drafted EDA code or prose, run through the four categories:

1. Does the code match the prose? (correctness)
2. Will it rerun in a fresh Codespace? (reproducibility)
3. Does any sentence overclaim? (interpretation)
4. Is the data and AI use respectful of the source and privacy? (stewardship)

## Where this goes next

[eda06](#) is the in-class studio where you apply these four categories to a deliberately flawed Table 1 starter.

# In-Class Studio: Planted-Error Audit

## Where this fits

You have produced a clean Table 1 ([eda03](#)), thought about what each statistic claims ([eda04](#)), and learned the four audit categories ([eda05](#)). This page is the in-class studio where you apply the audit to a starter file seeded with realistic AI-style mistakes. The studio drafts feed straight into Assignment 6.

## Before you start

Take 2 minutes to check that your Codespace is ready:

1. `library(readr); library(dplyr); library(knitr)` runs without error.
2. The cached data file resolves: `examples/nhanes-equity/data/nhanes_equity_v6.csv`.
3. Your `eda03-worked-example.qmd` renders cleanly so you have a clean reference.

If any step fails, ask your debugging partner before running the planted-error starter.

## Goal

Audit the planted-error starter, categorize at least five issues, and produce a corrected EDA note that another student could rerun.

## Inputs

- Worked example to compare against: `weeks/week07-eda-ai-audit/eda03-worked-example.qmd`
- Dataset: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Planted-error starter (the artifact you will audit):  
`weeks/week07-eda-ai-audit/instructor/planted-error-starter.qmd`

### Tester note

Capture: a screenshot of the rendered planted-error starter in the Codespaces Quarto preview, with the misleading interpretation paragraph clearly visible. This is the “before” state students will recognize during the studio.

## Activity Steps

1. Review the worked example and identify the cohort definition, missingness check, and Table 1 labels.
2. Open `instructor/planted-error-starter.qmd` and inspect both code and prose before running it.
3. Run or read the starter and annotate at least five issues.
4. Categorize each issue as **correctness**, **reproducibility**, **interpretation**, or **stewardship** (see [eda05](#) if you need a refresher).
5. Produce a corrected EDA note and Table 1.

## Required Outputs (studio draft)

These are the draft files you produce in class. The Assignment 6 deliverable list ([eda07](#)) is the same set, polished.

- `eda-note.qmd` — a rerunnable Quarto note with cohort definition, missingness table, and Table 1.
- `eda-note.html` — rendered output from the Quarto note.
- `table1.csv` — exported Table 1 or equivalent descriptive summary.

- `audit-note.md` — at least five detected issues with category, why it matters, and corrected approach.
- `provenance-stewardship-note.md` — a short note naming the data source, classroom-use context, and one privacy or stewardship caution.
- `ai-use-note.md` — if you used AI, explain what it helped with and how you verified the result.

## Minimum Table 1 Requirements

- Cohort: age 20-80.
- Stratifier: `IncomeGroup`.
- Variables: N, BMI mean (SD), and age mean (SD).
- Methods note: state that the table is descriptive, unweighted, and non-causal.

## Completion Check

You are done with the studio when another student can rerun your `eda-note.qmd`, see exactly who is included, understand how missingness was handled, and identify why the planted-error starter was misleading.

## Where this goes next

Polish the studio drafts into the Assignment 6 submission and the Milestone 2 deliverables. [eda07](#) walks through the A6 files one by one and shows which files double as M2 evidence.

# Assignment 6 Walkthrough and Reference

## Where this fits

You have produced the four artifacts in class. This page is the bridge from studio drafts to the Assignment 6 submission and the Milestone 2 preliminary analysis. The deliverables here mirror the [A6 README](#) exactly — this page explains what goes into each file and how to verify it.

## Deadline and link to M2

Assignment 6 is due the Monday after Week 7 class, 4 PM. Milestone 2 (Preliminary Analysis) is due the same Monday. The good news: most of the A6 files double as M2 deliverables, so you are doing the work once.

| A6 deliverable                 | Doubles as M2 deliverable?                     |
|--------------------------------|------------------------------------------------|
| eda-note.qmd                   | Yes — corresponds to preliminary-analysis.qmd  |
| eda-note.html                  | Yes — corresponds to preliminary-analysis.html |
| table1.csv                     | Yes — same file                                |
| audit-note.md                  | A6 only                                        |
| provenance-stewardship-note.md | Yes — feeds the M2 methods-note.md             |
| ai-use-note.md                 | Yes — corresponds to the M2 ai-audit-note.md   |

Your team can submit the same `.qmd` for both, with the appropriate file names in the respective `submission/` folders.

## The six A6 deliverables, explained

Each file goes in `assignments/assignment06-eda-ai-audit/submission/`.

### 1. `eda-note.qmd`

A rerunnable Quarto file that loads the CSV with a relative path, defines the analysis cohort, reports missingness, produces the Table 1, and includes a short methods note. Pattern after the [worked example](#).

- **Verify:** open it in a fresh Codespace, render top to bottom, no errors.

### 2. `eda-note.html`

The rendered output of the `.qmd` above.

- **Verify:** open the HTML in the Codespaces preview and confirm the cohort flow, missingness table, and Table 1 all appear.

### 3. `table1.csv`

The Table 1 exported as a CSV. Use `write_csv(table1, "table1.csv")` from inside the `.qmd` so it is generated by code, not by copy-paste.

- **Verify:** the CSV opens, has the same row counts as the rendered table, and contains N, BMI mean (SD), and age mean (SD) by `IncomeGroup`.



#### 4. audit-note.md

A short note listing at least five issues you identified in the planted-error starter ([studio page](#)). Use the structure from [eda05](#):

Issue 1. [What it is]. This is a [category] problem because [why it matters]. Corrected: [how]

- **Verify:** five issues, each tagged with one of the four categories (correctness, reproducibility, interpretation, stewardship), and each with a corrected approach.

#### 5. provenance-stewardship-note.md

A short note suitable for reuse in M2. Use the five-sentence template from [eda05](#): data source, source of source, access method, classroom-use boundary, one privacy caution.

- **Verify:** five named items, each in a complete sentence.

#### 6. ai-use-note.md

If you used AI to draft code or prose, name what it helped with and how you verified the result. If you did not use AI, write one sentence saying so.

- **Verify:** specific to chunks or prompts, not generic.

### Reproducibility checklist for A6

Before you click submit:

- The `.qmd` renders in a fresh Codespace with no manual edits.
- The data path is relative.
- `table1.csv` is written by code (`write_csv(...)`).
- Missingness is reported *before* the Table 1, not after.
- The methods note avoids causal language and names the data as descriptive and unweighted.
- All six files are committed and pushed.

## A methods-note template you can reuse

We analyzed the prepared NHANES Health Equity classroom CSV snapshot. The cohort included records with age 20-80. We summarized N, BMI mean (SD), and age mean (SD) by `IncomeGroup` using unweighted descriptive statistics. Records with missing BMI, age, or income were excluded after the missingness counts were reported. The summary describes the prepared classroom dataset only and should not be interpreted as causal effects or population estimates.

That is five sentences. Use it as a starting point, edit to match your team's choices.

## A provenance/stewardship sentence you can reuse

The dataset is a prepared classroom snapshot derived from public-use NHANES files at the CDC, cached in the course repository for offline classroom use. It supports descriptive learning activities only — not individual-level inference, small-cell claims, or upload of row-level records to external AI tools.

## Five quick knowledge checks

1. What is a cohort, in one sentence?
2. Why report missingness *before* dropping rows?
3. When is median (IQR) more honest than mean (SD)?
4. Name the four AI-audit categories.
5. Which A6 file feeds the M2 `methods-note.md`?

## EDA quick reference

| Task          | Code                                                                          | Note                       |
|---------------|-------------------------------------------------------------------------------|----------------------------|
| Load CSV      | <code>read_csv("examples/Relative-equity/data/nhanes_equality_v6.csv")</code> |                            |
| Cohort filter | <code>filter(Age &gt;= 20, Age &lt;= 80)</code>                               | Document the reason        |
| Cohort counts | <code>tibble(step, n)</code>                                                  | Flow diagram in text       |
| Missingness   | <code>vapply(df[vars], function(x) sum(is.na(x)), numeric(1))</code>          | Use the cohort denominator |
| Mean (SD)     | <code>sprintf("%.1f (%.1f)", mean(x, na.rm=TRUE), sd(x, na.rm=TRUE))</code>   | One readable string        |
| Group summary | <code>group_by(IncomeGroup) %&gt;% summarise(...)</code>                      | One row per group          |
| Export        | <code>write_csv(table1, "table1.csv")</code>                                  | Generated by code          |

## Where this goes next

Next week ([Week 8](#)) turns the audited Table 1 from this assignment into a small dashboard-style KT product. The cleaner your A6/M2 work is, the easier the dashboard will be.

## **Week 8: Dashboard Prototypes for KT**

# Week 8: Dashboard Prototypes for KT

## Overview

This is the first app-forward week. You will move from audited EDA outputs (Week 7) to a small dashboard-style Knowledge Translation product that helps a defined audience explore one descriptive question.

The **required core** pathway is the [Quarto starter dashboard](#): it renders reliably in every Codespace, makes the analysis visible, and is the file you will adapt for Assignment 7. The Shiny exemplar discussed on [dash05](#) is a **demonstration only** — your instructor will run it in class; you are not required to install Shiny, run the app, or produce A7 deliverables from it.

## Objectives

By the end of the week you can:

- explain the difference between a static report, a Quarto dashboard-style product, and a Shiny app;
- render the Quarto starter dashboard locally and adapt one audience-facing element;
- apply the 1-1-1-1-1 rule to translate one Week 7 finding into a dashboard prototype;
- distinguish text/UI changes (allowed for A7) from analytic changes (not allowed for A7);
- test a dashboard locally and record what was checked;
- apply privacy-safe publishing guidance before sharing any health-data product.

## Connection

Week 7 established an auditable descriptive foundation. Week 8 asks how a non-technical audience will encounter that evidence in a dashboard. Week 9 turns one dashboard finding into a concise scientific communication product.

## Dashboard Case Study: NHANES Equity

The **NHANES Health Equity Dashboard** is now the foreground case study. Use it as a small knowledge-translation product built on the cached NHANES equity dataset.

- App code: `examples/nhanes-equity/app/app.R`
- Cached data: `examples/nhanes-equity/data/nhanes_equity_v6.rds` and `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Optional rebuild script: `examples/nhanes-equity/scripts/build_nhanes_equity.R`

Run the dashboard from the repository root:

```
1 shiny::runApp("examples/nhanes-equity/app")
```

The app is offline-first: it loads the local RDS cache when available. Rebuilding from CDC uses `nhanesA`, requires internet access, and is optional unless assigned.

Companion materials:

- Week 10 report template: `weeks/week10-reporting-publishing/report-template.qmd`
- Week 11 static dashboard demo: `weeks/week11-portfolio-surgery/static-dashboard-ojs.qmd`

## Reading order

1. [dash01 — What a Dashboard Is, and Three Pathways](#)
2. [dash02 — From One Finding to a Prototype: The 1-1-1-1-1 Rule](#)
3. [dash03 — Starter Dashboard \(required core artifact\)](#)
4. [dash04 — Audience-Facing Text: What You Can and Cannot Change for A7](#)

5. [dash05](#) — Testing, Privacy-Safe Publishing, and the Shiny Demo
6. [dash06](#) — In-Class Studio: Dashboard Adaptation
7. [dash07](#) — Assignment 7 Walkthrough and Reference

## Class plan

1. Render the [Quarto starter dashboard](#) from the repository root.
2. Choose a stakeholder audience and write one question that audience could answer from the dashboard.
3. Identify one control, title, caption, help text, or interpretation note to revise.
4. Improve that audience-facing text without changing analytic logic.
5. Re-render the dashboard with a different `selected_income` value and record the test result.
6. Watch the optional Shiny exemplar demonstration ([dash05](#)) — for awareness only, not for A7.

## Student output

By the end of class each student has a local render note, an audience sentence, a before/after text change, a testing log entry, and a limitation statement — the studio drafts of every Assignment 7 file.

## Links

- Assignment: [Assignment 7 — Dashboard KT Prototype](#)

## Definition of done

- The Quarto starter dashboard renders locally from the cached NHANES data.

- The audience-facing change improves clarity without changing the analysis.
- The testing log records how the dashboard was checked in Codespaces.
- The privacy-safe publishing note avoids row-level data sharing, small-cell claims, and causal overreach.
- No A7 submission uses the Shiny app as its primary deliverable.

## **What students leave with**

A tested dashboard adaptation note and a decision rule for the term project: use the simplest pathway that serves the audience, with the Quarto pathway as the supported default and any other pathway requiring instructor approval.



# What a Dashboard Is, and Three Pathways

## Goal

Last week you produced an audited Table 1. This week turns one descriptive finding from that table into a small **audience-facing artifact** — a dashboard-style KT product. Before you build, get clear on what a dashboard is, what it is not, and which pathway you will use in this course.

## A dashboard is not a report

A short way to remember the difference:

- A **report** tells a fixed story to a known audience. It loads, renders, and stays the same.
- A **dashboard** lets a defined audience answer one question with one or two controls. The audience moves through the artifact and pulls out the slice that matters to them.
- A **data warehouse** is neither. It stores everything for someone else to query.

You are building the middle one. The audience is real (an analyst, a project team, a briefing group). The control is small (one dropdown, one filter, or one editable value). The interpretation and the limitation stay attached to whatever the audience sees.

A dashboard is **not** a free pass to be sloppy. It can mislead just as easily as a single figure can. Everything you learned in Weeks 6 and 7 about scale, captioning, missingness, and stewardship still applies.

## Three pathways, three roles in this course

| Pathway                            | What it is                                                                                                       | When it makes sense                                                   | Role in this course                                                |
|------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Quarto dashboard-style page</b> | A rerunnable .qmd with one or two editable values, a visualization, a table, an interpretation, and a limitation | Default; works in every Codespace; no server                          | <b>Required core</b> for A7                                        |
| <b>Shiny app exemplar</b>          | A live R server with reactive UI controls                                                                        | When real-time interactivity matters and you can host or run a server | <b>Demonstration only</b> — your instructor will run this in class |
| <b>Static HTML/JS dashboard</b>    | AI-generated self-contained files                                                                                | When a public-facing portfolio piece must run without a server        | <b>Optional demo</b> in Week 11                                    |

The **required** A7 deliverable is built on the Quarto pathway. Shiny and static HTML/JS are shown so you know they exist; they are not required of any student.

## Decision rule for your term project

When you choose a dashboard pathway for your team's final portfolio, use the simplest pathway that serves your audience:

- If a `.qmd` page with one editable value answers the audience question, that is the right choice.
- If real-time interactivity is essential and your team can run/host a server, the Shiny pathway is open — **with instructor approval**, since it is not a default course pathway.
- If a static, public-facing artifact is required, the Week 11 static HTML/JS demo path is the option to explore.

There is no rubric bonus for using a more complex pathway. The grading rubric measures audience fit, audit quality, and reproducibility — not technical flash.

## What “dashboard-style” means for Quarto

You will see this phrase throughout Week 8. It means a Quarto page that:

- loads the cached classroom dataset with a relative path,
- declares one **control value** as an editable R variable near the top of the file,
- shows one **table** and one **visualization** that update when the control changes,
- ends with one **interpretation** sentence and one **limitation** sentence.

That is what the [starter dashboard](#) does, and that is what your A7 adaptation will continue.

## What “demonstration only” means for Shiny

You will see the Shiny exemplar run live in class. You will be invited to open the URL the instructor shares and click around. You will **not** be required to:

- install or load Shiny in your own Codespace,
- run `shiny::runApp(...)` yourself,
- modify the Shiny app code,
- produce any A7 deliverable from the Shiny app.

If you find Shiny interesting and want to explore it for your term project, talk with your instructor before committing — Shiny is not a default approved A7 pathway and not a default approved final-portfolio pathway.

## **Where this goes next**

[dash02](#) shows the “1-1-1-1-1 rule” that lets you translate one Week 7 finding into a dashboard sketch.

# From One Finding to a Prototype: The 1-1-1-1-1 Rule

## Where this fits

You know what a dashboard is and which pathway is required ([dash01](#)). This page is the translation step: how do you take one Week 7 finding and turn it into a small dashboard prototype? The answer is a rule with five ones.

## The 1-1-1-1-1 rule

For your first dashboard prototype, commit to exactly:

- **1 audience** — name a real person or role (e.g., “a public-health analyst preparing a briefing”).
- **1 question** — phrased the way the audience would ask it.
- **1 control** — one thing the audience can change (one dropdown, one filter, one editable value).
- **1 visualization** — one chart that updates when the control changes.
- **1 limitation** — one sentence about what the artifact does *not* support.

Beginners try to build a dashboard with three controls, four charts, and a tab system. That dashboard is harder to make, harder to test, and easier to break. The 1-1-1-1-1 rule keeps you honest.

## A worked example using your Week 7 Table 1

Recall the Table 1 from [eda03](#): mean BMI by income group across NHANES cycles. Here is the 1-1-1-1-1 translation:

| Field         | Filled in                                                                                   |
|---------------|---------------------------------------------------------------------------------------------|
| Audience      | A public-health analyst preparing a short briefing on descriptive BMI patterns              |
| Question      | “Among adults age 20-80, how does mean BMI vary across NHANES cycles for one income group?” |
| Control       | A single <code>selected_income</code> value the analyst can change                          |
| Visualization | A line plot of mean BMI by cycle for the selected income group                              |
| Limitation    | The summary is descriptive and unweighted; it does not support causal or population claims  |

That is exactly what the [starter dashboard](#) implements.

## How to choose each field for your own project

### Audience

Be specific. “The public” is too broad. “A first-year MPH student doing a course assignment” is specific. “A regional health officer planning a fall briefing” is specific. The more specific the audience, the easier every later choice becomes.

## Question

Phrase it the way your audience would phrase it. Not “an exploration of the multivariate structure of NHANES BMI distributions” but “How does average BMI compare across income groups in this dataset?”. Plain language sharpens the rest of the design.

## Control

One control. If you find yourself wanting two, pick the one your audience would change first. The second can wait for a later version.

Reasonable Quarto controls for this course:

- An editable R variable near the top of the `.qmd` (the starter pattern — change the value, re-render).
- A Quarto parameter declared in the YAML header (`params:` block), updated at render time.
- A `selectInput`-style control inside a Shiny app — but only if you have approval for the Shiny pathway.

## Visualization

One figure. Apply everything from Week 6: honest scale, descriptive title, weighted/unweighted disclosure, missing-value note, caption template. The dashboard does not relax those standards — if anything, it raises them, because an audience-facing artifact reaches more readers.

## Limitation

One sentence. Use the templates from [eda05](#) on what descriptive comparison cannot establish. Place the sentence where the audience will actually see it — at the bottom of the figure caption, or under the table, or beside the control.

## What to write down before you build

Before you touch any code, write three lines in a note:

1. My audience is: \_\_\_\_\_
2. My audience would ask: \_\_\_\_\_
3. The control I will offer is: \_\_\_\_\_

If you cannot finish those three lines, the dashboard is not ready to build yet. Go back to Week 7 and pick a finding you can actually frame.

## A note on scope creep

Once you build your first prototype, you will be tempted to add a second control, a second figure, an “advanced settings” panel, or interactive tooltips. Resist. Every addition multiplies the surface area for misleading the audience. The A7 rubric does not reward extra controls; it rewards a clearly framed audience-facing artifact.

## Where this goes next

[dash03](#) is the working Quarto starter dashboard that implements the 1-1-1-1-1 worked example. You will adapt this file for A7.



# Starter Dashboard

## Where this fits

You named the three pathways on [dash01](#) and applied the 1-1-1-1 rule on [dash02](#). This page is the **required core artifact** for Week 8: the Quarto-rendered dashboard-style page you will adapt for Assignment 7. The Shiny exemplar discussed on [dash05](#) is a demonstration only — your A7 deliverable is built from the file on this page, not from the Shiny app.

## Goal

This starter shows how to turn the Week 7 descriptive summary into a dashboard-style KT product. It uses one editable control value, one visualization, one table, one interpretation, and one caution — the 1-1-1-1 rule made concrete.

## Audience Question

Audience: a public-health analyst preparing a short briefing on descriptive BMI patterns in the classroom NHANES equity dataset.

Question: among adults age 20-80, how does mean BMI vary across NHANES cycles for one selected income group?

## Setup

```

1 library(readr)
2 library(dplyr)
3 library(ggplot2)
4 library(knitr)
5
6 candidate_paths <- c(
7   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
8   "../examples/nhanes-equity/data/nhanes_equity_v6.csv"
9 )
10
11 data_path <- candidate_paths[file.exists(candidate_paths)][1]
12 if (is.na(data_path)) {
13   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
14 }
15
16 nhanes <- read_csv(data_path, show_col_types = FALSE)

```

## Control

In a Shiny app, this would become a dropdown. In this static starter, change the value of `selected_income` and render again.

```

1 available_income_groups <- sort(unique(na.omit(nhanes$IncomeGroup)))
2 selected_income <- "Low Income (<1.3)"
3
4 if (!selected_income %in% available_income_groups) {
5   selected_income <- available_income_groups[1]
6 }
7
8 selected_income
9 #> [1] "Low Income (<1.3)"

```

## Summary

```
1 dashboard_summary <- nhanes |>
2   filter(
3     Age >= 20,
4     Age <= 80,
5     IncomeGroup == selected_income,
6     !is.na(BMI),
7     !is.na(Cycle)
8   ) |>
9   group_by(Cycle) |>
10  summarise(
11    n = n(),
12    mean_bmi = round(mean(BMI), 1),
13    .groups = "drop"
14  )
15
16 kable(dashboard_summary)
```

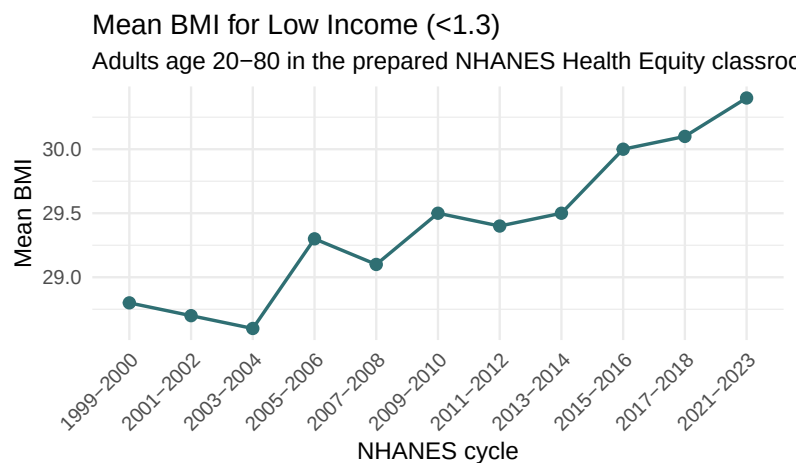
| Cycle     | n    | mean_bmi |
|-----------|------|----------|
| 1999-2000 | 1091 | 28.8     |
| 2001-2002 | 1089 | 28.7     |
| 2003-2004 | 1183 | 28.6     |
| 2005-2006 | 1091 | 29.3     |
| 2007-2008 | 1552 | 29.1     |
| 2009-2010 | 1817 | 29.5     |
| 2011-2012 | 1724 | 29.4     |
| 2013-2014 | 1762 | 29.5     |
| 2015-2016 | 1568 | 30.0     |
| 2017-2018 | 1275 | 30.1     |
| 2021-2023 | 1141 | 30.4     |

## Visualization

```

1 ggplot(dashboard_summary, aes(x = Cycle, y = mean_bmi, group = 1)) +
2   geom_line(linewidth = 0.8, color = "#2f6f73") +
3   geom_point(size = 2.4, color = "#2f6f73") +
4   labs(
5     title = paste("Mean BMI for", selected_income),
6     subtitle = "Adults age 20-80 in the prepared NHANES Health Equity classroom dataset",
7     x = "NHANES cycle",
8     y = "Mean BMI",
9     caption = "Descriptive, unweighted classroom summary. Missing BMI, income group, cycle, and out-of-cohort ages are excluded."
10  ) +
11  theme_minimal(base_size = 12) +
12  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



om summary. Missing BMI, income group, cycle, and out-of-cohort ages are excluded.

## Interpretation

For the selected income group, the table and figure describe the mean BMI pattern across available NHANES cycles in the prepared classroom dataset. The dashboard should help the audience ask better questions about pattern, timing, and context rather than treat the line as a causal explanation.

## Limitation

This starter is unweighted and descriptive. It does not account for NHANES survey design, does not estimate national prevalence, and should not be used to make causal claims about income and BMI.

## Testing

Before submitting a dashboard-style product:

1. Render this page from the repository root.
2. Change `selected_income` to another value from `available_income_groups`.
3. Render again and check that the table, title, and plot all update.
4. Verify that the caption still names the dataset, exclusions, and limitation.
5. Record any warning or error in your testing log.

## Privacy-Safe Publishing

- Use aggregate summaries, not row-level records.
- Avoid small-cell interpretations when `n` is low.
- State that outputs are descriptive and unweighted.
- Keep cached classroom data separate from any raw NHANES retrieval workflow.
- Do not upload row-level data to external AI tools.

## Where this goes next

[dash04](#) shows what counts as an audience-facing text change (allowed) versus an analytic change (not allowed for A7). [dash05](#) walks through how to test this page in Codespaces and shows the optional Shiny exemplar that your instructor will demo in class.

# Audience-Facing Text: What You Can and Cannot Change for A7

## Where this fits

[dash03](#) is the file you will adapt for Assignment 7. This page draws the line between **audience-facing text changes** (allowed for A7) and **analytic changes** (not allowed for A7). Knowing where that line is keeps the assignment focused and keeps your audit trail clean.

## The rule, in one sentence

For Assignment 7, you may change anything the audience reads. You may not change anything the analysis computes.

## Allowed changes (text-only)

These are the kinds of changes the A7 rubric rewards. They sharpen the audience experience without touching the analysis.

- **Rename a label.** “Cycle” becomes “Survey wave (NHANES cycle)” so a non-expert audience knows what to read.
- **Expand a caption.** A two-word caption becomes a one-sentence caption that names the statistic, source, exclusions, and limitation.

- **Add a help text line.** Beside the control, a sentence that says “This dashboard summarizes adults age 20-80 from the prepared classroom dataset.”
- **Add a limitation sentence.** Beside the figure, one sentence on what the artifact does not support.
- **Improve a README sentence.** A README that says “Dashboard for NHANES” becomes “Descriptive dashboard of unweighted mean BMI by income group across NHANES cycles, for classroom use.”
- **Adjust a title for the audience.** A neutral title becomes one that names the audience-facing question.

## Not allowed for A7 (analytic-only changes)

These changes touch the analysis. They are not part of A7. If you find a real analytic issue while testing, raise it with your instructor — do not silently change it.

- **Changing the cohort filter.** Switching from `Age >= 20` to `Age >= 18` is an analytic decision.
- **Adding or removing a survey weight.** Switching from unweighted to weighted is an analytic decision.
- **Adding a regression line.** Smoothing across cycles is an analytic decision (and usually inappropriate — see [viz04](#)).
- **Replacing `mean()` with `median()`.** Changing the central tendency statistic is an analytic decision.
- **Changing the grouping variable.** Switching from `IncomeGroup` to `Education` is an analytic decision.

If you are unsure whether a change is text or analytic, ask: “would a reader looking at the rendered HTML notice this change if I didn’t tell them?” If yes, it is probably text. If the *numbers* change, it is analytic.

## Before / after examples

### Before / after: a label

**Before:** `selectInput("strat_var1", "Primary Grouping (Rows):", ...)`

**After:** `selectInput("strat_var1", "Compare across (rows of the chart):", ...)`

The label is friendlier to a non-technical audience. The variable, choices, and computation are unchanged.

### Before / after: a caption

**Before:** `caption = "BMI by income"`

**After:** `caption = "Mean BMI by income group across five NHANES cycles. Adults age 20-80 in the prepared classroom dataset. Unweighted descriptive summary; records with missing BMI, income, or cycle are excluded."`

The caption now names the statistic, source, cohort, exclusions, and disclosure. The figure itself is unchanged.

### Before / after: a help text line

**Before:** (no help text near the control)

**After:** Right under the control: `> This dashboard uses adults age 20-80. To see a different income group, change the value and re-render.`

The audience now understands what the artifact covers and how to use it.



## Before / after: a README sentence

**Before:** # Dashboard

**After:** ‘# NHANES Health Equity Classroom Dashboard

Descriptive, unweighted mean BMI by income group across NHANES cycles. For classroom learning; not for population inference.’

The README now names the audience and the boundary.

## What to record in your A7

### before-after-text.md

A7 asks you to submit one before-after text change. For that file, paste:

- the **original text** verbatim,
- the **revised text** verbatim,
- one sentence on **why** the change helps the audience,
- one sentence confirming **no analytic logic was changed**.

That last sentence is what keeps the audit trail clean.

## Where this goes next

[dash05](#) covers how to test the starter dashboard locally, how to keep your publishing privacy-safe, and (briefly) the Shiny exemplar your instructor will demo in class.

# Testing, Privacy-Safe Publishing, and the Shiny Demo

## Where this fits

You have the starter dashboard ([dash03](#)) and you know what kind of changes are allowed ([dash04](#)). This page covers three things you do *before* you submit:

1. Test the starter dashboard in your Codespace.
2. Apply the privacy-safe publishing checklist.
3. Watch (but do not run) the optional Shiny exemplar your instructor will demo in class.

## Part 1 — Testing the starter dashboard

The A7 deliverable list asks for a `testing-log.md`. Use the routine below for the entry.

### A six-step test routine

1. **Render from the repository root.** Open `dash03-starter-dashboard.qmd` in your Codespace and click Render. Confirm there are no errors.
2. **Change the control.** Edit `selected_income` to a different value from `available_income_groups`. Save the file.
3. **Render again.** Confirm the title, table, and plot all updated.

4. **Verify the caption.** The caption should still name the dataset, exclusions, and limitation.
5. **Open the HTML preview.** Use the Codespaces preview pane and confirm the rendered page looks correct in the browser, not just in the editor.
6. **Record the test.** In `testing-log.md`, note the date, the command or render path used, the values you tried, and the result.

#### Tester note

Capture: a screenshot of the Codespaces Quarto preview showing the rendered starter dashboard for *two* different `selected_income` values, side by side or in sequence. This is the most common visual confirmation that the control “works.”

## Common failure modes and quick fixes

| Failure                          | What you see                                        | Quick fix                                                                                                |
|----------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Wrong working directory          | “Could not find examples/nhanes-equity/data/nhanes_ | Open the Quarto file via the <code>File Explorer</code> confirm Codespaces is rooted at the repo root    |
| Missing package                  | “there is no package called ‘ggplot2’ ”             | Run <code>library(ggplot2)</code> once; if it still fails, raise with your TA — do not install at random |
| Stale <code>_freeze</code> cache | Output reflects the old control value               | Delete the relevant <code>_freeze/</code> subfolder and re-render; or change the chunk slightly          |

| Failure            | What you see                         | Quick fix                                                                                          |
|--------------------|--------------------------------------|----------------------------------------------------------------------------------------------------|
| Preview pane blank | Render succeeds but preview is empty | Click the preview pane refresh icon, or open the HTML file from <code>docs/</code> (if configured) |

### Tester note

Capture: one screenshot of each failure state above, with the error visible. New students stall on these and screenshots in the Survival Guide help unblock faster.

## Part 2 — Privacy-safe publishing

Five rules apply every time you share a dashboard rendering, a screenshot, or a Quarto preview.

1. **No row-level data.** Your figures and tables show aggregate summaries. Never paste a row-level NHANES record into a screenshot, a chat with an external AI, or a public README.
2. **No small-cell interpretations.** If a group has very few records, do not interpret the cell. Combine, mark, or exclude the group (see [eda04](#)).
3. **State descriptive limits.** Every interpretation should say the artifact is descriptive and unweighted. No causal language. No “national prevalence” language.
4. **Keep cached data separate from raw retrieval.** The course uses a cached CSV intentionally. If you write code that retrieves raw NHANES from the CDC, document that as separate work — do not put a live API call in your dashboard.
5. **Do not upload row-level data to external AI tools.** Aggregate first. The AI conversation can see counts and means; it should not see individual records.

## What to write in `privacy-check.md`

A6 and A7 both expect a short privacy check. For A7, write three sentences:

- One naming the data source and the classroom-use boundary (reuse from [eda05](#)).
- One confirming the dashboard renders aggregate summaries only.
- One confirming you did not share row-level records with any external AI tool.

## Part 3 — The Shiny exemplar (demonstration only)

Your instructor will demo a Shiny app version of the NHANES dashboard in class. **You are not required to run this app, install Shiny, or produce anything from it.** Your A7 deliverable is built from the Quarto starter dashboard, not from this app.

What the demo will show you:

- How a Shiny app turns the same kind of analysis into a live, server-backed dashboard with reactive controls.
- The trade-offs: more interactivity, but it requires a server (or a Codespaces port forward), more packages, and more failure modes.
- Where the app code lives (`examples/nhanes-equity/app/app.R`) if you want to read it.

If, after watching the demo, you decide you want to use Shiny for your term-project final dashboard, that pathway is **not approved by default**. Talk with your instructor before committing. The Quarto pathway is the supported default for every student.

 Warning

Do **not** put `shiny::runApp(...)` in your A7 submission as your “dashboard.” A7 expects an adaptation of the Quarto starter. The Shiny app is shown to you; it is not what you build on.

## Where this goes next

[dash06](#) is the in-class studio where you produce draft versions of every A7 deliverable from the Quarto starter.

# In-Class Studio: Dashboard Adaptation

## Where this fits

You have everything you need: the worked Quarto starter ([dash03](#)), the rule for allowed changes ([dash04](#)), and the testing and privacy routines ([dash05](#)). This page is the in-class studio — about 60 minutes of structured work that produces draft versions of every Assignment 7 deliverable.

## Before you start

Take 2 minutes to confirm your Codespace is ready:

1. The cached CSV resolves: `examples/nhanes-equity/data/nhanes_equity_v6.csv`.
2. `dash03-starter-dashboard.qmd` renders cleanly with the default `selected_income`.
3. You have decided which `IncomeGroup` to set `selected_income` to for your adaptation (it does not have to be the default).

## Goal

Produce draft versions of every A7 file from the Quarto starter. Tomorrow's polish will make them assignment-ready; today's work makes sure the bones are there.

## The 60-minute studio sequence

You will move through six short steps. Each step produces one or two artifacts.

### Step 1 — Audience and question (10 min)

Write two short paragraphs in `dashboard-adaptation-note.md`:

- **Audience.** One specific role and what they need to decide.
- **Question.** Phrased the way the audience would phrase it.

Use the templates from [dash02](#). Be specific.

### Step 2 — Identify your control and visualization (5 min)

In the same note, add:

- **Control.** One control and the values it can take.
- **Visualization.** One chart and what it shows when the control changes.

For most students, this is the existing `selected_income` control and the line chart of mean BMI by cycle from the starter. That is fine.

### Step 3 — Pick one audience-facing change (10 min)

Read your starter file with the eyes of your stated audience. Find one element that could be clearer — a label, a caption, a help text line, a README sentence, or a limitation. Pick the one that would help the audience most.

Write `before-after-text.md`:

- The original text.
- The revised text.



- One sentence on why the change helps.
- One sentence confirming no analytic logic changed (see [dash04](#)).

## Step 4 — Apply the change and test (15 min)

Edit your local copy of the starter file with the revised text. Then run the six-step test routine from [dash05](#):

1. Render.
2. Change the control to a different value.
3. Render again.
4. Verify the caption.
5. Open the HTML preview.
6. Record the test in `testing-log.md`.

### Tester note

Capture: a screenshot of the rendered “after” state showing the revised audience-facing element clearly visible. Beginners need to see what a successful adaptation looks like before they attempt their own.

## Step 5 — Write the limitation and privacy check (10 min)

In `limitation-note.md`, write **one sentence** stating what the dashboard does *not* support. Use the pattern from [eda05](#) on descriptive-vs-causal.

In `privacy-check.md`, write the **three sentences** from [dash05](#):

- One naming the data source and classroom-use boundary.
- One confirming aggregate-only outputs.
- One confirming no row-level data was shared with external AI.

## Step 6 — Reproducibility and AI use (10 min)

In `reproducibility-note.md`, write one paragraph naming:

- The command or render path you used.
- The relative path to the dataset.
- A sentence on whether the dashboard renders in a fresh Codespace.

In `ai-use-note.md`, name what AI helped with (if anything) and how you verified the result.

## At the end of the studio

You should leave class with draft versions of all the files A7 expects. Polish at home and submit by the Monday 4 PM deadline.

## Required studio outputs (drafts)

- `dashboard-adaptation-note.md`
- `before-after-text.md`
- `testing-log.md`
- `limitation-note.md`
- `privacy-check.md`
- `reproducibility-note.md`
- `ai-use-note.md`
- One screenshot of the rendered “after” state (this becomes `dashboard-preview.png` after a final render)

## Completion check

You are done with the studio when a peer can read your `dashboard-adaptation-note.md` and identify your audience, your question, your control, your visualization, your text change, and your test result — without rerunning the dashboard first.

## Where this goes next

[dash07](#) walks through how each studio file becomes the corresponding A7 deliverable and what verification each file needs before submission.

# Assignment 7 Walkthrough and Reference

## Where this fits

You have studio drafts from [dash06](#). This page is the bridge from studio work to a submitted Assignment 7. The deliverables here mirror the [A7 README](#) exactly — this page explains what goes into each file, how to verify it, and how the work connects to your term-project final dashboard.

## Deadline and the decision rule

Assignment 7 is due the Monday after Week 8 class, 4 PM. The required pathway is the **Quarto starter dashboard** ([dash03](#)).

**Decision rule pinned at the top:** adapt the Quarto starter. The Shiny app is a demonstration only and is **not** an approved A7 submission pathway. If you are interested in using Shiny for your final-portfolio dashboard, talk with the instructor — Shiny is not a default-approved pathway and requires instructor approval.

## The eight A7 deliverables, explained

Each file goes in `assignments/assignment07-dashboard/submission/`.

### 1. dashboard-adaptation-note.md

Audience, question, control, visualization, the element you changed, and the KT rationale. Pattern after the studio Step 1 and Step 2 outputs.

- **Verify:** a peer can read this and identify your audience and question without seeing your code.

### 2. before-after-text.md

The original text, the revised text, why the change helps the audience, and a confirmation that no analytic logic changed. Pattern after the studio Step 3 output.

- **Verify:** the change is genuinely audience-facing text. If you cannot say “no analytic logic changed”, the change does not belong in A7 (see [dash04](#)).

### 3. testing-log.md

Date tested, the command or render path you used, the values you tried, and the result. Pattern after the studio Step 4 output.

- **Verify:** another student could reproduce your test by following your log.

### 4. limitation-note.md

One sentence on what the dashboard does not support. Use the descriptive-vs-causal patterns from [eda05](#).

- **Verify:** the sentence belongs *in or beside* the dashboard, not just in a separate file.

## 5. `privacy-check.md`

Three sentences from `dash05`: data source and classroom-use boundary, aggregate-only confirmation, no row-level data shared with AI.

- **Verify:** three sentences, each specific.

## 6. `dashboard-preview.png`

A screenshot or exported preview of the affected dashboard area (after the text change).

- **Verify:** the preview clearly shows the revised audience-facing element. No row-level table previews.

## 7. `reproducibility-note.md`

One paragraph naming the render command, the relative path, and whether the dashboard renders in a fresh Codespace.

- **Verify:** the path is relative, and the note matches what `testing-log.md` says.

## 8. `ai-use-note.md`

What AI helped with (or a sentence saying you did not use it) and how you verified the result.

- **Verify:** specific (which prompt? which file?), not vague.

# Reproducibility checklist for A7

Before you click submit:

- The adapted Quarto file renders in a fresh Codespace.
- The data path is relative (`examples/nhanes-equity/data/nhanes_equity_v6.csv`).
- The preview screenshot shows the revised text change clearly.

- The dashboard contains a visible limitation sentence (not just in a separate file).
- No row-level data appears in any screenshot or shared output.
- All eight files are committed and pushed.

## A pathway decision template you can reuse

When the term-project conversation comes up later in the term, your team will face the same pathway choice for the final portfolio. Use this short template to document the decision:

Our audience is \_\_\_\_\_. We need \_\_\_\_\_ kind of interactivity. The simplest pathway that serves this need is [**Quarto dashboard-style page / Shiny app / static HTML/JS**]. [If Shiny or static HTML/JS:] We have instructor approval recorded on \_\_\_\_\_.

Default to Quarto. Justify deviations.

## Privacy-safe publishing checklist

Reused from [dash05](#) for your reference:

- No row-level data in screenshots, AI chats, or public README.
- No small-cell interpretations.
- Descriptive limits stated; no causal language; no national-prevalence language.
- Cached classroom data kept separate from any raw retrieval workflow.
- Row-level data never uploaded to external AI tools.

## Paths you will use repeatedly

| What                                  | Where                                                                            |
|---------------------------------------|----------------------------------------------------------------------------------|
| Cached dataset (CSV)                  | <code>examples/nhanes-equity/data/nhanes_equity_v6.csv</code>                    |
| Cached dataset (RDS)                  | <code>examples/nhanes-equity/data/nhanes_equity_v6.rds</code>                    |
| Starter dashboard                     | <code>weeks/week08-dashboard-kt/dash03-starter-dashboard.qmd</code>              |
| Shiny exemplar (instructor demo only) | <code>examples/nhanes-equity/app/app.R</code>                                    |
| Adaptation template                   | <code>assignments/assignment07-dashboard/dashboard-adaptation-template.md</code> |
| Submission folder                     | <code>assignments/assignment07-dashboard/submission/</code>                      |

## Five quick knowledge checks

1. Name three differences between a static report and a dashboard.
2. Which pathway is the required core for A7?
3. In one sentence, why is the Quarto starter dashboard preferred over editing the Shiny app for A7?
4. Name two changes that count as “UI-text-only.” Name two that do not.
5. What three sentences go in `privacy-check.md`?

## Where this goes next

Next week ([Week 9](#)) takes one finding from your dashboard and turns it into a concise scientific communication product. The cleaner your A7 limitation sentence is, the easier the Week 9 plain-language summary will be.



## **Week 9: Communication of Scientific Findings**

# Dynamic Presentations

## Overview

Data analysis is useful only when the right audience can understand what the evidence does and does not say. This week turns one dashboard finding into a short, plain-language scientific communication product using Quarto slides.

## Connection

Week 8 made the dashboard audience-facing. This week turns one dashboard plot or table into a short slide story with a clear takeaway, evidence, and limitation. Next week you will expand that communication into a reproducible report.

## Dashboard Case Study: NHANES Equity

The **NHANES Health Equity Dashboard** is now the foreground case study. Use it as a small knowledge-translation product built on the cached NHANES equity dataset.

- App code: `examples/nhanes-equity/app/app.R`
- Cached data: `examples/nhanes-equity/data/nhanes_equity_v6.rds` and `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Optional rebuild script: `examples/nhanes-equity/scripts/build_nhanes_equity.R`

Run the dashboard from the repository root:

```
1 shiny::runApp("examples/nhanes-equity/app")
```

The app is offline-first: it loads the local RDS cache when available. Rebuilding from CDC uses `nhanesA`, requires internet access, and is optional unless assigned.

Companion materials:

- Week 10 report template: `weeks/week10-reporting-publishing/report-template.qmd`
- Week 11 static dashboard demo: `weeks/week11-portfolio-surgery/static-dashboard-ojs.qmd`

## Learning Objectives

By the end of this week, you should be able to:

- identify the audience and decision context for a result
- translate one technical finding into plain language
- build a short Quarto revealjs slide deck or slide outline
- cite the data source for a figure or table
- use peer feedback to reduce overclaiming and improve clarity

## In-Class Plan

1. Choose one plot or table from prior work.
2. Draft a three-slide story using [Starter Slides](#).
3. Rewrite one technical sentence in plain language.
4. Complete peer feedback using [Revealjs](#) and [Peer Feedback](#).
5. Revise the takeaway, caption, or limitation before submitting [Assignment 8](#).

## Plain-Language Translation

Start with a technical sentence such as:

Mean BMI differs by income group in the prepared NHANES classroom dataset.

Translate it for a nontechnical audience:

In this classroom dataset, average BMI is not the same across income groups, but this comparison is descriptive and does not prove income causes BMI differences.

Your translation should keep the evidence honest. Do not remove the limitation just to make the message sound stronger.

## Citation And Accessibility

Before submission, check that:

- the data source is named on the slide or in speaker notes
- any figure has a readable title, axis labels, and units
- colors are not the only way to understand groups
- text is large enough to read during a live presentation
- the limitation is visible, not hidden in tiny text
- AI-assisted wording has been checked against the actual output

## Student Output

Submit a 3-5 slide deck or slide outline, a plain-language summary, peer feedback notes, and an AI-use note. Assignment link: [Assignment 8](#).

## Definition Of Done

Week 9 is done when the slide story renders or is ready to render, the main message is understandable without code, and a peer can identify the evidence and limitation.

# Revealjs And Peer Feedback

## Goal

Turn one dashboard or report finding into a short scientific communication product and practice structured peer feedback.

## Three-Slide Structure

1. **Question and audience:** Who needs this finding, and why?
2. **Evidence:** Show one plot or table and state the descriptive pattern.
3. **Limitation and action:** Name what the data can and cannot support, then state the next decision or question.

Use [Starter Slides](#) if you want a renderable revealjs scaffold.

## Plain-Language Check

For each slide, ask:

- What is the one sentence the audience should remember?
- Which word or phrase would confuse someone outside the course?
- What limitation must remain visible to avoid overclaiming?

## Peer Feedback Form

Copy this table into your submission.

| Prompt                                     | Peer note |
|--------------------------------------------|-----------|
| Main message in one sentence               |           |
| Intended audience                          |           |
| Evidence shown                             |           |
| Limitation or caution shown                |           |
| Wording that needs plain-language revision |           |
| Accessibility issue, if any                |           |
| One concrete improvement before submission |           |

## Citation And Accessibility Checklist

- Data source is named.
- Figure or table has a clear title.
- Axes, units, and group labels are readable.
- Color choices work without relying on color alone.
- Slide text is brief enough to read during a talk.
- Limitation is visible.
- AI-assisted wording has been checked against the evidence.

## Student Output

Submit the rendered slide deck or slide outline, completed peer feedback form, plain-language summary, and AI-use note required by [Assignment 8](#).

## **Week 10: Writing & Publishing Reports**

# Reproducible Reporting

## Overview

This chapter simulates the final stretch of a research project: writing the manuscript. Students learn to assemble their analysis, code, and writing into a single “dynamic report” that automatically updates numbers and figures if the data changes. You will explore advanced publishing workflows to create professional, citation-ready portfolios that demonstrate your ability to execute end-to-end data science.

## Connection

Week 9 turned one result into a short slide story. This week expands the same evidence into a reproducible report with code-generated outputs, citations, and limitations. Next week you will stress-test whether the report and dashboard work in a fresh environment.

## Dashboard Case Study: NHANES Equity

The **NHANES Health Equity Dashboard** is now the foreground case study. Use it as a small knowledge-translation product built on the cached NHANES equity dataset.

- App code: `examples/nhanes-equity/app/app.R`
- Cached data: `examples/nhanes-equity/data/nhanes_equity_v6.rds` and `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Optional rebuild script: `examples/nhanes-equity/scripts/build_nhanes_equity.R`



Run the dashboard from the repository root:

```
1 shiny::runApp("examples/nhanes-equity/app")
```

The app is offline-first: it loads the local RDS cache when available. Rebuilding from CDC uses `nhanesA`, requires internet access, and is optional unless assigned.

Companion materials:

- Week 10 report template: `weeks/week10-reporting-publishing/report-template.qmd`
- Week 11 static dashboard demo: `weeks/week11-portfolio-surgery/static-dashboard-ojs.qmd`

## In-Class Activity

- Draft a short Quarto report that complements the dashboard rather than duplicating it.
- Include a data source paragraph, one figure or table, and a limitations note.
- Add a source note for CDC NHANES and the `nhanesA` retrieval pathway.
- Verify that the report uses relative paths and documents whether results came from the cached data.
- Use [Report Template](#) as the working model.
- Connect the draft to [M3: Project Update](#).

Student output: a rendered Quarto report with one figure, one table, citations, limitations, and an AI-use note. Assignment link: [Assignment 9](#).

## Publishing Check

For public GitHub Pages output, confirm that the report does not expose restricted data, private URLs, tokens, or sensitive examples. For private preview, submit the repository link on Canvas and grant course staff access.

## Grounding Audit

Before submission, choose three claims in the report and trace each one to a table, figure, code chunk, or citation. Revise any claim that cannot be grounded.

## Definition of Done

- The report renders from source.
- Citations resolve.
- Outputs are regenerated by code rather than pasted manually.
- The README explains how to reproduce the report.

# Report Template: NHANES Equity Summary

## Standalone YAML

If you copy this template into a project report outside the book, start the file with this YAML header:

```
1 ---
2 title: "NHANES Equity Summary"
3 format: html
4 bibliography: ../../references.bib
5 ---
```

## Purpose

This runnable template shows how to pair a dashboard with a short reproducible report. The dashboard supports exploration; the report fixes one question, documents the data, and cites the source.

## Question

How does mean BMI vary by income group among adults age 20-80 in the prepared NHANES Health Equity classroom dataset?

## Data Source

The report uses the cached CSV snapshot at `examples/nhanes-equity/data/nhanes_equity_v6.csv`. The source data are derived from CDC/NCHS NHANES public-use files (Centers for Disease Control and Prevention, National Center for Health Statistics 2026).

```
1 library(readr)
2 library(dplyr)
3 library(ggplot2)
4 library(knitr)
5
6 candidate_paths <- c(
7   "examples/nhanes-equity/data/nhanes_equity_v6.csv",
8   "../..examples/nhanes-equity/data/nhanes_equity_v6.csv"
9 )
10
11 data_path <- candidate_paths[file.exists(candidate_paths)][1]
12 if (is.na(data_path)) {
13   stop("Could not find examples/nhanes-equity/data/nhanes_equity_v6.csv")
14 }
15
16 nhanes <- read_csv(data_path, show_col_types = FALSE)
```

## Methods

We restricted the classroom dataset to adults age 20-80 and summarized BMI by income group. The summaries are un-weighted and descriptive. They should not be interpreted as causal effects or national population estimates.

```
1 analysis_df <- nhanes |>
2   filter(Age >= 20, Age <= 80)
3
4 missingness <- analysis_df |>
5   summarise(
6     rows = n(),
7     missing_bmi = sum(is.na(BMI)),
8     missing_income = sum(is.na(IncomeGroup)),
```

```

9     missing_age = sum(is.na(Age))
10   )
11
12 kable(missingness, caption = "Missingness check for key report variables.")

```

Table 17: Missingness check for key report variables.

| rows  | missing_bmi | missing_income | missing_age |
|-------|-------------|----------------|-------------|
| 57137 | 1009        | 5395           | 0           |

## Results

```

1 income_summary <- analysis_df |>
2   filter(!is.na(BMI), !is.na(IncomeGroup)) |>
3   group_by(IncomeGroup) |>
4   summarise(
5     n = n(),
6     mean_bmi = mean(BMI),
7     sd_bmi = sd(BMI),
8     .groups = "drop"
9   ) |>
10  mutate(
11    `BMI, mean (SD)` = sprintf("%.1f (%.1f)", mean_bmi, sd_bmi)
12  ) |>
13  select(IncomeGroup, n, `BMI, mean (SD)`)
14
15 kable(income_summary, caption = "Descriptive BMI summary by income group.")

```

Table 18: Descriptive BMI summary by income group.

| IncomeGroup        | n     | BMI, mean (SD) |
|--------------------|-------|----------------|
| High Income (>3.5) | 16330 | 28.6 (6.3)     |
| Low Income (<1.3)  | 15293 | 29.4 (7.5)     |
| Middle Income      | 19228 | 29.3 (7.0)     |

```

1 plot_df <- analysis_df |>
2   filter(!is.na(BMI), !is.na(IncomeGroup)) |>
3   group_by(IncomeGroup) |>
4   summarise(
5     n = n(),
6     mean_bmi = mean(BMI),
7     .groups = "drop"
8   )
9
10  ggplot(plot_df, aes(x = IncomeGroup, y = mean_bmi)) +
11    geom_col(fill = "#2C7FB8") +
12    geom_text(aes(label = paste0("n=", n)), vjust = -0.4, size = 3.5) +
13    labs(
14      title = "Mean BMI by income group",
15      subtitle = "Adults age 20-80; unweighted descriptive summary",
16      x = "Income group",
17      y = "Mean BMI"
18    ) +
19    expand_limits(y = 0) +
20    theme_minimal(base_size = 12)

```

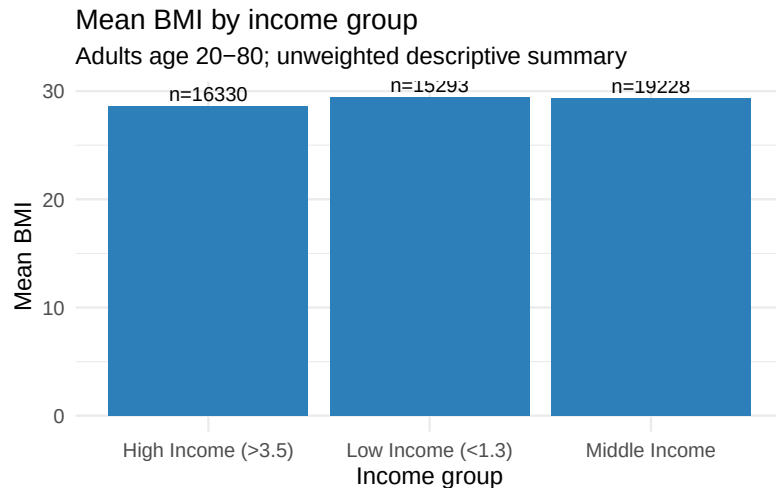


Figure 1: Mean BMI by income group among adults age 20-80 in the prepared NHANES Health Equity classroom dataset. Values are unweighted and descriptive.

## Limitations

- The table and figure use a prepared classroom dataset rather than raw NHANES files.
- The summaries are unweighted.
- The comparison is descriptive and does not establish causality.
- Missing BMI or income group values are excluded from the grouped summary.

## Plain-Language Summary

In this classroom dataset, mean BMI differs across income groups. The result is useful for practicing reproducible reporting and dashboard interpretation, but it should be described as a descriptive pattern rather than a causal finding.

## AI-Use and Audit Note

Document any AI assistance used to draft code, captions, or interpretation. State exactly which output you checked against the code, data dictionary, or rendered report.

## Publishing Checklist

- Report renders from source.
- Citations resolve.
- Figures and tables are generated by code.
- Public preview contains no restricted data or private links.
- Private preview access is granted to course staff when needed.
- Canvas submission includes the repository link, report path, and latest commit hash.

## Grounding Audit Prompt

Select three claims from the report. For each claim, write the supporting evidence source: code chunk, table, figure, or citation. Revise any claim that cannot be traced to evidence in the report.



## **Week 11: Portfolio Surgery**

# Advanced Topics

## Overview

Before the final submission, students must “bulletproof” their work. This chapter focuses on advanced project hygiene, dependency management (ensuring your code runs on someone else’s machine), and auditability. You will act as a peer reviewer to identify reproducibility failures in others’ code, fixing broken paths and missing seeds, to ensure your final project stands up to scientific scrutiny.

## Connection

Week 10 produced a report with code-generated outputs and citations. This week tests whether the report, dashboard, and instructions survive a fresh environment. The final portfolio should leave a reviewer able to rerun the core work without private setup knowledge.

## Dashboard Case Study: NHANES Equity

The **NHANES Health Equity Dashboard** is now the foreground case study. Use it as a small knowledge-translation product built on the cached NHANES equity dataset.

- App code: `examples/nhanes-equity/app/app.R`
- Cached data: `examples/nhanes-equity/data/nhanes_equity_v6.rds` and `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Optional rebuild script: `examples/nhanes-equity/scripts/build_nhanes_equity.R`

Run the dashboard from the repository root:

```
1 shiny::runApp("examples/nhanes-equity/app")
```

The app is offline-first: it loads the local RDS cache when available. Rebuilding from CDC uses `nhanesA`, requires internet access, and is optional unless assigned.

Companion materials:

- Week 10 report template: `weeks/week10-reporting-publishing/report-template.qmd`
- Week 11 static dashboard demo: `weeks/week11-portfolio-surgery/static-dashboard-ojs.qmd`

## In-Class Activity

- Run a fresh-session test: can the app launch from the repository root?
- Verify the cached RDS path works before attempting a CDC rebuild.
- Optionally run the rebuild script only if internet access and time are available.
- Optionally draft a static dashboard extension using a CSV snapshot and no Shiny server.
- Log any dependency, path, or README issues as reproducibility risks.
- Use [Reproducibility Stress Test](#) as the runbook.
- Check final submission expectations in [Final Portfolio](#).

Student output: a reproducibility test log, repaired README or path issues, and a final risk list. Assignment link: [Assignment 10](#).

## Static Dashboard Demo

The completed [static dashboard source file](#) demonstrates a no-server alternative that reads the NHANES CSV snapshot in the browser. Use it as an exemplar for optional portfolio extensions after the core Shiny and report workflows are reproducible.

## Definition of Done

- A peer can clone or open the repository in a fresh Codespace and reproduce the core outputs.
- Broken paths, missing dependencies, unclear READMEs, and ungrounded AI claims are fixed.
- The [Final Portfolio Checklist](#) is complete.

# Reproducibility Stress Test

## Goal

Test whether a project can run in a fresh environment and document every failure before final submission.

## Runbook

Complete these steps in a new Codespace or equivalent clean environment. Work from the repository root unless the project README says otherwise.

1. Open the repository from GitHub.
2. Read the README before running code.
3. Record the latest commit hash.
4. Install or restore dependencies using only documented instructions.
5. Render the final report.
6. Open or run the dashboard-style KT product.
7. Check that data paths are relative.
8. Check that citations resolve.
9. Check that figures and tables are generated from code.
10. Record every failure, warning, unclear instruction, and repair.
11. Commit repairs with clear messages.
12. Submit the updated repository link and commit hash on Canvas.

## NHANES Exemplar Checks

For the NHANES case study, use these checks:

- load `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- launch the Shiny exemplar from the repository root
- open the static dashboard source in `weeks/week11-portfolio-surgery/static-dashboard-ojs.qmd`
- render the Week 10 report template
- confirm the README explains cached data and optional CDC retrieval

## Sample Failure Log

| Check                                 | Result       | Evidence                               | Fix made                                  | Owner     |
|---------------------------------------|--------------|----------------------------------------|-------------------------------------------|-----------|
| Repository opens in fresh environment | Pass         | Codespace opened from GitHub           | None                                      | Group     |
| Packages install or load              | Needs repair | <code>library(tidyverse)</code> failed | Added dependency note to README           | Analyst   |
| Report renders                        | Pass         | <code>report.html</code> created       | None                                      | Writer    |
| Dashboard runs locally                | Needs repair | data path failed from repo root        | Replaced absolute path with relative path | Developer |
| Citations resolve                     | Pass         | reference list appears                 | None                                      | Writer    |
| README instructions are accurate      | Needs repair | missing dashboard command              | Added run command                         | Group     |

## Blank Failure Log

| Check                                 | Result | Evidence | Fix made | Owner |
|---------------------------------------|--------|----------|----------|-------|
| Repository opens in fresh environment |        |          |          |       |
| Packages install or load              |        |          |          |       |
| Report renders                        |        |          |          |       |
| Dashboard or KT product opens         |        |          |          |       |
| Data paths are relative               |        |          |          |       |
| Citations resolve                     |        |          |          |       |
| README instructions are accurate      |        |          |          |       |
| AI-use note is complete               |        |          |          |       |

## Peer Review Roles

- **Driver:** Runs the project exactly as documented.
- **Navigator:** Reads instructions aloud and logs ambiguity.
- **Auditor:** Checks paths, data provenance, citations, and AI-use notes.

Rotate roles after the first major failure or after 20 minutes.

## Optional Python Pathway

If the project uses Python, add a short dependency note with:

- how Python is launched
- required packages
- whether a notebook, script, or Quarto Python chunk is the primary workflow
- how outputs are regenerated

Keep the Python pathway documented, but do not add it to a project unless it supports the core audience-facing product.

## Final Portfolio Readiness

Before final submission, complete the [Final Portfolio Checklist](#) and review [Repository Sharing Guidance](#).

## Final Repair Checklist

- ☐ All required outputs render or run from a fresh environment.
- ☐ README instructions match the actual workflow.
- ☐ Data paths are relative.
- ☐ Required cached data files or data access instructions are present.
- ☐ Citations resolve.
- ☐ AI-use notes identify what was drafted and how it was checked.
- ☐ Remaining risks are documented in the failure log.



## **Weeks 12-13: Project Presentations**

# Week 12: Term Project Presentations, Part 1

## Overview

Week 12 is the first final presentation block for [M5: Final Presentation](#). Presentations should emphasize scientific clarity, knowledge translation, reproducibility, and auditability.

## Timing Rules

Each presentation uses this timing:

- 7 minutes for the presentation
- 3 minutes for questions
- 1 minute for transition

Presentation order is posted in Canvas. Groups presenting in Part 1 should arrive with rendered slides and the project repository link ready to share.

## Slide Submission

Before class, commit and sync:

```
1 milestones/m5-presentation/submission/slides.qmd
2 milestones/m5-presentation/submission/slides.html
```

If you render to PDF instead of HTML, submit `slides.pdf`. Submit the GitHub repository link on Canvas with the latest commit hash and the slide path.

## Presentation Rubric

Use this rubric while preparing and while giving peer feedback:

- audience and KT purpose are clear
- question is focused
- data source and limitations are transparent
- main visual is readable and evidence-based
- interpretation is accurate and avoids unsupported causal language
- privacy and stewardship risks are acknowledged
- timing fits the presentation window
- questions are answered with evidence and humility

## Peer Feedback Form

Complete one feedback note for each presentation you review.

| Prompt                                      | Feedback |
|---------------------------------------------|----------|
| Main message in one sentence                |          |
| Strongest evidence shown                    |          |
| One interpretation or limitation to clarify |          |
| One reproducibility or source question      |          |
| One audience-facing improvement             |          |

## Slide Checklist

Before presenting, check that:

- slides render on the presenting machine
- fonts and figures are readable from a distance
- every chart has a title, units, and limitation cue
- source and data notes are visible
- the final slide gives the audience a clear takeaway
- backup notes include data source, cohort, and AI-use details

## Portfolio Reminder

After presenting, use peer questions to revise the final report, dashboard-style KT product, and [Final Portfolio Checklist](#).

## Definition Of Done

Week 12 is done when Part 1 presentations are complete, peer feedback has been recorded, and presenting groups know which revisions to prioritize before the final portfolio.

# Week 13: Term Project Presentations, Part 2

## Overview

Week 13 completes [M5: Final Presentation](#) and turns presentation feedback into final portfolio revision. The last step is not more drafting; it is making the project clear, reproducible, and ready for assessment.

## Timing Rules

Each presentation uses this timing:

- 7 minutes for the presentation
- 3 minutes for questions
- 1 minute for transition

Presentation order is posted in Canvas. Groups presenting in Part 2 should submit slides before class using the same M5 submission folder as Week 12.

## Slide Submission

Commit and sync:

```
1 milestones/m5-presentation/submission/slides.qmd
2 milestones/m5-presentation/submission/slides.html
```

If you render to PDF instead of HTML, submit `slides.pdf`. Submit the GitHub repository link on Canvas with the latest commit hash and the slide path.

## Closing Reflection

After presenting or reviewing, write a short note for your group:

- What claim felt strongest?
- What limitation needs to be clearer in the final portfolio?
- What reproducibility issue still needs one final check?
- What audience-facing improvement would make the KT product easier to use?

Use the note to guide final revisions.

## Final Portfolio Reminder

The final portfolio package is described in [Final Portfolio](#). Before submission, complete the [Final Portfolio Checklist](#) and confirm the repository sharing choice using [Repository Sharing Guidance](#).

## Final Revision Priorities

Focus the remaining time on:

- rendering the final report
- checking dashboard or KT product access
- fixing broken paths
- clarifying data provenance and limitations
- completing the AI-use note
- submitting the repository link and commit hash on Canvas

## Definition Of Done

Week 13 is done when all presentations are complete, peer feedback has been returned, and each group has a clear final portfolio checklist for the last submission.

## **Assignments and Milestones**

# Assignments

Weekly assignments are scaffolded practice. Assignments 1-3 are foundational gates graded Complete/Incomplete. Assignments 4-10 support the term portfolio and use the best-5-of-7 grading policy described in the syllabus.

## Assignment Map

| Assignment   | Week | Focus                                                     | Grading mode        | Brief        |
|--------------|------|-----------------------------------------------------------|---------------------|--------------|
| Assignment 1 | 2    | Modern work-flows: Codespaces, Quarto, com-mit/sync       | Complete/Incomplete | Assignment 1 |
| Assignment 2 | 3    | R with AI: import, inspect, trans-form, audit             | Complete/Incomplete | Assignment 2 |
| Assignment 3 | 4    | Git collabora-tion: branches, pull requests, repo hygiene | Complete/Incomplete | Assignment 3 |



| Assignment    | Week | Focus                                            | Grading mode | Brief                     |
|---------------|------|--------------------------------------------------|--------------|---------------------------|
| Assignment 4  | 5    | Polyglot awareness and R deepening               | Best 5 of 7  | <a href="#">A4 brief</a>  |
| Assignment 5  | 6    | Visualization and AI visual audit                | Best 5 of 7  | <a href="#">A5 brief</a>  |
| Assignment 6  | 7    | EDA, Table 1, planted-error audit                | Best 5 of 7  | <a href="#">A6 brief</a>  |
| Assignment 7  | 8    | Dashboard-style KT prototype                     | Best 5 of 7  | <a href="#">A7 brief</a>  |
| Assignment 8  | 9    | Scientific communication and peer feedback       | Best 5 of 7  | <a href="#">A8 brief</a>  |
| Assignment 9  | 10   | Reproducible report, citations, publishing check | Best 5 of 7  | <a href="#">A9 brief</a>  |
| Assignment 10 | 11   | Portfolio surgery and reproducibility test       | Best 5 of 7  | <a href="#">A10 brief</a> |

## Grading Logic

Assignments 1-3 must be completed to pass the course. They are not numerically weighted, but they establish the workflow students need for every later assignment.

Assignments 4-10 are scored on Canvas. Only the best 5 of these 7 scores count toward the final grade, so students have flexibility while still practicing consistently after the foundation weeks.

## Common Reproducibility Contract

Every assignment must:

- run or render in GitHub Codespaces;
- use relative paths rather than machine-specific paths;
- keep source files and rendered outputs in the requested submission folder;
- commit and sync the final work before submitting the repository link on Canvas;
- document packages or dependencies when adding anything beyond the course defaults.

## Common AI-Use Expectation

AI may help students brainstorm, debug, translate, refactor, or improve wording. Students remain responsible for checking every code path, column name, statistic, citation, and interpretation. When AI is used, the submission must include an AI-use note explaining what the tool helped with, what changed, and how the result was verified.

# Assignment Template

Use this structure for weekly applied assignments. Keep the student-facing brief short enough to act on, but specific enough that a beginner can submit the right files without private instructions.

## Title

Use:

```
1 # Assignment X: Short Descriptive Title
```

## Purpose

Explain why the assignment matters, how it connects to the week, and how it prepares students for the next week or a project milestone.

## Learning Objectives

Use 3-5 concise bullets.

## Inputs

List week pages, data files, starter files, examples, templates, or milestone pages. Use relative paths only.

## Tasks

Provide a numbered workflow a beginner can follow.

## AI-Use Expectations

State what AI may help with and what students must verify. Require an AI-use or audit note when code, interpretation, debugging, writing, or translation used AI support.

## Reproducibility Requirements

State that the work must run or render in GitHub Codespaces, use relative paths, and declare dependencies when new packages are added.

## What To Submit

Give exact filenames and the exact folder:

```
1 assignments/assignmentXX-short-name/submission/
```

## Submission Route

Students submit their GitHub repository link on Canvas after committing and syncing the required files.

## Grading Checklist

For Assignments 1-3, use Complete/Incomplete. For Assignments 4-10, use a practical 7-point checklist or a Complete/Needs Work checklist with the scored Canvas rubric.

## Definition of Done

End with a short checklist students can use before submitting.

# Assignment 1: Modern Workflows

## Purpose

This assignment checks that your fork-based course workflow is working before the analysis weeks begin. You will fork the course template repository, launch or reopen a GitHub Codespace from your fork, edit a Quarto document, render output, commit and sync your work, and stop the Codespace when you are finished.

## Learning Objectives

- Fork the course template repository under your GitHub account.
- Launch or reopen the correct Codespace from your fork.
- Edit a Quarto document using YAML, Markdown, and a simple code chunk.
- Render a Quarto document and locate the output file.
- Commit and sync a small reproducible change.
- Stop the Codespace to conserve free monthly hours.

## Inputs

- Week 2 page: `weeks/week02-modern-workflows/index.qmd`
- Course data example: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Submission folder: `assignments/assignment01-workflows/submission/`

## Tasks

1. Open the course template repository linked from Canvas.
2. Click **Fork** and create a personal fork under your GitHub account.
3. Launch Codespaces from your forked repository.
4. Create the folder `assignments/assignment01-workflows/submission/`.
5. Create a Quarto file named `workflow-check.qmd` in that folder.
6. Add a YAML title, your name, and the date.
7. Add one short paragraph explaining why relative paths matter in this course.
8. Add one code chunk that prints the current working directory or lists the files in the submission folder.
9. Render `workflow-check.qmd` to HTML.
10. Use the Source Control panel to inspect the diff.
11. Commit with a clear message such as `Complete Assignment 1 workflow check`.
12. Sync/push the commit to your fork on GitHub.
13. Stop the Codespace after confirming the commit appears on GitHub.
14. Write a short reflection naming one workflow step that felt solid and one step you want to practice.

## AI-Use Expectations

AI may help you understand an error message or explain Quarto syntax. You must personally verify that the file renders and that the commit is visible on GitHub. If AI helped, include one sentence in `reflection.md` saying what it helped with and how you checked the result.

## Reproducibility Requirements

- The Quarto file must render in GitHub Codespaces.
- Paths must be relative to the repository.
- The rendered HTML must be generated from the submitted `.qmd` file.

- The final files must be committed and synced to your fork.

## What To Submit

Place all submission files in:

```
1 assignments/assignment01-workflows/submission/
```

Submit these files:

- `workflow-check.qmd`: edited Quarto source.
- `workflow-check.html`: rendered output.
- `reflection.md`: 150-250 words on the workflow, including any AI-use sentence if relevant.
- `commit-evidence.md`: the commit hash or GitHub URL showing the final synced commit.

## Submission Route

Submit your forked GitHub repository link on Canvas after committing and syncing the required files.

## Grading Checklist

Complete:

- Personal fork was created or located successfully.
- Codespace was launched or reopened from the fork.
- `workflow-check.qmd` exists in the required submission folder.
- `workflow-check.html` renders from the source file.
- Reflection names one solid step and one step to practice.
- Commit evidence is included.
- Work is committed and synced to the fork on GitHub.
- Codespace was stopped after work was complete.



Incomplete:

- Required files are missing.
- Rendered output does not match the source file.
- Work is not committed or not synced.
- Reflection or commit evidence is missing.
- Submission uses machine-specific paths.

## Definition of Done

You are done when the fork repository link on Canvas points to a synced commit containing `workflow-check.qmd`, `workflow-check.html`, `reflection.md`, and `commit-evidence.md` in the Assignment 1 submission folder.

# Assignment 2: R With AI

## Purpose

This assignment turns the Week 3 R basics into a small auditable analysis note. You will load data with a relative path, inspect it before summarizing, use `dplyr` verbs, write one custom function, and document how any AI-generated code was checked.

## Learning Objectives

- Load R packages and import a CSV with a relative path.
- Inspect rows, columns, variable types, and missingness before analysis.
- Use `filter()`, `mutate()`, `group_by()`, and `summarise()` for a descriptive summary.
- Write one small custom function and test it on the dataset.
- Audit AI-generated code for wrong column names, missingness mistakes, and overclaiming.

## Inputs

- Week 3 page: `weeks/week03-r-with-ai/index.qmd`
- Starter note: `assignments/assignment02-r-with-ai/starter-analysis.qmd`
- Dataset: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Submission folder: `assignments/assignment02-r-with-ai/submission/`

## Tasks

1. Create the folder `assignments/assignment02-r-with-ai/submission/`.
2. Copy the starter structure into `analysis-note.qmd`.
3. Load `readr`, `dplyr`, and any other package used in the file.
4. Import the NHANES CSV using a relative path from the repository root.
5. Inspect the dataset with row count, column names, variable types, and missingness.
6. Create one cleaned or analysis-ready table using at least three `dplyr` verbs.
7. Write one custom function, such as a function that returns mean and SD for a numeric variable.
8. Save a small output file named `output/clean_data.csv` or `output/summary_table.csv`.
9. Add AI audit comments near code that was drafted, debugged, or revised with AI assistance.
10. Render the Quarto note to HTML.
11. Commit and sync the source, rendered output, saved output file, and AI-use note.

## AI-Use Expectations

AI may help draft code, explain an error, or suggest a function structure. You must verify that all column names exist, missing values are handled intentionally, outputs match the code, and interpretations remain descriptive. Include `ai-use-note.md` even if AI was not used.

## Reproducibility Requirements

- The analysis must render in GitHub Codespaces.
- The data path must be relative.
- Output files must be created by code, not manually edited.
- New packages must be named in the Quarto file and in `dependency-note.md`.

- `renv` awareness is introductory here: run `renv::status()` and note whether you used the existing course environment or added packages. Use `renv::snapshot()` only if you intentionally added a package.

## What To Submit

Place all submission files in:

```
1 assignments/assignment02-r-with-ai/submission/
```

Submit these files:

- `analysis-note.qmd`: Quarto source with data import, inspection, cleaning, function, and AI audit comments.
- `analysis-note.html`: rendered output.
- `output/clean_data.csv` or `output/summary_table.csv`: code-generated output.
- `ai-use-note.md`: what AI helped with and how you checked it, or a sentence saying AI was not used.
- `dependency-note.md`: packages used and whether any were added beyond the course environment.

## Submission Route

Submit your GitHub repository link on Canvas after committing and syncing the required files.

## Grading Checklist

Complete:

- CSV is imported with a relative path.
- Dataset inspection includes size, column names/types, and missingness.
- `dplyr` workflow uses at least three relevant verbs.
- One custom function is defined and used.

- Output CSV is generated by code.
- AI-use note and dependency note are present.
- Quarto file renders in Codespaces.

Incomplete:

- Data path is absolute or machine-specific.
- Code does not render.
- Missingness is ignored or silently dropped.
- Function is absent or unused.
- Output file is missing or manually created.
- AI-use or dependency note is missing.

## Definition of Done

You are done when `analysis-note.qmd` renders, the output CSV is regenerated by code, and the Canvas submission links to the synced repository folder.

# Assignment 3: Git Collaboration

## Purpose

This assignment checks that you can make a small, reviewable change using Git. You will create a branch, make a documentation-only edit, inspect the diff, open a pull request, and connect the workflow to M0 group repository readiness.

## Learning Objectives

- Make meaningful commits with clear messages.
- Read a diff before committing.
- Use `.gitignore` to keep generated or sensitive files out of version control.
- Create a branch and open a pull request.
- Document a limited change so collaborators can review it.

## Inputs

- Week 4 page: `weeks/week04-git-collaboration/index.qmd`
- M0 connection: `milestones/m0-group-formation/README.md`
- Submission folder: `assignments/assignment03-git/submission/`

## Tasks

1. Create the folder `assignments/assignment03-git/submission/`.
2. Create a new branch with a short name such as `docs/git-audit-note`.
3. Make one documentation-only edit in your repository. Good choices include clarifying a README sentence, adding a path note, or improving a caption.
4. Inspect the diff and confirm the change is limited to documentation or text.
5. Check `.gitignore` and add a safe pattern if a generated or temporary file is being tracked.
6. Commit with a message that explains the change.
7. Push the branch and open a pull request.
8. In the pull request description, explain what changed and how you verified the scope.
9. Add a short note on whether your group repository is ready for M0: members, access, communication plan, and Codespaces access.

## AI-Use Expectations

AI may help summarize a diff or draft a pull request description. You must verify the diff yourself and make sure the pull request does not include unrelated file changes. Include an AI-use sentence in `git-audit-note.md`.

## Reproducibility Requirements

- All changes must be visible in Git history.
- The pull request must show a focused diff.
- `.gitignore` must not hide source files needed to reproduce the work.
- Evidence files must use relative paths or GitHub URLs.

## What To Submit

Place all submission files in:

```
1 assignments/assignment03-git/submission/
```

Submit these files:

- `pull-request-evidence.md`: pull request URL, branch name, and final commit hash. If a pull request cannot be opened, include screenshot filenames and a written explanation.
- `documentation-change.md`: the before/after text or a link to the changed documentation file.
- `git-audit-note.md`: diff summary, `.gitignore` check, AI-use sentence, and M0 readiness note.

## Submission Route

Submit your GitHub repository link on Canvas after committing and syncing the required files.

## Grading Checklist

Complete:

- Branch was created for the work.
- Documentation-only change is focused and reviewable.
- Diff was inspected and summarized.
- Commit message is clear.
- Pull request evidence or equivalent text evidence is included.
- `.gitignore` was checked.
- M0 readiness note names repository access and team communication status.

Incomplete:



- Work was committed directly without evidence of a branch or review step.
- Diff includes unrelated changes.
- Pull request evidence is missing.
- `.gitignore` check is absent.
- M0 readiness note is missing.
- Work is not synced to GitHub.

## Definition of Done

You are done when the Canvas submission links to a synced repository with pull request evidence, a documentation change summary, and a Git audit note in the Assignment 3 submission folder.

# Assignment 4: Polyglot Parity

## Purpose

This assignment strengthens R first, then translates a small descriptive task to Python/pandas and checks whether both languages produce the same answer. The goal is careful bilingual reading, not Python mastery.

## Learning Objectives

- Build a trusted R `dplyr` summary before translating it.
- Refactor repeated R logic into a small function.
- Translate a simple summary to Python/pandas with AI support if useful.
- Run parity checks for row counts, group labels, and rounded statistics.
- Keep notebook outputs and dependency notes clean for review.

## Inputs

- Week 5 page: `weeks/week05-polyglot-r-deepening/index.qmd`
- R deepening page: `weeks/week05-polyglot-r-deepening/r-deepening-parity.qmd`
- Dataset: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Parity template: `assignments/assignment04-polyglot/parity-table-template.csv`
- Submission folder: `assignments/assignment04-polyglot/submission/`

## Tasks

1. Create or use the folder `assignments/assignment04-polyglot/submission/`.
2. Create `polyglot-parity.qmd`.
3. In R, load the NHANES CSV with a relative path.
4. Produce a descriptive BMI summary by `IncomeGroup` and `Gender`.
5. Refactor the R summary into one small reusable function.
6. Create `translation.ipynb` and translate the same summary to Python/pandas.
7. Create `helpers.py` with at least one reusable Python helper called from the notebook.
8. Complete a parity table comparing row counts, filtered row counts, grouping labels, and rounded summary values.
9. Write a dependency note naming Python packages used. Include `requirements.txt` only if you add packages beyond the course defaults.
10. Clear bulky notebook outputs before committing.
11. Render the Quarto file to HTML.
12. Commit and sync the assignment folder.

## AI-Use Expectations

AI may help translate R to Python, explain tracebacks, or suggest helper functions. You must verify that the translated code answers the same question, uses the same filters, handles missingness the same way, and returns matching grouped results after rounding. Include `ai-use-note.md`.

## Reproducibility Requirements

- R and Python files must run in GitHub Codespaces.
- Paths must be relative to the repository.
- The notebook must not contain large stale outputs.
- Dependencies must be documented in `dependency-note.md`.
- Differences between R and Python results must be explained before changing code.

## What To Submit

Place all submission files in:

```
1 assignments/assignment04-polyglot/submission/
```

Submit these files:

- `polyglot-parity.qmd`: R analysis, R function, parity discussion, and rendered-output instructions.
- `polyglot-parity.html`: rendered Quarto output.
- `translation.ipynb`: Python/pandas translation notebook.
- `helpers.py`: reusable Python helper script called from the notebook.
- `parity-table.csv`: completed parity table.
- `dependency-note.md`: Python/R dependency note, plus `requirements.txt` if extra Python packages were added.
- `ai-use-note.md`: what AI helped translate or debug and how you verified the result.

## Submission Route

Submit your GitHub repository link on Canvas after committing and syncing the required files.

## Grading Checklist

Canvas applies a 7-point rubric. This assignment is part of the best-5-of-7 set for Assignments 4-10:

| Criterion                                                  | Points |
|------------------------------------------------------------|--------|
| R summary is correct, clear, and uses a relative data path | 1      |
| R function reproduces the summary logic                    | 1      |
| Python translation matches the same analytic question      | 1      |
| Parity table checks row counts, groups, and rounded values | 1.5    |
| Dependency and notebook hygiene are documented             | 1      |

| Criterion                                                  | Points |
|------------------------------------------------------------|--------|
| AI-use note explains generation, changes, and verification | 1      |
| Files render/run from the submission folder and are synced | 0.5    |

## Definition of Done

You are done when the R summary, Python translation, and parity table agree or any differences are clearly explained, and the required files are committed in the Assignment 4 submission folder.

# Assignment 5: Visualization Audit

## Purpose

This assignment turns a descriptive summary into an honest visual explanation. You will create a `ggplot2` visualization, audit a flawed plot, revise your own design, and write a caption that states the statistic, data source, exclusions, and limitations.

## Learning Objectives

- Create a clear `ggplot2` visualization from the NHANES classroom dataset.
- Identify misleading scale, title, grouping, missingness, and caption choices.
- Revise a plot so the visual design matches the evidence.
- Explain the descriptive and unweighted limits of the figure.
- Connect visualization readiness to the M1 project proposal.

## Inputs

- Dataset: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Worked example: `weeks/week06-visualization/viz03-worked-example.qmd`
- Studio activity: `weeks/week06-visualization/viz06-studio-visual-audit.qmd`
- Walkthrough and reference: `weeks/week06-visualization/viz07-assignment-a5-and-reference.qmd`
- Flawed plot starter: `weeks/week06-visualization/instructor/flawed-plot-code.R`
- Visual audit template: `assignments/assignment05-visualization/visual-audit-note-template.md`
- M1 connection: `milestones/m1-proposal/README.md`

## Tasks

1. Create or use the folder `assignments/assignment05-visualization/submission/`.
2. Review the Week 6 worked example.
3. Choose one descriptive comparison from the NHANES CSV snapshot.
4. Draft an initial plot using R and `ggplot2`.
5. Audit the flawed plot starter and your own draft for scale choices, labels, missing context, aggregation, and overclaiming.
6. Revise your plot so it supports a descriptive interpretation.
7. Export the corrected plot as `corrected-plot.png`.
8. Write a caption that includes data source, statistic, exclusions, and limitation.
9. Add an optional `plotly` preview only if it helps you inspect the figure; the required submission is still the static plot.
10. Write a short M1 link note naming how the visualization standard affects your project dataset, audience, or feasibility.

## AI-Use Expectations

AI may help draft `ggplot2` code, suggest labels, or identify possible visual problems. You must verify the code against real column names, check that the scale does not exaggerate the result, and remove causal or population-level claims that the plot cannot support. Include `ai-use-note.md`.

## Reproducibility Requirements

- The Quarto file must render in GitHub Codespaces.
- The dataset path must be relative.
- The plot image must be exported by code or a documented render step.

- Captions must state that the analysis is descriptive and unweighted when using the classroom NHANES snapshot.

## What To Submit

Place all submission files in:

```
1 assignments/assignment05-visualization/submission/
```

Submit these files:

- `visualization-audit.qmd`: rerunnable Quarto file that loads the CSV, creates the final plot, and explains the revision.
- `visualization-audit.html`: rendered output.
- `corrected-plot.png`: exported final plot.
- `visual-audit-note.md`: at least three issues checked or corrected.
- `caption.md`: final figure caption.
- `m1-link-note.md`: two or three sentences connecting the visualization standard to M1.
- `ai-use-note.md`: what AI helped with and how you checked it, or a sentence saying AI was not used.

## Submission Route

Submit your GitHub repository link on Canvas after committing and syncing the required files.

## Grading Checklist

Canvas applies a 7-point rubric. This assignment is part of the best-5-of-7 set for Assignments 4-10:



| Criterion                                                                     | Points |
|-------------------------------------------------------------------------------|--------|
| Plot is generated by rerunnable <code>ggplot2</code> code with relative paths | 1.5    |
| Visual audit identifies at least three meaningful issues                      | 1      |
| Corrected plot improves scale, labeling, grouping, or caption clarity         | 1.5    |
| Caption states statistic, source, exclusions, and limitations                 | 1      |
| Interpretation stays descriptive and non-causal                               | 0.75   |
| AI-use note and M1 link note are specific                                     | 0.75   |
| Required files are committed and synced                                       | 0.5    |

## Definition of Done

You are done when a peer can render `visualization-audit.qmd`, see the corrected plot, understand the main comparison from the caption, and identify how you checked the plot for misleading design.

# Assignment 6: EDA and AI Audit

## Purpose

This assignment builds a transparent preliminary analysis. You will define a cohort, check missingness, create a Table 1-style descriptive summary, audit planted errors, and revise a provenance/stewardship note for reuse in M2.

## Learning Objectives

- Define an analysis cohort before summarizing results.
- Report missingness for key variables.
- Create a Table 1-style summary with N, mean, and SD.
- Detect planted correctness, reproducibility, interpretation, and stewardship errors.
- Write descriptive methods text that avoids causal overclaiming.

## Inputs

- Dataset: `examples/nhanes-equity/data/nhanes_equity_v6.csv`
- Worked example: `weeks/week07-eda-ai-audit/eda03-worked-example.qmd`
- Studio activity: `weeks/week07-eda-ai-audit/eda06-studio-planted-errors.qmd`
- Walkthrough and reference: `weeks/week07-eda-ai-audit/eda07-assignment-a6-and-reference.qmd`
- Planted-error starter: `weeks/week07-eda-ai-audit/instructor/planted-error-starter.qmd`
- Audit template: `assignments/assignment06-eda-ai-audit/audit-note-template.md`
- M2 connection: `milestones/m2-preliminary-analysis/README.md`

## Tasks

1. Create or use the folder `assignments/assignment06-eda-ai-audit/submission/`.
2. Load the NHANES CSV snapshot with a relative path.
3. Define and document an analysis cohort using age 20-80.
4. Create a missingness table for key variables.
5. Produce a Table 1-style summary by `IncomeGroup`.
6. Include N, BMI mean (SD), and age mean (SD).
7. Write a short methods note stating that the analysis is descriptive, unweighted, and non-causal.
8. Audit the planted-error starter and list at least five issues.
9. Categorize each issue as correctness, reproducibility, interpretation, or stewardship.
10. Revise the provenance/stewardship statement so it names the data source, classroom-use context, and one privacy or stewardship caution.

## AI-Use Expectations

AI may help draft code, explain an error, or improve the organization of the audit note. You must verify cohort filters, missingness handling, labels, summary statistics, and interpretation against the code and output. Include `ai-use-note.md`.

## Reproducibility Requirements

- The EDA note must render in GitHub Codespaces.
- Paths must be relative.
- `table1.csv` must be generated by code.
- Missingness handling must be stated before interpreting summaries.
- The methods note must avoid causal and population-level claims unsupported by the unweighted classroom example.

## What To Submit

Place all submission files in:

```
1 assignments/assignment06-eda-ai-audit/submission/
```

Submit these files:

- `eda-note.qmd`: rerunnable Quarto file with cohort definition, missingness table, Table 1, and methods note.
- `eda-note.html`: rendered output.
- `table1.csv`: exported Table 1 or equivalent descriptive summary.
- `audit-note.md`: at least five planted issues with category, why it matters, and corrected approach.
- `provenance-stewardship-note.md`: short revised note suitable for reuse in M2.
- `ai-use-note.md`: what AI helped with and how you checked it, or a sentence saying AI was not used.

## Submission Route

Submit your GitHub repository link on Canvas after committing and syncing the required files.

## Grading Checklist

Canvas applies a 7-point rubric. This assignment is part of the best-5-of-7 set for Assignments 4-10:

| Criterion                                           | Points |
|-----------------------------------------------------|--------|
| Cohort and exclusions are documented before results | 1      |
| Missingness table is complete for key variables     | 1      |

| Criterion                                                            | Points |
|----------------------------------------------------------------------|--------|
| Table 1 includes N, BMI mean (SD), and age mean (SD) by group        | 1.5    |
| Methods note is descriptive, unweighted, and non-causal              | 1      |
| Audit note identifies at least five planted issues across categories | 1.25   |
| Provenance/stewardship and AI-use notes are specific                 | 0.75   |
| Required files render, export, commit, and sync                      | 0.5    |

## Definition of Done

You are done when `eda-note.qmd` renders, `table1.csv` is re-generated by code, and the audit note explains at least five planted errors and corrected approaches.

# Assignment 7: Dashboard KT Prototype

## Purpose

This assignment introduces dashboard-style Knowledge Translation. You will run or render a small dashboard product, identify the audience and question, document one control or interaction, improve one audience-facing element, test the product, and state a limitation.

## Learning Objectives

- Explain how a dashboard differs from a static report.
- Identify an audience-specific KT question.
- Document one control, filter, or dashboard-style interaction.
- Connect one visualization to a plain-language interpretation.
- Test a dashboard product in Codespaces and record the result.

## Inputs

- Week 8 index: `weeks/week08-dashboard-kt/index.qmd`
- Required core artifact (the file you adapt): `weeks/week08-dashboard-kt/dash03-starter-dashboard.qmd`
- Walkthrough and reference: `weeks/week08-dashboard-kt/dash07-assignment-a7-and-reference.qmd`
- Shiny exemplar (instructor demonstration only; not an A7 submission pathway): `examples/nhanes-equity/app/app.R`
- Cached data: `examples/nhanes-equity/data/nhanes_equity_v6.rds`
- CSV snapshot: `examples/nhanes-equity/data/nhanes_equity_v6.csv`

- Adaptation template: `assignments/assignment07-dashboard/dashboard-adaptation-template.md`

## Tasks

1. Create or use the folder `assignments/assignment07-dashboard/submission/`.
2. From the repository root, render the Quarto starter dashboard (`weeks/week08-dashboard-kt/dash03-starter-dashboard.qmd`).  
The Shiny exemplar is a demonstration and is not an approved A7 submission pathway without prior instructor approval.
3. Choose one audience, such as a public-health analyst, student project team, or briefing group.
4. Write one audience-specific dashboard question.
5. Identify one control, filter, or dashboard-style interaction and the visualization it affects.
6. Identify one label, caption, help text, interpretation note, or README sentence that could be clearer for that audience.
7. Make one documentation-only or UI-text-only improvement. Do not change analytic logic for this assignment.
8. Re-render the starter dashboard with a different control value and record the test result.
9. Write one limitation statement that belongs in or beside the dashboard.
10. Complete the privacy-safe publishing check before sharing screenshots or rendered outputs.

## AI-Use Expectations

AI may help improve wording for labels, captions, or limitation statements. You must verify that the revised language still matches the dashboard output and does not imply causality, population inference, or access to row-level data. Include `ai-use-note.md`.

## Reproducibility Requirements

- The dashboard or starter dashboard must run or render in GitHub Codespaces.
- Paths must be relative.
- The testing log must name the command or render path used.
- Screenshots must avoid exposing row-level data.
- The submitted adaptation must not require private credentials or local-only files.

## What To Submit

Place all submission files in:

```
1 assignments/assignment07-dashboard/submission/
```

Submit these files:

- **dashboard-adaptation-note.md**: audience, question, control/interaction, visualization, changed element, and KT rationale.
- **before-after-text.md**: original text and revised text.
- **testing-log.md**: command or render path used, date tested, and result.
- **limitation-note.md**: one limitation or caution suitable for the dashboard.
- **privacy-check.md**: evidence that no row-level data, small-cell overclaim, or causal overclaim is shared.
- **dashboard-preview.png**: screenshot or exported preview of the affected dashboard area.
- **reproducibility-note.md**: how the dashboard was run or rendered in Codespaces.
- **ai-use-note.md**: what AI helped with and how you checked it, or a sentence saying AI was not used.



## Submission Route

Submit your GitHub repository link on Canvas after committing and syncing the required files.

## Grading Checklist

Canvas applies a 7-point rubric. This assignment is part of the best-5-of-7 set for Assignments 4-10:

| Criterion                                                                | Points |
|--------------------------------------------------------------------------|--------|
| Audience and dashboard question are specific                             | 1      |
| Control/filter or interaction and visualization are described accurately | 1      |
| Audience-facing change improves clarity without changing analytic logic  | 1.25   |
| Interpretation and limitation are descriptive and non-causal             | 1      |
| Testing and reproducibility notes show a successful local check          | 1      |
| Privacy-safe publishing check is complete                                | 0.75   |
| Preview, AI-use note, and required files are committed and synced        | 1      |

## Definition of Done

You are done when the submitted notes let a reviewer identify the audience, control, visualization, text improvement, test re-

sult, and limitation without rerunning the app first.

# Assignment 8: Scientific Communication

## Purpose

This assignment turns one result from your analysis or dashboard into a short audience-specific communication product. You will create a small slide deck or slide outline, write a plain-language summary, cite the data source, and use peer feedback to improve the claim.

## Learning Objectives

- Translate one technical result into a clear audience-facing takeaway.
- Build a short Quarto revealjs deck or structured slide outline.
- Use one figure or table from prior work as evidence.
- State a limitation and source note beside the claim.
- Give and apply peer feedback on clarity, evidence, and KT value.

## Inputs

- Week 9 page: `weeks/week09-science-communication/index.qmd`
- Peer feedback page: `weeks/week09-science-communication/revealjs-peer-feedback.qmd`
- Renderable Week 9 slide starter: `weeks/week09-science-communication/starter-slides.qmd`
- Slide starter: `assignments/assignment08-communication/slide-outline-template.qmd`
- Peer feedback template: `assignments/assignment08-communication/peer-feedback-template.md`
- Prior work from Assignment 5, 6, or 7
- Submission folder: `assignments/assignment08-communication/submission/`

## Tasks

1. Create or use the folder `assignments/assignment08-communication/submission/`.
2. Choose one figure, table, or dashboard finding from prior work.
3. Define the audience in one sentence.
4. Write one plain-language takeaway that a non-technical reader can understand.
5. Create either `slides.qmd` as a revealjs deck or `slide-outline.qmd` as a structured slide outline.
6. Include 3-5 slides or slide sections: context, evidence, takeaway, limitation, and next step.
7. Add one source note or citation for the dataset.
8. Exchange peer feedback using the template.
9. Revise at least one sentence or slide based on feedback.
10. Document any AI assistance used for simplification, slide wording, or feedback synthesis.

## AI-Use Expectations

AI may help simplify jargon, suggest slide structure, or tighten wording. You must verify that the simplified text still matches the evidence, citation, and limitation. AI must not invent findings, sources, or audience needs. Include `ai-use-note.md`.

## Reproducibility Requirements

- The deck or outline must render in GitHub Codespaces.
- Any figure or table must be traceable to prior code or a submitted output.
- Source notes and citations must be visible.
- Paths must be relative.
- Peer feedback must be specific enough to act on.

## What To Submit

Place all submission files in:

1 `assignments/assignment08-communication/submission/`

Submit these files:

- `slides.qmd` or `slide-outline.qmd`: source file for the communication product.
- `slides.html` or `slide-outline.html`: rendered output.
- `plain-language-summary.md`: 150-250 word audience-facing summary.
- `peer-feedback.md`: feedback received and one revision made.
- `source-note.md`: dataset/source note and citation information.
- `ai-use-note.md`: what AI helped with and how you checked it, or a sentence saying AI was not used.

## Submission Route

Submit your GitHub repository link on Canvas after committing and syncing the required files.

## Grading Checklist

Canvas applies a 7-point rubric. This assignment is part of the best-5-of-7 set for Assignments 4-10:

| Criterion                                                   | Points |
|-------------------------------------------------------------|--------|
| Audience and KT purpose are specific                        | 1      |
| Slide deck or outline has a clear 3-5 part story            | 1.25   |
| Figure/table evidence is traceable and accurately described | 1      |
| Plain-language summary is clear and avoids overclaiming     | 1      |
| Source note and limitation are visible                      | 1      |
| Peer feedback is specific and revision is documented        | 1      |

| Criterion                                             | Points |
|-------------------------------------------------------|--------|
| AI-use note and render/sync requirements are complete | 0.75   |

## Definition of Done

You are done when a peer can read the deck or outline, identify the main takeaway, see the evidence and limitation, and understand what changed after feedback.

# Assignment 9: Reproducible Report

## Purpose

This assignment expands a project finding into a short reproducible Quarto report. The report should regenerate a table and figure from code, cite the data source, state limitations, and prepare the work for M3 Project Update review.

## Learning Objectives

- Write a Quarto report that combines narrative, code, tables, figures, and citations.
- Generate at least one table and one figure from code.
- Use BibTeX citations and a visible source note.
- Include limitations and a grounding audit for claims.
- Prepare a preview or publishing check suitable for M3 review.

## Inputs

- Week 10 page: `weeks/week10-reporting-publishing/index.qmd`
- Report template: `weeks/week10-reporting-publishing/report-template.qmd`
- Bibliography example: `references.bib`
- M3 connection: `milestones/m3-project-update/README.md`
- Publishing checklist template: `assignments/assignment09-reporting/publishing-checklist-template`
- Grounding audit template: `assignments/assignment09-reporting/grounding-audit-template.md`

## Tasks

1. Create or use the folder `assignments/assignment09-reporting/submission/`.
2. Start from the Week 10 report template structure.
3. Define one descriptive project question. If your project dataset is not ready, use the NHANES classroom dataset and explain that this is a practice report.
4. Load the data with a relative path.
5. Generate at least one table and one figure from code.
6. Add citations using a local `references.bib` file or the course bibliography workflow.
7. Write a limitations section that avoids causal overclaiming.
8. Complete a grounding audit: each claim should point to code output, a citation, or a stated limitation.
9. Complete a publishing check: GitHub Pages preview if available, or a private rendered HTML preview in the repository.
10. Write an M3 update note explaining what is ready for peer review and what feedback would be useful.

## AI-Use Expectations

AI may help outline the report, debug Quarto, improve wording, or identify unclear claims. You must verify every generated number, citation, and source claim against the rendered output and bibliography. Include `ai-use-note.md`.

## Reproducibility Requirements

- `report.qmd` must render in GitHub Codespaces.
- Tables and figures must be generated by code.
- Paths must be relative.
- Bibliography entries must resolve.
- Publishing preview must not require private local files.
- The report must complement the dashboard-style product rather than merely duplicate it.



## What To Submit

Place all submission files in:

```
1 assignments/assignment09-reporting/submission/
```

Submit these files:

- `report.qmd`: rerunnable Quarto report.
- `report.html`: rendered report.
- `references.bib`: bibliography entries used by the report.
- `outputs/`: generated table or figure files if the report writes external outputs.
- `publishing-checklist.md`: GitHub Pages or private preview check.
- `grounding-audit.md`: claim-by-claim check against code output, citation, or limitation.
- `m3-update-note.md`: what is ready for M3 review and what feedback is requested.
- `ai-use-note.md`: what AI helped with and how you checked it, or a sentence saying AI was not used.

## Submission Route

Submit your GitHub repository link on Canvas after committing and syncing the required files.

## Grading Checklist

Canvas applies a 7-point rubric. This assignment is part of the best-5-of-7 set for Assignments 4-10:

| Criterion                                        | Points |
|--------------------------------------------------|--------|
| Report renders in Codespaces with relative paths | 1      |

| Criterion                                                             | Points |
|-----------------------------------------------------------------------|--------|
| Includes one<br>code-generated table and<br>one code-generated figure | 1.25   |
| Citations resolve through<br>BibTeX workflow                          | 1      |
| Limitations are accurate<br>and visible                               | 1      |
| Grounding audit checks<br>key claims against<br>evidence              | 1      |
| Publishing checklist and<br>M3 update note are<br>complete            | 1      |
| AI-use note and required<br>files are committed and<br>synced         | 0.75   |

## Definition of Done

You are done when `report.qmd` renders to `report.html`, citations resolve, outputs are generated by code, and the M3 update note tells reviewers exactly what feedback you need.

# Assignment 10: Portfolio Surgery

## Purpose

This assignment stress-tests your project before the final portfolio. You will run the work in a fresh Codespace, repair dependency or path failures, improve the README, complete peer review, and document remaining risks for M4 and final submission.

## Learning Objectives

- Test a report and dashboard product in a clean environment.
- Diagnose broken paths, missing dependencies, unclear README steps, and stale outputs.
- Document failures and repairs in a reproducible audit trail.
- Give and receive peer feedback on clarity, auditability, and KT value.
- Create a final risk list before portfolio submission.

## Inputs

- Week 11 page: `weeks/week11-portfolio-surgery/index.qmd`
- Reproducibility protocol: `weeks/week11-portfolio-surgery/reproducibility-test.qmd`
- Static dashboard exemplar: `weeks/week11-portfolio-surgery/static-dashboard-ojs.qmd`
- M4 connection: `milestones/m4-peer-review/README.md`
- Failure log template: `assignments/assignment10-portfolio-surgery/failure-log-template.md`
- Peer review template: `assignments/assignment10-portfolio-surgery/peer-review-template.md`

## Tasks

1. Create or use the folder `assignments/assignment10-portfolio-surgery/submission/`.
2. Open your project repository in a fresh Codespace or equivalent clean environment.
3. Follow only the README and documented setup instructions.
4. Render the report or book.
5. Run or render the dashboard-style product.
6. Check relative paths, package loading, citations, data availability, output freshness, and AI-use notes.
7. Log each failure before repairing it.
8. Repair dependency, path, README, citation, or stale-output issues.
9. Commit repairs with clear messages.
10. Exchange peer review focused on reproducibility, auditability, clarity, and KT value.
11. Write a final risk list naming remaining issues or stating that no known risks remain.

## AI-Use Expectations

AI may help interpret error messages, suggest README wording, or organize the failure log. You must verify every repair by rerunning the affected output in the clean environment. Include `ai-use-note.md`.

## Reproducibility Requirements

- The test must be performed in a fresh Codespace or clean equivalent environment.
- Repairs must be committed and synced.
- README instructions must match the tested workflow.
- Relative paths must work from the repository root.
- Any remaining risk must be documented rather than hidden.

## What To Submit

Place all submission files in:

1 `assignments/assignment10-portfolio-surgery/submission/`

Submit these files:

- `reproducibility-test.md`: completed fresh-environment test protocol.
- `failure-log.md`: failures found, evidence, fix attempted, and result.
- `peer-review.md`: feedback received or given for M4 readiness.
- `repair-summary.md`: files changed and why.
- `final-risk-list.md`: remaining risks or a statement that no known risks remain.
- `ai-use-note.md`: what AI helped with and how you checked it, or a sentence saying AI was not used.

## Submission Route

Submit your GitHub repository link on Canvas after committing and syncing the required files. Include the final repair commit or pull request link in the Canvas comments.

## Grading Checklist

Canvas applies a 7-point rubric. This assignment is part of the best-5-of-7 set for Assignments 4-10:

| Criterion                                            | Points |
|------------------------------------------------------|--------|
| Fresh-environment reproducibility test is documented | 1.25   |
| Failure log records errors before repairs            | 1      |

| Criterion                                                              | Points |
|------------------------------------------------------------------------|--------|
| Repairs address paths, dependencies, README, citations, or outputs     | 1.25   |
| Report and dashboard product are rerun or failure is clearly explained | 1      |
| Peer review is specific, actionable, and tied to evidence              | 1      |
| Final risk list is honest and useful for final portfolio planning      | 1      |
| AI-use note and synced repair evidence are complete                    | 0.5    |

## Definition of Done

You are done when a reviewer can open the submission folder, see what failed, see what was repaired, see what still needs attention, and follow the README to reproduce the core project outputs.

## Milestones

# Milestones

Milestones move the term project from team setup to final portfolio. They are short checkpoints, not separate side projects. Each one should make the next stage easier to complete and easier for another person to review.

## Submission Contract

For every milestone:

- commit and sync the required files before submitting
- submit the GitHub repository link on Canvas
- include the latest commit hash in the Canvas comment box
- use relative paths
- document any package or data access requirements
- disclose and audit AI use

## Milestone Map

| Milestone | Timing | Focus                                 | Link                                 |
|-----------|--------|---------------------------------------|--------------------------------------|
| M0        | Week 4 | Group formation and repository access | <a href="#">M0: Group Formation</a>  |
| M1        | Week 6 | Project proposal and data intake      | <a href="#">M1: Project Proposal</a> |



| Milestone       | Timing         | Focus                                          | Link                                     |
|-----------------|----------------|------------------------------------------------|------------------------------------------|
| M2              | Week 7         | Preliminary analysis and Table 1-style summary | <a href="#">M2: Preliminary Analysis</a> |
| M3              | Week 10        | Draft report and dashboard-style update        | <a href="#">M3: Project Update</a>       |
| M4              | Week 11        | Peer review and reproducibility feedback       | <a href="#">M4: Peer Review</a>          |
| M5              | Weeks 12-13    | Final presentation                             | <a href="#">M5: Final Presentation</a>   |
| Final Portfolio | Final deadline | Complete project record                        | <a href="#">Final Portfolio</a>          |

## Reproducibility Expectations

Milestone work should be reviewable in GitHub Codespaces or a documented local setup. Keep data access clear, use relative paths, and avoid manual edits to generated outputs unless the edit is explicitly documented.

## AI-Use Expectations

AI tools may help with drafting, code review, debugging, and clarity checks. Students remain responsible for correctness. Each milestone includes an AI-use note so reviewers can see what was AI-assisted and what was verified by the team.

## Definition Of Done

A milestone is ready when the required files are present, the repository is synced, the Canvas submission points to the correct commit, and another student could understand what changed without private explanation.

# Milestone Template

Use this template for project milestones M0-M5 and the Final Portfolio.

## Deliverable

- **Milestone:**
- **Due:**
- **Submit:**
- **Main artifact:**

## Required Evidence

- Repository link.
- Rendered report, dashboard, slides, or review document.
- Commit history showing meaningful progress.
- AI-use note.
- Reproducibility check result.

## Definition of Done

- The artifact opens or renders in the supported workflow.
- Data provenance and stewardship decisions are documented.
- The work avoids unsupported causal or predictive claims.
- A peer or TA can rerun the project from a fresh environment.

# M0: Group Formation

## Purpose

M0 makes sure every project team has a working collaboration setup before analytic work begins. By the end of this milestone, each group should know who is on the team, where the shared repository lives, how communication will happen, and how the team will handle reproducibility and AI use.

## Due Timing

Submit by the Week 4 deadline posted in Canvas. This milestone is complete/incomplete and is meant to unblock the project work that follows.

## Required Deliverables

Create a `submission/` folder in this milestone directory:

```
1 milestones/m0-group-formation/submission/
```

Add these files:

- `group-formation.md`
- `repo-access-evidence.md`
- `communication-plan.md`
- `codespace-check.md`
- `ai-use-note.md`

## What To Submit

Submit your GitHub repository link on Canvas after committing and syncing the files above. In the Canvas comment box, include the latest commit hash for the submission.

## File Guidance

`group-formation.md` should include:

- group name
- student names
- GitHub usernames
- one-sentence project area or dataset interest
- primary repository owner

`repo-access-evidence.md` should include:

- shared repository link
- confirmation that all group members can view the repo
- confirmation that all group members can open the repo in Codespaces or a local equivalent

`communication-plan.md` should include:

- primary communication channel
- expected response window
- weekly check-in plan
- how the group will document decisions

`codespace-check.md` should include:

- who tested the repository
- what command or render was tested
- whether the test passed
- any setup issue and the planned fix

`ai-use-note.md` should include:

- whether AI tools were used to draft the group plan
- what the group verified independently
- what the group will not outsource to AI

## Reproducibility Expectations

Use relative paths and keep project files inside the shared repository. If a group member needs a package, script, or external file to reproduce work, document it in the repository README before the next milestone.

## AI-Use Expectations

AI tools may help draft communication norms or check clarity. The team remains responsible for verifying repository access, Codespaces setup, project scope, and all submitted claims.

## Grading Checklist

This milestone is marked complete when:

- all group members are listed with GitHub usernames
- shared repository access is documented
- communication norms are specific enough to use
- Codespaces or local setup has been tested
- AI-use expectations are acknowledged
- files are committed, synced, and linked through Canvas

## Definition Of Done

M0 is done when every group member can reach the shared repository, knows the team communication plan, and can explain how the project will be kept reproducible from this point forward.

# M1: Project Proposal

## Purpose

M1 turns a broad project idea into a scoped, feasible, and ethically grounded analytic plan. The proposal should make the audience, data source, question, reproducibility plan, and first visualization direction clear enough for feedback.

## Due Timing

Submit by the Week 6 deadline posted in Canvas. This milestone prepares the group for Week 7 preliminary analysis and Week 8 dashboard prototyping.

## Required Deliverables

Create or update:

```
1 milestones/m1-proposal/submission/
```

Include these files:

- `proposal.md`
- `data-intake-card.md`
- `initial-visual-plan.md`
- `reproducibility-plan.md`
- `ai-use-note.md`

## What To Submit

Submit your GitHub repository link on Canvas after committing and syncing the files above. In the Canvas comment box, include the latest commit hash and the folder path for the milestone.

## Proposal Guidance

`proposal.md` should include:

- project title
- group members
- audience and KT purpose
- one primary question
- one secondary question
- expected data source
- planned analytic output
- known limitations

`data-intake-card.md` should summarize:

- data source and steward
- access method
- file format
- unit of analysis
- key variables
- privacy and stewardship considerations
- Indigenous Data Sovereignty or CARE/OCAP awareness when relevant

`initial-visual-plan.md` should include:

- one planned figure
- what the figure should help the audience decide or understand
- one risk of misinterpretation

`reproducibility-plan.md` should include:

- expected repository folders



- data access plan
- package/dependency notes
- how another student could rerun the work

## Reproducibility Expectations

Use relative paths. Do not commit restricted, private, or identifiable data. If public data are too large for GitHub, commit a small sample or documented access script and explain how the full file is obtained.

## AI-Use Expectations

AI may help brainstorm wording, search terms, or code structure. Students must verify data-source claims, privacy considerations, citations, and feasibility. Include an `ai-use-note.md` describing what AI helped with and what was checked manually.

## Grading Checklist

This milestone is marked complete when:

- the question is specific and feasible
- the audience and KT purpose are named
- data provenance and access are documented
- privacy, stewardship, and equity risks are acknowledged
- at least one visualization direction is described
- reproducibility plan is concrete
- AI use is disclosed and audited

## Definition Of Done

M1 is done when another group can read the proposal and understand what you will analyze, why it matters, what data you will use, and how the work will be rerun.

## M2: Preliminary Analysis

### Purpose

M2 checks that the project can move from proposal to evidence. The milestone should show a documented cohort or analytic sample, a missingness check, a descriptive summary, and a careful interpretation that avoids overclaiming.

### Due Timing

Submit by the Week 7 deadline posted in Canvas. This milestone prepares the group for Week 8 dashboard prototyping and Week 9 communication work.

### Required Deliverables

Create or update:

1 `milestones/m2-preliminary-analysis/submission/`

Include these files:

- `preliminary-analysis.qmd`
- `preliminary-analysis.html`
- `table1.csv`
- `missingness-summary.csv`
- `methods-note.md`
- `ai-audit-note.md`

## What To Submit

Submit your GitHub repository link on Canvas after committing and syncing the files above. In the Canvas comment box, include the latest commit hash and the folder path for the milestone.

## Analysis Requirements

Your preliminary analysis should include:

- a clear analytic sample or cohort definition
- a short explanation of inclusion and exclusion decisions
- a missingness check for key variables
- a Table 1-style descriptive summary
- at least one figure or table generated by code
- a methods note that states what the summary can and cannot claim

## Reproducibility Expectations

The `.qmd` file must render in GitHub Codespaces or a documented local environment. Use relative paths, document packages, and avoid manual edits to generated tables or figures.

## AI-Use Expectations

AI may help draft code, suggest summaries, or review language. Students must check row counts, filters, missingness handling, labels, and interpretation. The `ai-audit-note.md` must identify at least three checks performed by the group.

## Grading Checklist

This milestone is marked complete when:

- cohort definition is explicit
- missingness is reported
- Table 1-style output includes counts and descriptive statistics
- code-generated output is reproducible
- interpretation is descriptive and non-causal unless justified
- AI-generated help is disclosed and audited
- files are committed, synced, and submitted through Canvas

## Definition Of Done

M2 is done when a peer can rerun the analysis, see the same descriptive outputs, and understand the limits of the claims being made.

# M3: Project Update

## Purpose

M3 turns the preliminary analysis into a project draft that is ready for feedback. The update should show the current evidence, the intended KT product, and the main decisions that still need peer or instructor critique.

## Due Timing

Submit by the Week 10 deadline posted in Canvas. This milestone supports report drafting, publishing checks, and the Week 11 reproducibility review.

## Required Deliverables

Create or update:

1 `milestones/m3-project-update/submission/`

Include these files:

- `project-update.md`
- `report-draft.qmd`
- `report-draft.html`
- `dashboard-preview.md`
- `reproducibility-note.md`
- `feedback-request.md`
- `ai-use-note.md`

## What To Submit

Submit your GitHub repository link on Canvas after committing and syncing the files above. In the Canvas comment box, include the latest commit hash, the folder path, and the rendered report link if one is available.

## Update Requirements

`project-update.md` should include:

- current project title and audience
- refined research or KT question
- data source and analytic sample
- one finding that appears stable
- one finding or choice that needs review
- next steps before peer review

`dashboard-preview.md` should include:

- link or screenshot path for the dashboard-style product
- intended audience
- one interaction, filter, or decision-support feature
- one limitation or privacy caution

`feedback-request.md` should ask for targeted feedback on:

- correctness
- interpretation
- visual clarity
- reproducibility
- audience fit

## Reproducibility Expectations

The report draft must render in GitHub Codespaces or a documented local environment. Relative paths are required. Generated figures and tables should come from code, not manual copying.

## AI-Use Expectations

AI may help revise prose, identify unclear sections, or suggest checks. Students must verify citations, numeric results, interpretations, and any code suggested by AI. The `ai-use-note.md` should name the main AI-assisted tasks and the verification steps.

## Grading Checklist

This milestone is marked complete when:

- report draft renders
- current findings are supported by code-generated evidence
- dashboard-style product is visible or clearly previewed
- feedback request is specific
- reproducibility note explains how to rerun the work
- AI use is disclosed and audited

## Definition Of Done

M3 is done when another student can open the repo, render the draft report, review the dashboard preview, and provide useful feedback without guessing what the project is trying to do.

# M4: Peer Review

## Purpose

M4 gives each team structured feedback on correctness, reproducibility, interpretation, and audience fit before final submission. The review should be specific, respectful, and useful enough that the receiving group can act on it.

## Due Timing

Submit by the Week 11 deadline posted in Canvas. Peer review supports the final presentation and final portfolio revision.

## Required Deliverables

Create or update:

1 `milestones/m4-peer-review/submission/`

Include these files:

- `peer-review.md`
- `reproducibility-check.md`
- `auditability-check.md`
- `kt-feedback.md`
- `reviewer-reflection.md`
- `ai-use-note.md`



## What To Submit

Submit your GitHub repository link on Canvas after committing and syncing the files above. In the Canvas comment box, include the reviewed group name and the latest commit hash for your submission.

## Review Requirements

`peer-review.md` should include:

- reviewed project title
- two strengths
- three prioritized revision suggestions
- one question for the group
- one issue that would block final reproducibility if unfixed

`reproducibility-check.md` should document:

- whether the report rendered
- whether paths were relative
- whether data access was clear
- whether figures and tables were generated from code
- any error messages or setup blockers

`auditability-check.md` should document:

- whether data provenance is clear
- whether AI use is disclosed
- whether claims are supported by evidence
- whether limitations are stated

`kt-feedback.md` should focus on:

- audience fit
- plain-language clarity
- visual clarity
- actionability
- privacy-safe communication

## Reproducibility Expectations

Run the reviewed project in GitHub Codespaces when possible. If a full rerun is not possible, document exactly where the process stopped and what evidence you reviewed instead.

## AI-Use Expectations

AI may help organize feedback or identify unclear writing. Reviewers must independently inspect the files, outputs, and claims. Do not use AI to invent results or infer evidence that is not in the reviewed repository.

## Grading Checklist

This milestone is marked complete when:

- feedback is specific and actionable
- reproducibility check is documented
- auditability and KT feedback are included
- at least one blocking risk is named or the absence of blockers is justified
- AI use is disclosed
- files are committed, synced, and submitted through Canvas

## Definition Of Done

M4 is done when the reviewed group can use your comments to improve the final portfolio without needing a follow-up meeting to understand the feedback.

# M5: Final Presentation

## Purpose

M5 is the oral KT version of the final project. The presentation should explain the audience, question, data, one or two main findings, limitations, and the practical meaning of the work without overclaiming.

## Due Timing

Presentation order and date are posted in Canvas. Submit slides before your presentation block. The standard timing is 7 minutes of presentation, 3 minutes of questions, and 1 minute for transition.

## Required Deliverables

Create or update:

1 [milestones/m5-presentation/submission/](#)

Include these files:

- `slides.qmd`
- `slides.html` or `slides.pdf`
- `source-and-limitations-note.md`
- `presentation-checklist.md`
- `peer-feedback-received.md`
- `ai-use-note.md`

## What To Submit

Submit your GitHub repository link on Canvas after committing and syncing the files above. In the Canvas comment box, include the latest commit hash, the slide path, and any rendered slide link.

## Slide Requirements

Use 5 to 7 slides:

1. Title, team, audience, and KT purpose
2. Question and why it matters
3. Data source, analytic sample, and stewardship note
4. Main figure or table
5. Interpretation and limitation
6. Practical implication or next step
7. Backup or Q&A slide if useful

## Source And Limitations Note

`source-and-limitations-note.md` should include:

- data source and citation
- key inclusion or exclusion choices
- whether results are descriptive, predictive, or causal
- one privacy or stewardship caution
- one limitation that should be stated aloud

## Reproducibility Expectations

Slides should render in GitHub Codespaces or a documented local environment. Figures and tables should come from the project repository whenever possible. Screenshots may be used for dashboards, but the source page or app path must be documented.

## AI-Use Expectations

AI may help improve slide wording, draft speaker notes, or identify jargon. Students must verify all numbers, citations, screenshots, and claims. The `ai-use-note.md` should state what AI helped with and what was checked manually.

## Presentation Rubric

Use this checklist while preparing:

- audience and KT purpose are clear
- question is focused
- data source and limitations are transparent
- main visual is readable from the back of the room
- interpretation is accurate and non-causal unless justified
- privacy and stewardship risks are acknowledged
- timing fits the presentation window
- questions are answered with evidence and humility

## Definition Of Done

M5 is done when the slides are committed, rendered, submitted through Canvas, and ready to present without relying on local-only files.

# Final Portfolio

## Purpose

The final portfolio is the complete project record: report, code, data documentation, dashboard-style KT product, presentation materials, reproducibility evidence, and AI-use audit. It should let another person understand what was done, why it matters, and how to rerun or inspect the work.

## Due Timing

Submit by the final portfolio deadline posted in Canvas. The syllabus deadline is the course final submission deadline unless Canvas states a more specific time.

## Required Deliverables

Create or update:

```
1 milestones/final-portfolio/submission/
```

Include these files:

- final-portfolio-checklist.md
- repository-access-note.md
- final-risk-list.md
- ai-use-note.md

Your project repository should also include:

- final report source and rendered output

- dashboard-style KT product or static preview
- data provenance and stewardship documentation
- reproducibility instructions
- presentation slides or presentation summary
- clear README for the whole project

## What To Submit

Submit your GitHub repository link on Canvas after committing and syncing all final materials. In the Canvas comment box, include:

- latest commit hash
- final report path
- dashboard or KT product path
- final portfolio checklist path
- rendered public link if the repository is public and safe to publish

## Public And Private Repository Guidance

Use a public repository only when the data and outputs are allowed to be public. Do not publish restricted, private, identifiable, or culturally sensitive data. If the repository must stay private, keep it private, grant course staff access, and submit the private repository link on Canvas.

Public repositories should avoid raw restricted data and should use documented public data, simulated data, or aggregate outputs that are safe to share. Private repositories still require clear documentation and reproducibility instructions for course staff. See [Repository Sharing Guidance](#) before changing repository visibility.

## Reproducibility Checklist

Use the [Final Portfolio Checklist](#) to confirm:

- repository opens from a fresh clone or Codespace
- README explains the project, audience, and file structure
- all paths are relative
- required packages are documented
- report renders from source
- dashboard or KT product can be opened or rerun
- generated figures and tables can be traced to code
- data provenance and access rules are documented
- privacy and stewardship risks are addressed
- AI use is disclosed and audited
- final commit is synced before Canvas submission

## AI-Use Expectations

AI may help polish writing, check code, create revision plans, or test explanations. Students must verify all outputs, citations, code, numeric results, and interpretations. The final `ai-use-note.md` should describe major AI uses across the project and the checks used to protect correctness.

## Grading Checklist

The final portfolio is ready for assessment when:

- report is complete and evidence-based
- dashboard-style KT product matches the intended audience
- data provenance and limitations are transparent
- repository is organized and reproducible
- final outputs are committed and synced
- presentation materials align with the final project
- AI-use audit is complete
- repository access route is clear



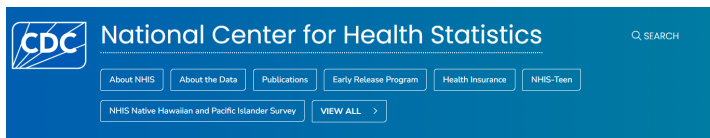
## Definition Of Done

The final portfolio is done when course staff can open the submitted repository, locate the final artifacts, rerun or inspect the work, and understand the project without asking for missing files.

## **Reference: Data Sources**

# NHIS

The National Health Interview Survey (NHIS) is a cross-sectional household interview survey conducted by the US Centers for Disease Control (CDC). The survey collects information on the health status, health care access, and health behaviors of the US population.



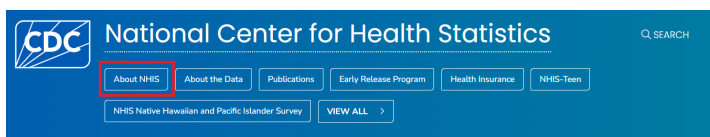
## National Health Interview Survey



[CDC website for NHIS](#)

## About NHIS

You can find more information about the NHIS survey by clicking on [About NHIS](#):



## About NHIS

For Everyone  
NOVEMBER 20, 2024

On the **About NHIS** page, you will find information about what the survey collects, data and documentation, the instructions on how to use the data, and information on related NHIS surveys.

### Overview

The National Health Interview Survey (NHIS) monitors the health of people across the country—the U.S. population—by collecting and analyzing data on a broad range of health topics. NHIS focuses on the health of U.S. children and adults.

CDC's [National Center for Health Statistics](#) (NCHS) conducts NHIS. It is the nation's largest and oldest national health survey. NHIS began collecting data in 1957.

Each year, NHIS collects data from about 27,000 adults through confidential, face-to-face interviews throughout the year. Many of these adults also provide health information about

#### ON THIS PAGE

##### Overview

[What's collected](#)

[Data and documentation](#)

[How the data are used](#)

[Related NHIS surveys](#)

You can click on each of the links or scroll down to find information on each of these topics one by one. For example, if we click on the **How the data are used** link, we will find information as follows:

### How the data are used

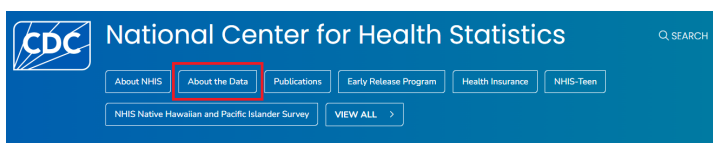
NCHS staff use NHIS data to produce official national health estimates. One of NHIS's key strengths is its ability to produce estimates for many subgroups of the U.S. population. NHIS can do this because it collects data from such a large group of people.

NHIS health statistics provide health information by—

- Demographic characteristics, like age and marital status
- Geographic location, like region and metro area size
- Social and economic circumstances, like employment status and poverty level

## About the Data

If you click on **About the Data** link, you will find information about the NHIS questionnaires, datasets, and documentation:



Under the **Overview** tab, you will find information about different NHIS cycles and the questionnaires, datasets, and documentation available in each cycle. As an example, let's click on the **2023 NHIS Questionnaires, Datasets, and Documentation** link to see the details for the 2023 NHIS:

[NHIS Data, Questionnaires and Related Documentation](#)

## Overview

Resources on this page provide information about the [National Health Interview Survey](#) (NHIS) data collection, its participants, and its protocols. It also provides analytic notes and references. They will help you understand, analyze, and use NHIS data.

Find survey instruments, data, and other resources by year at—

- [2025 NHIS Questionnaires, Datasets, and Documentation](#)
- [2024 NHIS Questionnaires, Datasets, and Documentation](#)
- [2023 NHIS Questionnaires, Datasets, and Documentation](#)
- [2022 NHIS Questionnaires, Datasets, and Documentation](#)
- [2021 NHIS Questionnaires, Datasets, and Documentation](#)
- [2020 NHIS Questionnaires, Datasets, and Documentation](#)
- [2019 NHIS Questionnaires, Datasets, and Documentation](#)

## ON THIS PAGE

### Overview

[Before 2019](#)

[Data linked to NHIS data](#)

[Other NHIS survey data](#)

[Related surveys](#)

[Related pages](#)

## NHIS Questionnaires, Datasets, and Documentation

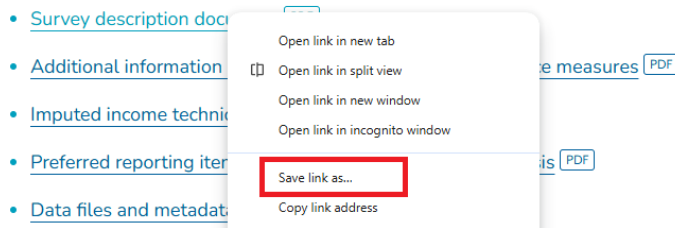
To view the description of the 2023 NHIS, click on the **Survey description document** link:

## Using our data

- [Survey description document](#) [PDF](#)

This will open the PDF of the 2023 NHIS survey description. Alternatively, you can right-click on the link, click on **Save link as...** and save the PDF file on your computer for future reference.

## Using our data



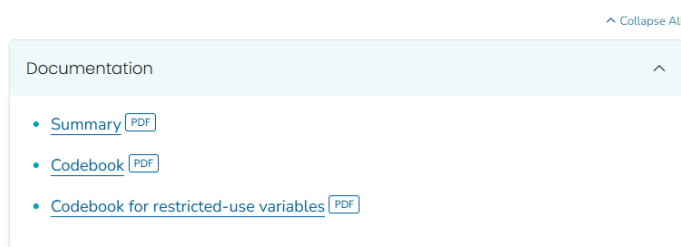
Similarly, you can open and save the questionnaires and related materials under the **Questionnaires and survey materials** tab.

In the recent NHIS, data are available in 5 categories: interview data for adults, interview data for children, data for parent-child pairs, imputed income, and paradata. You can click each link for documentation to see a summary of the data and the codebook.

Data are available in 5 categories:

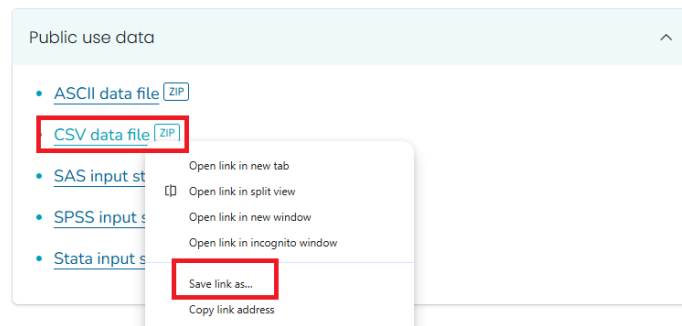
- Sample adult interview
- Sample child interview
- Parent-child pair data
- Imputed income
- Paradata

## Sample adult interview

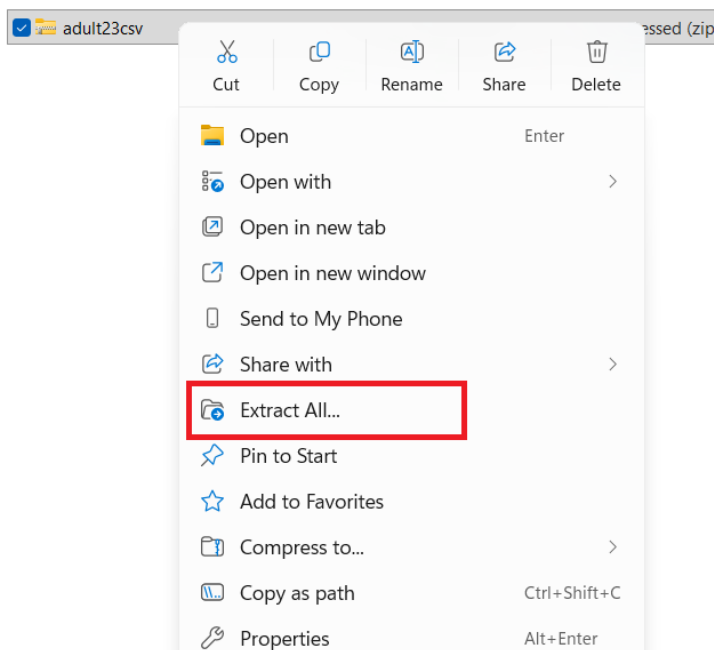


## NHIS Data

The data files are stored on the NHIS website in different formats. You can save the dataset in any statistical package that supports these file formats, e.g., ASCII, CSV, SAS. Let us save the `adult` dataset in the CSV format, which is in a zip file.



The next step is to unzip the file to extract the CSV. Depending on your computer, you can unzip the file with or without software. For example, if you are using Windows, you can right-click the zip file and select **Extract All...** to unzip it.



Once unzipped, you will find two files: the CSV file for the dataset and a readme file with some instructions. Below is the example for the `adult` dataset:

|                                                                                           |                       |           |
|-------------------------------------------------------------------------------------------|-----------------------|-----------|
|  adult23 | Microsoft Excel Co... | 28,738 KB |
|  readme  | Text Document         | 3 KB      |

To open the datafile in Excel, - you can double click on the CSV file, or - right click on the CSV file and click on **Open**.

Later, you will learn how to open the data file in different statistical software. For Excel, it would take a few seconds/minutes because of the large size of the dataset. Once you open the data file, you will see the data in a tabular format with rows and columns.

| URBRRL | RATCAT_A | INCTCFLG_A | IMPINCFLG_A | LANGSPECR_A | LANGSOC_A | LANGDOC_A | LANGMED_A | LANGHM_A | PPSU | PSTRAT |
|--------|----------|------------|-------------|-------------|-----------|-----------|-----------|----------|------|--------|
| 3      | 4        | 0          | 0           |             |           |           |           |          | 2    | 103    |
| 4      | 8        | 0          | 0           |             |           |           |           |          | 2    | 122    |
| 4      | 14       | 0          | 0           |             |           |           |           |          | 2    | 122    |
| 4      | 10       | 0          | 0           |             |           |           |           |          | 2    | 122    |
| 4      | 5        | 0          | 0           |             |           |           |           |          | 2    | 122    |
| 4      | 14       | 0          | 0           |             |           |           |           |          | 2    | 122    |
| 3      | 12       | 0          | 0           |             |           |           |           |          | 2    | 103    |

- Each row represents a person.
- Each column represents a variable.
- The first row contains the variable names. The remaining rows contain the data for each person.

## Use of the codebook

To understand the variables in the dataset, you can refer to the codebook. The codebook provides information about the variable names, labels, and values. For example, we want to understand the variable `SEX_A`:



| FQ      | FR    | FS     | FT    | FU       | FV       |
|---------|-------|--------|-------|----------|----------|
| SRVY_YR | SEX_A | AGEP_A | AGE65 | ASTATNEV | MENTHOLC |
| 2023    | 1     | 67     |       | 1        |          |
| 2023    | 1     | 73     |       | 1        |          |
| 2023    | 1     | 48     |       | 1        |          |
| 2023    | 2     | 42     |       | 1        |          |
| 2023    | 2     | 50     |       | 1        |          |
| 2023    | 2     | 46     |       | 1        |          |
| 2023    | 2     | 36     |       | 1        |          |
| 2023    | 1     | 44     |       | 1        |          |

We can look for `SEX_A` in the codebook, which will provide us with the variable label and the values for that variable (e.g., 1 for Male, 2 for Female, and so on).

17 / 653 | - 117% + | | | | | | |

---

2023 NATIONAL HEALTH INTERVIEW SURVEY (NHIS)  
Codebook for Sample Adult File (Version: 24 June 2024)  
PUBLIC USE

---

**Variable Information:**

Variable: **SEX\_A**

Module: Roster

Section: HHC : Household Composition

File(s): Adult

Data Type: Numeric

Length: 1

Question Text:

Description: Sex of Sample Adult

Recode: Yes

Universe: HHSTAT\_A=1

Universe Description: Sample adults 18+

Sources:

Question ID:

Keywords:

Notes:

Evaluation Report:

**Unweighted frequencies:**

| SEX_A | Sex of Sample Adult | Frequency | Percent |
|-------|---------------------|-----------|---------|
| 1     | Male                | 13457     | 45.58   |
| 2     | Female              | 16059     | 54.40   |
| 7     | Refused             | 4         | 0.01    |
| 8     | Not Ascertained     | 0         | 0.00    |
| 9     | Don't Know          | 2         | 0.01    |

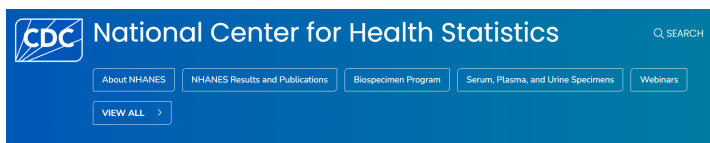
Frequency Missing:

There are so many variables in the dataset, and it is not possible to understand all of them at once. You can refer to the codebook to understand the variables that you are interested in for your analysis. For example, the `WTFA_A` variable is the sample adult weight variable used to calculate weighted estimates

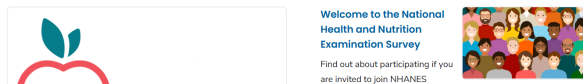
for the US adult population. Similarly, the `HYPEV_A` variable is the hypertension ever variable, indicating whether a doctor or other health professional has ever told the person they have hypertension. Again, you can refer to the codebook to understand the variables that you are interested in.

# NHANES

The National Health and Nutrition Examination Survey (NHANES) is a cross-sectional survey conducted by the US Centers for Disease Control (CDC). The survey collects information on the health and nutritional status of the US population. The survey includes interviews, physical examinations, and laboratory tests.



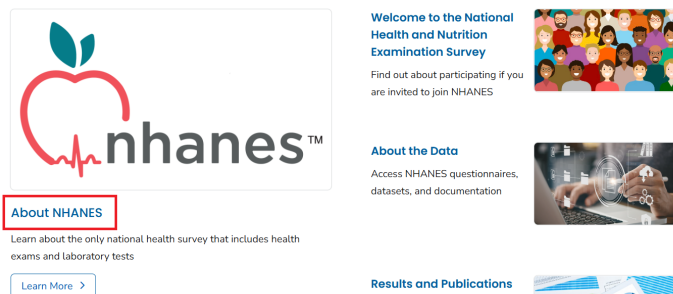
## National Health and Nutrition Examination Survey



[CDC website for NHANES](#)

## About NHANES

You can find more information about the NHANES survey by clicking on [About NHANES](#):



## About NHANES

For Everyone  
DECEMBER 18, 2024

### AT A GLANCE

- NHANES is the National Health and Nutrition Examination Survey.
- NHANES is a national survey that measures the health and nutrition of adults and children in the United States.
- NHANES is the only national health survey that includes health exams and laboratory tests for participants of all ages.



On the **About NHANES** page, you will find information about what the survey collects, data and documentation, the instructions on how to use the data, and information on related NHANES surveys.

## Overview

The National Health and Nutrition Examination Survey (NHANES) collects data about the health of adults and children in the United States. We also collect data about what participants eat, drink, and take as supplements to determine how many nutrients are in their diet. These dietary interviews and blood tests help us measure the nutritional status of U.S. adults and children.

CDC's [National Center for Health Statistics](#) (NCHS) conducts NHANES. NHANES is the only national health survey that includes health exams, laboratory tests, and dietary interviews for participants of all ages.

### ON THIS PAGE

- Overview
- What's collected
- Data and documentation
- How the data are used
- Program director
- Contact NHANES

You can click on each of the links or scroll down to find information on each of these topics one by one. For example, if we click on the **Data and documentation** link, we will find information as follows:

## Data and documentation

NCHS [publishes reports](#) featuring analysis of NHANES data. Selected estimates also are available as part of [interactive data visualizations](#) that provide tables and charts.

### Data files

[NHANES data files](#) and [related documentation](#) are publicly available to download and analyze. The related documentation can help researchers understand and use NHANES data. All potentially identifiable information has been removed to ensure the confidentiality of participants and their households.

[NHANES restricted data files](#) are available through the NCHS Research Data Center (RDC) for a fee. To access restricted NHANES data, researchers must submit requests to the

### RELATED PAGES

- What NHANES Covers and How It Works
- Who Participates In NHANES
- How People Use NHANES Data
- What NHANES Data Have Achieved
- Ethics Review Board Approval

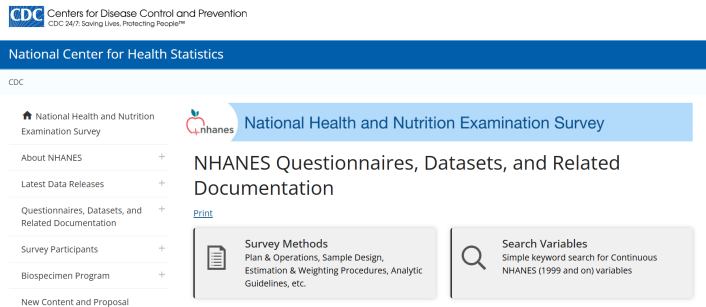
[VIEW ALL](#)  
National Health and Nutrition  
Examination Survey

## Survey Methods and Analytic Guidelines

If you click on **NHANES data files and related documentation** link, you will find information about the NHANES questionnaires, datasets, and documentation:

## Data files

[NHANES data files and related documentation](#) are publicly available to download and analyze. The related documentation can help researchers understand and use NHANES data. All potentially identifiable information has been removed to ensure the confidentiality of participants and their households.

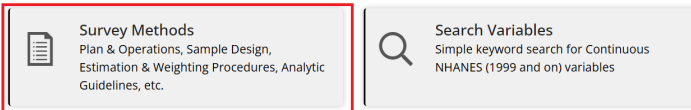


You can click **Survey Methods** to find information on survey methods and analytic guidelines. The **Survey Methods and Analytic Guidelines** page provides information about the plan and operation of the NHANES survey, the sampling design, methods for calculating weights and variance estimation, and guidelines for analyzing NHANES data, along with links to related resources.

[NHANES data files and related documentation](#)

## NHANES Questionnaires, Datasets, and Related Documentation

[Print](#)



## Search Variables

If you click on **Search Variables** link, you will find a search tool to search for variables in the NHANES datasets (1999 and on).

## NHANES Questionnaires, Datasets, and Related Documentation

[Print](#)

**Survey Methods**  
Plan & Operations, Sample Design,  
Estimation & Weighting Procedures, Analytic  
Guidelines, etc.

**Search Variables**  
Simple keyword search for Continuous  
NHANES (1999 and on) variables

You can search for variables by entering keywords in the search box. For example, if you enter **diabetes** in the search box, you will find a list of variables related to diabetes in the NHANES datasets.

### Search Variables

[Print](#)

Search for words and "quoted phrases" in selected fields (Data File Name, SAS Label, Variable Description, Variable Name) of Continuous NHANES (1999 and on) variables that have published documentation. *Multiply imputed Dual Energy X-Ray Absorptiometry variables are not included.*

Refer to the [Survey Content Brochure](#) for the full content of Continuous NHANES.

|                         |                                            |
|-------------------------|--------------------------------------------|
| Search Term             | <input type="text" value="diabetes"/>      |
| Fields to Search        | <input type="text" value="All"/>           |
| Sort By                 | <input type="text" value="Variable Name"/> |
| Limited Access          | <input type="text" value="Exclude"/>       |
| Release Cycle           | <input type="text" value="All"/>           |
| Search Result Page Size | <input type="text" value="50"/>            |

Search

Number of variables found: 459

| Variable Name | SAS Label                            | Variable Description                                                                                                                            | Data File Name   | Component Link                          |
|---------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------|
| DID040        | Age when first told you had diabetes | How old (was SP/were you) when a doctor or other health professional first told (you/him/her) that (you/he/she) had diabetes or sugar diabetes? | Diabetes (DIQ_D) | <a href="#">2005-2006 Questionnaire</a> |
| DID040        | Age when first told you had diabetes | How old (was SP/were you) when a doctor or other health professional first told (you/him/her) that (you/he/she) had diabetes or sugar diabetes? | Diabetes (DIQ_E) | <a href="#">2007-2008 Questionnaire</a> |
| DID040        | Age when first told you had diabetes | How old (was SP/were you) when a doctor or other health professional first told (you/him/her) that (you/he/she) had diabetes or sugar diabetes? | Diabetes (DIQ_F) | <a href="#">2009-2010 Questionnaire</a> |
| DID040        | Age when first told you had          | How old (was SP/were you) when a doctor or other health professional first told (you/him/her) that                                              | Diabetes (DIQ_G) | <a href="#">2011-2012 Questionnaire</a> |

You can restrict your search to a specific NHANES cycle by selecting the cycle from the drop-down menu. For example, if you select the **2017–2018** cycle, you will find a list of variables related to diabetes in the 2017-2018 NHANES dataset.

Limited Access

Exclude

Release Cycle

2017-2018

Search Result Page Size

50

Search

Number of variables found: 58

| Variable Name | SAS Label                                | Variable Description                                                                                    | Data File Name              | Component Link                          |
|---------------|------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------|
| DIQ175C       | Age                                      | Why (Do you/Does SP) think (you are/he is/she is) at risk for diabetes or prediabetes? [Anything else?] | Diabetes (DIQ_J)            | <a href="#">2017-2018 Questionnaire</a> |
| RHQ163        | Age told you had diabetes while pregnant | How old (were you/was SP) when (you were/she was) first told (you/she) had diabetes during a pregnancy? | Reproductive Health (RHQ_J) | <a href="#">2017-2018 Questionnaire</a> |

## NHANES Questionnaires, Datasets, and Documentation

Let's explore the questionnaires, datasets, and documentation for the 2017-2018 NHANES cycle. You need to click on **NHANES 2017-2018** under the **Continuous NHANES** tab:

### NHANES Questionnaires, Datasets, and Related Documentation

[Print](#)



**Survey Methods**  
Plan & Operations, Sample Design, Estimation & Weighting Procedures, Analytic Guidelines, etc.



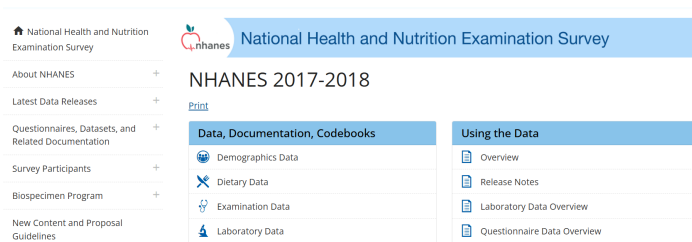
**Search Variables**  
Simple keyword search for Continuous NHANES (1999 and on) variables

[» Guidelines for High Quality Analyses of NHANES Data](#)

#### Continuous NHANES

|                  |                        |                                          |                  |
|------------------|------------------------|------------------------------------------|------------------|
| NHANES 2025-2026 | NHANES 08/2021-08/2023 | NHANES 2017-March 2020 Pre-Pandemic Data |                  |
| NHANES 2019-2020 | NHANES 2017-2018       | NHANES 2015-2016                         | NHANES 2013-2014 |

This will open the page for the 2017-2018 NHANES cycle, where you can find information on the survey description, questionnaires, datasets, and documentation.



There are six categories of data available for the 2017-2018 NHANES cycle: Demographics, Dietary, Examination, Laboratory, Questionnaire, and Limited Access data. You can click each category to find the datasets and documentation available in that category. For example, if you click on **Demographics Data**, you will find a list of datasets and documentation related to demographics data in the 2017-2018 NHANES cycle.

Data are available in 6 categories:

- Demographics
- Dietary
- Examination
- Laboratory
- Questionnaire
- Limited Access

## NHANES 2017-2018

[Print](#)

| Data, Documentation, Codebooks | Using the Data                         |
|--------------------------------|----------------------------------------|
| Demographics Data              | Overview                               |
| Dietary Data                   | Release Notes                          |
| Examination Data               | Laboratory Data Overview               |
| Laboratory Data                | Questionnaire Data Overview            |
| Questionnaire Data             | Examination Data Overview              |
| Limited Access Data            | Survey Methods and Analytic Guidelines |

## 2017-2018 Demographics Data - Continuous NHANES

[Print](#)

- [NHANES 2017-2018 Demographics Variable List](#)
- [Questionnaire Instruments](#)
- [Exam Procedure Manuals](#)
- [SAS Universal Viewer](#)
- [Data User Agreement](#)

| Data File Name                           | Doc File                   | Data File                                  | Date Published |
|------------------------------------------|----------------------------|--------------------------------------------|----------------|
| Demographic Variables and Sample Weights | <a href="#">DEMO_1.Doc</a> | <a href="#">DEMO_1 Data [XPT - 3.3 MB]</a> | February 2020  |

There is a **Doc File** available for each dataset that provides information about the dataset, including the variables it contains, their coding, and any special considerations for using it. It is important to read the **Doc** file before using the dataset to understand the data structure and how to use it properly. The **Data File** contains the dataset data in **.XPT** format, which is a SAS Transport file, not a CSV.



## 2017-2018 Demographics Data - Continuous NHANES

[Print](#)

- [NHANES 2017-2018 Demographics Variable List](#)
- [Questionnaire Instruments](#)
- [Exam Procedure Manuals](#)
- [SAS Universal Viewer](#)
- [Data User Agreement](#)

| Data File Name                           | Doc File                   | Data File                                  | Date Published |
|------------------------------------------|----------------------------|--------------------------------------------|----------------|
| Demographic Variables and Sample Weights | <a href="#">DEMO_J.Doc</a> | <a href="#">DEMO_J Data [XPT - 3.3 MB]</a> | February 2020  |

## Data

You need to click on the [DEMO\\_J Data \[XPT - 3.3 MB\]](#) link to download the dataset. Note that you cannot easily open an .XPT file in Excel or Google Sheets. In the real world, data isn't always in a friendly format. For this week, let's download the `NHANES_Demo_Converted.csv` file from our Canvas page to look at the rows. Later, we will use an AI co-pilot to write R code that reads .XPT files automatically.

Let's open the `NHANES_Demo_Converted.csv` file in Excel and see the first few rows:

|    | A     | B        | C        | D        | E        | F        | G        |
|----|-------|----------|----------|----------|----------|----------|----------|
| 1  | SEQN  | SDDSRVYR | RIDSTATR | RIAGENDR | RIDAGEYR | RIDAGEMN | RIDRETH1 |
| 2  | 93703 | 10       | 2        | 2        | 2        | NA       | 5        |
| 3  | 93704 | 10       | 2        | 1        | 2        | NA       | 3        |
| 4  | 93705 | 10       | 2        | 2        | 66       | NA       | 4        |
| 5  | 93706 | 10       | 2        | 1        | 18       | NA       | 5        |
| 6  | 93707 | 10       | 2        | 1        | 13       | NA       | 5        |
| 7  | 93708 | 10       | 2        | 2        | 66       | NA       | 5        |
| 8  | 93709 | 10       | 2        | 2        | 75       | NA       | 4        |
| 9  | 93710 | 10       | 2        | 2        | 0        | 11       | 3        |
| 10 | 93711 | 10       | 2        | 1        | 56       | NA       | 5        |
| 11 | 93712 | 10       | 2        | 1        | 18       | NA       | 1        |
| 12 | 93713 | 10       | 2        | 1        | 67       | NA       | 3        |
| 13 | 93714 | 10       | 2        | 2        | 54       | NA       | 4        |
| 14 | 93715 | 10       | 2        | 1        | 71       | NA       | 5        |

- Each row represents a person.
- Each column represents a variable.
- The first row contains the variable names. The remaining rows contain the data for each person.

For example, `SEQN` is the unique identifier for each person in the dataset, and `RIAGENDR` is the variable for gender.

## Use of the Codebook

To understand the variables in the dataset, you can refer to the codebook. The codebook provides information about the variable names, labels, and values. The Doc File explained earlier contains the codebook for the dataset. You need to click on the DEMO\_J Doc link to open the codebook on a new page. Let's we want to understand the variables SEQN and RIAGENDR:

### National Health and Nutrition Examination Survey

2017-2018 Data Documentation, Codebook, and Frequencies

Demographic Variables and Sample Weights (DEMO\_J)

Data File: DEMO\_J.xpt

First Published: February 2020

Last Revised: NA

#### Component Description

The demographics file provides individual, family, and household-level information on the following topics:

- Survey participant's household interview and examination status;

#### TABLE OF CONTENTS

- Component Description
- Eligible Sample
- Interview Setting and Mode of Administration
- Quality Assurance & Quality Control
- Data Processing and Editing
- Analytic Notes
- References
- Codebook
  - SEQN - Respondent sequence number
  - SDDSRVYR - Data release cycle
  - RIDSTATR - Interview/Examination status
  - RIAGENDR - Gender
  - RIDAGEYR - Age in years at screening
  - RIDAGEMN - Age in months at screening - 0 to 24 mos

You need to click on the **SEQN - Respondent sequence number** link or scroll down to find the information about the SEQN variable.

#### SEQN - Respondent sequence number

**Variable Name:** SEQN  
**SAS Label:** Respondent sequence number  
**English Text:** Respondent sequence number.  
**Target:** Both males and females 0 YEARS - 150 YEARS

#### SDDSRVYR - Data release cycle

**Variable Name:** SDDSRVYR  
**SAS Label:** Data release cycle  
**English Text:** Data release cycle

#### TABLE OF CONTENTS

- Component Description
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- Data Processing and Editing
- Analytic Notes
- References
- Codebook
  - SEQN - Respondent sequence number
  - SDDSRVYR - Data release cycle
  - RIDSTATR - Interview/Examination status
  - RIAGENDR - Gender

Similarly, you can click on the **RIAGENDR - Gender** link or scroll down to find the information about the RIAGENDR variable.

#### RIAGENDR - Gender

**Variable Name:** RIAGENDR  
**SAS Label:** Gender  
**English Text:** Gender of the participant.  
**Target:** Both males and females 0 YEARS - 150 YEARS

| Code or Value | Value Description | Count | Cumulative | Skip to Item |
|---------------|-------------------|-------|------------|--------------|
| 1             | Male              | 4557  | 4557       |              |
| 2             | Female            | 4697  | 9254       |              |
| .             | Missing           | 0     | 9254       |              |

#### RIDAGEYR - Age in years at screening

**Variable Name:** RIDAGEYR

#### TABLE OF CONTENTS

- Component Description
- Eligible Sample
- Interview Setting and Mode of Administration
- Quality Assurance & Quality Control
- Data Processing and Editing
- Analytic Notes
- References
- Codebook
  - SEQN - Respondent sequence number
  - SDDSRVYR - Data release cycle
  - RIDSTATR - Interview/Examination status
  - RIAGENDR - Gender
  - RIDAGEYR - Age in years at screening
  - RIDAGEMN - Age in months at screening - 0 to 24 mos

As you can see, the **RIAGENDR** variable has two values: 1 for Male and 2 for Female, without any missing values. You can also see the frequency of each value in the dataset. These numbers/frequencies should exactly match the frequencies in your dataset. You can use the codebook to understand other variables in the dataset. Similarly, you can use the other data files (e.g., Dietary) and their codebook to understand the variables in those datasets. Unlike the demographic file, which uses only one dataset and one codebook, the other files use multiple datasets and codebooks. As an example, if you click on the **Questionnaire Data** link, you will find multiple datasets and codebooks for questionnaire data:

## 2017-2018 Questionnaire Data - Continuous NHANES

[Print](#)

- [NHANES 2017-2018 Questionnaire Variable List](#)
- [Questionnaire Instruments](#)
- [2017-2018 Questionnaire Data Overview](#)
- [SAS Universal Viewer](#)
- [Data User Agreement](#)

| Data File Name                    | Doc File                     | Data File                                   | Date Published |
|-----------------------------------|------------------------------|---------------------------------------------|----------------|
| Acculturation                     | <a href="#">ACO.J.Doc</a>    | <a href="#">ACO.J.Data [XPT - 396.3 KB]</a> | February 2020  |
| Alcohol Use                       | <a href="#">ALO.J.Doc</a>    | <a href="#">ALO.J.Data [XPT - 434.4 KB]</a> | December 2020  |
| Audiometry                        | <a href="#">AUO.J.Doc</a>    | <a href="#">AUO.J.Data [XPT - 3.9 MB]</a>   | July 2020      |
| Blood Pressure & Cholesterol      | <a href="#">BPO.J.Doc</a>    | <a href="#">BPO.J.Data [XPT - 531.8 KB]</a> | February 2020  |
| Cardiovascular Health             | <a href="#">CDO.J.Doc</a>    | <a href="#">CDO.J.Data [XPT - 518.7 KB]</a> | February 2020  |
| Consumer Behavior                 | <a href="#">CBO.J.Doc</a>    | <a href="#">CBO.J.Data [XPT - 435.4 KB]</a> | April 2021     |
| Consumer Behavior Phone Follow-up | <a href="#">CBOPFA.J.Doc</a> | <a href="#">CBOPFA.J.Data [XPT - 2.7]</a>   | June 2021      |

# BRFSS

The Behavioral Risk Factor Surveillance System (BRFSS) is a cross-sectional survey conducted by the US Centers for Disease Control (CDC). The survey collects information on health-related risk behaviors, chronic health conditions, and use of preventive services among US adults. BRFSS completes more than 400,000 adult interviews each year, making it the largest continuously conducted health survey system in the world.



[CDC website for BRFSS](#)

## About BRFSS

You can find more information about the BRFSS survey by clicking on [BRFSS FAQs](#):



On the **BRFSS FAQs** page, you will find information about the BRFSS survey, including what the survey collects, how the survey is conducted, and how to use the data.

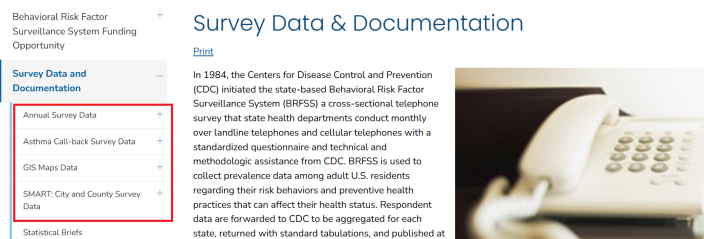
## Survey Data & Documentation

If you click on **Survey Data & Documentation** link, you will find information about the BRFSS questionnaires, datasets, and documentation:



There are four types of data files available for download: 1) annual data, 2) asthma call-back data, 3) GIS map data, and 4) SMART data. The annual data files include information on health-related risk behaviors, chronic health conditions, and use of preventive services among US adults. The asthma call-back data files include information on asthma-related health outcomes and healthcare utilization among adults with asthma. The GIS map data files include information on the geographic distribution of the data. The SMART (Selected Metropolitan/Micropolitan Area Risk Trends) data provide prevalence rates for selected conditions and behaviors for cities and their surrounding counties.

## BRFSS Survey Data & Documentation



For the annual data, you can click on the **Annual Survey Data** link to find information about the annual data files, including how to access the data, how to use the data, and how to analyze the data. There are two types of annual data files available for download: 1) the raw data files and 2) the prevalence and trends database.

## Annual Survey Data

There are separate links for each year of data, and you can click on each link to find information about the data for that year. For example, if you click on the **2023 Annual Survey Data** link, you will find information about the 2023 annual data files, including how to access the data, how to use the data, and how to analyze the data.

### Annual Survey Data

| 2017 - 2024 Annual Survey Data          |                                         |
|-----------------------------------------|-----------------------------------------|
| <a href="#">2024 Annual Survey Data</a> | <a href="#">2020 Annual Survey Data</a> |
| <a href="#">2023 Annual Survey Data</a> | <a href="#">2019 Annual Survey Data</a> |
| <a href="#">2022 Annual Survey Data</a> | <a href="#">2018 Annual Survey Data</a> |
| <a href="#">2021 Annual Survey Data</a> | <a href="#">2017 Annual Survey Data</a> |

## 2023 BRFSS Survey Data and Documentation

[Print](#)

The 2023 BRFSS data continue to reflect the changes initially made in 2011 for weighting methodology (raking) and adding cell-phone-only respondents. The aggregate BRFSS combined landline and cell phone data set is built from the landline and cell phone data submitted for 2023 and includes data from 48 states, the District of Columbia, Guam, Puerto Rico, and the US Virgin Islands. During 2023, Kentucky and Pennsylvania were unable to collect enough data to meet the minimum requirements to be included in this public data set. The data set has been modified to comply with President Trump's executive orders. There are some missing values which may appear to be inconsistencies in the data based on a respondents' answers to questions that were removed.

### 2023 Survey Data Information

[2023 BRFSS Overview CDC](#)  [PDF – 213 KB]

The 2023 BRFSS Survey Data and Documentation page provides information about three resources: 1) data information such as the codebook and sampling weights, 2) data files in ASCII and SAS XPT transport formats, and 3) SAS resources. To download/save the data files, click the specific format you want. For example, if you click on the 2023 BRFSS Data (SAS Transport Format) link, the annual survey data for 2023 will be downloaded in a ZIP file that contains the dataset in a .XPT format.

### Data Files

There are 433,323 records for 2023. More information on participation is available in the [states conducting surveillance, by year table](#). The data files are provided in ASCII and SAS Transport formats. The September update includes updated race/ethnicity calculated variables for Mississippi. The February update includes changes to the data set which has been modified to comply with President Trump's executive orders. There are some missing values which may appear to be inconsistencies in the data based on a respondents' answers to questions that were removed.

[2023 BRFSS Data \(ASCII\)](#)  [ZIP – 41.5 MB]

February, 2025

This file for the combined landline and cell phone data set is in ASCII format. It has a fixed record length of 2111 positions.

[2023 BRFSS Data \(SAS Transport Format\)](#) 

[ZIP – 64.3 MB]

February, 2025

This file for the combined landline and cell phone data set was exported from SAS V9.4 in the XPT transport

You need to unzip the file to extract the data file in .XPT format. However, opening the data file in Excel or Google Sheets is a bit tricky. The SAS XPT (or ASCII) data file needs to be converted/formatted appropriately before it can be opened in Excel or Google Sheets. Many statistical software packages, such as R, SAS, or STATA, can read the data file in .XPT format. Below is the data file, where the column names are the variable names, and the rows are the observations (i.e., survey respondents):

|    | STATE FIPS CODE | FILE MONTH | INTERVIEW DATE | INTERVIEW MONTH | INTERVIEW DAY | INTERVIEW YEAR | FINAL DISPOSITION | ANNUAL SEQUENCE NUMBER | PRIMARY SAMPLING UNIT | CORRECT TELEPHONE NUMBER? | PRIVATE RESIDENCE? | DO YOU LIVE IN COLLEGE HOUSING? | RESIDENT OF STATE | CELLULAR TELEPHONE | ARE YOU 18 YEARS OF AGE OR OLDER? |
|----|-----------------|------------|----------------|-----------------|---------------|----------------|-------------------|------------------------|-----------------------|---------------------------|--------------------|---------------------------------|-------------------|--------------------|-----------------------------------|
| 1  | 1               | 1          | 1/03/2023      | 03              | 01            | 2023           | 1100              | 2023000001             | 2023000001            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 2  | 1               | 1          | 1/03/2023      | 01              | 06            | 2023           | 1100              | 2023000002             | 2023000002            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 3  | 1               | 1          | 1/03/2023      | 03              | 08            | 2023           | 1100              | 2023000003             | 2023000003            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 4  | 1               | 1          | 1/03/2023      | 03              | 06            | 2023           | 1100              | 2023000004             | 2023000004            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 5  | 1               | 1          | 1/03/2023      | 01              | 06            | 2023           | 1100              | 2023000005             | 2023000005            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 6  | 1               | 1          | 1/03/2023      | 01              | 09            | 2023           | 1100              | 2023000006             | 2023000006            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 7  | 1               | 1          | 1/03/2023      | 03              | 21            | 2023           | 1100              | 2023000007             | 2023000007            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 8  | 1               | 1          | 1/03/2023      | 01              | 06            | 2023           | 1100              | 2023000008             | 2023000008            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 9  | 1               | 1          | 1/03/2023      | 03              | 15            | 2023           | 1100              | 2023000009             | 2023000009            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 10 | 1               | 1          | 1/03/2023      | 01              | 08            | 2023           | 1100              | 2023000010             | 2023000010            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 11 | 1               | 1          | 1/03/2023      | 03              | 08            | 2023           | 1100              | 2023000011             | 2023000011            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 12 | 1               | 1          | 1/03/2023      | 01              | 08            | 2023           | 1100              | 2023000012             | 2023000012            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 13 | 1               | 1          | 1/03/2023      | 03              | 01            | 2023           | 1100              | 2023000013             | 2023000013            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 14 | 1               | 1          | 1/03/2023      | 01              | 06            | 2023           | 1100              | 2023000014             | 2023000014            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 15 | 1               | 1          | 1/03/2023      | 03              | 21            | 2023           | 1100              | 2023000015             | 2023000015            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |
| 16 | 1               | 1          | 1/03/2023      | 03              | 13            | 2023           | 1100              | 2023000016             | 2023000016            | 1                         | 1                  | -                               | 1                 | 2                  |                                   |

To understand the variables in the dataset, you can refer to the codebook:

### 2023 Survey Data Information

[2023 BRFSS Overview CDC](#) [PDF – 213 KB]

Provides information on the background, design, data collection and processing, and the statistical and analytical issues for the combined landline and cell phone data set.

[2023 BRFSS Codebook CDC](#) [ZIP – 3 MB]

Codebook for the file showing variable name, location, and frequency of values for all reporting areas combined for the combined landline and cell phone data set.

You need to unzip the file to extract the codebook in HTML format. The codebook provides information about the variable names, labels, and values. For example, if you want to understand the variable **Drinking and Driving**, you can find the variable name, label, and values in the codebook:

| Label: Drinking and Driving<br>Section Name: Calculated Variables<br>Module Number: 15<br>Question Number: 3<br>Column: 2110<br>Type of Variable: Num<br>SAS Variable Name: _DRNKDRV<br>Question Prologue:<br>Question: Drinking and Driving (Reported having driven at least once when perhaps had too much to drink) |                                                |           |            |                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-----------|------------|---------------------|
| Value                                                                                                                                                                                                                                                                                                                  | Value Label                                    | Frequency | Percentage | Weighted Percentage |
| 1                                                                                                                                                                                                                                                                                                                      | Have driven after having too much to drink     | 6,192     | 1.43       | 1.36                |
| 2                                                                                                                                                                                                                                                                                                                      | Have not driven after having too much to drink | 205,541   | 47.43      | 45.81               |
| 9                                                                                                                                                                                                                                                                                                                      | Don't know/Not Sure/Refused/Missing            | 221,590   | 51.14      | 52.83               |

You can similarly download the asthma call-back, GIS map, and SMART datasets. To understand the variables in those datasets, you can refer to the codebook for each dataset.

[Asthma Call-back Data](#)

[GIS Data](#)

[SMART Data](#)



## Prevalence and Trends Database

The BRFSS prevalence and trends tool allows users to view and download prevalence estimates through charts, graphs, and maps. You can explore the estimates by selecting the topic or location. For example, if you select by **topic**, e.g., for alcohol consumption for Alabama for the year 2024, you need to select the options from the dropdown menu and click on the **View Results** button to see the results:

**CDC** BRFSS Prevalence & Trends Data Search

Per a court order, HHS is required to restore this website to its version as of 12:00 AM on January 29, 2025. Information on this page may be modified and/or removed in the future subject to the terms of the court's order and implemented consistent with applicable law. Any information on this page promoting gender ideology is extremely inaccurate and disconnected from truth. The Trump Administration rejects gender ideology due to the harms and divisiveness it causes. This page does not reflect reality and therefore the Administration and this Department reject it.

[Print](#)

The enhanced BRFSS Prevalence & Trends Tool allows you to explore prevalence data by location or topic.

- Just like the original tool, this version allows you to view and download prevalence estimates through charts, graphs, and maps.
- In addition, this version also provides access to prevalence estimates from the BRFSS core data at the state level as well as data from Selected Metropolitan/Micropolitan Area Risk Trends (SMART).

These prevalence estimates have been updated to include both crude and age-adjusted prevalence.

The tool will continue to be updated as new functions and data become available.



**Explore by Location** Explore by Topic

Explore bar and trendline graphs for survey questions by selecting a state, territories or Metropolitan/Micropolitan Statistical Areas (MMSAs).

| Dataset    | Location         | Class               | Topic          | Year |
|------------|------------------|---------------------|----------------|------|
| Nationwide | States - Alabama | Alcohol Consumption | Binge Drinking | 2024 |

Question: Binge drinkers (males having five or more drinks on one occasion, females having four or more drinks on one occasion) (variable calculated from one or more BRFSS questions)

**View Results** Clear Filters

Prevalence and Trends Database

The results will show the (i) prevalence estimates for alcohol consumption in Alabama for the year 2024, and (ii) the trend of alcohol consumption in Alabama over time.

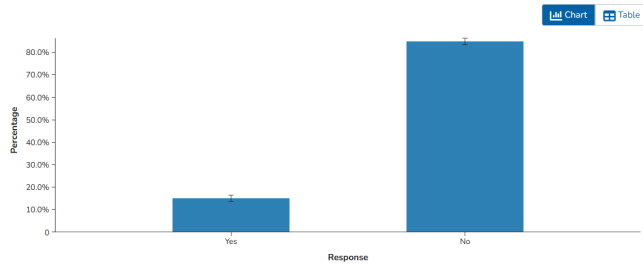
[View Results](#) [Clear Filters](#)

Binge drinkers (males having five or more drinks on one occasion, females having four or more drinks on one occasion) (variable calculated from one or more BRFSS questions)

*Alcohol Consumption: Binge Drinking*

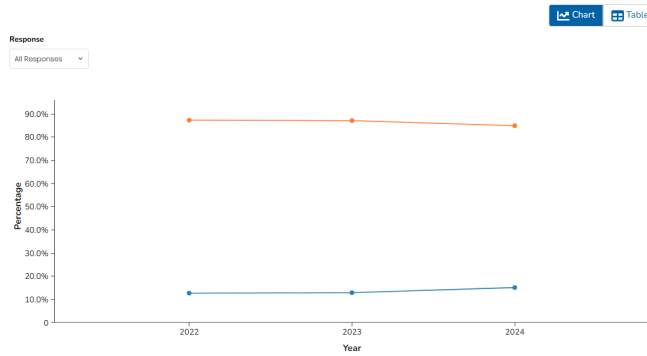
View By:  Data Type:

2024, Alabama



The ordering of subcategory rows in the table view may vary across responses.

All Available Years, Alabama



# CCS

The Canadian Cannabis Survey (CCS) is an annual survey conducted by Health Canada to gather information on cannabis use among people aged 16 years and older living in all Canadian provinces and territories. The survey collects data on various aspects of cannabis consumption, including their knowledge, attitudes and behaviours towards cannabis use.

Government of Canada / Gouvernement du Canada

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## Canadian Cannabis Survey (CCS) 2024 Public Use Microdata File (PUMF)

Have your say

\*\*\*\*\*

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The 2024 Canadian Cannabis Survey (CCS) recruited people aged 16 years and older living in all Canadian provinces and territories. Respondents completed a survey on patterns of cannabis use for medical and non-medical purposes, as well as their knowledge, attitudes and behaviours towards cannabis use. A total of 11,666 respondents completed online surveys between April 4 and July 2, 2024. Summary results are available here: <https://www.canada.ca/en/health-canada/services/drugs-medication/cannabis/research-data/canadian-cannabis-survey-2024-summary.html>

Additional

The Office Cannabis Science and Surveillance releases the PUMF for CCS 2024 to allow

More than 11,000 respondents participated in the 2024 survey. The study provides valuable insights into cannabis use patterns, the cannabis market, and issues of public safety. The survey results are published on Health Canada's website:

[Health Canada website for CCS](#)



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Additional

The 2024 Canadian Cannabis Survey (CCS) recruited people aged 16 years and older living in all Canadian provinces and territories. Respondents completed a survey on patterns of cannabis use for medical and non-medical purposes, as well as their knowledge, attitudes and behaviours towards cannabis use. A total of 11,666 respondents completed online surveys between April 4 and July 2, 2024. Summary results are available here:

<https://www.canada.ca/en/health-canada/services/drugs-medication/cannabis/research-data/canadian-cannabis-survey-2024-summary.html>

The Office Cannabis Science and Surveillance releases the PUMF for CCS 2024 to allow



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Drugs and medication
Cannabis
Cannabis research and data

Canadian Cannabis Survey 2024: Summary

On this page

- Introduction
- Results by theme
- Methods
- Considerations

## Open Licence

The data from the Canadian Cannabis Survey is available under the Open Government Licence - Canada (OGL-Canada). This licence allows users to freely use, modify, and distribute the data, provided that they attribute the source of the data and do not use it for commercial purposes. The OGL-Canada promotes transparency and encourages the use of government data for research, analysis, and public benefit. You can follow the [Open Government Licence - Canada](#) link for more information on the terms and conditions of the licence.

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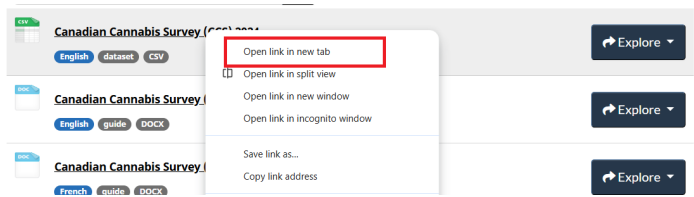
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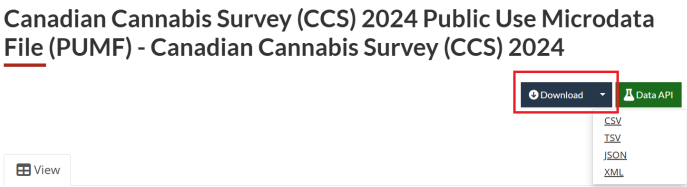
The **Data and Resources** tab on the Health Canada website provides access to the public use data, user guides, and the data dictionary.

The CCS 2024 data is available to download in various formats, including CSV, TSV, JSON, and XML. The data can be accessed through the Health Canada website, where you can choose the format that best suits your needs. You need to

open the data link (say, in a new tab in your browser) and select the desired format to download the data.



Under the **Download** tab, you can find the data in different formats. For example, you can download the data in CSV format by clicking on the **CSV** link:



After downloading, you can open the data in Excel, Google Sheets, or any statistical software. The data is structured in a tabular format, with each row representing an individual respondent and each column representing different variables:

| 1  | A      | B    | C             | D     | E   | F               | G              | H    | I          | J               | K             | L          | M            |
|----|--------|------|---------------|-------|-----|-----------------|----------------|------|------------|-----------------|---------------|------------|--------------|
| 2  | seq_id | wave | sample_weight | age65 | sex | sexuality_recod | community_size | prov | race_white | health_physical | health_mental | education3 | income_recod |
| 3  | 1      | 7    | 2622.409      | 2     | 2   | 1               | 1              | 7    | 1          | 2               | 2             | 3          | 3            |
| 4  | 2      | 7    | 1517.9545     | 4     | 1   | 1               | -7             | 5    | 1          | 2               | 3             | 3          | 2            |
| 5  | 3      | 7    | 1547.9545     | 1     | 1   | 1               | 1              | 1    | 0          | 2               | 4             | 1          | 1            |
| 6  | 4      | 7    | 3090          | 1     | 2   | 1               | 2              | 7    | 1          | 4               | 3             | 1          | 2            |
| 7  | 5      | 7    | 2479.0746     | 4     | 1   | 2               | 2              | 2    | 0          | 2               | 4             | 3          | 4            |
| 8  | 6      | 7    | 2306.2096     | 6     | 1   | 1               | 2              | 2    | 1          | 1               | 1             | 2          | 3            |
| 9  | 7      | 7    | 4030.9425     | 4     | 1   | 1               | 1              | 7    | 0          | 4               | 3             | 1          | 2            |
| 10 | 8      | 7    | 1933.1395     | 1     | 2   | 1               | 2              | 9    | 1          | 1               | 1             | 1          | 3            |
| 11 | 9      | 7    | 4273.64       | 3     | 1   | 2               | 9              | 9    | 0          | 4               | 3             | 3          | -8           |
| 12 | 10     | 7    | 1512.9487     | 5     | 1   | -8              | -8             | 6    | -8         | 3               | 2             | 1          | -8           |
| 13 | 11     | 7    | 2438.4091     | 4     | 2   | 1               | 2              | 1    | 0          | 3               | 4             | 2          | 3            |
| 14 | 12     | 7    | 1960.8133     | 1     | 1   | 1               | 2              | 5    | 1          | 3               | 4             | 1          | 1            |

The **Canadian Cannabis Survey (CCS) 2024 PUMF User Guide** gives detailed information on the survey methodology, data collection process, and how to use the public use microdata file (PUMF). You can also find the variable names and their descriptions in this file.

[2024 Public Use Microdata File](#)

## Data and Resources

Search resources...

- Canadian Cannabis Survey (CCS) 2024
  - English dataset CSV
  - Explore
- Canadian Cannabis Survey (CCS) 2024 PUMF User Guide
  - English guide DOCK
  - Explore
- Canadian Cannabis Survey (CCS) 2024 PUMF User Guide
  - French guide DOCK
  - Explore
- Public Use Microdata Files (PUMF) Derived Variable Specifications
  - English specification DOCK
  - Explore
- Public Use Microdata Files (PUMF) Derived Variable Specifications
  - French specification DOCK
  - Explore

Let's find a variable (`access_ease_legal`) in the Excel file and check its description in the data dictionary. You can use the Find function in Excel (Ctrl + F) to search for the variable name:

| cansirisk_cantli | risk_cand | risk_cane | access_ease_legal | access_eewarn_see | warn_crec | warn_kno | warr |
|------------------|-----------|-----------|-------------------|-------------------|-----------|----------|------|
| 3                | 3         | 3         | 3                 | 5                 | 4         | -7       | -7   |
| 2                | 2         | 2         | 4                 | 2                 | 4         | -7       | -7   |
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| 2                | 3         | 5         | 4                 | 5                 | 1         | 4        | 4    |
| 5                | 5         | 5         | 4                 | 5                 | 4         | -7       | -7   |
| 2                | 2         | 2         | 4                 | 4                 | 3         | 3        | 2    |
| 4                | 4         | 4         | 1                 | 3                 | 4         | -7       | -7   |
| 3                | 3         | 3         | 4                 | 3                 | 4         | -7       | -7   |
| 2                | 2         | 2         | 4                 | 4                 | 5         | 7        | -7   |
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|                                                        |                                                                                                                                                                                                                                               |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>access_ease_legal</b>                               | How easy or difficult is it for you to get legal cannabis?                                                                                                                                                                                    |
| <i>[variable added in 2024 to replace access_ease]</i> | <input type="radio"/> 1 Very difficult<br><input type="radio"/> 2 Fairly difficult<br><input type="radio"/> 3 Fairly easy<br><input type="radio"/> 4 Very easy<br><input type="radio"/> 5 I don't know<br><input type="checkbox"/> -7 Missing |

In this dataset, -7 generally represents missing. Always check the data dictionary for the specific coding of missing values, “don’t know,” or “refused” responses, as they can vary between datasets.



# CADS

The Canadian Alcohol and Drugs Survey (CADS) is a national survey conducted by Statistics Canada that collects data on the use of substances, such as alcohol, cannabis and other drugs. The study was conducted by Statistics Canada in 2019 with the cooperation and support of Health Canada. The target population for the survey was all persons aged 15 years and over living in Canada, excluding residents of the Yukon, Northwest Territories, and Nunavut, full-time residents of institutions, and residents of Native reserves.



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### Canadian Alcohol and Drugs Survey: Public Use Microdata File

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This public use microdata file (PUMF) is from the Canadian Alcohol and Drugs Survey and includes information about the use of substances, including alcohol, cannabis and other drugs. This product is for users who prefer to do their own analysis by focusing on specific subgroups in the population or by cross-classifying variables that were not the focus of our release in The Daily.

Additional

Publisher - Current Organization Name: Statistics Canada

[CADS 2019 dataset on the CADS website](#)

## Open Licence

Similar to the [Canadian Cannabis Survey](#), the CADS is also licensed under the [Open Government Licence - Canada](#). This means that the data can be freely used, modified, and shared by anyone for any purpose, as long as they provide proper attribution to the source of the data.

## Canadian Alcohol and Drugs Survey: Public Use Microdata File

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**Publisher - Current Organization Name:** Statistics Canada

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## Data and Resources

The **Data** and **Resources** tab provides access to the public use data, user guides, and the data dictionary.

### Data and Resources





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[English](#) [French](#) [dataset](#) [CSV](#)

 **Explore** ▾



**Canadian Alcohol and Drugs Survey: Public Use Microdata File**  
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 **Explore** ▾



**Canadian Alcohol and Drugs Survey: Public Use Microdata File**  
[French](#) [website](#) [HTML](#)

 **Explore** ▾

The public use data file is available in a ZIP format. You need to open the **Canadian Alcohol and Drugs Survey: Public Use Microdata File** link, and click on **Go to resource** to download the ZIP file.

## Data and Resources

CSV

Canadian Alcohol and Drugs Survey: Public Use Microdata File

English French dataset CSV

Explore

DATA

Canadian Alcohol and Drugs Survey: Public Use Microdata File

English French dataset SAS

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HTML

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English website HTML

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HTML

Canadian Alcohol and Drugs Survey: Public Use Microdata File

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### Canadian Alcohol and Drugs Survey: Public Use Microdata File - Canadian Alcohol and Drugs Survey: Public Use Microdata File

Go to resource

The ZIP file contains the following files: an instruction on how to cite CADS 2019, data documentation, and the dataset in CSV format.

|                                             |                    |                    |        |
|---------------------------------------------|--------------------|--------------------|--------|
| How to cite CADS2019_Comment citer ECAD2019 | 27-Jul-21 10:42 AM | Adobe Acrobat D... | 388 KB |
| Documentation                               | 05-Aug-21 6:56 AM  | File folder        |        |
| Data_donnees                                | 05-Aug-21 6:46 AM  | File folder        |        |

## Data Documentation

The data documentation folder includes three sub-folders: Codebook\_Livre des codes, Guide, and Questionnaires.

|                          |                   |             |
|--------------------------|-------------------|-------------|
| Codebook_Livre des codes | 05-Aug-21 6:56 AM | File folder |
| Guide                    | 05-Aug-21 6:56 AM | File folder |
| Questionnaires           | 05-Aug-21 6:56 AM | File folder |

The **Guide** folder contains user guides (in English and French) that provide an overview of the survey, including the methodology, data collection process, data processing, data quality, and sampling weights.

|                                                                                                |                |                   |                    |          |
|------------------------------------------------------------------------------------------------|----------------|-------------------|--------------------|----------|
|  ECAD2019_GuF | <b>French</b>  | 14-Jul-21 6:47 AM | Adobe Acrobat D... | 1,653 KB |
|  CADS2019_UgE | <b>English</b> | 14-Jul-21 6:45 AM | Adobe Acrobat D... | 1,306 KB |

The **Codebook\_Livre des codes** folder contains codebooks (in English and French) that provide detailed information about the variables in the dataset, including variable names, labels, and values. There are two types of codebooks: those with and those without variable frequencies. But both codebooks provide the same information about the variables in the dataset.

**Codebook including frequencies for variables**

|                                                                                                               |                |                    |                    |          |
|---------------------------------------------------------------------------------------------------------------|----------------|--------------------|--------------------|----------|
|  ECAD2019_FMGD_lvCd          | <b>French</b>  | 22-Dec-20 11:50 AM | Adobe Acrobat D... | 1,948 KB |
|  CADS2019_PUMF_Cdbk          |                | 22-Dec-20 11:41 AM | Adobe Acrobat D... | 1,925 KB |
|  ECAD2019_FMGD_lvCd_zerofreq |                | 22-Dec-20 11:24 AM | Adobe Acrobat D... | 1,885 KB |
|  CADS2019_PUMF_Cdbk_zerofreq | <b>English</b> | 20 11:16 AM        | Adobe Acrobat D... | 1,863 KB |

**Codebook excluding frequencies for variables**

The **Questionnaires** folder contains the survey questionnaires (in English and French) that were used to collect the data.

## Data

The **Data\_donnees** folder contains the dataset in CSV format. You can open the dataset in Excel or any statistical software to explore the data. The row represents the respondent, and the column represents the variable:

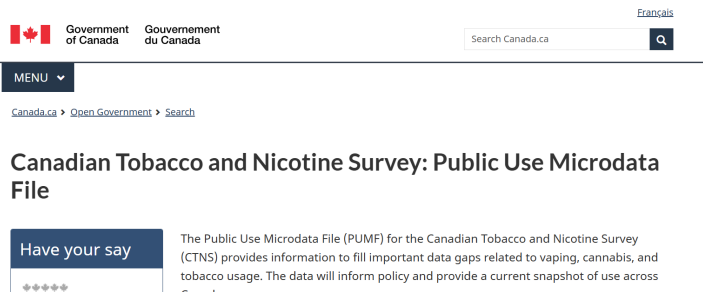
| CADS                                                                              |        |         |         |      |         |          |     |        |         |        |        |        |        |         |       |   |
|-----------------------------------------------------------------------------------|--------|---------|---------|------|---------|----------|-----|--------|---------|--------|--------|--------|--------|---------|-------|---|
| File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat |        |         |         |      |         |          |     |        |         |        |        |        |        |         |       |   |
| A1 PUMFID                                                                         |        |         |         |      |         |          |     |        |         |        |        |        |        |         |       |   |
|                                                                                   | A      | B       | C       | D    | E       | F        | G   | H      | I       | J      | K      | L      | M      | N       | O     | P |
|                                                                                   | PUMFID | REFYEAR | REFMONT | PROV | DVURBAN | MHHTYPEB | SEX | DV_AGE | HHLI_01 | HDSIZE | HWB_05 | HWB_10 | ALC_05 | ALC_10P | ALC_J |   |
| 1                                                                                 | 1000   | 2019    | 12      | 46   | 2       | 3        | 1   | 8      | 1       | 2      | 2      | 2      | 2      | 1       | 16    |   |
| 2                                                                                 | 1001   | 2019    | 9       | 46   | 2       | 6        | 1   | 5      | 1       | 4      | 1      | 1      | 1      | 1       | 17    |   |
| 3                                                                                 | 1002   | 2019    | 6       | 35   | 1       | 3        | 2   | 8      | 1       | 2      | 3      | 2      | 2      | 1       | 18    |   |
| 4                                                                                 | 1003   | 2019    | 8       | 11   | 1       | 3        | 1   | 8      | 1       | 3      | 2      | 2      | 2      | 1       | 18    |   |
| 5                                                                                 | 1004   | 2019    | 6       | 12   | 1       | 3        | 1   | 9      | 1       | 3      | 2      | 3      | 1      | 16      |       |   |
| 6                                                                                 | 1005   | 2019    | 10      | 12   | 1       | 3        | 1   | 8      | 1       | 2      | 2      | 2      | 1      | 1       | 99    |   |
| 7                                                                                 | 1006   | 2019    | 10      | 35   | 1       | 3        | 1   | 6      | 1       | 1      | 1      | 1      | 1      | 1       | 18    |   |
| 8                                                                                 | 1007   | 2019    | 12      | 48   | 1       | 3        | 1   | 7      | 1       | 2      | 4      | 4      | 1      | 1       | 16    |   |
| 9                                                                                 | 1008   | 2019    | 10      | 46   | 1       | 6        | 1   | 9      | 1       | 3      | 1      | 1      | 1      | 1       | 17    |   |
| 10                                                                                | 1009   | 2019    | 6       | 47   | 1       | 3        | 2   | 9      | 1       | 2      | 2      | 2      | 2      | 1       | 16    |   |
| 11                                                                                | 1010   | 2019    | 11      | 24   | 1       | 8        | 2   | 7      | 1       | 2      | 3      | 2      | 1      | 1       | 16    |   |

You can refer to the codebook to understand the variables that you are interested in for your analysis. For example, if you are interested in the variable DVURBAN, you can refer to the codebook to find out that the variable (e.g., for the English version of the codebook with frequencies):

|                   |                                                                                                                    |           |                    |           |    |
|-------------------|--------------------------------------------------------------------------------------------------------------------|-----------|--------------------|-----------|----|
| Variable Name:    | DVURBAN                                                                                                            | Length:   | 1.0                | Position: | 14 |
| Question Name:    |                                                                                                                    |           |                    |           |    |
| Concept:          | Characteristics of community where the respondent lives                                                            |           |                    |           |    |
| Question Text:    |                                                                                                                    |           |                    |           |    |
| Universe:         | All respondents                                                                                                    |           |                    |           |    |
| Note:             | The category 'Population centre' is historically comparable to 'Urban' which was used in previous years for CTUMS. |           |                    |           |    |
| Source:           |                                                                                                                    |           |                    |           |    |
| Answer Categories | Code                                                                                                               | Frequency | Weighted Frequency | %         |    |
| Population centre | 1                                                                                                                  | 7,208     | 21,321,106         | 68.8      |    |
| Rural             | 2                                                                                                                  | 2,438     | 5,122,120          | 16.5      |    |
| Not stated        | 9                                                                                                                  | 647       | 4,552,224          | 14.7      |    |
| Total             |                                                                                                                    | 10,293    | 30,995,450         | 100.0     |    |

# CTNS

The Canadian Tobacco and Nicotine Survey (CTNS) is a cross-sectional survey that collects information on tobacco and nicotine use among Canadians aged 15 years and older. The survey is conducted by Statistics Canada on behalf of Health Canada and provides valuable insights into the prevalence of cigarette smoking, vaping, cannabis, and alcohol use.



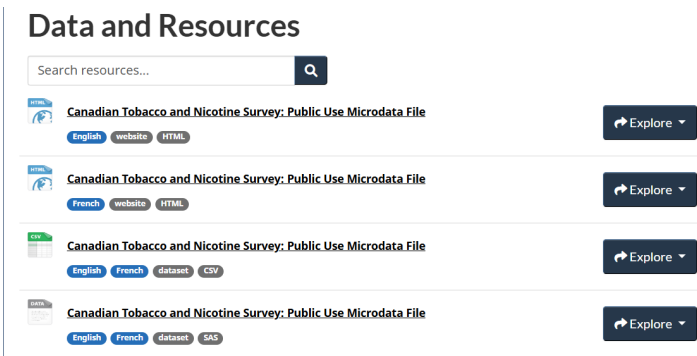
[CTNS 2022 dataset on file for CTNS](#)

## Open Licence

The CTNS is licensed under the [Open Government Licence - Canada](#). This means that the data can be freely used, modified, and shared by anyone for any purpose, as long as they provide proper attribution to the source of the data.

## Data and Resources

The **Data and Resources** tab provides access to the public use data, user guides, and the data dictionary.



The public use data file is available in a ZIP format. You can click on the [Canadian Tobacco and Nicotine Survey: Public Use Microdata File](#) link, followed by [Go to resource](#) link to download the ZIP file.



[Canada.ca](#) > [Open Government](#) > [Search](#) > [Canadian Tobacco and Nicotine Survey: Public Use Microdata File](#)

## Canadian Tobacco and Nicotine Survey: Public Use Microdata File - Canadian Tobacco and Nicotine Survey: Public Use Microdata File

[Go to resource](#)

Similar to the [Canadian Alcohol and Drugs Survey \(CADS\)](#) tutorial, the ZIP file for CTNS contains the following files: citation instructions, data documentation, and the dataset in CSV format. The data documentation, questionnaire, and codebook are also the same as those for CADS. The only difference is that the data documentation and questionnaire for CTNS are labelled **CTNS** rather than **CADS**.

# CSADS

The Canadian Student Alcohol and Drugs Survey (CSADS) is a national survey that collects data on the alcohol and drug use patterns of Canadian students in grades 7 to 12. The survey is conducted every two years and provides valuable insights into the prevalence and trends of substance use among Canadian youth.

## Canadian Student Alcohol and Drugs Survey (CSADS): Public Use Microdata File

Have your say

★★★★★

[Rate this dataset](#)

[Provide feedback](#)

Every two years, Health Canada conducts the Canadian Student Alcohol and Drugs Survey (CSADS), a cross-sectional survey of students in grades 7 to 12 across the Canadian provinces.

With tens of thousands of participants each cycle, it is one of the largest youth surveys in the world. For over 30 years, the survey—previously known as the Youth Smoking Survey (YSS; pre-2014) and the Canadian Student Tobacco, Alcohol and Drugs Survey (CSTADS;

[CSADS Canada website files CTNS](#)

## Open Licence

The CTNS is licensed under the [Open Government Licence - Canada](#). This means that the data can be freely used, modified, and shared by anyone for any purpose, as long as they provide proper attribution to the source of the data.

## Data and Resources

The **Data and Resources** tab provides access to the public use data, user guides, the data dictionary, and information on bootstrap weights.



## Data and Resources

Search resources...

|                                                                                   |                                                   |                         |
|-----------------------------------------------------------------------------------|---------------------------------------------------|-------------------------|
|  | <b>2021-2022 CSTADS: PUMF — User Guide</b>        | <a href="#">Explore</a> |
|  | <b>2021-2022 ECTADE : FMGD — Mode d'emploi</b>    | <a href="#">Explore</a> |
|  | <b>2021-2022 CSTADS: PUMF — Data</b>              | <a href="#">Explore</a> |
|  | <b>2021-2022 CSTADS: PUMF — Bootstrap Weights</b> | <a href="#">Explore</a> |
|  | <b>2023-2024 CSADS: PUMF — User Guide</b>         | <a href="#">Explore</a> |
|  | <b>2023-2024 ECADE : FMGD — Mode d'emploi</b>     | <a href="#">Explore</a> |
|  | <b>2023-2024 CSADS: PUMF — Data</b>               | <a href="#">Explore</a> |
|  | <b>2023-2024 CSADS: PUMF — Bootstrap Weights</b>  | <a href="#">Explore</a> |

To download the 2021–2022 CSTADS, you can click on the [2021-2022 CSTADS: PUMF – Dataset in CSV format](#) link. The data is available to download in various formats. Under the **Download** tab, you can find the formats. For example, you can download the data in CSV format by clicking on the [CSV](#) link:

**MENU**

[Canada.ca](#) > [Open Government](#) > [Search](#) > [Canadian Student Alcohol and Drugs Survey \(CSADS\): Public Use Microdata File](#)

### Canadian Student Alcohol and Drugs Survey (CSADS): Public Use Microdata File - 2021-2022 CSTADS: PUMF — Data

[Download](#)
[Data API](#)

[View](#)

[CSV](#)  
[TSV](#)  
[JSON](#)  
[XML](#)

After downloading, you can open the data in Excel, Google Sheets, or any statistical software. The data is structured in a tabular format, with each row representing an individual respondent and each column representing different variables:

|    | A     | B      | C     | D        | E       | F     | G        | H       | I        | J      | K      | L      | M      | N      | O      |
|----|-------|--------|-------|----------|---------|-------|----------|---------|----------|--------|--------|--------|--------|--------|--------|
| 1  | SEQID | PROVID | GRADE | DVGENDEI | DVURBAN | DVRES | DVORIENT | DVDESCR | WTPUMF   | GH_010 | GH_020 | SS_010 | SS_020 | TS_011 | SS_030 |
| 2  | 18338 | 11     | 9     | 2        | 1       | 3     | 2        | 1       | 1.164716 | 2      | 3      | 2      | 96     | 4      | 96     |
| 3  | 16111 | 24     | 10    | 1        | 1       | 1     | 2        | 4       | 36.14117 | 3      | 3      | 2      | 96     | 4      | 96     |
| 4  | 20587 | 10     | 7     | 2        | 1       | 1     | 2        | 1       | 1.448578 | 2      | 1      | 2      | 96     | 4      | 96     |
| 5  | 54568 | 12     | 8     | 99       | 2       | 1     | 1        | 1       | 3.03666  | 4      | 5      | 2      | 96     | 3      | 96     |
| 6  | 40991 | 10     | 7     | 1        | 2       | 1     | 2        | 1       | 3.745233 | 2      | 3      | 2      | 96     | 4      | 96     |
| 7  | 5254  | 12     | 7     | 1        | 1       | 2     | 2        | 1       | 10.0715  | 3      | 3      | 2      | 96     | 4      | 96     |
| 8  | 10440 | 35     | 9     | 99       | 1       | 1     | 99       | 1       | 75.0031  | 5      | 4      | 2      | 96     | 4      | 96     |
| 9  | 38149 | 48     | 12    | 1        | 1       | 1     | 2        | 1       | 25.83337 | 1      | 3      | 1      | 16     | 4      | 2      |
| 10 | 32243 | 59     | 11    | 2        | 1       | 1     | 2        | 1       | 47.16534 | 1      | 3      | 1      | 16     | 3      | 2      |
| 11 | 5513  | 35     | 11    | 1        | 2       | 1     | 99       | 1       | 99.52681 | 2      | 5      | 2      | 96     | 3      | 96     |
| 12 | 37146 | 48     | 10    | 1        | 1       | 2     | 2        | 4       | 24.62732 | 3      | 2      | 2      | 96     | 4      | 96     |
| 13 | 21979 | 46     | 9     | 1        | 2       | 1     | 2        | 1       | 65.69607 | 1      | 4      | 2      | 96     | 4      | 96     |
| 14 | 33925 | 12     | 7     | 1        | 2       | 3     | 1        | 1       | 3.60877  | 3      | 4      | 2      | 96     | 4      | 96     |
| 15 | 19919 | 94     | 7     | 2        | 1       | 2     | 00       | 00      | 76.14944 | 9      | 9      | 9      | 00     | 4      | 00     |

The user guide gives detailed information on the survey methodology, data collection process, and how to use the public use microdata file (PUMF). You can also find the variable names and their descriptions in this DOCX file.

**2021-2022 CSTADS: PUMF — User Guide**

[English](#)
[guide](#)
[DOCX](#)

Explore

**2021-2022 ECTADE : FMGD — Mode d'emploi**

[French](#)
[guide](#)
[DOCX](#)

Explore

**2021-2022 CSTADS: PUMF — Data**

[English](#)
[French](#)
[dataset](#)
[CSV](#)

Explore

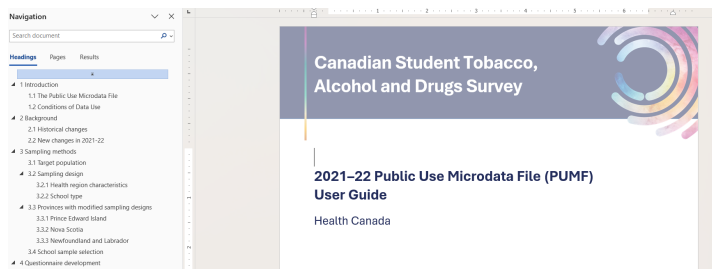
**2021-2022 CSTADS: PUMF — Bootstrap Weights**

[English](#)
[French](#)
[dataset](#)
[CSV](#)

Explore

## Canadian Student Alcohol and Drugs Survey (CSADS): Public Use Microdata File - 2021-2022 CSTADS: PUMF — User Guide

Download



To find a variable description, you are referred to the **User Guide**. For example, the variable GH\_010 is described as follows:

| Navigation                                      |       |         |   |
|-------------------------------------------------|-------|---------|---|
| 04.010                                          |       | WTPUMP  | 9 |
| 1 result                                        |       |         |   |
| Headings                                        | Pages | Results |   |
| 6.2.1 Skip patterns and inconsistent responses  |       |         |   |
| 6.2.2 Imputations of core smoking questions     |       |         |   |
| 6.2.3 Reducing underestimation risk in the PLME |       |         |   |
| 6.2.4 Standard codes                            |       |         |   |
| 6.3 Survey weights                              |       |         |   |

|        |    |                                                                                                                                                                   |                                                                                                            |
|--------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
|        |    | Unknown: if not selected, was assigned to the perturbation to maintain privacy; more information on how these weights were generated can be found in section 6.3. |                                                                                                            |
| 04.010 | 10 | In general, would you say your physical health is excellent, very good, good, fair or poor?                                                                       | 1 = Excellent<br>2 = Very good<br>3 = Good<br>4 = Fair<br>5 = Poor<br>6 = I do not know<br>99 = Not stated |
|        |    |                                                                                                                                                                   | 1 = Excellent<br>2 = Very good                                                                             |

# PHAC Health Infobase

The Health Infobase is a comprehensive digital platform maintained by the Public Health Agency of Canada (PHAC). The Infobase provides access to a wide range of health-related data tools, infographics, charts and blogs. There are 182 data sources available on the Infobase. The Health Infobase serves as a central hub for national surveillance data, interactive data tools, and public health reports. The platform is frequently updated with the latest surveillance data.



[PHAC Health Infobase](#)

## Data Categories

The Infobase provides access to health data under six categories:

- **Diseases and disorders:** Comprehensive tracking of chronic conditions such as cancer, cardiovascular diseases, and COVID-19.
- **Drugs and vaccines:** Data on prescription drugs, vaccine coverage, and immunization rates across the country.

- **Mental health, healthy living and environment:** Information on mental health indicators, physical activity levels, nutrition, and environmental factors such as wildfires.
- **Injuries and death:** Data on death, suicide, violence, and drowning across regions.
- **Population groups and demography:** Data on birth, maternity, children, family, and disabilities.
- **Substance use:** Data on substance use such as alcohol, cannabis use, opioids, and smoking.

Under each category, users can select the data for specific health topics. The three top categories are **Respiratory diseases**, **Opioids** and **Mental health**.

### Find a data product

Use filters to find specific data products. You can filter by **category**, by **format**, or by typing in **keywords**.

▼ Categories

Diseases and disorders

+

Drugs and vaccines

+

Mental health, healthy living and environment

+

Injuries and death

+

Population groups and demography

+

Substance use

+

▼ Filter

Type in keywords to filter results as you type



[Healthy Canadians and Communities Fund](#)

Data exploration tool

Inequalities

Interactive map of funded interventions across Canada that support Canadians who experience health inequalities and are at greater risk of developing chronic disease.

Last updated: 2026-04-07

### Find a data product

Use filters to find specific data products. You can filter by **category**, by **format**, or by typing in **keywords**.

▼ Categories

Diseases and disorders

—

☐ All diseases and disorders

☒ Blood and metabolic diseases

☐ Chronic diseases

☐ Cancer

☐ COVID-19

▼ Filter

Type in keywords to filter results as you type

Blood and metabolic diseases X

Showing 5 filtered from 182 total products

Clear all filters



[Canadian Chronic Disease Surveillance System \(CCDSS\)](#)

Data exploration tool

Chronic diseases

Cardiovascular diseases

Musculoskeletal disorders

Neurological diseases

Respiratory diseases

Mental health

Blood and metabolic diseases

Autism

You can filter the data by category, by format, or by typing in keywords.

## Find a data product

Use filters to find specific data products. You can filter by **category**, by **format**, or by typing in **keywords**.

▼ Categories

Diseases and disorders

+

Drugs and vaccines

+

Mental health, healthy living and environment

+

Injuries and death

+

Population groups and demography

+

Substance use

+

Filter covid

Showing 27 filtered from 182 total products

[Clear all filters](#)



[COVID-19 Vaccine mistrust among Black Communities in Canada](#)

Interactive report

COVID-19

Vaccines

Interactive report on COVID-19 vaccine mistrust among Black communities in Canada, based on data from the 2022 CanBlack Vax survey.

You can also filter the data by type, such as blog, infographic, or report.

▼ Type

☐ Blog

☐ Data exploration tool

☐ Indicator framework

☐ Infographic

☐ Interactive report

at greater risk of developing chronic disease.

Last updated: 2026-04-07



[Transfusion and transplantation-related adverse events](#)

Interactive report

Injuries

Summary of national surveillance systems for transfusion errors, transfusion-transmitted injuries, and cells, tissues and

## Data Availability

The level of information provided in the data sources varies. For some data sources/products, you can find blogs, infographics, and reports along with individual-level data. However, for some data sources/products, you may only find the summary statistics without individual-level data, i.e., only aggregated-level data. For example, the **Human Papillomavirus Indicator Framework** product provides quick statistics, data tools and a list of publications:

▼ Categories

Diseases and disorders

—

☐ All diseases and disorders

☐ Blood and metabolic diseases

☐ Chronic diseases

☒ Cancer

☐ COVID-19

☐ Cardiovascular diseases

☐ Developmental

Filter Type in keywords to filter results as you type

Cancer X

Showing 10 filtered from 182 total products

[Clear all filters](#)



[Human Papillomavirus Indicator Framework \(HPVIF\)](#)

Data exploration tool

Infections

Vaccines

Cancer

The Human Papillomavirus Indicator Framework (HPVIF) presents HPV vaccination coverage and HPV outcome indicators for Canada.

Last updated: 2025-12-15

506

## Human Papillomavirus Indicator Framework (HPVIF)

The Human Papillomavirus Indicator Framework (HPVIF) presents HPV vaccination coverage and HPV outcome indicators for Canada. Sub-national estimates, sociodemographic breakdowns, and trends over time are presented, when possible, as well as international comparisons. Since HPV vaccination can prevent HPV-associated cancers, this framework will serve as a valuable resource for monitoring Canada's progress against HPV.

Last updated: 2025-12-15

|             |           |              |
|-------------|-----------|--------------|
| Quick stats | Data tool | Publications |
|-------------|-----------|--------------|

The **Drug and alcohol use in Canada: Postsecondary students** product provides summary statistics as well as links to download data in CSV format:

Substance use

☐

All substance use

☒

Alcohol

☐

Cannabis

☐

Opioids

☐

Smoking and vaping

☐

Stimulants



Drug and alcohol use in Canada: Postsecondary students

Data exploration tool | Alcohol | Cannabis  
Opioids | Prescription drugs | Smoking and vaping  
Stimulants

Explore Health Canada's latest data on drug and alcohol use by postsecondary students in Canada.

[Canada.ca](#) > [Health](#) > [Health science, research and data](#) > [Health Infobase](#)

# Postsecondary students

## Drug and alcohol use in Canada

Explore Health Canada's latest data on drug and alcohol use by postsecondary students in Canada.

Last updated: 2025-12-31

General population

Youth

Postsecondary students

### About the postsecondary survey

Health Canada monitors the prevalence of substance use and its impact on postsecondary students through the Canadian Postsecondary Education Alcohol and Drug Use Survey (CPADS). Governments and non-governmental organizations use this information to shape policies and programs that address substance use and support students.

In the 2024–25 school year, 29,371 students from 9 provinces and 2 territories completed the survey. The results focus on



[Read](#)  
the 2024-2025 summary report



[Read](#)  
previous summary reports



[Download](#)  
dashboard data (.csv)

We'd love your feedback! Please take a minute to complete this short and anonymous survey.

Next

# CCDSS

The Canadian Chronic Disease Surveillance System (CCDSS) is a collaborative network of provincial and territorial chronic disease surveillance systems in Canada. The CCDSS is supported by the Public Health Agency of Canada (PHAC) and provincial/territorial governments. The CCDSS was established to provide the estimates of the prevalence and incidence of chronic diseases in Canada, as well as to monitor trends over time for more than 20 chronic diseases to help inform health policies and resource planning. The CCDSS is part of the PHAC Health Infobase that we covered in the previous tutorial on [Health Infobase](#). The CCDSS uses health administrative data, such as hospital discharge records, physician billing claims, and prescription drug data.



[CCDSS website](#)

## Key Indicator Categories

The CCDSS website provides a variety of indicators related to chronic diseases, such as autism, cardiovascular diseases,



chronic respiratory diseases, diabetes, mental health disorders, musculoskeletal disorders, neurological disorders, and multimorbidity.

Diseases, conditions and indicators

Table 1: Diseases, conditions and indicators included in the CCDSS

| Morbidity and mortality                                                                                                                                                                        | Health events and complications                                                                                        | Use of health services                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>incidence/rate of newly identified cases</li><li>prevalence (cumulative and active for select diseases/conditions)</li><li>all-cause mortality</li></ul> | <ul style="list-style-type: none"><li>annual prevalence</li><li>all-cause mortality following a health event</li></ul> | <ul style="list-style-type: none"><li>annual prevalence</li></ul> |

Table 2: CCDSS case definitions and surveillance period reported

| Disease, condition and indicator | Age  | Case definition summary                                                  | First reported year | Last reported year |
|----------------------------------|------|--------------------------------------------------------------------------|---------------------|--------------------|
| Autism                           |      |                                                                          |                     |                    |
| Autism                           | 1-19 | One or more hospital separation records, or two or more physician claims | 2000-2001           | 2023-2024          |
| Cardiovascular diseases          |      |                                                                          |                     |                    |
| Acute myocardial infarction      | 20+  | One or more hospital inpatient admission records                         | 2000-2001           | 2023-2024          |

Data Tool

The CCDSS website provides a data tool that allows users to explore and visualize the data on chronic diseases in Canada. The data tool provides interactive charts and tables that allow users to explore the data by province/territory.

About the Canadian Chronic Disease Surveillance System (CCDSS)

The CCDSS is a collaborative network of provincial and territorial surveillance systems, supported by the Public Health Agency of Canada (PHAC). The system collects data on all residents who are eligible for provincial or territorial health insurance. It can generate national estimates and trends over time for over 20 chronic diseases and conditions, and other selected health outcomes. To identify people with chronic diseases and conditions, provincial and territorial health insurance registry records are linked using a unique personal identifier to the corresponding physician claims, hospital separation records and prescription drug records.

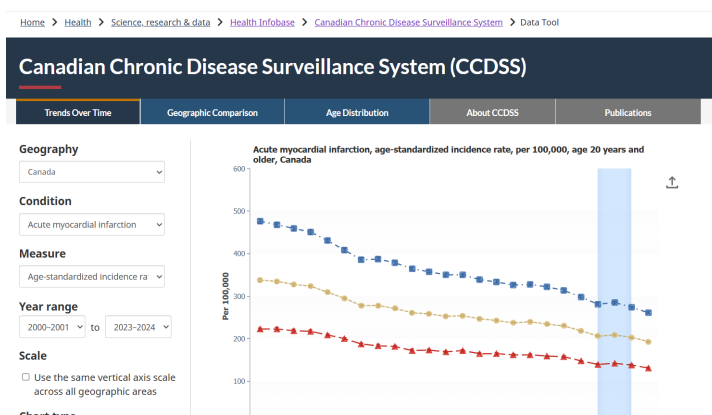
Key information about the CCDSS methods and case definitions is presented below. More detailed documents are available here:

[CCDSS summary of methods](#); [CCDSS case definitions](#); [CCDSS COVID-19 guidance for data users](#).

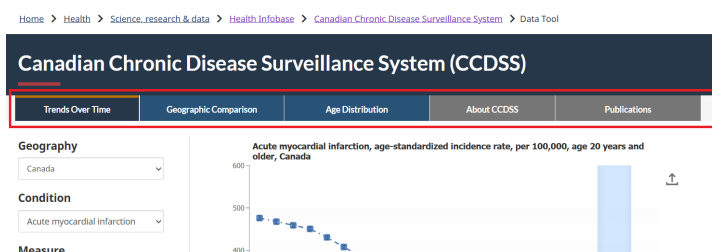
Use the CCDSS Data Tool

Users can use the data tool to explore the prevalence and incidence of cases, mortality rates, and health service utilization, and to track individuals living with two or more chronic conditions.

[CCDSS Data Tool](#)

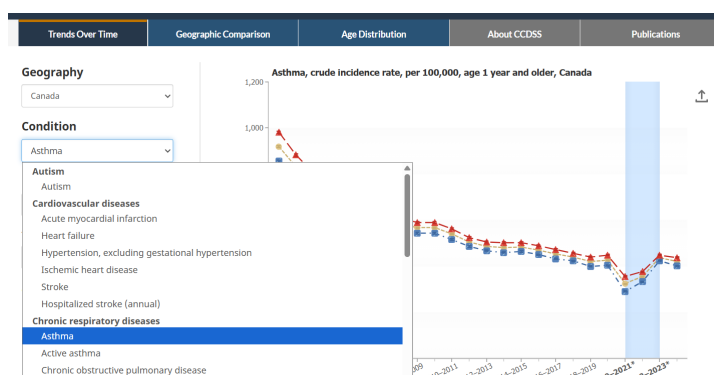


Users can explore trends over time, compare data across provinces/territories, and view the distribution of cases by age group. Under the **Publication** tab, you can find information on the repository for data-driven insights, fact sheets, peer-reviewed publications, and reports.



## Interactive chart example

Consider **Asthma** under **Chronic respiratory diseases** as an example.



You can explore different summary measures by filtering by geography, measure, and year. Again, you can further explore trends over time, compare data across provinces/territories, and view the distribution of cases by age group.



You can also download the summary data in CSV format for further analysis. To do that, you need to scroll down to the bottom of the **Summary Table**, and click on the **Download** button.

|       |            |     |      |         |      |        |
|-------|------------|-----|------|---------|------|--------|
| Males | 2019-2020  | 404 | 1.54 | 401-407 | 0.38 | 68,605 |
| Males | 2020-2021* | 292 | 1.31 | 290-295 | 0.45 | 49,690 |
| Males | 2021-2022* | 333 | 1.39 | 331-336 | 0.42 | 57,745 |
| Males | 2022-2023* | 425 | 1.55 | 422-428 | 0.36 | 75,625 |
| Males | 2023-2024  | 402 | 1.48 | 399-405 | 0.37 | 73,400 |

Download Detailed Table: Asthma, crude incidence rate

Save Summary Table

Once you open the CSV file in Excel or other software, you will find that the data is not at the individual or person

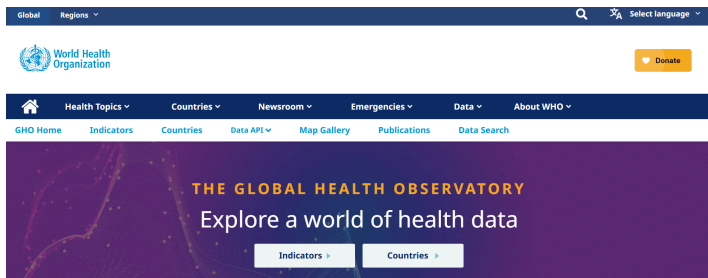
level. Instead, the data are aggregated at the country and province/territory levels, and summary measures are provided for both sexes and for years. For example, see row 4. The **Geography** column says *Canada* and the **Rate (per 100,000)** column says *919*. This is an example of aggregate data.

|    | A                                                     | B          | C              | D         | E           | F                  | G              | H            | I            | J           | K           | L          |
|----|-------------------------------------------------------|------------|----------------|-----------|-------------|--------------------|----------------|--------------|--------------|-------------|-------------|------------|
| 1  | Data Table: Asthma, Crude incidence rate, per 100,000 |            |                |           |             |                    |                |              |              |             |             |            |
| 2  |                                                       |            |                |           |             |                    |                |              |              |             |             |            |
| 3  | Geography                                             | Sex        | Age group type | Age group | Fiscal year | Rate (per 100,000) | Standard Error | Lower 95% CI | Upper 95% CI | Rate CV (%) | Case counts | Population |
| 4  | Canada                                                | Both sexes | Total          | 1+        | 2000-2001   | 919                | 1.8            | 916          | 923          | 0.2         | 260730      | 28367890   |
| 5  | Canada                                                | Females    | Total          | 1+        | 2000-2001   | 961                | 2.6206         | 976          | 986          | 0.27        | 140225      | 14289240   |
| 6  | Canada                                                | Males      | Total          | 1+        | 2000-2001   | 856                | 2.4657         | 851          | 861          | 0.29        | 120505      | 14078655   |
| 7  | Canada                                                | Both sexes | Total          | 1+        | 2001-2002   | 833                | 1.7103         | 830          | 837          | 0.21        | 237405      | 28488200   |
| 8  | Canada                                                | Females    | Total          | 1+        | 2001-2002   | 883                | 2.4814         | 878          | 886          | 0.26        | 126365      | 14337965   |
| 9  | Canada                                                | Males      | Total          | 1+        | 2001-2002   | 763                | 2.3526         | 779          | 788          | 0.3         | 110820      | 14150240   |
| 10 | Canada                                                | Both sexes | Total          | 1+        | 2002-2003   | 762                | 1.6318         | 759          | 765          | 0.21        | 218155      | 28623720   |
| 11 | Canada                                                | Females    | Total          | 1+        | 2002-2003   | 804                | 2.3637         | 800          | 809          | 0.29        | 115840      | 14399350   |
| 12 | Canada                                                | Males      | Total          | 1+        | 2002-2003   | 719                | 2.2487         | 715          | 724          | 0.31        | 102310      | 14224360   |
| 13 | Canada                                                | Both sexes | Total          | 1+        | 2003-2004   | 727                | 1.5893         | 724          | 730          | 0.22        | 209140      | 28774260   |
| 14 | Canada                                                | Females    | Total          | 1+        | 2003-2004   | 773                | 2.3117         | 769          | 778          | 0.3         | 111860      | 14468175   |
| 15 | Canada                                                | Males      | Total          | 1+        | 2003-2004   | 680                | 2.1801         | 676          | 684          | 0.32        | 97275       | 14306080   |
| 16 | Canada                                                | Both sexes | Total          | 1+        | 2004-2005   | 696                | 1.5513         | 693          | 699          | 0.22        | 201445      | 28932495   |
| 17 | Canada                                                | Females    | Total          | 1+        | 2004-2005   | 738                | 2.2531         | 734          | 743          | 0.31        | 107350      | 14541550   |
| 18 | Canada                                                | Males      | Total          | 1+        | 2004-2005   | 654                | 2.1316         | 650          | 658          | 0.33        | 94095       | 14398940   |

Because this data is not at the person-level, and the summary measures are aggregated, you do not need a codebook to translate columns. However, you cannot use this data to calculate the prevalence or incidence of asthma in Canada because it is already aggregated. You can only use this data to explore the trends over time, compare data across provinces/territories, and view the distribution of cases by age group and sex. The individual patients have already been erased to protect their privacy.

# GHO

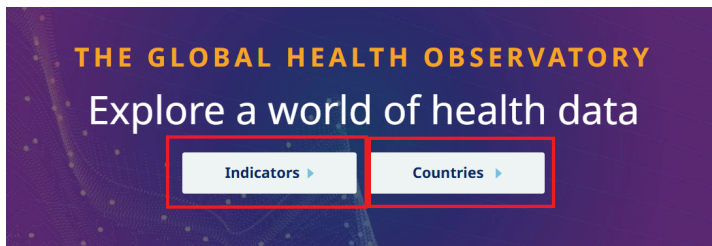
The Global Health Observatory (GHO) is the World Health Organization's (WHO) main health statistics repository. The GHO provides data on a wide range of health topics, such as mortality, morbidity, risk factors, health systems, and health equity. The GHO collects data from member states, international organizations, and other sources to provide a comprehensive picture of global health. The GHO is a valuable resource for researchers, policymakers, and public health professionals who are interested in understanding global health trends and disparities. The GHO provides various ways to interact with its data, including interactive maps and downloadable CSV files.



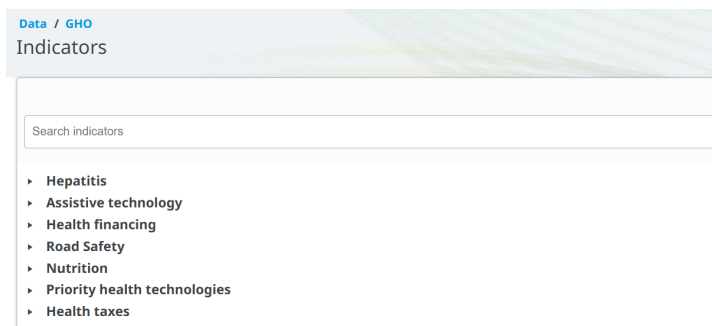
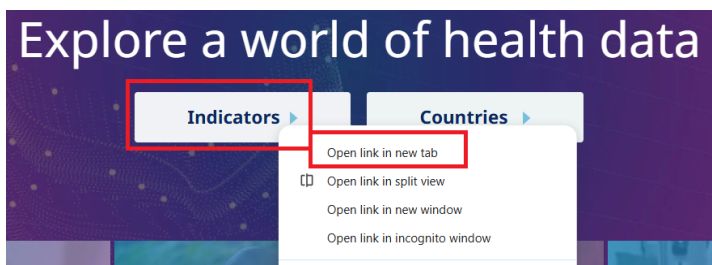
[WHO Global Health Observatory](#)

## World Health Data

You can browse information by broader health indicators or by specific countries.



To explore the data by indicators, click on the **Indicators** tab. You can find a list of indicators across categories such as Hepatitis, Nutrition, Mental health, and so on.



You can also use the search bar to search for specific indicators. For example, if you search for **Life expectancy at birth**, you will find a list of indicators related to life expectancy, such as **Life expectancy at birth (years)**.

Data / GHO
Indicators

Life expectancy at birth

- World Health Statistics
  - Life expectancy and life tables
    - Life expectancy at birth (years)
- World Health Statistics
  - Life expectancy at birth (years)
- Global Health Estimates: Life expectancy and leading causes of death and disability
  - Mortality and global health estimates
    - Life expectancy at birth (years)

To explore the data by country, click on the **Countries** tab. You can find a list of countries and regions. You can click the country name to see the data for that country.

World Health Organization

Data
Indicators
Countries
Dashb

WHO Glob

All

A

B

C

D

E

F

G

H

I

J

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L

M

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V

W

X

Y

Z

A

Afghanistan
Albania
Algeria
Andorra
Angola
Antigua and Barbuda
Argentina
Armenia
Australia
Austria
Azerbaijan

B

Bahamas
Bahrain
Bangladesh
Barbados

Australia

Health data overview for Australia

Population
26 451 124 (2023)

Current health expenditure (% of GDP)
10.54 (2021)

WHO region
Western Pacific

World Bank income level
High income (HIC)

Population, Australia

In Australia, the current population is 26,451,124 as of 2023 with a projected increase of 23% to 32,506,969 by 2050.

Population growth rate

Australia, 2023

0.95%

+0.012 percentage points change since 2022

rate

Australia

On this page

Population

Life expectancy

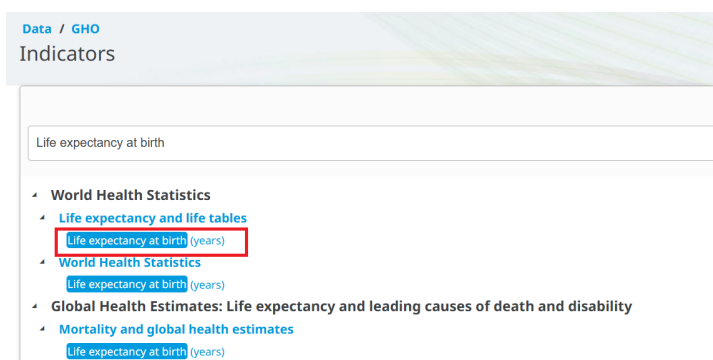
Leading causes of death

Health statistics

515

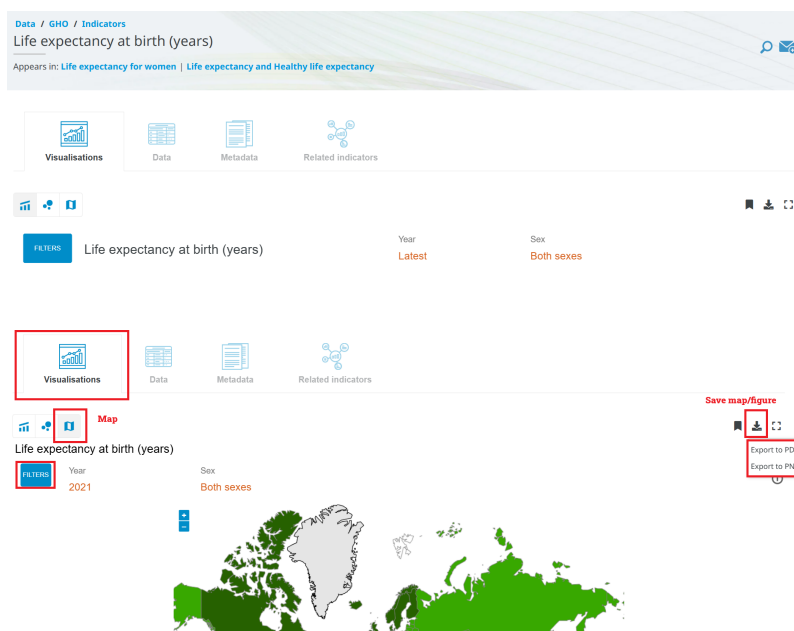
## Data by indicator example

Let's click on the Life expectancy at birth (years) indicator to see the details:



You can find different visualization options, such as charts and maps. You can also download the chart and map to your computer.

[GHO Indicators](#)





To see the data table, you can check the **Data** section. You can also download the data in CSV format for further analysis. To do that, you need to click on the **Download** button at the top right corner of the data table.

Visualisations

Data

Metadata

Related indicators

FILTERS

Last updated: 2024-08-02

Indicator

Location

Life expectancy at birth (years)

Both sexes

Male

Female

Life expectancy at age 60 (years)

Both sexes

Male

Female

Afghanistan

2021

2020

2019

2018

2017

EXPORT DATA in CSV format

Right-click here & Save link

Similar to the [Canadian Chronic Disease Surveillance System \(CCDSS\) database](#), the life expectancy data is also at an aggregated level, which means that the data is summarized at the country level, and does not contain individual-level data:

| IndicatorCode | Indicator     | ValueType                        | ParentLocationCode | ParentLocation | Location type         | SpatialDimValueCode | Location | Period type     | Period | Dim1       | FactValue  |       |
|---------------|---------------|----------------------------------|--------------------|----------------|-----------------------|---------------------|----------|-----------------|--------|------------|------------|-------|
| 1             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | LSO      | Lesotho         | Year   | 2021       | Male       | 48.73 |
| 2             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | CAF      | Central Af Year | 2021   | Male       | 49.57      |       |
| 3             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | LSO      | Lesotho         | Year   | 2021       | Both sexes | 51.48 |
| 4             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | SWZ      | Eswatini        | Year   | 2021       | Male       | 51.64 |
| 5             | WHOSIS_000001 | Life expectancy at birth (years) | text               | EMR            | Eastern Mediterranean | Country             | SOM      | Somalia         | Year   | 2021       | Male       | 51.75 |
| 6             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | CAF      | Central Af Year | 2021   | Both sexes | 52.31      |       |
| 7             | WHOSIS_000001 | Life expectancy at birth (years) | text               | EMR            | Eastern Mediterranean | Country             | SOM      | Somalia         | Year   | 2021       | Both sexes | 53.95 |
| 8             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | LSO      | Lesotho         | Year   | 2021       | Female     | 54.6  |
| 9             | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | SWZ      | Eswatini        | Year   | 2021       | Both sexes | 54.59 |
| 10            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | MOZ      | Mozambique      | Year   | 2021       | Male       | 54.8  |
| 11            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | CAF      | Central Af Year | 2021   | Female     | 55.38      |       |
| 12            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | GNB      | Guinea-Bi Year  | 2021   | Male       | 56.08      |       |
| 13            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | ZWE      | Zimbabwe        | Year   | 2021       | Male       | 56.19 |
| 14            | WHOSIS_000001 | Life expectancy at birth (years) | text               | EMR            | Eastern Mediterranean | Country             | SOM      | Somalia         | Year   | 2021       | Female     | 56.53 |
| 15            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | SSD      | South Sud Year  | 2021   | Male       | 56.48      |       |
| 16            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | NAM      | Namibia         | Year   | 2021       | Male       | 57.27 |
| 17            | WHOSIS_000001 | Life expectancy at birth (years) | text               | AFR            | Africa                | Country             | NAM      | Namibia         | Year   | 2021       | Male       | 57.27 |

The data on the website, as well as the downloaded data in CSV format, both provide the point estimate with 95% confidence interval:

Visualisations

Data

Metadata

Related indicators

FILTERS

Last updated: 2024-08-02

Indicator

Location

Life expectancy at birth (years)

Both sexes

Male

Female

Life expectancy at age 60 (years)

Both sexes

Male

Female

Afghanistan

2021

2020

2019

2018

EXP

Right

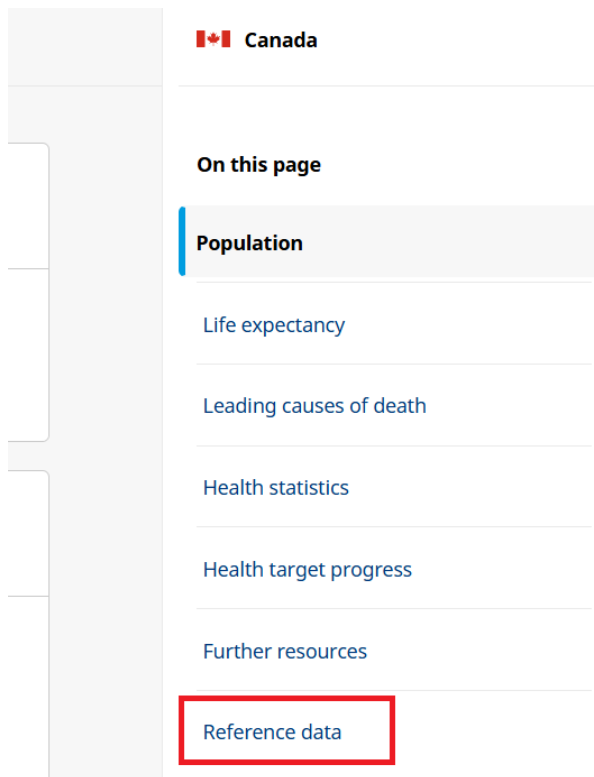
## Data by country example

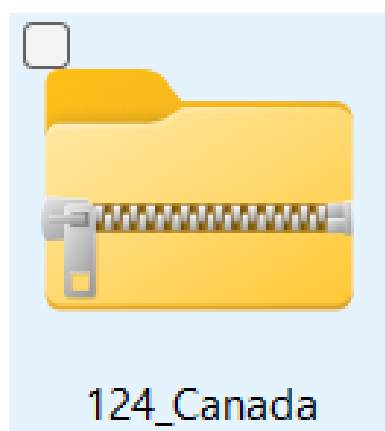
Let's click on **Canada** to explore the data for Canada.



You can explore different indicators for Canada. You can also download all indicators as a ZIP file by clicking **Reference data**.

[GHO Data by Country](#)





Once you unzip the folder, you will find several folders. Each folder contains a codelist, data, a data dictionary, along with a license and terms of use for different indicators.

|                                                                                         |                    |             |
|-----------------------------------------------------------------------------------------|--------------------|-------------|
| 1F96863_3.4.1 - cardiovascular disease, cancer, diabetes or chronic respiratory disease | 03-Mar-26 10:33 AM | File folder |
| 2D6FBE4_3.3.5 - Neglected tropical diseases                                             | 03-Mar-26 10:33 AM | File folder |
| 5C8435F_3.C.1 - Health worker density and distribution                                  | 03-Mar-26 10:33 AM | File folder |
| 6A64C9A_7.1.2 - Clean fuels                                                             | 03-Mar-26 10:33 AM | File folder |
| 12EE54A_6.2.1 - Safely managed sanitation & hand-washing                                | 03-Mar-26 10:33 AM | File folder |
| 168BF41_3.4.2 - Suicide                                                                 | 03-Mar-26 10:33 AM | File folder |
| 45CA7C8_3.C.1 - Health worker density and distribution                                  | 03-Mar-26 10:33 AM | File folder |
| 75DDA77_3.A.1 - Age-standardized prevalence of tobacco use                              | 03-Mar-26 10:33 AM | File folder |
| 77D059C_3.3.1 - HIV infections                                                          | 03-Mar-26 10:33 AM | File folder |
| 84FD3DE_3.9.3 - Unintentional poisoning                                                 | 03-Mar-26 10:33 AM | File folder |
| 608DE39                                                                                 | 03-Mar-26 10:33 AM | File folder |
| 1548EA3_6.1.1 - Safely managed drinking water                                           | 03-Mar-26 10:33 AM | File folder |
| 217795A_3.C.1 - Health worker density and distribution                                  | 03-Mar-26 10:33 AM | File folder |
| 361734E_16.1.1 - Intentional homicide                                                   | 03-Mar-26 10:33 AM | File folder |
| 1772666_3.1.2 - Births attended by skilled health personnel                             | 03-Mar-26 10:33 AM | File folder |

For example, the 77D059C\_3.3.1 - HIV infections folder contains the following files:

|                                                                                                                                 |                    |
|---------------------------------------------------------------------------------------------------------------------------------|--------------------|
|  124_77D059C_Code list_2025-03-26              | 13-Apr-26 11:43 PM |
|  124_77D059C_Dataset_2025-03-26                | 13-Apr-26 11:43 PM |
|  124_77D059C_License & Terms of Use_2025-03-26 | 13-Apr-26 11:43 PM |
|  124_77D059C_Metadata_2025-03-26               | 13-Apr-26 11:43 PM |
|  DDS - Data Dictionary 2024-12-12              | 13-Apr-26 11:43 PM |

Metadata

The GHO provides metadata for each indicator, providing important context for understanding the data and its limitations. For example, the metadata for the **Hepatitis - new infections** indicator includes information on the definition of the indicator, the methods used to calculate the indicator, data sources, and other relevant information.

[Indicator Metadata Registry List](#)

Data / GHO

Indicators

Search indicators

Hepatitis

Hepatitis

Hepatitis - new infections

Hepatitis - number of persons living with chronic hepatitis

Hepatitis - deaths caused by chronic hepatitis, number

Hepatitis - prevalence of chronic hepatitis among the general population

Hepatitis - deaths caused by chronic hepatitis B and C (per 100 000)

Data / GHO / Indicators

Hepatitis - new infections

Appears in: [Hepatitis cases and deaths](#)

Data

Metadata

Related indicators

Indicator name:

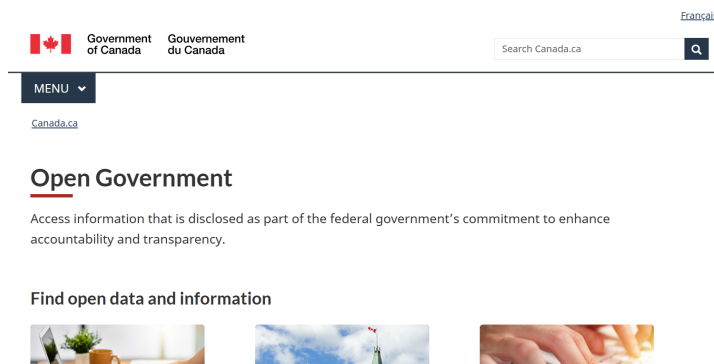
Hepatitis - (chronic HBV and HCV) incidence

Short name:

Chronic hepatitis incidence number

# Open Portal

Open Government Portal Canada is a platform that provides access to searching for a wide range of public data. Users can explore and download data in different formats under the [Open Government Licence - Canada](#).

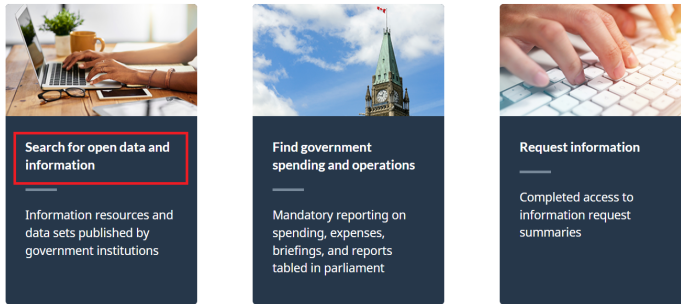


[Open Government Portal Canada](#)

## Search for Open Data

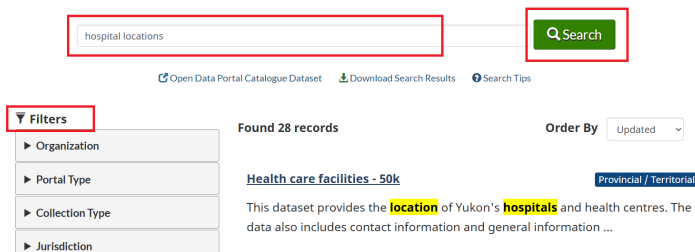
To search for open data, you can use the search bar on the Open Government Portal Canada homepage. For example, if you are interested in finding data on “hospital locations”, you can enter that term in the search bar and click the search icon. There are many filter options to narrow your search results, such as format and organization.

## Find open data and information



[Canada.ca](#) > [Open Government](#)

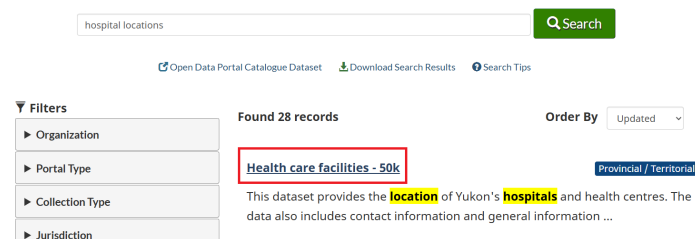
## Open Government Portal



To explore a specific dataset, click its title. This will take you to a page with more information about the dataset, including its description, format options, and download links. For example, if you click on the “Health care facilities - 50k” dataset, you will find detailed information about the location of Yukon’s hospitals and health centres.

[Canada.ca](#) > [Open Government](#)

## Open Government Portal



## Health care facilities - 50k

### Have your say

★★★★★

[Rate this dataset](#)  
[Provide feedback](#)

This dataset provides the location of Yukon's hospitals and health centres. The data also includes contact information and general information about each facility.

Distributed from GeoYukon by the Government of Yukon. Discover more digital map data and interactive maps from Yukon's digital map data collection. For more information: [geomatics.help@yukon.ca](mailto:geomatics.help@yukon.ca)

Under the **Data** and **Resources** tab, you can find the available formats for downloading the location of Yukon's hospitals and health centres.

## Data and Resources

View on Map

Search resources...

**ArcGIS Online layers**

English web service ESRI REST

Explore

**ArcGIS Online layers**

English web service ESRI REST

Explore

**Shape files**

English dataset HTML

Explore

**XML metadata**

English dataset HTML

Explore

**Original metadata (<https://open.yukon.ca>)**

English dataset HTML

Explore

Similarly, say we are interested in “opioid use” and want to find open data related to that topic. We can enter “opioid use” into the search bar and filter by **Open data**:

opioid use

Search

[Open Data Portal Catalogue Dataset](#)
[Download Search Results](#)
[Search Tips](#)

Filters

Portal Type - Open Data

Organization

Portal Type

☒ Open Data 10
 ☐ Open Information 34

Collection Type

Found 10 records

Order By Updated

Opioid agonist treatment archives

Federal

Correctional Service Canada provides Opiate Agonist Treatment to patients with an **opioid use** disorder. Archives of these quarterly web pages ...

Record Modified: Apr 9, 2026 Record Released: Nov 28, 2025

Publisher: Correctional Service of Canada

Formats: [CSV](#) [PDF](#)

Keywords: CSC OAT OUD opiate treatment Suboxone Methadone Sublocade health drugs institutions

To explore the data on opioid agonist treatment, we can click the dataset title. This will take us to a page with more information about the dataset, including its description, format options, and download links.

opiod use Search

[Open Data Portal Catalogue Dataset](#) [Download Search Results](#) [Search Tips](#)

**Filters**

Portal Type: Open Data

Organization

Portal Type

☒ Open Data 10

☐ Open Information 34

Collection Type

**Found 10 records**

Order By Updated

**Opioid agonist treatment archives** Federal

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Formats: [CSV](#) [PDF](#)

Keywords: CSC OAT OUD opiate treatment Suboxone Methadone Sublocade health drugs institutions

## Data and Resources

Search resources... Search

**Opioid Agonist Treatment (OAT) in CSC: July 2019** Explore

English dataset CSV

**Opioid Agonist Treatment (OAT) in CSC: July 2019** Explore

French dataset CSV

**Opioid Agonist Treatment: December 2019** Explore

English dataset CSV

**Opioid Agonist Treatment: December 2019** Explore

French dataset CSV

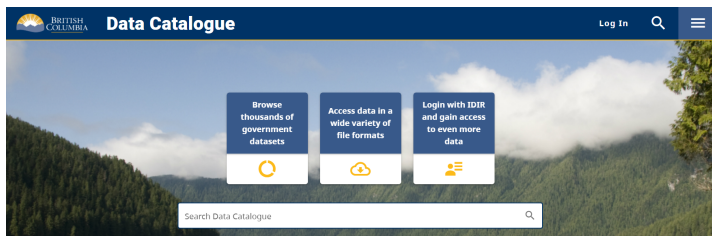
**Opioid Agonist Treatment: March 2020** Explore

English dataset CSV



# BC Data Catalogue

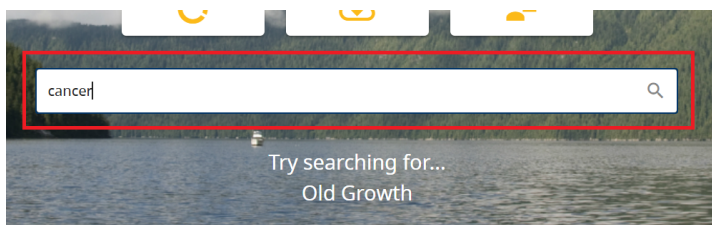
The British Columbia Data Catalogue is a central hub provided by the Government of British Columbia (BC). Its primary purpose is to provide public access to thousands of datasets managed by the provincial government and its partners. Essentially, it is a digital library designed to make government information transparent and usable for citizens, researchers, and developers.



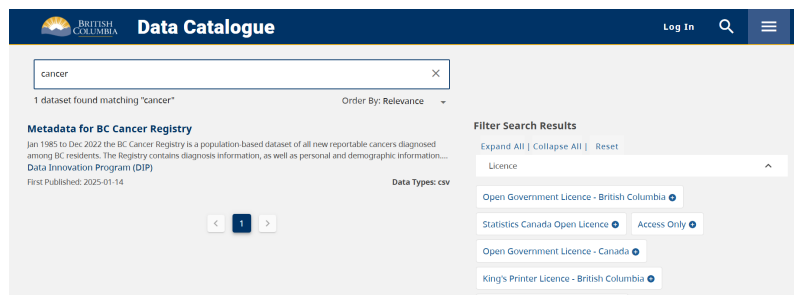
[British Columbia Data Catalogue](#)

## Search for Data

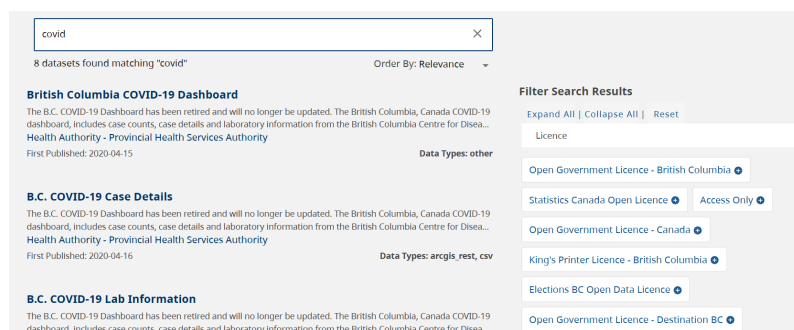
To search for data, you can use the search bar on the BC Data Catalogue homepage. For example, if you are interested in finding data on “cancer”, you can enter that term in the search bar and click the search icon.



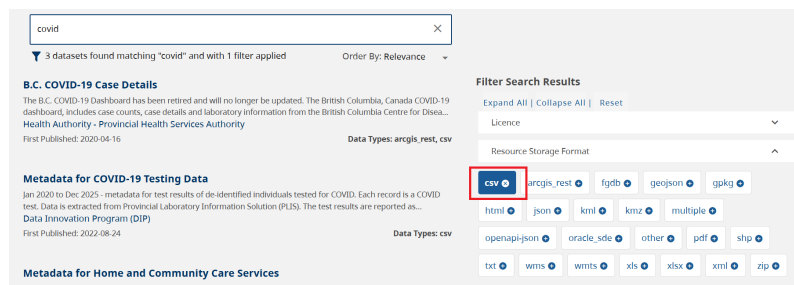
[Search Data Catalogue](#)



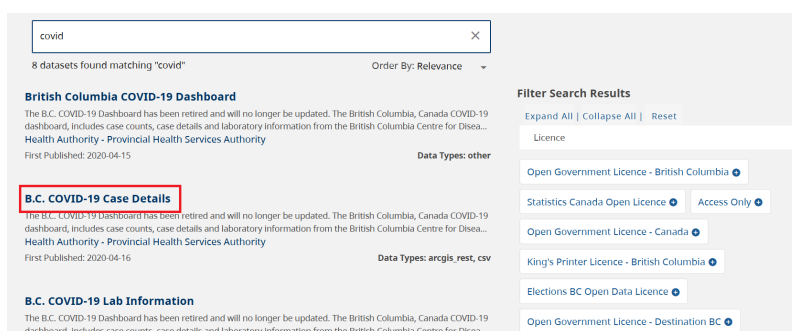
Similarly, say we are interested in “COVID-19” and want to find open data related to that topic. We can enter “covid” into the search bar:



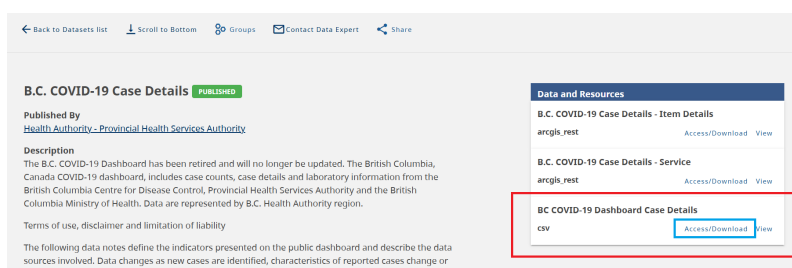
You can filter your search by license, format, group, and so on:



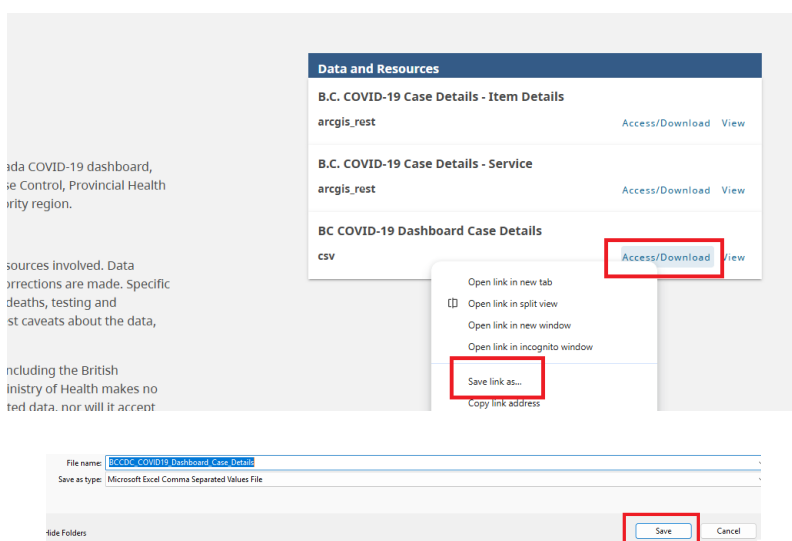
Let’s explore the “COVID-19 Case Details” dataset. To do that, we can click the dataset title. This will take us to a page with more information about the dataset, including its description and related links.



You will find information on COVID-19 case details as well as a link to download the data:



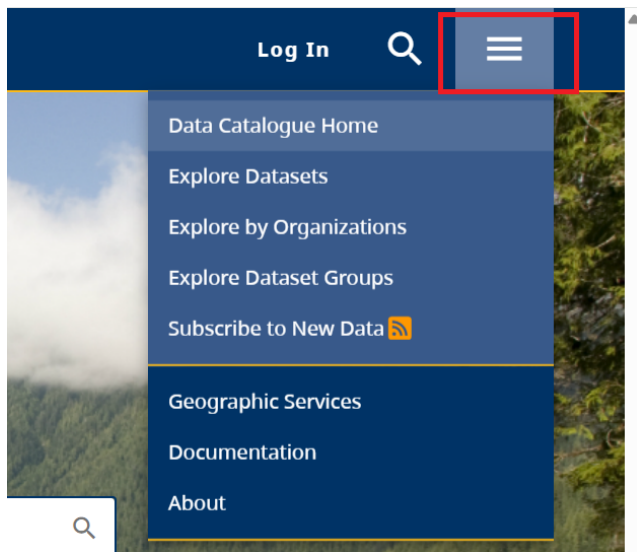
To download the dataset, right-click on the **Access/Download** link and select **Save link as...** This will allow you to save the dataset to your computer.



After downloading, you can open the data in Excel, Google Sheets, or any statistical software. The data is structured in a tabular format, with each row representing an individual case and each column representing different variables.

| 1  | Reported_Date | HA                | Sex | Age_Group | Classification_Reported |
|----|---------------|-------------------|-----|-----------|-------------------------|
| 2  | 29-01-20      | Out of Canada     | M   | 40-49     | Lab-diagnosed           |
| 3  | 06-02-20      | Vancouver Coastal | F   | 50-59     | Lab-diagnosed           |
| 4  | 10-02-20      | Out of Canada     | M   | 30-39     | Lab-diagnosed           |
| 5  | 10-02-20      | Out of Canada     | F   | 20-29     | Lab-diagnosed           |
| 6  | 18-02-20      | Interior          | F   | 30-39     | Lab-diagnosed           |
| 7  | 24-02-20      | Fraser            | F   | 30-39     | Lab-diagnosed           |
| 8  | 24-02-20      | Fraser            | M   | 40-49     | Lab-diagnosed           |
| 9  | 03-03-20      | Fraser            | M   | 50-59     | Lab-diagnosed           |
| 10 | 03-03-20      | Vancouver Coastal | F   | 60-69     | Lab-diagnosed           |
| 11 | 03-03-20      | Vancouver Coastal | F   | 30-39     | Lab-diagnosed           |
| 12 | 03-03-20      | Vancouver Coastal | F   | 80-89     | Lab-diagnosed           |
| 13 | 03-03-20      | Vancouver Coastal | M   | 60-69     | Lab-diagnosed           |
| 14 | 05-03-20      | Out of Canada     | F   | 50-59     | Lab-diagnosed           |
| 15 | 05-03-20      | Vancouver Coastal | M   | 60-69     | Lab-diagnosed           |
| 16 | 05-03-20      | Fraser            | F   | 50-59     | Lab-diagnosed           |
| 17 | 05-03-20      | Vancouver Coastal | M   | 30-39     | Lab-diagnosed           |

To learn more about BC Data Catalogue, click on the three bars on the right side of the page:



Centers for Disease Control and Prevention, National Center for Health Statistics. 2026. “National Health and Nutrition Examination Survey.” <https://www.cdc.gov/nchs/nhanes/>.